

SECTION **10.0**

ORTHOPEDIC SURGERY

CHAPTER 10.1

Hand Surgery

SURGEONS

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DARRACH PROCEDURE

SURGICAL CONSIDERATIONS

Description

The **Darrach procedure** ([Fig. 10.1-1](#)) is a resection of the distal ulna. The distal 2 cm of the ulna is resected subperiosteally, and local soft tissues are used to stabilize and cover the remaining ulna. It is commonly performed in patients who have had a disruption of the distal radioulnar joint with subluxation of the ulna. It also is indicated for patients who have had a malunion of a distal radius fracture such that the radius has shortened relative to the ulna or is abnormally angulated, resulting in dorsal subluxation of the ulna and impingement of the ulnar head upon the carpus. This malposition causes painful motion of the wrist and forearm and posttraumatic degenerative arthritis of the ulnar head, carpus, and sigmoid notch of the distal radius. Disorders of the distal radioulnar joint and degeneration of the ulnar head, which may lead to attrition rupture of the overlying extensor tendons, are common in rheumatoid arthritis. This dorsal prominence of the ulnar head is treated by **Darrach resection**, combined with a soft tissue procedure to stabilize the remaining ulna. Osteoarthritic degeneration of the distal radioulnar joint, either 2° trauma (see above) or due to idiopathic osteoarthritis, responds well to this procedure.

Variant procedure or approaches

The **Sauvé-Kapandji procedure** retains the articular portion of the distal ulna and fuses it to the sigmoid notch. The area of distal ulna that is resected is proximal to the joint; instability of the proximal stump may become problematic. Hemiresection with tendon interposition and distal ulnar arthroplasty are less common techniques for managing distal radioulnar arthritis.

Usual preop diagnosis

Arthritis or derangement of the distal radioulnar joint; rheumatoid arthritis; ulnar impingement syndrome; malunion of Colles fracture or other fracture of the distal radius

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

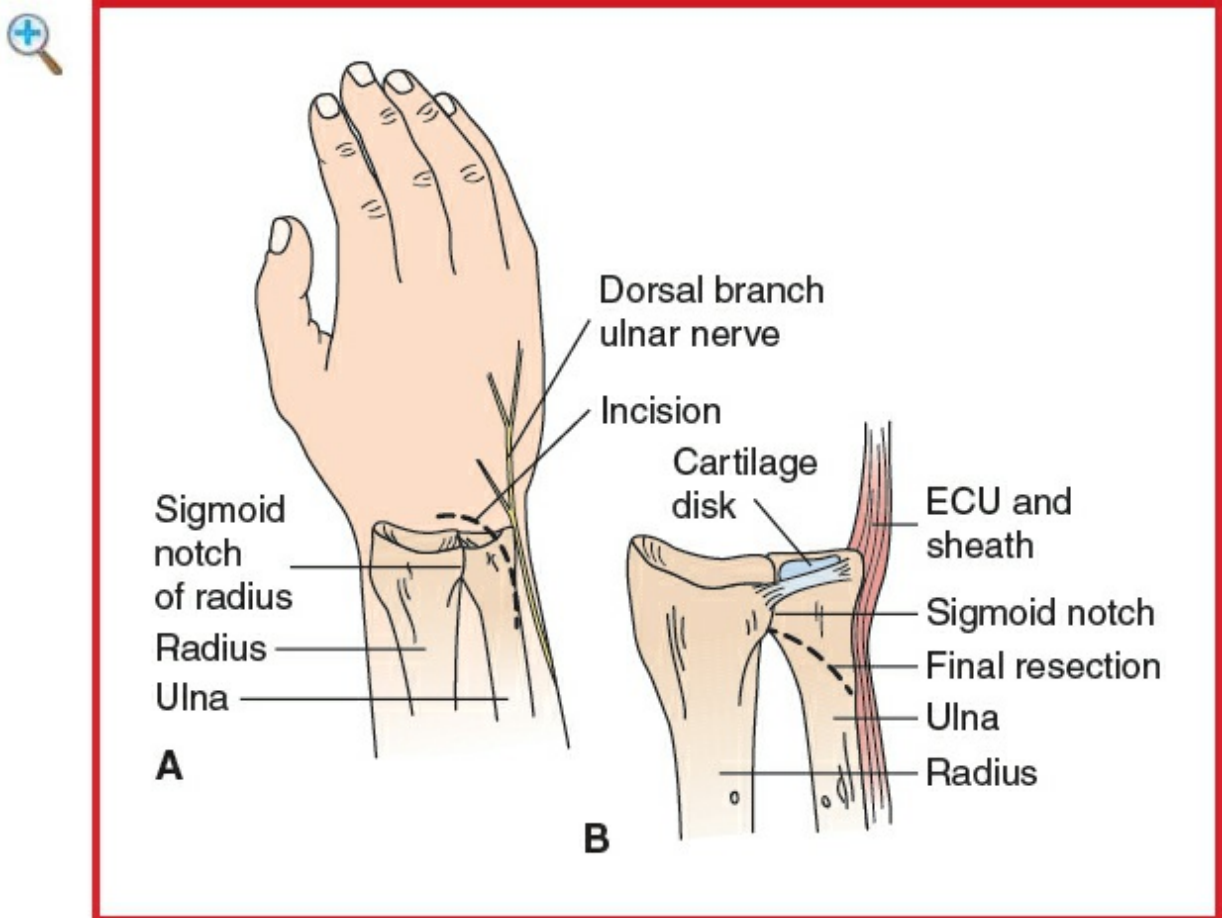


Figure 10.1-1 The Darrach procedure. **A:** Skin incision. Avoid dorsal cutaneous branch of the ulnar nerve. **B:** The distal ulna is resected at the radioulnar articulation just proximal to the sigmoid notch. ECU, extensor carpi ulnaris. (Reproduced with permission from Chapman MW. Chapman's Orthopaedic

■ SUMMARY OF PROCEDURES

Position	Supine, with arm extended on hand surgery table
Incision	Dorsal-ulnar, over distal ulna
Special instrumentation	Pneumatic tourniquet
Antibiotics	Cefazolin 2 g iv
Surgical time	1–2.5 h, depending on associated procedures
Tourniquet	150 mm Hg above SBP; max time = 120 min
Closing considerations	Routine skin closure; postop splint placed at conclusion of procedure
EBL	Minimal; performed under tourniquet control
Postop care	PACU → home
Mortality	None associated with procedure
Morbidity	Ulnar nerve injury: rare Postop swelling (rarely requires specific treatment)
Pain score	5–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adult population
Male:Female	Slight predominance of females, due to incidence of malunion of Colles fractures in women with senile osteoporosis as well as incidence of inflammatory arthropathies in women
Incidence	Not uncommon
Etiology	See Usual preop diagnosis above.
Associated conditions	Rheumatoid arthritis

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Wrist Procedures, [p. 1037](#).

Suggested Reading

1. Grawe B, Heincelman C, Stern P: Functional results of the Darrach procedure: long-term outcome study. J Hand Surg Am 2012; 37(12):2475–80.

DORSAL STABILIZATION AND EXTENSOR SYNOVECTOMY OF THE RHEUMATOID WRIST

SURGICAL CONSIDERATIONS

Description

This procedure is indicated for patients with rheumatoid arthritis and extensor tenosynovitis refractory to medical treatment, as well as extensor tendon ruptures and/or intercarpal synovitis. The procedure is performed under tourniquet control through a straight dorsal incision over the wrist. A **radical tenosynovectomy** of the extensor tendons in all six extensor compartments is carried out. Tendon ruptures or impending ruptures are repaired with tendon grafts or side-to-side anastomoses. Bone spurs are removed, and a synovectomy of the distal radioulnar joint is carried out. A **modified Darrach procedure**, with resection or osteoplasty of the distal ulna, is usually performed. If there is evidence of synovitis within the wrist joint, a synovectomy is performed through a dorsal arthrotomy. A flap of the extensor retinaculum is transposed beneath the extensor tendons to reinforce the dorsal wrist ligaments and, thus, stabilize the wrist to prevent volar subluxation of the carpus. A posterior interosseous neurectomy is carried out at the same time. The remaining extensor retinaculum is divided into two transverse strips, and one is used to stabilize the distal ulna. The second strip is placed dorsal to the extensor tendons, so they will not bowstring during wrist extension.

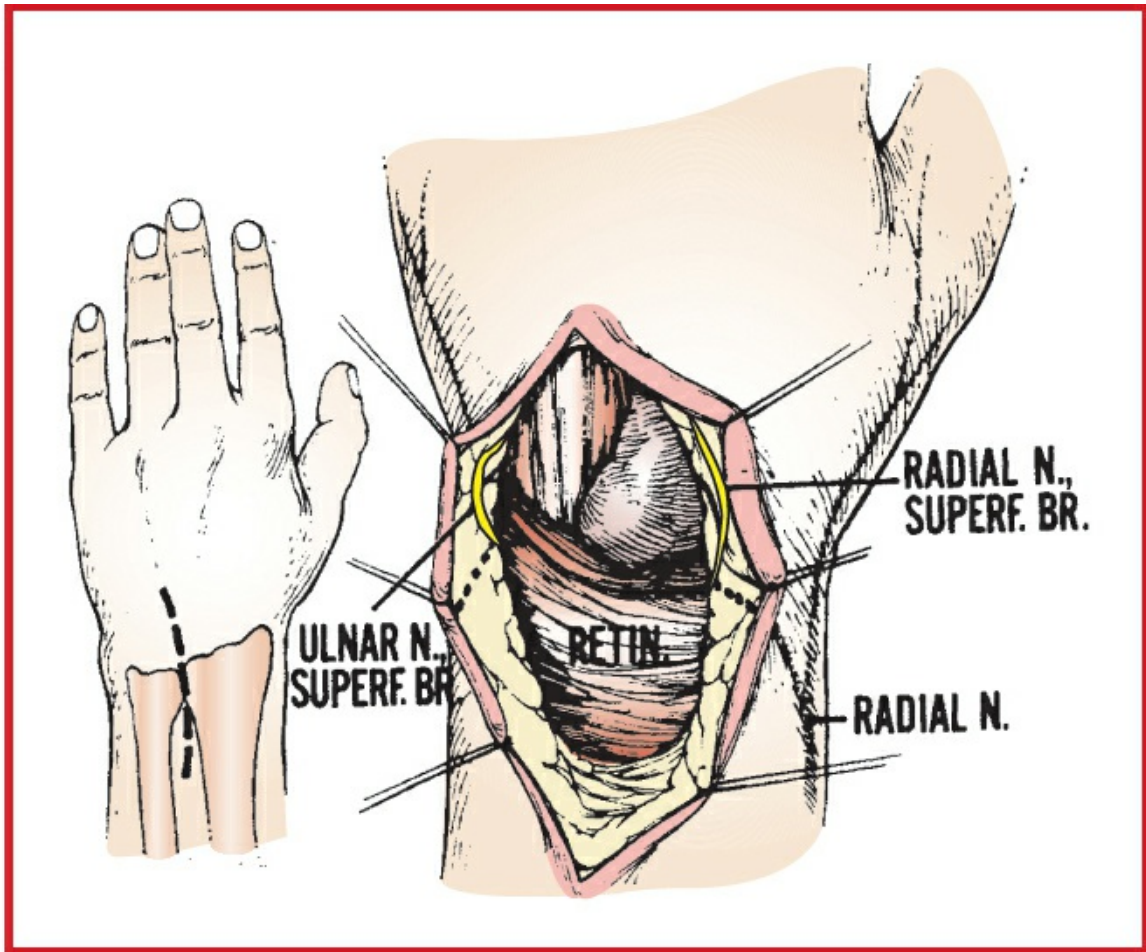


Figure 10.1-2 Incision and exposure for dorsal tenosynovectomy. Note superficial branches of radial and ulnar nerves protected in skin flaps. (Illustration by Elizabeth Roselius, 1988. Reprinted from Green DP, Hotchkiss RN, Pederson WC, eds *Operative Hand Surgery*, 2nd edition. Churchill Livingstone, London: 1988. Copyright © 1988 Elsevier. With permission.)

Usual preop diagnosis

Rheumatoid arthritis with extensor tendon tenosynovitis; extensor tendon rupture; distal radioulnar joint synovitis and/or subluxation

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Supine, with arm extended on hand surgery table
Incision	Dorsal wrist (Fig. 10.1-2)

Special instrumentation	Pneumatic tourniquet
Antibiotics	Cefazolin 2 g iv
Surgical time	2 h
Tourniquet	150 mm Hg above SBP; max time = 120 min
Closing considerations	Postop splint
EBL	Minimal; tourniquet used until dressing in place
Postop care	PACU → home
Mortality	None associated with procedure
Morbidity	Extremity swelling (typically does not require treatment) Delayed healing 2° immunosuppression and steroid use Wound infection: rare (unless patient is immunosuppressed)
Pain score	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Procedure uncommon before 4th decade
Male:Female	As in all patients with rheumatoid arthritis, females more common
Incidence	Not uncommon
Etiology	Connective tissue disorder; rheumatoid arthritis or variant
Associated conditions	All conditions associated with connective tissue disorders, including active rheumatoid arthritis, steroid dependency, immunosuppressive therapy, and/or skin fragility

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Wrist Procedures [p. 1037](#).

Suggested Reading

- l. Chim HW, Reese SK, Toomey SN, et al: Update on the surgical treatment for rheumatoid arthritis of the wrist and hand. J Hand Ther 2014; 27:134–41.

METACARPOPHALANGEAL AND INTERPHALANGEAL JOINT ARTHROPLASTY AND ARTHRODESIS

SURGICAL CONSIDERATIONS

Description

Joint replacement (arthroplasty) in the hand is most commonly indicated in patients with rheumatoid arthritis with severe joint destruction → pain and dysfunction. Another option in these patients is arthrodesis or joint fusion, which involves removing the degenerative cartilaginous surfaces and using implanted metal to secure two opposing bone surfaces. Joint arthroplasty is rarely indicated in patients with osteoarthritis. The most common prostheses, made of silicone rubber and popularized by Swanson, differ from total joint replacement in the hip or knee in that they do not function as true joints but rather as spacers in a **resection arthroplasty**. Most of the stability and motion of these joints depend on meticulous soft tissue reconstructions involving tendon and ligament transfers, as well as intensive postop physical therapy. To obtain good results, patients must be well motivated and understand their disease process and what will be asked of them during the recovery period. The results for **metacarpophalangeal (MP) arthroplasty** are far better than those obtained in the proximal interphalangeal joints. **Proximal interphalangeal joint arthroplasty** is sometimes complicated by soft tissue deformities in rheumatoid arthritis such as swan-neck and boutonniere deformities. A swan-neck deformity consists of hyperextension of the PIP joint and flexion of the DIP joint. Conversely, a boutonniere deformity involves flexion at the PIP joint and hyperextension of the DIP joint. Arthroplasty is generally not indicated for the index finger due to the high stresses associated with pinch that may compromise the longevity or stability of an arthroplasty. **Distal interphalangeal (DIP) arthroplasty** is rarely performed because these patients do well with fusions. The procedure for the MP joints is performed through a dorsal transverse incision under tourniquet control. The metacarpal heads are removed with an oscillating saw and the intramedullary canals reamed to accept the stems of the prostheses. After they have been placed with a no-touch technique, the capsule is closed, and the supporting ligaments are reconstructed with centralization of extensor tendons. Technique for PIP joints is similar, although

some surgeons will use a longitudinal or volar approach. A splint with support for each finger is placed at the conclusion of surgery. Reconstructive procedures of the wrist and fingers can be combined with arthroplasty.

Variant procedure or approaches

Some surgeons favor longitudinal incisions rather than a transverse incision for the approach to the MP joints.

Usual preop diagnosis

Osteoarthritis or rheumatoid arthritis or other connective tissue disorder

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES	
Position	Supine, with arm extended on hand surgery table
Incision	Dorsal hand, transverse, or longitudinal
Special instrumentation	MP or PIP joint prostheses and associated instruments for preparing the medullary canal; power for bone cuts; K-wires, plates, or wire loop for fusion; pneumatic tourniquet
Antibiotics	Cefazolin 2 g iv
Surgical time	2.5 h
Tourniquet	150 mm Hg above SBP; max time = 120 min
Closing considerations	Critical postop splinting
EBL	Minimal; tourniquet used throughout procedure
Postop care	PACU → home
Mortality	None associated with procedure
Morbidity	Swelling (may require early splitting of dressing) Wound infection: rare (if it occurs, requires removal of prosthesis)

Prosthesis infection: rare

Pain score 3–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	>50 yr
Male:Female	As in rheumatoid arthritis, females predominate.
Incidence	Uncommon
Etiology	Rheumatoid arthritis or other connective tissue disorder
Associated conditions	As in rheumatoid arthritis (e.g., skin fragility, steroid dependency, immunosuppression)

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Wrist Procedures, [p. 1037](#).

ARTHRODESIS OF THE WRIST

■ SURGICAL CONSIDERATIONS

Description

A variety of arthrodeses can be performed about the wrist. These include **radiopancarpal arthrodesis (total wrist fusion)** and radiolunate, radioscapolunate, and intercarpal arthrodeses (partial wrist fusions). Radiopancarpal arthrodesis is generally performed as a salvage procedure for wrist pathology that cannot be treated with a procedure that preserves wrist motion. These indications include posttraumatic degenerative arthritis following fractures, dislocations and ligamentous injuries, idiopathic osteoarthritis, and/or rheumatoid arthritis. Predictable patterns of arthritis occur as a result of both scapholunate interosseous ligament injury (scapholunate advanced collapse) and scaphoid nonunion (scaphoid nonunion advanced collapse). As arthritic changes progress, patients are no longer candidates for either reconstruction of the ligaments or fixation of the scaphoid. Another alternative to fusion is a proximal row carpectomy, which is a motion-preserving procedure that may be used in patients without advanced capitolunate arthritis. In the nonrheumatoid patient, an effective procedure is an arthrodesis using an iliac crest bone graft fixed with plate and screws. More recent techniques with improved plate designs may not require iliac crest bone

graft. As another option, local bone from the distal radius may be utilized. It has been shown that a position of fusion in 10–15° of dorsiflexion provides the greatest grip strength. In the rheumatoid patient, a technique using intramedullary fixation with a large-diameter Steinmann pin is preferred. Rheumatoid bone is osteoporotic 2° disuse and chronic steroid administration. Screw fixation is not ideal in this soft bone. Bone graft is obtained locally in these patients, usually from the resected ulnar head. **Radiolunate fusion** also is indicated in rheumatoid patients who have progressive ulnar translation of the carpus. The lunate acts to block this translation of the carpus. **Radioscapholunate fusion** is indicated in patients with radiocarpal arthritis. This procedure preserves about 50% of wrist motion that occurs at the midcarpal joint. Bone graft is necessary and is easily obtained from the distal radius through the same incision. There are a variety of **intercarpal arthrodeses**, including **triscaphe** (scaphotrapezial trapezoid), **scaphocapitate**, **lunotriquetral**, and **four corner** (capitate-hamate-triquetral-lunate). These procedures are indicated for the treatment of intercarpal arthritis, carpal instabilities due to intercarpal ligament tears, and Kienböck disease (aseptic necrosis of the lunate). Special instrumentation (e.g., **Cloward cervical spine instrumentation**) is useful for obtaining a bicortical plug of bone from the iliac crest with minimal dissection. Local bone graft from the distal radius may be obtained using curettes or larger core needles.

Usual preop diagnosis

Posttraumatic arthritis; osteoarthritis or rheumatoid arthritis; Kienböck disease; carpal instability

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Supine, with arm extended on hand surgery table. The iliac crest may be prepped and elevated with a sandbag beneath the ipsilateral buttock.
Incision	Dorsal wrist, transverse or longitudinal
Special instrumentation	Pneumatic tourniquet; possibly power for preparation of opposing bone surfaces; K-wires, screws, or special instrumentation for fusion

Unique considerations	Bone-graft donor site
Antibiotics	Cefazolin 2 g iv
Surgical time	2 h
Tourniquet	150 mm Hg above SBP; max time = 120 min
Closing considerations	Immobilization with splints
EBL	Minimal; procedure is performed under tourniquet control. If iliac crest bone graft is used, there may be increased blood loss.
Postop care	PACU → home, sometimes admission for pain control and elevation
Mortality	None associated with procedure
Morbidity	Nonunion of fusion: ≤20%
Pain score	5–7, if no iliac graft; 6–9, if iliac graft used

■ PATIENT POPULATION CHARACTERISTICS

Age range	>40 yr
Male:Female	Females predominate in rheumatoid arthritis and males in posttraumatic arthritis.
Incidence	Common
Etiology	Trauma (common); rheumatoid arthritis (common)
Associated conditions	Typical for rheumatoid arthritis (e.g., skin fragility, steroid dependency, immunosuppression)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Wrist Procedures, [p. 1037](#).

Suggested Readings

1. Wei DH, Feldon P: Total wrist arthrodesis: indications and clinical outcomes. *J Am Acad Orthop Surg* 2017; 25(1):3–11.

2. Wysocki RW, Cohen MS: Complications of limited and total wrist arthrodesis. *Hand Clin* 2010; 26(2):221–8.

TOTAL WRIST REPLACEMENT

SURGICAL CONSIDERATIONS

Description

The major indication for this procedure is rheumatoid arthritis of the wrist. **Total wrist replacement (TWR)** is often recommended in patients with bilateral wrist disease. An arthrodesis will be carried out on the nondominant helping hand and a TWR on the dominant hand to preserve dexterity. Many surgeons prefer to avoid **bilateral wrist arthrodesis**, although some patients with bilateral fusions have been able to function relatively well. Currently available prostheses are suitable only for low-demand patients and are not indicated for high-demand patients with posttraumatic arthritis, especially younger patients. High-demand patients will do better with wrist arthrodesis. Silastic wrist prostheses are associated with a high failure rate and silicone synovitis, and their use has been abandoned by many surgeons. The most commonly used prostheses today are metal on ultrahigh-molecular-weight polyethylene articulations that are fixed with methyl methacrylate cement or bone ingrowth into porous stems. The distal radius articular surface is resected to accept the implant, the proximal carpal row is resected, and the distal carpus is prepared to accept the distal implant. All of these prostheses depend on intact, normally functioning wrist extensor tendons, especially the extensor carpi radialis brevis, for balance and function. Absence of this tendon is felt by many to be an absolute contraindication to this procedure. Because these tendons are so commonly affected by rheumatoid arthritis, the patient population for this procedure is limited. In addition to functioning tendons, meticulously accurate placement of the components in relation to the centers of rotation of the wrist is critical for success. If the centers of rotation of the prosthesis do not duplicate those of the normal wrist, early component loosening and failure are likely. Intraop radiographs are useful in verifying component position. These patients frequently have other upper extremity deformities that will require reconstruction. Because of the complexity of TWR, other reconstructive procedures are not carried out at the same time.

Usual preop diagnosis

Rheumatoid arthritis

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist

should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Supine, with arm extended on hand surgery table
Incision	Dorsal wrist
Special instrumentation	TWR instrumentation; power for bone cuts; pneumatic tourniquet
Antibiotics	Cefazolin 2 g iv
Surgical time	2 h
Tourniquet	150 mm Hg above SBP; max time = 120 min
Closing considerations	Postop splint
EBL	Minimal; procedure performed under tourniquet control
Postop care	PACU → home vs admission for pain control and elevation
Mortality	None associated with procedure
Morbidity	Infection: rare (but requires removal of prosthesis and methyl methacrylate cement, if used) Poor wound healing
Pain score	4–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Rare before 4th decade; most in 6th and 7th decades
Male:Female	Females outnumber males, as in rheumatoid arthritis.
Incidence	Rare
Etiology	Rheumatoid arthritis
Associated conditions	Rheumatoid arthritis

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Wrist Procedures, [p. 1037](#).

Suggested Reading

1. Halim A, Weiss AC: Total wrist arthroplasty. J Hand Surg Am 2017; 42(3):198–209.

THUMB CARPOMETACARPAL JOINT FUSION/ARTHROPLASTY/STABILIZATION

SURGICAL CONSIDERATIONS

Description

Patients with degenerative arthritis of the carpometacarpal (CMC) joint of the thumb present with subluxation, pain, and synovitis of the joint. A **synovectomy and ligament reconstruction** to restore stability will treat pain and may delay further degeneration. This procedure is performed through a curvilinear incision over the joint. A distally attached graft of the radial 1/2 of the flexor carpi radialis tendon is passed through a drill hole in the base of the metacarpal and woven into the joint capsule. In the later stages of degeneration, patients must be treated with either an arthroplasty or an arthrodesis. For more progressive arthritis, a variety of **arthroplasty techniques** are available to the surgeon; most involve removal of degenerated articular surfaces, soft tissue interposition, and often K-wire fixation to suspend the metacarpal which is used. Cemented arthroplasty techniques were initially associated with a high loosening rate and fell out of favor. Newer designs and techniques do not have long-term follow-up yet. The status of the MCP joint of the thumb must be considered prior to CMC intervention. Hyperextension at the MCP joint will increase forces across any CMC procedure.

Variant procedure or approaches

Some surgeons choose to address CMC arthritis arthroscopically with joint debridement and sometimes material interposition.

Usual preop diagnosis

Osteoarthritis of CMC joint; basal joint arthritis; synovitis of CMC joint; CMC joint dislocation; trauma

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Supine, with arm extended on hand surgery table
Incision	Curvilinear over joint at base of thumb. If soft tissue interposition is used, additional incisions may be required.
Special instrumentation	Pneumatic tourniquet; power if K-wire suspension is to be used
Antibiotics	Cefazolin 2 g iv
Surgical time	1.5–2 h
Tourniquet	150 mm Hg above SBP; max time = 120 min
EBL	Minimal; procedures performed under tourniquet control
Postop care	PACU → home
Mortality	None associated with procedure
Morbidity	Nonunion of arthrodesis: ≤20% Particulate synovitis (silicone rubber prostheses) Dislocation (newer prostheses)
Pain score	8–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	Joint stabilization in 3rd–5th decades; arthroplasty and arthrodesis in 5th–8th decades
Male:Female	Basal joint instability and arthritis much more prevalent in women
Incidence	Common
Etiology	Trauma may play a role in producing instability. Intra-articular fracture of the base of the first metacarpal with a nonanatomic reduction → incongruence of the joint → posttraumatic arthritis.

	Rheumatoid arthritis → instability and degeneration of the joint. Congenital ligamentous laxity → unstable basal joints and arthritis in many patients
Associated conditions	Rheumatoid arthritis; osteoarthritis; carpal tunnel syndrome (CTS)

ANESTHETIC CONSIDERATIONS FOR WRIST PROCEDURES

(Procedures covered: Darrach procedure; dorsal stabilization and extensor synovectomy of the rheumatoid wrist; metacarpophalangeal and interphalangeal joint arthroplasty; arthrodesis of the wrist; total wrist replacement; thumb carpometacarpal joint fusion/arthroscopy/stabilization)

PREOPERATIVE

Airway	Rheumatoid involvement of the C-spine, TMJ, and cricoarytenoid joint (CAJ) are common in this patient population. Erosion of cervical vertebrae → unstable C-spine (e.g., atlantoaxial subluxation) necessitates extreme care in head and neck manipulation. C-spine fusion (↓ neck ROM), TMJ arthritis (↓ mouth opening), and CAJ arthritis (laryngeal narrowing, hoarseness, DOE, stridor) portend difficult intubation and may necessitate awake fiberoptic intubation. In the case of CAJ arthritis, the use of a smaller ETT may be required.
Respiratory	Rheumatoid patients may exhibit Sx of pleural effusion (✓ CXR) or pulmonary fibrosis (dyspnea, diffuse rales, ↓ diffusing capacity, honeycomb appearance in CXR). Tests: Consider CXR, PFTs, and ABGs in affected patients.
Cardiovascular	Rheumatoid patients may suffer from pericarditis, myocarditis, valvular disease, and cardiac conduction defects. Because of the physical limitations imposed by the disease process, it may prove difficult to evaluate these patients' cardiovascular status;

	hence, cardiology consultation, ECG, and ECHO may be useful in preparing for surgery. Tests: Consider ECG and ECHO, especially in patients with severe rheumatoid arthritis.
Neurologic	Rheumatoid patients may have cervical or lumbar radiculopathies that should be documented carefully preop. In addition, peripheral neuropathy with consequent sensory/motor defects may be present. Tests: Consider C-spine radiographs to r/o occult subluxations in rheumatoid patients with neck pain or upper extremity radiculopathy.
Musculoskeletal	Bony deformities or muscle contractures may necessitate special attention to positioning.
Hematologic	Anemia, eosinophilia, and thrombocytosis may be present. Venous access may be difficult 2° vasculitis and ↑ skin fragility (steroid-induced). Virtually all of these patients will be on some type of anti-inflammatory medication that may result in anemia or Plt inhibition. Ideally, patients should discontinue NSAIDs at least 5 d preop and aspirin, 7 d preop.
Endocrine	Rheumatoid patients may be on oral corticosteroids and require supplemental periop steroids (e.g., 100 mg hydrocortisone q8h iv) to treat adrenal suppression, although the routine use of “stress-dose steroids” has been questioned.
Laboratory	Hb/Hct serves as a minimum in otherwise healthy rheumatoid patients. Severe rheumatoid patients may require more extensive testing to screen for drug effects, etc. (e.g., serum electrolytes; glucose; kidney function tests; LFTs).
Premedication	Mild-to-moderate premedication (e.g., in adults, midazolam 1–2 mg iv, fentanyl 50–100 mcg iv, titrated to effect) is often desirable before placement of a regional block.
ERAS	Check compliance with presurgical ERAS

recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, $\pm$ celecoxib 200 mg po, with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

Regional anesthesia, GA, or a combination of the two can be used. A brachial plexus block via the supraclavicular, axillary, or infraclavicular approach is excellent for this procedure; it is a means of avoiding tracheal intubation for GA if airway difficulty is anticipated. Furthermore, it decreases admission rate, speeds discharge from PACU in day surgery setting, and increases patient satisfaction. Intravenous regional anesthesia (Bier block) is most useful for short procedures (<1 h). If regional anesthesia is contraindicated, rheumatoid patients may require awake fiberoptic intubation.

Regional anesthesia

1.5% mepivacaine, 30 mL for routine, superficial cases; 0.5% bupivacaine, or 0.5% ropivacaine, 30 mL for procedures >2.5 h or if extended analgesia is desired.

Ultrasound-guided brachial plexus block

Ultrasound-guided brachial plexus using supraclavicular, infraclavicular, or axillary approaches are all suitable for surgery below the shoulder. The infraclavicular approach is the most effective for placement of peripheral nerve catheters. The supraclavicular approach is associated with phrenic nerve blockade in 50% of patients and should be avoided in patients with respiratory compromise.

Bier block

The Bier block (intravenous regional anesthesia) is a technique for short (<60 min) wrist and hand surgeries. 40–50 mL of 0.5% lidocaine is commonly used. For short procedures (<30 min), consider reducing the dose of iv anesthetic (e.g., 25 mL of 0.5% lidocaine) and using a forearm tourniquet instead of an upper arm tourniquet. The OR staff should be alerted that iv regional anesthesia is being used, so that all are ready to proceed after the tourniquet is inflated. Tourniquet pain and postop

pain are reduced by adding ketorolac (20 mg) or clonidine (1 mcg/kg) to the lidocaine solution.

General anesthesia

Induction	Standard induction in patients with normal airways	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance	
Emergence	Skin closure frequently is followed by application of a splint, and the patient should remain anesthetized during the splinting procedure. Cases with difficult airways require awake extubation. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 18 ga × 1 BSS @ 1.5–3 mL/kg/h	IV placed in the contralateral upper extremity. See Restrictive Fluid Management, Appendix K , p. K-4.
Monitoring	Standard monitors	
Positioning	Special handling required ✓ and pad pressure points ✓ eyes ✓ C-spine instability	As with nearly all orthopedic cases, positioning is a subtle yet crucial aspect of anesthetic management. Rheumatoid patients may have contractures that require special attention. Steroid-dependent patients require special handling because of fragile skin.
Brachial plexus block complications	Local anesthetic systemic toxicity Inadequate block Pneumothorax Nerve injury	Pneumothorax is very rare. Transient nerve injury <1% Permanent nerve injury risk ~1:2500
Bier block	Local anesthetic toxicity	Proper exsanguination and

complications	Inadequate block Thrombophlebitis	tourniquet function are critical. Upper arm tourniquet should remain inflated a minimum of 25 min. Systemic toxic reaction to the local anesthetic may occur as a result of tourniquet leak or inadvertent premature (<20 min) upper arm tourniquet release. Treatment is supportive. Sz are controlled with midazolam, with appropriate airway protection. Rx severe toxicity with Intralipid® 20% 1.5 mL/kg iv. Even with a functioning tourniquet, it is possible to overcome tourniquet pressure by injecting anesthetic too vigorously. Care must be taken when switching from proximal to distal upper arm tourniquet; never deflate proximal tourniquet until verifying that distal tourniquet is working.
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▼ POSTOPERATIVE

Multimodal pain management	Multimodal therapy in combination with regional block. Continuous peripheral nerve block should be considered.	Regional or combined regional-general anesthetic techniques are excellent for wrist procedures, especially with respect to postop pain management (see p. 1037).
ERAS	PONV protocol (see Appendix B, p. B-4 and p.	Encourage early enteral nutrition and mobilization.

Suggested Readings

1. Arslanian B, Mehrzad R, Kramer T, et al: Forearm Bier block: a new regional anesthetic technique for upper extremity surgery. *Ann Plast Surg* 2014; 73(2):156–7.
2. Chandrasoma J, Harrison TK, Ching H, et al: Peripheral nerve blocks for hand procedures. *N Engl J Med* 2018; 379(10):e15.
3. Falcão LF, Perez MV, de Castro I, et al: Minimum effective volume of 0.5% bupivacaine with epinephrine in ultrasound-guided interscalene brachial plexus block. *Br J Anaesth* 2013; 110(3):450–5.
4. Mian A, Chaudhry I, Huang R, et al: Brachial plexus anesthesia: a review of the relevant anatomy, complications, and anatomical variations. *Clin Anat* 2014; 27(2):210–21.
5. Seyoum D, Khan W: Perioperative considerations in rheumatoid arthritis patients. *Curr Rheumatol Rev* 2016; 12(3):185–9.
6. Steiner MM, Calandruccio JH: Use of wide-awake local anesthesia no tourniquet in hand and wrist surgery. *Orthop Clin North Am* 2018; 49(1):63–8.
7. Ustuner E, Yilmaz A, Ozgencil E, et al: Ultrasound anatomy of the brachial plexus nerves in the neurovascular bundle at the axilla in patients undergoing upper-extremity block anesthesia. *Skeletal Radiol* 2013; 42(5):707–13.

EXCISION OF GANGLION OF THE WRIST

SURGICAL CONSIDERATIONS

Description

Ganglion cysts about the wrist most commonly occur dorsally, originating from the scapholunate joint. The second most common site is volar to the scaphotrapezial joint. To prevent recurrence, these synovial fluid-filled outpouchings of the joint capsule must be excised completely. This requires isolating the stalk of the cyst to its origin and excising a small cuff of normal joint capsule with the cyst. The joint, therefore, must be entered for a complete excision. For patients with dorsal ganglia who have considerable preoperative pain, a posterior interosseous neurectomy may be done at the same time as the excision. Older studies found that the recurrence rate was decreased by the use of GA, as opposed to local or regional anesthetics. This was due to the fact that a more complete excision was performed when the patient was under GA. Hand

specialists today feel that regional anesthetics are quite acceptable for this procedure, as long as the surgeon performs a meticulous excision. More recently, hand surgeons are using lidocaine with epinephrine which eliminates the need for a tourniquet. This field block works well for simple cases such as a dorsal ganglion excision. Volar wrist ganglions commonly are near the radial artery, which is at risk during excision. A preop Allen test should be performed to ensure that, if the radial artery is interrupted, there will not be ischemia in the hand. Primary small dorsal ganglions may be approached arthroscopically.

Usual preop diagnosis

Ganglion cyst, primary or recurrent

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES	
Position	Supine, with arm extended on hand surgery table
Incision	Longitudinal or transverse directly over cyst
Special instrumentation	Pneumatic tourniquet—unnecessary if the surgeon plans to use a field block with lidocaine and epinephrine
Antibiotics	Cefazolin 2 g iv
Surgical time	0.5–1.5 h
Tourniquet	150 mm Hg above SBP; max time = 120 min
Closing considerations	Routine skin closure. Large recurrent cysts may require a repair of the wrist capsule. Splint applied in OR
EBL	Minimal; performed under tourniquet control or with an epinephrine-containing field block
Postop care	PACU → home
Mortality	None associated with procedure
Morbidity	Injury to radial artery: rare (Because of the vascular interconnections between radial and ulnar arteries, loss of

	radial artery flow rarely → complications.)
Pain score	2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adults
Male:Female	1:1
Incidence	Very common
Etiology	Unknown. Trauma has been associated with ~50% of ganglion cysts. Underlying carpal instabilities, such as scapholunate instability, have been implicated.
Associated conditions	Carpal instability; trauma (wrist sprains and strains)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Repair of Flexor Tendon Laceration, [p. 1044](#).

PALMAR AND DIGITAL FASCIECTOMY

SURGICAL CONSIDERATIONS

Description

This procedure is indicated for the treatment of Dupuytren contractures of the digits, which produces a neoplastic thickening of the palmar and digital fascia. These pathologic cords (whose active cell is the myofibroblast) contract and, through their connections with the skin, tendon sheath, and phalangeal bone, cause flexion contractures of the MP, proximal interphalangeal, and DIP joints. The disease is progressive, and the common treatment is surgical excision of the fascia. Surgical options for treatment include **needle aponeurectomy**, which incises the diseased fascia but does not remove it, and collagenase (Xiaflex®), which dissolves away the abnormal fascia. In addition to the pathologic changes in the fascia of the hands, many patients also have thickening of the plantar fascia of the foot (Ledderhose disease) and the dorsal fascia of the penis (Peyronie disease). Patients with severe contractures that have been neglected may require amputation. Because the pathologic fascia is so intimately connected to the skin, it is sometimes necessary to excise the skin and replace it with

full-thickness skin grafts. The groin is an excellent donor site for these grafts.

Usual preop diagnosis

Dupuytren contracture

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES	
Position	Supine, with arm extended on hand surgery table
Incision	Transverse or longitudinal palmar. Groin may be used as a full-thickness skin graft donor site.
Antibiotics	Cefazolin 2 g iv
Surgical time	1–3 h
Tourniquet	150 mm Hg above SBP; max time = 120 min. Because of the need to deflate tourniquet so that hemostasis can be obtained, Bier block is not suitable.
Closing considerations	Must obtain meticulous hemostasis. Z-plasties and skin grafts used frequently
EBL	Minimal; dissection done under tourniquet control. A small amount of blood loss occurs when tourniquet is released and hemostasis is obtained.
Postop care	PACU → home
Mortality	None associated with procedure
Morbidity	Hematoma (usually requires operative intervention) Skin necrosis (may require secondary skin grafting) Digital nerve and artery injury (Vascular injuries typically recognized immediately. An operating microscope may be needed, and the procedure will be prolonged.) Reflex-sympathetic dystrophy (so-called sympathetic flare reaction; requires prompt treatment, including stellate ganglion blockade)

Recurrence rate: 40–80%

Pain score 4–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Typically, 40–60 yr
Male:Female	More common in males (1.7:1)
Incidence	Common (U.S.: 21% @ 65 yo); R ring finger most frequently involved
Etiology	Definite heritance—associated with strong family Hx. More common in Scandinavians. Almost never seen in Blacks. Experimental studies suggest that microhematomas 2° repetitive trauma may be important in the disease process.
Associated conditions	Alcoholism; epilepsy; occupational hand trauma; diabetes

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Repair of Flexor Tendon Laceration, [p. 1044](#).

REPAIR OF LACERATED TENDONS/NERVES

■ SURGICAL CONSIDERATIONS

Description

The prognosis and difficulty of a flexor tendon repair depend on the anatomic site of the laceration. There are five zones of injury in the upper extremity ([Fig. 10.1-3](#)). Zone I is distal to the flexor digitorum superficialis (FDS) tendon insertion and involves only the flexor digitorum profundus (FDP) tendon. Zone II extends from the entrance to the fibro osseous sheath at the metacarpal head to the FDS insertion. Lacerations usually involve both the FDS and FDP. These are the most difficult to repair and have the worst prognosis, as the tendons are apt to become scarred to each other and limit gliding. Zone III is the palm; zone IV is at the level of the carpal canal; and zone V is in the forearm. Lacerations in these areas are easier to repair and have good prognoses for restoration of tendon gliding and, thus, digit motion. Associated injuries to the neural structures are common. Digital nerve lacerations are seen in zone II and median or ulnar nerve injuries

in zone IV and proximal. Lacerations to the dorsal side of the hand involving the extensor tendons may be repaired in the emergency room as they often do not involve neurovascular structures, and the extensor digitorum communis tendon as well as the junctura tendinum may prevent retraction of the proximal tendon into the forearm. The exception to this is the thumb and radial dorsal hand, where the extensor and abductor tendons may retract, and the dorsal sensory branch of the radial nerve is at risk.

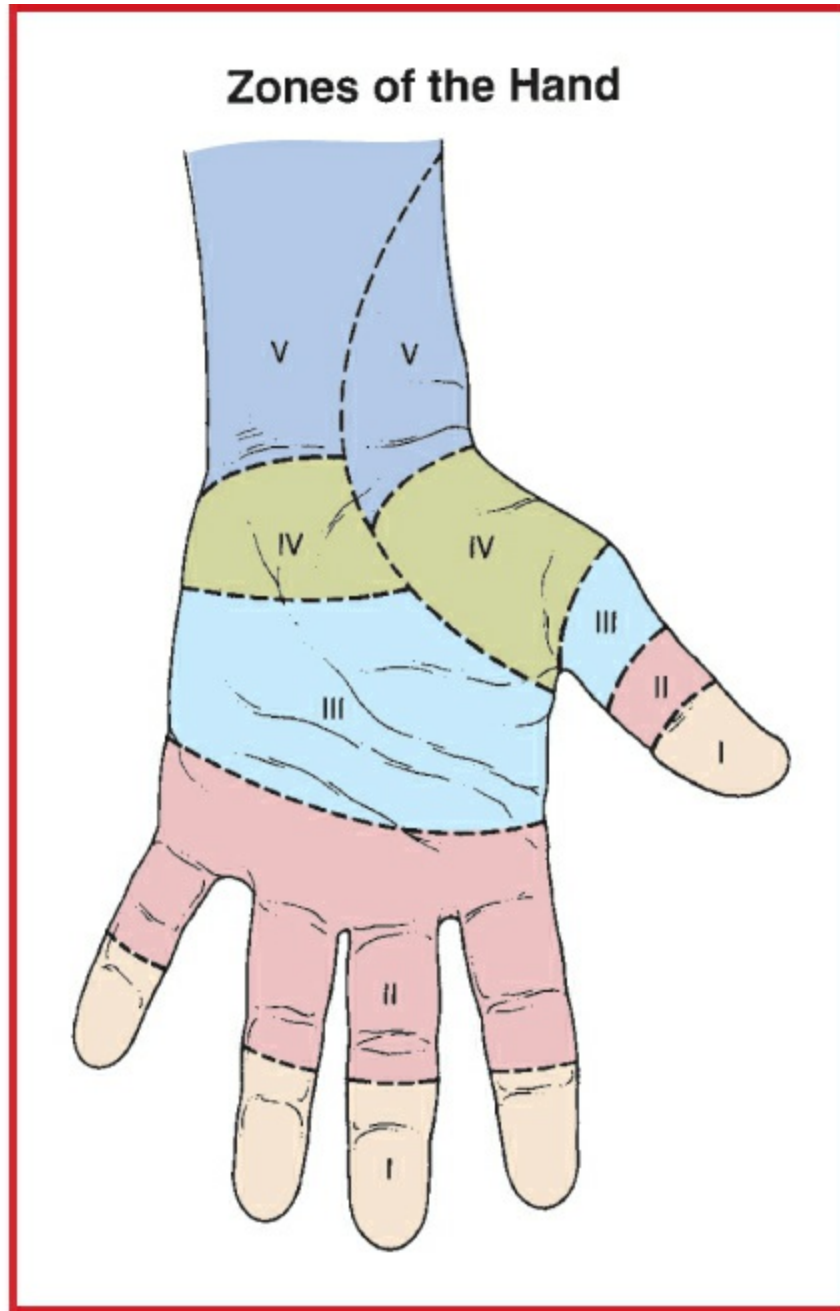


Figure 10.1-3 Zone classification of flexor tendon injuries. (Reproduced with permission from Scott-Conner CEH, Dawson DLOperative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

In general, nerve injuries are repaired at the time of the tendon repair. Depending on the surgeon, an operating microscope or loupe magnification may be used. Some surgeons will use a field block with local anesthetic and epinephrine to allow the patient to test the repair and ensure smooth gliding of the tendon intraop.

Tendons lacerated in the finger are often pulled back into the palm by muscular contraction. A palmar incision is required to retrieve the tendon, which must then be threaded carefully through the pulleys in the digit. Suture techniques for tendon repair create a juncture that is far weaker than an intact tendon. For this reason, the juncture must be protected from mechanical stress for a period of 8 wk or more. This is done by splinting the hand with the wrist and digits flexed so that the pull on the tendon by its muscle is limited. It is important that the patient emerges gently from anesthesia to limit the stress on the repair. The best results are obtained when repair is carried out within 7 d of the injury, although primary repair can be performed up to 3 wk. After 7 d, the muscle begins to undergo irreversible contracture. If the flexor tendon is advanced after this has occurred, a flexion contracture results. If a flexor tendon laceration is neglected, a palm-to-fingertip tendon graft should be performed. If the tendon bed is suitable for gliding, the graft can be accomplished in one stage. If not, a Silastic tendon spacer (rubber rod) must be placed at the first stage; 6–8 wk later, a palm-to-fingertip graft is placed in the bed prepared with the Silastic rod. Tendon graft donor sites include the palmaris longus tendon and toe extensors.

A variant of the sharp flexor tendon laceration is the FDP avulsion from its insertion in zone I. This is the so-called jersey finger—initially named for the classic mechanism of someone grasping the jersey of a ball carrier. This injury occurs during forceful grasp and most commonly affects the ring finger. The FDP tendon retracts to various predictable levels per the classification system of Leddy. Type I retracts in to the palm; this injury is most likely to disrupt the blood supply to the tendon via the vincula. Type II retracts to the level of the PIP joint, and type III retracts to the level of the DIP joint. In general, these injuries should be explored and repaired within 7 d. Repair may require minisuture anchors or a button/pullout suture through the nail/distal phalanx because there is no sufficient tendon distally for a primary repair. If neglected, these patients should be treated with a **distal interphalangeal (DIP) arthrodesis**. A flexor tendon graft through an intact FDS tendon usually is not indicated because tendon adhesions will commonly interfere with the function of the FDS, leading to decreased overall active motion of the digit. The most common complication is the development of tendon adhesions, which limit tendon gliding and digit motion. If these patients fail to improve within a 3- to 6-mo course of physical therapy, they require an operative tendolysis to lyse the adhesions.

Usual preop diagnosis

Flexor tendon laceration; FDP avulsion (“jersey finger”); digital nerve laceration; median nerve laceration

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Supine, with arm extended on hand surgery table. The foot may be prepped for a graft.
Incision	Zig-zag (Brunner) within the digits, extensile approach hand or wrist
Special instrumentation	Pneumatic tourniquet (unnecessary if using a local anesthetic field block containing epinephrine), ± operating microscope for nerve injuries
Antibiotics	Cefazolin 2 g iv
Surgical time	1–2 h; may be extended for nerve repair and treatment of associated injuries
Closing considerations	Tendon and nerve repairs must be protected with splints before emergence from GA. Smooth extubation (see Emergence, below)
EBL	Minimal; procedure performed under tourniquet control
Postop care	PACU → home
Mortality	None associated with procedure
Morbidity	Tendon adhesions: 25% Rupture of tendon repair: <5% Infection: rare
Pain score	2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adult population, rarely children
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Male:Female	Slight male predominance, due to occupational injuries
Incidence	Not uncommon
Etiology	Trauma

ANESTHETIC CONSIDERATIONS

(Procedures covered: excision of ganglion of the wrist; palmar and digital fasciectomy; repair of flexor tendon laceration)

PREOPERATIVE

The majority of patients presenting for these procedures are usually otherwise healthy. Many of them present for elective surgery as a result of progressive functional impairment and pain, and preop workup is routine.

Neurologic	If regional anesthesia is contemplated, preexisting sensory or motor defects should be documented carefully.
Laboratory	Hb/Hct (healthy patients); otherwise, as indicated from H&P
Premedication	Mild-to-moderate premedication (e.g., in adults, midazolam 1–2 mg iv, fentanyl 50–100 mcg iv, titrated to effect) is often desirable before placement of a regional block.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, ± celecoxib 200 mg po with sip of water. Consider aprepitant 40 mg po for GA patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Regional anesthesia (most common), GA, or a combination of the two may be used. Intravenous regional anesthesia (Bier block) is most useful for short procedures that last

for <1 h (see Anesthetic Considerations). Some surgeons prefer to place a local anesthetic field block so that the patient is able to participate and move their finger during surgery. Sedation should be titrated so that the patient can participate in testing.

A brachial plexus block via the axillary, infraclavicular, or supraclavicular approach is another good choice for this procedure. Because most of these procedures are done on an outpatient basis, brachial plexus block without GA is usually preferred to promote early “street-readiness.”

Regional anesthesia

1.5% mepivacaine 30cc for routine, superficial cases; 0.5% bupivacaine or 0.5% ropivacaine 30 mL for procedures >2.5 h or if extended analgesia is desired.

Ultrasound-guided brachial plexus block	Ultrasound-guided brachial plexus using supraclavicular, infraclavicular, or axillary approaches are all suitable for surgery below the shoulder. The infraclavicular approach is the most effective for placement of peripheral nerve catheters. The supraclavicular approach is associated with phrenic nerve blockade in 50% of patients and should be avoided in patients with respiratory compromise.
Bier block	The Bier block (intravenous regional anesthesia) is useful for short (<60 min), wrist and hand surgeries. 40–50 mL of 0.5% lidocaine is commonly used. For short procedures (<30 min), consider reducing the dose of iv anesthetic (e.g., 25 mL 0.5% lidocaine) and using a forearm tourniquet instead of an upper arm tourniquet. The OR staff should be alerted that iv regional anesthesia is being used, so that all are ready to proceed after the tourniquet is inflated. Tourniquet pain and postop pain are reduced by adding ketorolac (20 mg) or clonidine (1 mcg/kg) to the lidocaine solution.

General anesthesia

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see

	Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance	
Emergence	Standard emergence. Skin closure is frequently followed by application of a splint; patient should remain anesthetized during splinting procedure. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 18 ga × 1 BSS @ 1.5–3 mL/kg/h	An 18-ga iv catheter placed in the contralateral upper extremity should be adequate. See Restrictive Fluid Management, Appendix K , p. K-4.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	
Supraclavicular block complications	Inadequate block Intravascular injection Persistent paresthesia Local anesthetic toxicity Horner syndrome Pneumothorax Phrenic nerve block	Ulnar nerve sparing—may require separate block Rx severe LA toxicity with Intralipid® 20% 1.5 mL/kg iv initially Sx: cough, pleuritic pain, SOB Respiratory impairment possible
Infraclavicular block complications	Inadequate block Intravascular injection Pneumothorax Phrenic nerve block Persistent paresthesia Local anesthetic toxicity	Pneumothorax is very rare. Usually asymptomatic. Incidence ≤50%
Field block complications	Local anesthetic toxicity Inadequate block	An accidental intravascular injection, although unlikely, can cause CNS toxicity. Sz should be treated with midazolam titrated to effect,

		accompanied by airway control. Rx severe toxicity with Intralipid® 20% 1.5 mL/kg iv, initially. If there is any question of a full stomach, intubation should be accomplished rapidly.
Bier block complications	Local anesthetic toxicity Inadequate block Thrombophlebitis	Proper exsanguination and tourniquet function are critical. Upper arm tourniquet should remain inflated a minimum of 25 min. Systemic toxic reaction to the local anesthetic may occur as a result of upper arm tourniquet leak or inadvertent premature (<20 min) upper arm tourniquet release. Treatment is supportive. Sz are controlled with midazolam, with appropriate airway protection. Even with a functioning tourniquet, it is possible to overcome tourniquet pressure by injecting too vigorously. Care must be taken when switching from proximal to distal upper arm tourniquet; never deflate proximal tourniquet until verifying that distal tourniquet is working.

POSTOPERATIVE

Multimodal pain management	Oral analgesics are usually sufficient.	The lingering analgesia of the brachial plexus block is often
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		sufficient for pain relief in the recovery room; oral analgesic therapy can be instituted prior to discharging patient to home.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization.
Tests	None routinely indicated	

Suggested Readings

1. Arslanian B, Mehrzad R, Kramer T, et al: Forearm Bier block: a new regional anesthetic technique for upper extremity surgery. *Ann Plast Surg* 2014; 73(2):156–7.
2. Chandrasoma J, Harrison TK, Ching H, et al: Peripheral nerve blocks for hand procedures. *N Engl J Med* 2018; 379(10):e15.
3. Mian A, Chaudhry I, Huang R, et al: Brachial plexus anesthesia: a review of the relevant anatomy, complications, and anatomical variations. *Clin Anat* 2014; 27(2):210–21.
4. Steiner MM, Calandruccio JH: Use of wide-awake local anesthesia no tourniquet in hand and wrist surgery. *Orthop Clin North Am* 2018; 49(1):63–8.
5. Ustener E, Yilmaz A, Ozgencil E, et al: Ultrasound anatomy of the brachial plexus nerves in the neurovascular bundle at the axilla in patients undergoing upper-extremity block anesthesia. *Skeletal Radiol* 2013; 42(5):707–13.

WRIST ARTHROSCOPY/REPAIR OF TRIANGULAR FIBROCARILAGE COMPLEX TEARS

SURGICAL CONSIDERATIONS

Description

Wrist arthroscopy may be performed for either diagnostic or therapeutic indications. A smaller diameter version of the standard arthroscope is used for visualizing the wrist joint. The primary entry portals are on the dorsum of the wrist and course between the extensor compartments. Irrigation is used during the procedure. Unlike the knee joint, where visualization is obtained by distention of the joint, in-the-wrist visualization is obtained by distraction. The digits are placed in finger traps, and up to 10 lb of traction

can be placed on the wrist. Specialized instrumentation is available to resect and débride intra-articular structures and to place sutures to repair torn ligaments. The **triangular fibrocartilage complex** is a source of ulnar-sided wrist pain. It is a term given to the soft tissues spanning the distal radioulnar joint. It is composed of the articular disk, radioulnar ligaments, ulnocarpal ligaments, extensor carpi ulnaris tendon sheath, and meniscus homologue. Advances in both imaging and arthroscopic techniques in the recent past have led to our ability to better treat patients with injuries to the triangular fibrocartilage complex (TFCC). The classification of these injuries credited to Palmer involves both traumatic and degenerative tears, which are associated with ulnar-positive variance and ulnar impaction or impingement syndrome. If an open procedure, such as repair of an intercarpal ligament or an ulnar shortening procedure, is contemplated following diagnostic arthroscopy, either GA or regional block is preferred.

Variant procedure or approaches

The standard approach is to suspend the forearm vertically in traction. The forearm also can be placed horizontally on the hand table in traction.

Usual preop diagnosis

Internal derangement of the wrist of unknown etiology; tears of the TFCC; intercarpal instability due to intercarpal ligament tears; scapholunate dissociation; lunotriquetral dissociation; fracture of distal radius; ulnar impingement syndrome

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Supine, with arm extended on hand surgery table
Incision	Small incisions are made on the dorsum of the wrist for instrument insertion.
Special instrumentation	2.7-mm-diameter arthroscope, 0 or 30° field of view; light source and video camera; television monitor; surgical power shaver; joint irrigation system; traction device for forearm; pneumatic tourniquet
Unique	Patient is often awake and observes surgery on monitor.

considerations	
Antibiotics	Cefazolin 2 g iv
Surgical time	30 min to 2 h
Tourniquet	150 mm Hg above SBP; max time = 120 min
Closing considerations	Arthroscopic portals are each closed with a single skin suture.
EBL	Minimal; procedure performed with tourniquet control
Postop care	PACU → home
Mortality	None associated with procedure
Morbidity	Infection: <1% Swelling (2° to irrigation fluid): Common Nerve and artery damage: Uncommon
Pain score	1–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adults
Male:Female	1:1
Incidence	Least common form of arthroscopy
Etiology	See Usual preop diagnosis above.
Associated conditions	Antecedent trauma

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Carpal Tunnel Release.

Suggested Readings

1. Ahsan ZS, Yao J: Complications of wrist and hand arthroscopy. *Hand Clin* 2017; 33(4):832–8.
2. Wolf JM, Dukas A, Pensak M: Advances in wrist arthroscopy. *J Am Acad Orthop Surg* 2012; 20(11):725–34.

CARPAL TUNNEL RELEASE

SURGICAL CONSIDERATIONS

Description

This is the most commonly performed procedure in hand surgery. It consists of the transection of the transverse carpal ligament through either an open-palmar or an endoscopic approach (Fig. 10.1-4). In patients with severe synovitis, as in rheumatoid arthritis, a synovectomy should be performed at the same time. If there is advanced thenar atrophy and weakness of thumb opposition, a tendon transfer may be done at that time. The most common transfer is the **Camitz opponensplasty**, in which the palmaris longus tendon is prolonged with palmar fascia and transferred to the thumb. Transfers of the extensor indicis proprius and superficial flexor tendons also can be performed. Surgeons differ in their approach to anesthesia for this procedure: some prefer local infiltration, whereas others prefer regional as there is less fluid edema.

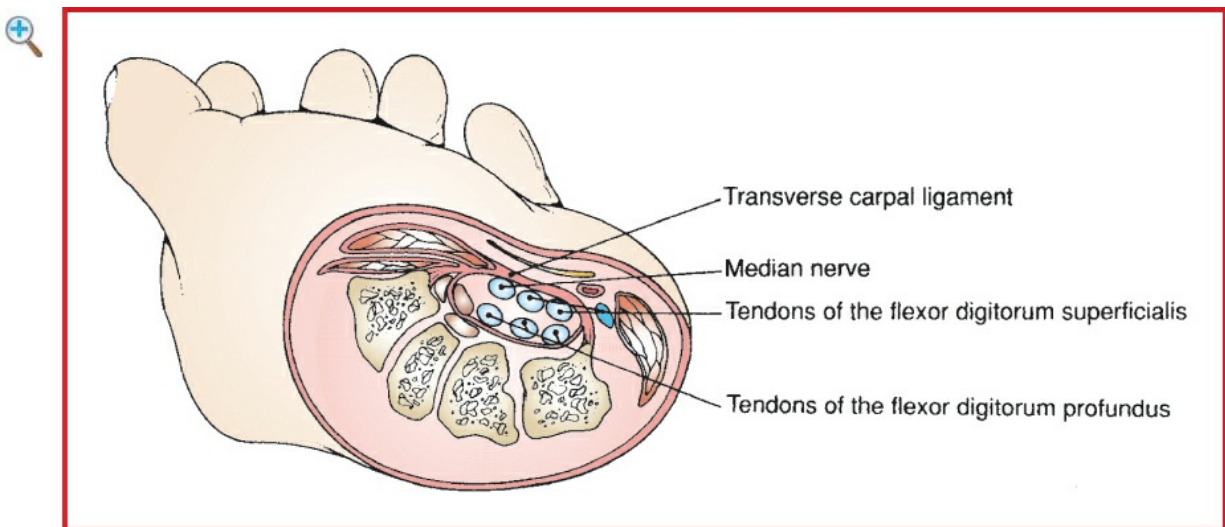


Figure 10.1-4 Exposure of the carpal tunnel. (Reproduced with permission from Scott-Conner CEH, Dawson DL: Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

Usual preop diagnosis

Carpal tunnel syndrome (CTS); median nerve compression at the wrist

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURES

Open Carpal

Endoscopic Carpal Tunnel

	Tunnel Release	Release
Position	Supine	⇐
Incision	Longitudinal incision in palm; may extend to forearm	2 cm transverse at proximal flexion crease of wrist; some use a distal palmar incision also.
Special instrumentation	Pneumatic tourniquet	Endoscopic carpal tunnel release system: endoscope, sheath and trocar, special cutting tools
Unique considerations	Although rare, damage to neurovascular structures	Danger of intraop injury to digital nerves, median nerve, tendons, superficial vascular arch
Antibiotics	±Cefazolin 2 g iv	⇐
Surgical time	30 min	30 min
Tourniquet	150 mm Hg above SBP; max time = 120 min	⇐
Closing considerations	Simple skin closure; some surgeons will splint	⇐
EBL	Minimal; performed under tourniquet control	Minimal
Postop care	PACU → home	⇐
Mortality	None associated with procedure	⇐
Morbidity	Overall complication rate: 4%	⇐
	Complex regional pain syndrome (CRPS): <5%	Rare

	Hematoma: rare	Complications of ulnar nerve palsy <5%
	Infection: uncommon	Tendon and nerve laceration: rare (<1%)
		Vascular injury: less common than nerve injury
Pain score	1–2	1–2

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–80 yr; 50% are 40–60 yr
Male:Female	1:2
Incidence	Common
Etiology	Compression of the median nerve within the carpal tunnel by synovitis; mass effect 2° tumor or fracture fragments, peripheral neuropathy, gout, anomalous structures, thrombosis of a persistent median artery, and idiopathic; repetitive trauma (e.g., computer use)
Associated conditions	Rheumatoid arthritis; thyroid imbalance; diabetes; amyloidosis; multiple myeloma; alcoholism; hemophilia; pregnancy; menopause; gout; fractures of the distal radius; Kienböck disease

ANESTHETIC CONSIDERATIONS

(Procedures covered: tendolysis of flexor or extensor tendon; wrist arthroscopy; carpal tunnel release)

PREOPERATIVE

In general, there are two patient populations involved: (a) healthy patients with Hx of wrist trauma and (b) rheumatoid patients. (See Anesthetic Considerations for Wrist Procedures, for discussion of preop concerns in the rheumatoid patient.)

Laboratory	None; otherwise, as indicated from H&P
Premedication	Mild-to-moderate premedication (e.g., in adults,

	midazolam 1–2 mg iv, fentanyl 50–100 mcg iv, titrated to effect) is often desirable before placement of a regional block.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, ± celecoxib 200 mg po, with sip of water. Consider aprepitant 40 mg po with sip of water for GA patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Intravenous regional anesthesia is suitable only for procedures that are <1 h. For longer procedures, supraclavicular and infraclavicular brachial plexus blocks are good alternatives. Regional techniques are especially appropriate for outpatients. Some of these procedures can be performed with local infiltration of anesthetic.

Intravenous regional block	The Bier block (intravenous regional anesthesia) is for short (<60 min), superficial wrist and hand surgeries. 40–50 mL of 0.5% lidocaine is commonly used. A very brief operative procedure may be an indication to reduce the dose of iv anesthetic agent by, for example, having the surgeon use a forearm tourniquet instead of an upper arm tourniquet. The OR staff should be alerted that iv regional anesthesia is being used, so that all are ready to proceed after the tourniquet is inflated. Tourniquet pain and postop pain are reduced by adding ketorolac (20 mg) or clonidine (1 mcg/kg) to the lidocaine solution.	
Blood and fluid requirements	Minimal blood loss IV: 20 ga × 1 BSS @ 1.5–2 mL/kg/h	A 20-ga iv catheter placed in the nonoperative upper extremity should be adequate. See Restrictive Fluid Management, Appendix K, p. K-4.

Monitoring Positioning	Standard monitors ✓ and pad pressure points ✓ eyes	
Intravenous regional block complications	Local anesthetic toxicity Inadequate block Thrombophlebitis	Proper exsanguination and tourniquet function are critical. A tourniquet should remain inflated a minimum of 25 min. Systemic toxic reaction to the local anesthetic may occur as a result of tourniquet leak or inadvertent premature (<20 min) tourniquet release. Treatment is supportive. Sz are controlled with midazolam, with appropriate airway protection. Even with a functioning tourniquet, it is possible to overcome tourniquet pressure by injecting too vigorously. Care must be taken when switching from proximal to distal tourniquet; never deflate proximal tourniquet until verifying that distal tourniquet is working.

POSTOPERATIVE

Multimodal pain management	Oral analgesics are usually sufficient.	Residual analgesia with iv regional anesthesia is minimal unless ketorolac or clonidine is used.
ERAS	PONV protocol (see Appendix B, p. B-4 and p.	Encourage early enteral nutrition and mobilization.

Tests

None routinely indicated

Suggested Readings

1. Chandrasoma J, Harrison TK, Ching H, et al: Peripheral nerve blocks for hand procedures. *N Engl J Med* 2018; 379(10):e15.
2. Maga JM, Cooper L, Gebhard RE: Outpatient regional anesthesia for upper extremity surgery update (2005 to present) distal to shoulder. *Int Anesthesiol Clin* 2012; 50(1):47–55.
3. Steiner MM, Calandruccio JH: Use of wide-awake local anesthesia no tourniquet in hand and wrist surgery. *Orthop Clin North Am* 2018; 49(1):63–8.

FIXATION OF FRACTURES AND DISLOCATIONS OF THE WRIST AND HAND

SURGICAL CONSIDERATIONS

Description

Patients with fractures of the distal radius, distal ulna, carpus, metacarpals, and phalanges that cannot be treated adequately with closed methods require fixation. Difficult fractures may require a combination of techniques such as percutaneous pinning or external fixation in addition to **open reduction and internal fixation (ORIF)**. The criteria for adequate treatment include anatomic reduction of the fracture fragments with focus on articular congruity and stable fixation for maintenance of this reduction. Cast/splint treatment or fixation of any kind is an effort to decrease motion at the fracture site and allow the body to produce a bony union with acceptable length, alignment, and rotation. Patients with carpal/wrist fractures or dislocations may have signs of neurologic compromise, specifically the median nerve. Often a closed reduction will improve these symptoms, but if not, urgent operative treatment is indicated. Most dislocations within the hand are reducible in the emergency room setting with adequate anesthesia; there are irreducible varieties that warrant immediate operative treatment. Vascular compromise of the hand associated with these injuries is rare, usually occurring in patients with severe crush or high-energy injuries. The devascularized hand is a surgical emergency, and revascularization must be carried out as soon as possible. High-energy injuries such as gunshot wounds may have significant soft tissue components that must be treated. The possibility of a coexisting compartment

syndrome should be considered, and a fasciotomy may be needed at the time of surgery. Open fractures are, by definition, contaminated and should be irrigated and débrided within 8 h of the injury. Surgical approaches depend on the nature and location of the fracture but typically are longitudinal. Some fractures of the articular surface of the distal radius are amenable to percutaneous pin fixation using fluoroscopy and possibly arthroscopic assistance. Screw fixation of the scaphoid also can be accomplished via percutaneous methods using image guidance. This injury may be associated with a picture of perilunar instability. Damage to intercarpal ligaments may require open repair or reduction and percutaneous pinning. Intraoperative fluoroscopy often is utilized, as are standard portable radiographs to monitor and assess the quality of the reduction of the fracture.

Soft tissue coverage of these injuries may be problematic, and **local flaps** or **free microvascular tissue transfers** may be indicated. Rarely are these procedures done with the initial operative debridement; an antibiotic bead pouch or wound VAC may be used to temporize open wounds. Free transfers can come from the same limb (radial forearm flap, based on the radial artery; lateral upper arm flap, based on the posterior radial collateral artery) or a remote site (latissimus dorsi muscle, based on the thoracodorsal artery; or scapular skin flap, based on the circumflex scapular artery). Remote flaps require special patient positioning and draping. Microsurgical tissue transfers also may need special pharmacologic considerations, including the administration of heparin or dextran to prevent thrombosis of the anastomosis, and avoidance of vasoconstrictors. These procedures also significantly increase operative time and equipment needs.

Usual preop diagnosis

Fractures of the distal radius (Colles', Barton's, Smith's are common eponyms); wrist dislocations (perilunate, lunate); fractures of the carpus, metacarpals, or phalanges; GSW; crush injuries; blast injuries

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Supine, with arm extended on hand surgery table
Incision	Usually volar, longitudinal for distal radius; longitudinal

	either midaxial, midlateral or dorsal for phalanges; dorsal for metacarpals and carpal fractures; a volar incision is used if symptoms of acute carpal tunnel syndrome are present.
Special instrumentation	Fluoroscopy; internal and external fixation devices and power tools; pneumatic tourniquet
Unique considerations	Associated injuries and soft tissue problems in high-energy fractures
Antibiotics	Cephalosporin prophylaxis (e.g., cefazolin 2 g) is indicated for the treatment of closed fractures. Open fractures may require the addition of an aminoglycoside or penicillin depending on the amount and type of contamination.
Surgical time	30 min to 3 h (longer if free tissue transfer is required)
Tourniquet	150 mm Hg above SBP; max time = 120 min
Closing considerations	Some injuries require local flaps or free microsurgical tissue transfers for closure. Fasciotomy wounds may be left open. Splint applied at surgery
EBL	Minimal for fracture treatment, as tourniquet control is used
Postop care	PACU → home unless significant soft tissue injuries requiring additional procedures
Mortality	Usually 2° associated injuries
Morbidity	Loss of reduction, requiring repair Nonunion (requires additional surgical procedure) Infection
Pain score	3–9 (Great variation, probably depending on degree of median, ulnar, or dorsal radial sensory nerve involvement. Postop pain management is often problematic. Early use of stellate ganglion blocks may be beneficial in preventing the development of sympathetic mediated pain syndromes.)

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages. More conservative approaches are used with elderly patients.
Male:Female	1:1
Incidence	Very common
Etiology	Trauma
Associated conditions	Traumatic injuries

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The majority of patients presenting for these procedures are relatively young and healthy. Most present for elective repair of a traumatic injury, and preop workup is routine. Reimplantation and some injuries, such as irreducible dislocations or open fracture, require immediate attention and necessitate emergency surgery and full-stomach considerations.

Neurologic	If regional anesthesia is contemplated, preexisting sensory or motor defects should be documented carefully preop.
Laboratory	As indicated from H&P
Premedication	Mild-to-moderate premedication (e.g., in adults, midazolam 1–2 mg iv, fentanyl 50–100 mcg iv, titrated to effect) is often desirable before placement of a regional block.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, ±celecoxib 200 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for GA patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique: GETA, regional anesthesia, or a combination is commonly used. A brachial plexus block is excellent for short (1–2 h) procedures on the wrist and hand. Ultrasound-guided block may be performed with a supraclavicular infraclavicular or axillary approach. Regional anesthesia alone is a means of avoiding the risk of aspiration pneumonitis associated with GA in the patient with a full stomach whose operation must be done emergently. A combination of regional and general anesthesia is appropriate for prolonged cases and those that require the use of bone grafts harvested from the iliac crest or a free-tissue transfer.

General anesthesia

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance
Emergence	Skin closure is frequently followed by application of a splint; patient should remain anesthetized during splinting procedure. Consider iv acetaminophen if po dose not given. PONV prophylaxis should include ondansetron.

Regional anesthesia

1.5% mepivacaine (30 mL) for routine superficial cases; 0.5% bupivacaine or 0.5% ropivacaine, 30 mL for procedures >2.5 h or if extended analgesia is desired.

Ultrasound-guided brachial plexus block	Ultrasound-guided brachial plexus using supraclavicular, infraclavicular, or axillary approaches are all suitable for surgery below the shoulder. The infraclavicular approach is the most reliable for placement of peripheral nerve catheters. The supraclavicular approach is associated with phrenic nerve blockade in 50% of patients and should be avoided in patients with respiratory compromise.	
Blood and fluid requirements	Minimal-to-moderate blood loss IV: 18 ga × 1	An 18-ga iv catheter placed in the nonoperative upper extremity should be

	BSS @ 1.5–3 mL/kg/h	adequate. See Restrictive Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	
brachial plexus block complications	Local anesthetic toxicity Inadequate block Pneumothorax Persistent paresthesia Phrenic nerve block Horner syndrome	Rx severe LA toxicity with Intralipid® 20% 1.5 mL/kg iv initially Usually asymptomatic. Incidence ≤50%

POSTOPERATIVE

Multimodal pain management	Continuous nerve block can be considered in appropriate cases. Multimodal therapy in combination with regional block	Regional or combined regional-general anesthetic techniques are excellent for wrist procedures, especially with respect to postop pain management.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization.
Tests	None routinely indicated	

Suggested Readings

1. Coleman AC: Perioperative pain management for upper extremity surgery. *Orthop Clin North Am* 2017; 48(4):487–94.
2. Jones NF, Jupiter JB, Lalonde DH: Common fractures and dislocations of the hand. *Plast Reconstr Surg* 2012; 130(5):722e–36e.
3. Madison SJ, Humsl J, Loland VJ, et al: Ultrasound-guided root/trunk (interscalene block for hand and forearm anesthesia. *Reg Anesth Pain Med* 2013; 38(3):226–32.
4. Park SK, Lee SY, Kim WH, et al: Comparison of supraclavicular and infraclavicular

brachial plexus block: a systemic review of randomized controlled trials. *Anesth Analg* 2017; 124(2):636–44.

5. Ustuner E, Yilmaz A, Özgencil E, et al: Ultrasound anatomy of the brachial plexus nerves in the neurovascular bundle at the axilla in patients undergoing upper-extremity block anesthesia. *Skeletal Radiol* 2013; 42(5):707–13.

DIGIT AND HAND REPLANTATION

SURGICAL CONSIDERATIONS

Description

Patients with traumatic amputations of digits and the hand are candidates for emergency microsurgical replantation of these parts. In children, replantation is attempted for essentially all amputations. In the adult, replantation is carried out for amputations of the thumb, multiple digits, and amputations through the palm or proximal. In general, amputations of a single digit are not candidates for replantation because of the minimal loss of function in relation to the long rehabilitation period and expected outcome. Certainly, a single digit amputated proximal to the insertion of the FDS tendon (zone II [Fig. 10.1-3](#)) should not be replanted. The condition of the amputated part plays an important role in the decision to proceed with replantation. A severely crushed, contaminated, or burned part cannot be expected to survive and function. In addition to the condition of the amputated part, the method of preservation and the time from initial injury play a role in deciding on whether or not replantation is attempted. The patients also may have associated traumatic injuries (i.e., intra-abdominal bleeding with a positive peritoneal lavage, chest injuries), which will take precedence over replantation. The patient's overall health status must be assessed and may play a role in deciding whether or not to proceed with replantation.

There are a variety of reasons that people suffer amputations. Some studies of these patients have shown an increased incidence of psychopathology, such as alcohol or substance abuse. More often than not, these injuries occur in industrial accidents or in a population of people using power tools at home. Rarely children will gain access to dangerous objects, and self-inflicted injuries can occur. Because these procedures are emergent, patients often arrive at the hospital with full stomachs. Although regional anesthesia techniques provide peripheral vasodilation through their sympatholytic effect, surgeons prefer GA because of the unpredictable length of the procedures. While the patient is being readied for induction, the surgeon prepares the amputated part in the OR. At this time, the structures to be repaired are tagged, which saves a great deal of

anesthetic time. When the patient is prepped and draped, the hand is irrigated and débrided, and the corresponding structures are tagged in similar manner. The amputated part is brought to the field, and the actual replantation is performed. Once arterial blood flow is reestablished, the patient must be kept warm to prevent vasospasm. As with other microsurgical procedures, pharmacologic intervention is indicated to prevent thrombosis; iv heparin and dextran are normally administered. Skin grafts for soft tissue coverage and vein grafts to replace segmental vascular defects are commonly used. Vein grafts can be obtained from the ipsilateral upper extremity or from the lower extremity, especially the dorsum of the foot. Either the lateral thigh or abdomen is an excellent donor site for split-thickness skin grafts. Rarely is an immediate microsurgical free tissue transfer indicated for soft tissue coverage.

Usual preop diagnosis

Traumatic amputation of the digits or hand

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Supine, with arm extended on hand surgery table
Incision	Incisions extend both proximal and distal to traumatic amputation to explore neurovascular structures as well as define the zone of injury. Lower extremity prepped and draped as donor site for vein grafts from dorsum of foot, split-thickness skin graft from the thigh
Special instrumentation	Operating microscope; microsurgical instrumentation
Unique considerations	Emergency procedure
Antibiotics	Cefazolin 2 g iv
Surgical time	3–12 h
Tourniquet	150 mm Hg above SBP; max time = 120 min
Closing	Routine volar hand splint

considerations	
EBL	<500 mL
Postop care	ICU → requires monitoring in intensive nursing environment. Should be kept pain-free for extended period postop to minimize vessel spasm. Patients will benefit from postop sedation.
Mortality	Minimal for digit and hand. Mortality becomes an important issue when large amounts of muscle are part of the reattachment.
Morbidity	Loss of replanted part (vessel thrombosis): 10% Infection: rare
Pain score	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages, typically adults of working age
Male:Female	1:1
Incidence	Uncommon
Etiology	Trauma
Associated conditions	Substance abuse; alcoholism

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

In general, there are two patient populations for hand replantation: (a) isolated hand injury patients (common) and (b) multiple trauma victims (rare).

Respiratory	As suggested by coexisting disease or acute trauma injuries. Evidence of occult chest injury, including pneumothorax and pulmonary contusion, should be sought. Tests: Consider CXR; ABGs in victims of significant trauma.
Cardiovascular	As suggested by coexisting disease or acute trauma injuries. ✓ for evidence of occult cardiac or

	mediastinal injuries, such as myocardial contusion or great vessel rupture. Tests: Consider CXR (with NG tube in place to assess mediastinal widening) and ECG in victims of significant trauma.
Neurologic	As suggested by coexisting disease or acute trauma injuries. The possibility of closed head injury should be addressed in multiple trauma victims. Verify integrity of C-spine. Tests: Consider head CT prior to beginning a long procedure under GA in a patient with evidence of head trauma; C-spine x-ray.
Gastrointestinal	All patients should be considered to have a full stomach and to be at risk for aspiration pneumonitis. In general, they should receive preop medication to reduce stomach volume and acidity (e.g., metoclopramide 10 mg iv and ranitidine 50 mg iv).
Hematologic	Multiple trauma victims are likely to suffer from acute blood loss. Although blood loss from these procedures is generally modest, a preop T&C for several U PRBCs is wise for trauma patients. Tests: CBC
Metabolic	~50% of trauma victims are intoxicated. Anesthesia-related implications of ethanol intoxication include decreased anesthetic requirements, diuresis, vasodilation, and hypothermia.
Laboratory	As suggested by coexisting disease or acute trauma injuries. In general, most victims of significant trauma are best served by obtaining a wide variety of baseline lab studies to screen for unrecognized injury. These studies normally include ABGs, UA, renal function tests, LFTs, serum amylase, and tox screen.
Premedication	Full-stomach precautions: consider Na citrate 0.3 M 30 mL po, metoclopramide 10–20 mg iv; H₂-blocker
ERAS	Check compliance with presurgical ERAS

recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, $\pm$ celecoxib 200 mg po, $\pm$ gabapentin 300 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

GETA, after rapid sequence induction. Because of the unpredictable length of these procedures and the possible need for bone and/or vessel grafts, regional anesthesia is not feasible as the primary technique. A concurrent, continuing brachial plexus block, however, will provide sympathetic blockade, as well as postop analgesia, and catheter placement should be considered before inducing GA. The hand injury repair may be done concurrently with other procedures in multiple trauma victims.

Induction	Rapid sequence induction is mandatory in emergency cases, unless awake intubation is performed. C-spine fracture patients and those with facial injuries may require awake fiberoptic intubation. Hemodynamically unstable, acute trauma patients may be induced more safely with etomidate or ketamine.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance for stable patients. Hemodynamically unstable, acute trauma victims undergoing emergency surgery may be better served by using a combination of medications designed to have minimal hemodynamic consequences (e.g., fentanyl for analgesia, rocuronium for muscle relaxation, and scopolamine or midazolam for amnesia). N ₂ O is best avoided in the trauma patient.
Emergence	Difficult airway or full-stomach cases require awake extubation. Trauma victims who have undergone a prolonged procedure or who have significant associated cardiopulmonary injuries should remain intubated for postop mechanical ventilation. Consider iv

	acetaminophen if po dose given >6 h ago. PONV prophylaxis may include ondansetron and dexamethasone.	
Blood and fluid requirements	Significant blood loss IV: 16 ga × 1 BSS @ 1.5–3 mL/kg/h + replacement of blood loss Fluid/blood warmers, heating blanket, warmed circuit humidifier	A 16-ga iv catheter in nonoperative upper extremity should be adequate in hemodynamically stable patients. Acute trauma victims who are unstable require a minimum of two large-bore iv catheters or large-bore central lines. Keeping the patient well hydrated will ↓ incidence of small vessel thrombosis. See Restrictive Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors	Invasive hemodynamic monitoring and TEE should be considered in acute, multiple trauma victims.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Hemodynamic instability	Previously unrecognized injuries (e.g., pneumothorax, cardiac tamponade, intracranial bleeding) should be considered as a cause of unexplained intraop hemodynamic instability in all acute trauma victims.

POSTOPERATIVE

Complications	Sepsis ARDS	
Multimodal pain management	PCA. Regional analgesia	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization.
Tests	None routinely indicated	

Suggested Readings

1. Barbary S, Dap F, Dautel G: Finger replantation: surgical technique and indications Chir Main 2013; 32(6):363–72.
2. MacKay BJ, Nacke E, Posner M: Hand transplantation—a review.Bull Hosp Jt Dis 2014; 72(1):76–88.
3. Scheker LR, Becker, GW: Distal finger replantation.J Hand Surg Am 2011; 36(3):521–8.

Suggested Viewing

Links are available online to the following videos:

Hand Ganglion Cyst Excision (Dorsal) with Bier Block: https://www.youtube.com/watch?v=_7ihlyW7LxQ

Wrist Arthroscopy Surgery: <https://www.youtube.com/watch?v=Wg1GwRBMiZ4>

Carpal Tunnel Surgery: Inside the Operating Room: <https://www.youtube.com/watch?v=JdHrRf7JinA>

Trigger Finger Surgery: Inside the Operating Room: <https://www.youtube.com/watch?v=DISCFr0YeiM>

Dupuytren's Contracture Release: https://www.youtube.com/watch?v=yjYRlo_Zupw

CHAPTER 10.2

Shoulder/Arm Surgery

SURGEONS

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ANESTHESIOLOGIST

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SURGEON'S PERSPECTIVE: REGIONAL ANESTHESIA IN SHOULDER SURGERY

Anesthetic options in arthroscopic and open shoulder surgery include GETA and/or interscalene brachial plexus block (ISB). For more complex surgical procedures that require paralysis for adequate exposure, such as anatomic and reverse shoulder arthroplasty, GETA is typically utilized. ISB, frequently in conjunction with MAC, can be an effective option for arthroscopic surgery procedures including rotator cuff and labral repair/debridement, subacromial decompression, and biceps tenodesis/tenotomy.

Each anesthetic modality has advantages and disadvantages. GETA will predictably render the patient insensible to pain and unconscious, while regional anesthesia ($\pm$ MAC) is inherently dependent on the anesthesiologist's expertise and the patient's anatomy. If the patient does not experience adequate analgesia, conversion to GETA is sometimes required. Potential side effects of GA may include airway complications, short-term cognitive impairment, PONV, and delayed recovery. For ISB, potential side effects include symptomatic diaphragmatic hemiparesis, pneumothorax, permanent neurologic injury, Horner syndrome, or severe hypotension and bradycardia.

For outpatient rotator cuff repair, many benefits of ISB have been documented (ability to bypass PACU, decreased hospital admissions, decreased pain and incidence of sore

throat, less PONV, quicker time to ambulation, quicker food and water intake, quicker discharge time, and improved patient satisfaction). In April 2018, the FDA approved bupivacaine liposome injectable suspension (Exparel®) for the use in ISB following shoulder surgery in adults. An RCT demonstrated decreased postsurgical opioid consumption, increased opioid-free patients, and increased time to first opioid rescue through 48 h. Compared to previous local anesthetic agents, the analgesic effects of Exparel® can approach 48–72 h and may be superior to a continuous ISB (fewer complications and lower cost).

Suggested Readings

1. Hadzic A, Williams BA, Karaca PE, et al: For outpatient rotator cuff surgery, nerve block anesthesia provides superior same-day recovery over general anesthesia. *Anesthesiology* 2005; 102:1001–7.
2. Malik O, Kaye AD, Kaye A, et al: Emerging roles of liposomal bupivacaine in anesthesia practice. *J Anaesthesiol Clin Pharmacol* 2017; 33(2):151–6.
3. Sabesan VJ, Shahriar R, Petersen-Fitts GR, et al: A prospective randomized controlled trial to identify the optimal postoperative pain management in shoulder arthroplasty: liposomal bupivacaine versus continuous interscalene catheter. *J Shoulder Elbow Surg* 2017; 26(10):1810–7.

ARTHROSCOPIC SHOULDER SURGERY

SURGICAL CONSIDERATIONS

Description

The role of arthroscopy in shoulder surgery has advanced tremendously in the past decade and is now routinely performed by most shoulder surgeons. Arthroscopic procedures include **subacromial decompression (SAD)**, **distal clavicle resection (Mumford procedure)**, **debridement** (for labral tear, infection, or synovitis), **rotator cuff (RC) repair**, **anterior capsule-labral repair** for recurrent dislocation (**Bankart repair**), **capsular plication** for multidirectional instability (MDI), **capsular release for frozen shoulder**, and **repair of SLAP lesions** (superior labral anterior/posterior tears).

Procedures done arthroscopically are less painful postoperatively than open procedures because they produce less trauma to normal tissues. Rehabilitation is, therefore, facilitated. Interscalene block has been shown to provide good postop analgesia of shorter duration, but its clinical application with arthroscopic procedures is surgeon dependent because the postop pain is usually moderate to mild. Some surgeons prefer

only a GETA for shoulder arthroscopy. The use of indwelling intra-articular pain catheters has fallen out of favor in the past few years due to multiple case reports of chondrolysis, a devastating complication characterized by end-stage arthrosis of the glenohumeral joint.

Arthroscopic shoulder surgery may be performed in the beach chair or lateral decubitus position. Beach chair positioners are available with a trough for the head and a breakaway shoulder pad to provide important access to the posterior shoulder. The lateral decubitus position utilizes distal traction of 5–10 lb, with the arm abducted 30–45°. Both are safe positions for the brachial plexus because the shoulder is not excessively abducted. The “down” arm in the lateral position is placed in forward flexion, and an axillary roll is placed underneath the upper chest wall.

Initially, an 18-ga spinal needle is inserted into the glenohumeral joint, passing through the posterior deltoid and infraspinatus muscle and the posterior capsule of the joint (see shoulder anatomy, [Fig. 10.2-1](#)). Placement is verified by injecting saline to inflate the joint capsule. A stab incision is made using a No. 11 blade in the direction previously defined by the finder needle. Sharp and then blunt trocars are used to gain access to the joint and permit insertion of the arthroscopic device. Improper insertion of the instruments can injure the axillary or suprascapular nerves and the cartilage of the glenohumeral joint. Initial diagnostic arthroscopy is carried out through the posterior portal. Bupivacaine 0.5% with epinephrine 1:200,000 often is infiltrated into portals and the joint or subacromial space at the onset of surgery to help with analgesia and hemostasis. An anterior portal is used for instrumentation within the glenohumeral joint. After joint arthroscopy, the scope is placed into the subacromial space, where a direct lateral portal is used for instrumentation. Accessory portals are established as needed, depending on the procedure performed. Joint debridement and anterior capsulolabral stabilization are usually performed through anterior portals. RC repair, subacromial bursectomy, acromioplasty, and distal clavicle resection are done within the subacromial space (deep to the deltoid and superficial to the RC). Epinephrine (1 mg/3 L) in the irrigation fluid and maintaining MAP < 80 mm Hg help control bleeding, thus enhancing visualization during surgery.

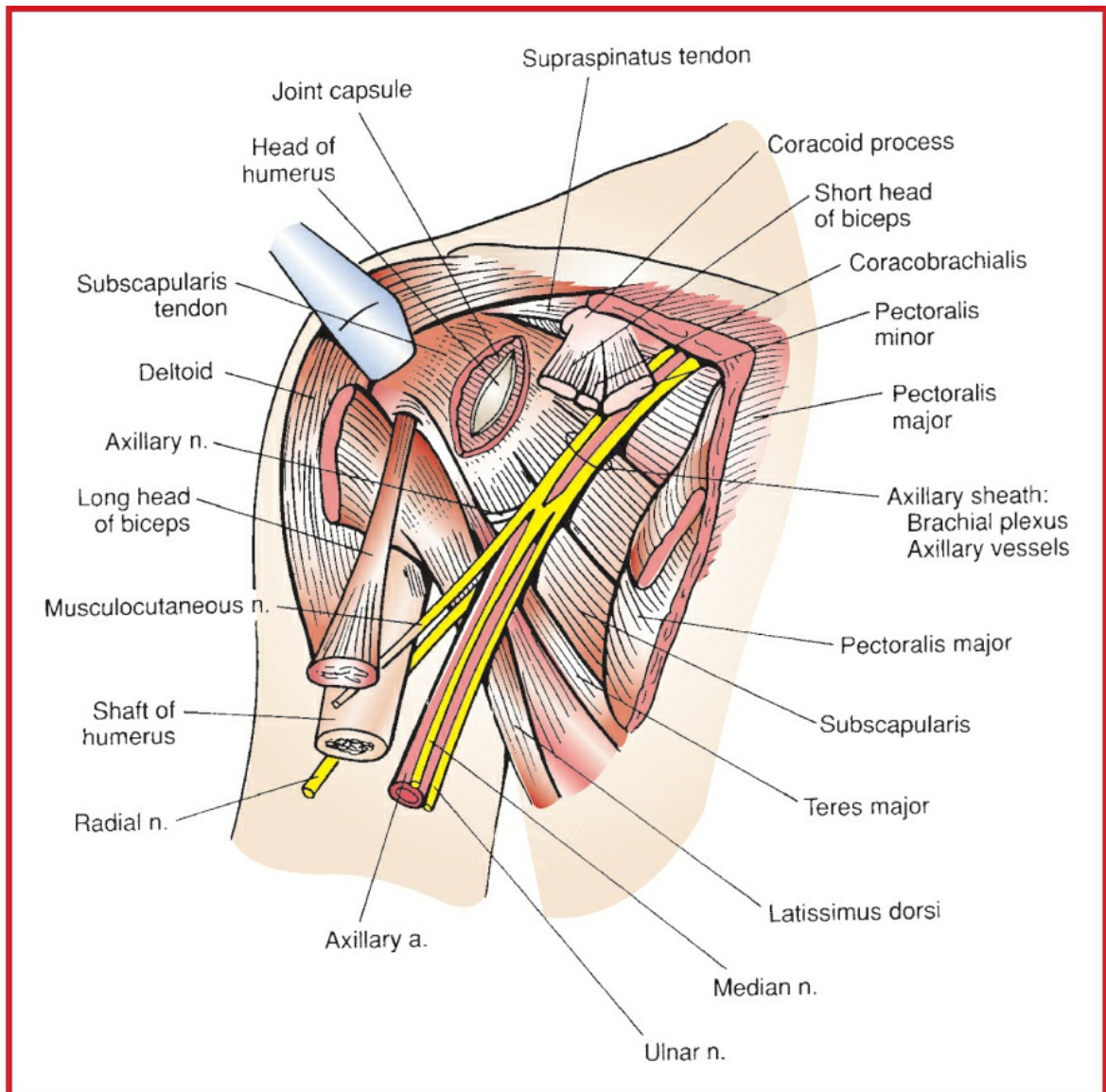


Figure 10.2-1 Anatomy of the shoulder joint, anterior. (Reproduced with permission from Hoppenfeld S, de Boer P: *Surgical Exposures in Orthopaedics: The Anatomic Approach* 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 1994.)

Usual preop diagnosis

Rotator cuff tear; subacromial impingement; glenohumeral instability; AC arthritis; labral tear

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plans (see [Appendix K: Enhanced Recovery After Surgery](#), p. K-1).

■ SUMMARY OF PROCEDURE

Position	Lateral decubitus or beach chair
Incision	Posterior arthroscopic portal, anterior instrumentation portal, lateral instrumentation portal for visualizing subacromial bursa; superior portal for semisitting position
Special instrumentation	Arthroscope; power burrs; arthroscopic shavers; suture-passing instruments; bone anchors; radiofrequency cautery
Unique considerations	Rigid eye patch over ipsilateral eye, to prevent corneal abrasion, suggested. Positioning of the head with appropriate support, removing upper section of operating table, if possible, for better access with semisitting position. ETT taped to opposite side of face. MAP $\leq$ 80
Antibiotics	Cefazolin 2 g iv preop, particularly if bone work performed
Surgical time	Positioning the patient is time intensive; can add as much as 45 min Reconstructive: 1–4 h Diagnostic: <1–1.5 h
EBL	Minimal: <200 mL (less with use of epinephrine, electrocautery, and laser)
Postop care	Frequently outpatient; may be overnight if interscalene or supraclavicular block is given or reconstructive procedure performed. Intra-articular pain catheter commonly used
Mortality	Rare
Morbidity	VAE possible Extravasation of fluid (NS or LR): >50% Brachial plexus injury (lateral decubitus position): rare Breakage of instruments: <1% Infection: 0–3%

Pain score 4 (diagnostic); 5–7 (reconstruction)

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–40 yr (instability); 35–75 yr (rotator cuff and acromial pathology)
Male:Female	2:1–4:1
Incidence	Very common: >50,000/yr
Etiology	Young patients: usually sports related Older patients: cuff and acromial pathology; age-wear phenomenon Rotator cuff pathology and acromial impingement (frequently coexist)
Associated conditions	Cervical arthritis and radiculopathy with rotator cuff pathology

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Surgery for Shoulder Instability.

Suggested Readings

1. Dattani R, Smith CD, Patel VR: The venous thromboembolic complications of shoulder and elbow surgery: a systematic review. *Bone Joint J* 2013; 95-B(1):70–4.
2. Farmer KW, Wright TW: Shoulder arthroscopy: the basics. *J Hand Surg Am* 2015; 40(4):817–21.
3. Noud PH, Esch J: Complications of arthroscopic shoulder surgery. *Sports Med Arthrosc Rev* 2013; 21(2):89–96.
4. Sabesan VJ, Shahriar R, Petersen-Fitts GR, et al: A prospective randomized controlled trial to identify the optimal postoperative pain management in shoulder arthroplasty: liposomal bupivacaine versus continuous interscalene catheter. *J Shoulder Elbow Surg* 2017; 26(10):1810–7.
5. Salazar D, Hazel A, Tauchen AJ, et al: Neurocognitive deficits and cerebral desaturation during shoulder arthroscopy with patient in beach-chair position: a review of the current literature. *Am J Orthop (Belle Mead NJ)* 2016; 45(3):E63–8.

SURGERY FOR ACROMIAL IMPINGEMENT, ROTATOR CUFF TEARS, AND

ACROMIOCLAVICULAR JOINT ARTHRITIS

SURGICAL CONSIDERATIONS

Description

Subacromial impingement is a common degenerative condition of middle age. It may be related to extrinsic anatomic factors (e.g., hooked acromion), a traumatic episode, or intrinsic factors (e.g., tissue degeneration). The subacromial space (space between the acromion and humeral head) is occupied by the supraspinatus (superior rotator cuff [RC]) muscle and tendon and the bursa, which allows for smooth gliding of the cuff tendon under the acromion. Trauma produces hemorrhage and inflammation in the bursa; swelling of the bursa decreases the space available under the acromion. These tissues may then be “pinched” between the greater tuberosity of the humerus and the lateral aspect of the acromion ([Fig. 10.2-2](#)) with forward elevation or abduction of the arm. This further increases the inflammation, producing a vicious cycle.

Impingement of the anterolateral acromion on the insertion of the supraspinatus (along with poor vascularity of this part of the cuff) is a leading hypothesis for the etiology of degenerative **RC tears**.

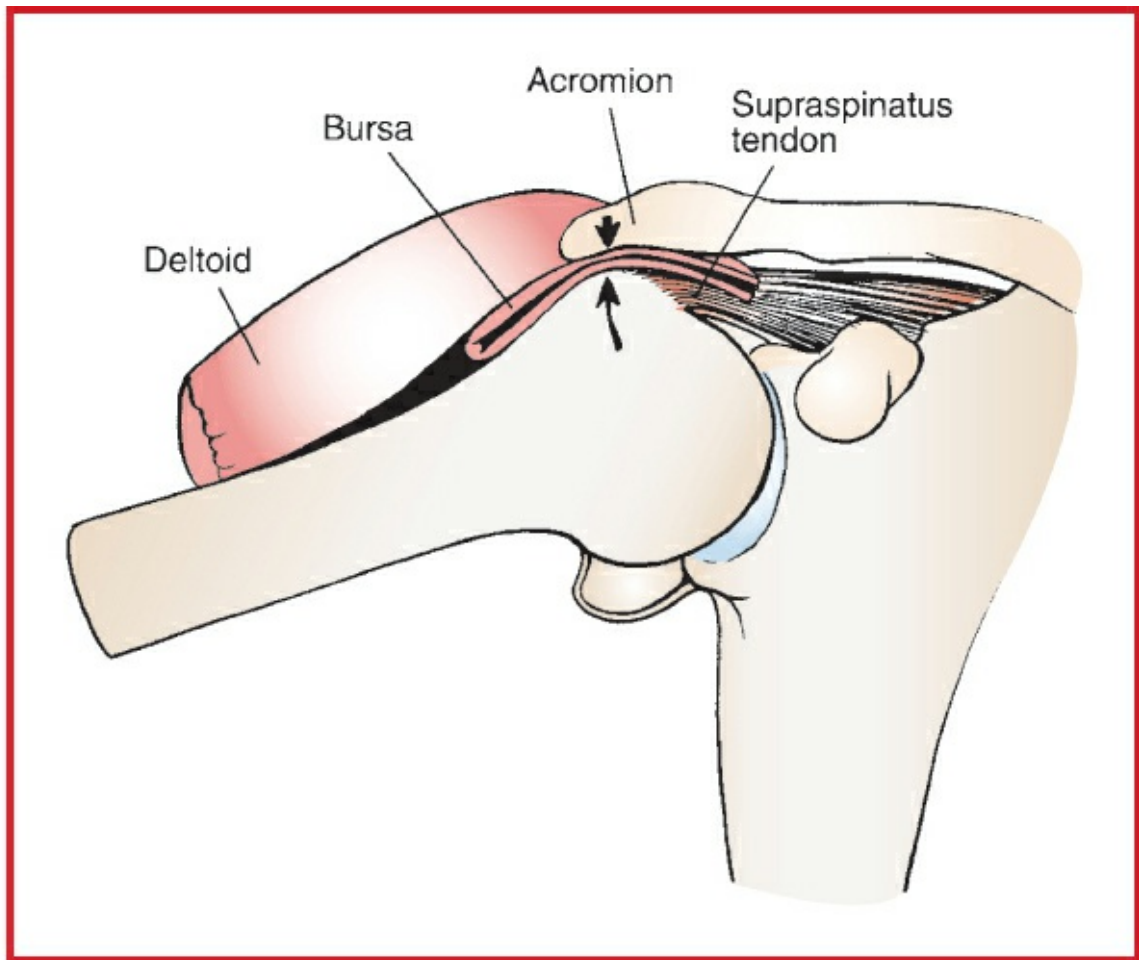


Figure 10.2-2 Abduction of the arm can impinge the subacromial bursa between the greater tuberosity and the undersurface of the acromion and coracoacromial ligament (between arrows). (Reproduced with permission from Hoppenfeld S, de Boer P: *Surgical Exposures in Orthopaedics: The Anatomic Approach*, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 1994.)

AC joint arthritis is a common radiographic finding in adults, but it is often asymptomatic. Distal clavicle excision is performed for clinically symptomatic AC joint arthritis.

Subacromial impingement

Surgical treatment of subacromial impingement is indicated when nonoperative treatment (e.g., cortisone injection, physical therapy) fails. Surgery involves shaving of the anterolateral aspect of the undersurface of the acromion (creating a flat surface and providing more room in the subacromial space). This may be accomplished with open techniques in combination with open RC repair, but it is more commonly done arthroscopically. The bursa is usually inflamed and quite vascular. Bleeding may obscure arthroscopic visualization and is controlled with electrocautery, with epinephrine in the irrigant, and by maintaining relative ↓ BP (MAP < 80 mm Hg).

Rotator cuff tears

RC ([Fig. 10.2-3](#)) repair may be performed using the **direct lateral open approach**, the **miniopen (deltoid-splitting) approach** in conjunction with arthroscopy, or all arthroscopically. If a deltoid-splitting incision is used, care is taken not to extend the split more than 5 cm distal to the acromion because of possible injury to the axillary nerve, which innervates the deltoid 5 cm or more from the lateral aspect of the acromion. If the beach chair position is used for open RC surgery or arthroscopy, the upper limb is draped free. The arm is manipulated, and traction is frequently applied. It is important that the head is secured (the head may be taped to the table or special beach chair positioner), the eyes are protected, and the anesthesiologist frequently checks to see that the surgeon is not pulling the patient off the table (not always apparent from the surgeon's side of the drape). Traction on the brachial plexus is more likely in the lateral decubitus position 2° arm traction.

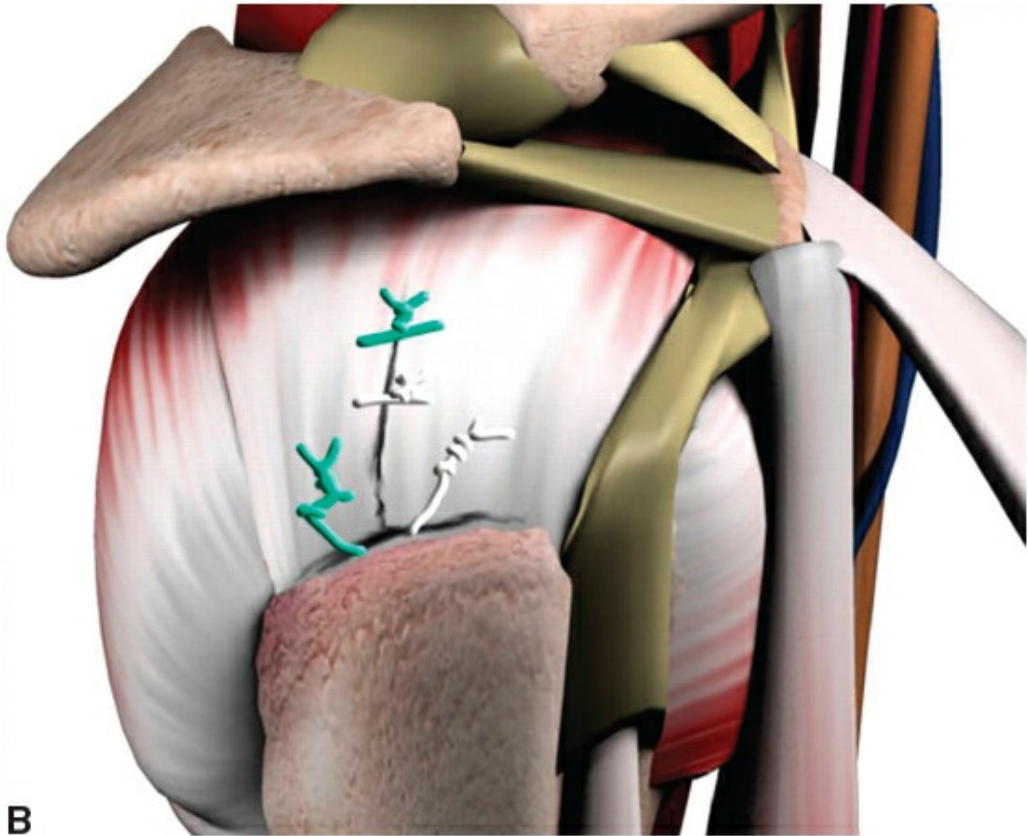
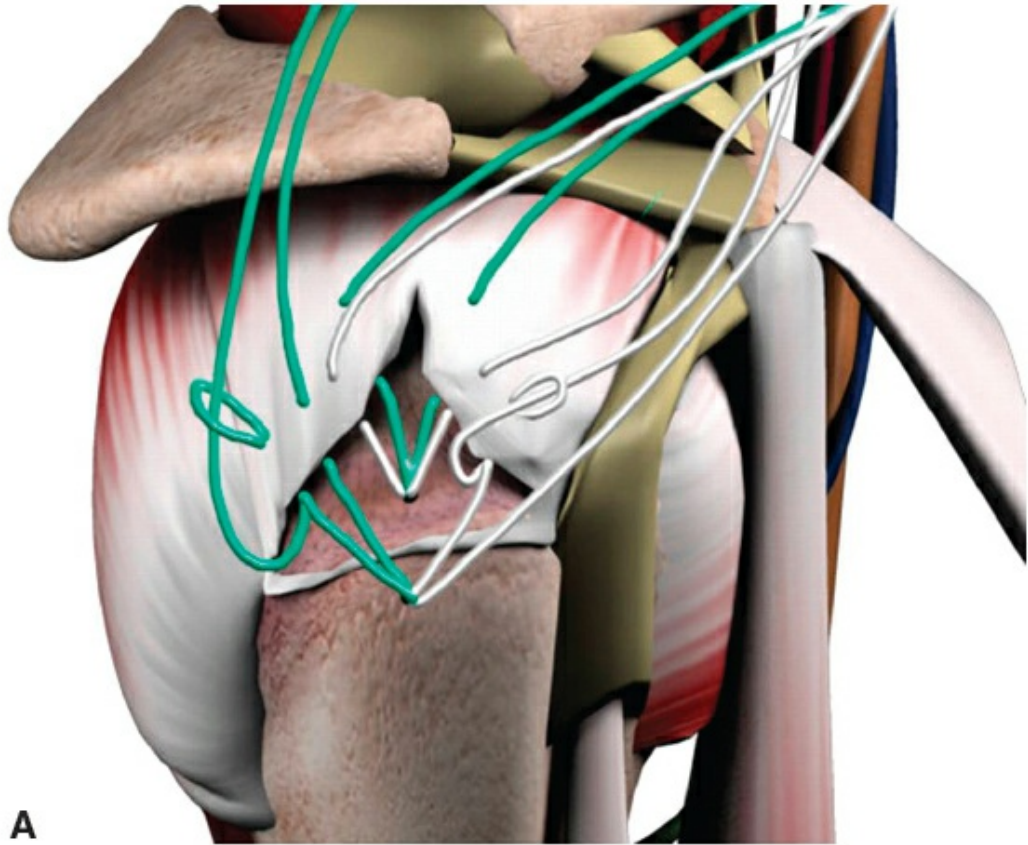


Figure 10.2-3 Lateral view of the right shoulder showing a rotator cuff repair. **A:** Rotator cuff tear. **B:** Rotator cuff repair with suture anchors. (Reproduced with permission from Lafosse L, Brozka R, Toussaint B, et al: The outcome and structural integrity of arthroscopic rotator cuff repair with use of the double-row suture anchor technique. *J Bone Joint Surg* 2007; 89(7):1533–41.)

Open RC repair involves suturing the cuff insertion back to the greater tuberosity through drill holes or with suture anchors. Arthroscopic repair requires percutaneous anchor placement and arthroscopic suture passing and knot tying. Bleeding is minimal with the arthroscopic technique but may approach 400 mL with an open procedure. Both require that the patient remains relaxed until all dressings are applied and he/she is fitted with an abduction sling. Patients typically are admitted for 24 h for pain control or if a drain is used.

AC joint disease

Surgery for **AC joint arthritis** is usually performed in conjunction with SAD and/or RC repair and may be done open or arthroscopically. It involves simple resection of the distal 5 mm of the clavicle through an incision directly over the joint or through an accessory anterior portal. Again, relative ↓ BP is required for the arthroscopic procedure.

Repair of AC joint dislocation (“shoulder separation”) is uncommon. Most AC separations are treated nonoperatively because long-term functional results are the same or better than those treated surgically. Severe AC separations occasionally require surgery, when the dislocated clavicle is buttonholed posteriorly through the trapezius, the deltoid origin has been avulsed from the clavicle, or the clavicle is displaced inferiorly below the cricoid process. Repair is performed in the beach chair position with the incision carried out over the AC joint and distal third of the clavicle. The clavicle is reduced and held in place with a large screw into the base of the coracoid, a large suture wrapped around the coracoid, or with K-wires across the AC joint. The coracoclavicular ligament often is repaired or reconstructed with tendon graft, or the coracoacromial ligament is transferred from the edge of the acromion to the clavicle. Following reduction and fixation, the deltoid is reattached to the clavicle if it has been avulsed, and the patient is placed in an immobilizer after skin closure. The operation is technically challenging, and the brachial plexus and subclavian vessels are at risk with screw placement and with inferior dislocations.

Usual preop diagnosis

RC tears (partial or complete); AC arthritis; impingement; bursitis; bicipital tendinitis;
AC separation

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plans (see [Appendix K, p. K-1](#)).

■ SUMMARY OF PROCEDURE

Position	Beach chair; semisitting, ~40–70°; or lateral decubitus
Incision	Oblique, saber-type incision anteriorly over distal acromion; lateral or deltopectoral incision for wider exposure; deltoid-splitting incision for RC tears
Special instrumentation	Power equipment for bone work; self-retaining retractors for cuff repairs; suture anchors
Unique considerations	Rigid eye protection for ipsilateral eye and careful head positioning
Antibiotics	Cefazolin 2 g iv preop
Surgical time	1–3 h
Closing considerations	Muscle relaxation when mobilizing cuff and during closure. Arm in sling and swathe, or abduction pillow for large tears. Immobilizer should be positioned prior to awakening patient to minimize potential for rupture of repair.
EBL	200–400 mL
Postop care	Maintenance of position in sling and swathe; no active motion of shoulder girdle for 2 d to 6 wk, depending on procedure
Mortality	Rare
Morbidity	Infection: 1–5% Axillary nerve damage: <2% Musculocutaneous nerve damage: <2% Breakage of instruments: <1% Brachial plexus damage (position dependent) Suprascapular nerve at risk for RC mobilization
Pain score	5–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Rotator cuff tears >40 yr (younger for athletes)
Male:Female	3:1
Incidence²	5–30% of general population affected by RC and acromial conditions, depending on age, thickness of tear, associated conditions
Etiology	Age related; trauma (70% involved in light work)
Associated conditions	Bursitis; tendinitis; impingement; RC disease (especially in first-time glenohumeral dislocations >40 yr); diabetes and renal failure; hypermobility

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Surgery for Shoulder Instability.

Suggested Readings

1. Audigé L, Flury M, Müller AM: Complications associated with arthroscopic rotator cuff tear repair: definition of a core event set by Delphi consensus process. *J Shoulder Elbow Surg* 2016; 25(12):1907–17.
2. Chalmers PN, Van Thiel GS, Trenhaile SW: Surgical exposures of the shoulder. *J Am Acad Orthop Surg* 2016; 24(4):250–8.
3. Mahon HS, Christensen JE, Brockmeier SF: Shoulder rotator cuff pathology: common problems and solutions. *Clin Sports Med* 2018; 37(2):179–96.

SURGERY FOR SHOULDER INSTABILITY

■ SURGICAL CONSIDERATIONS

Description

Shoulder instability is classified as multidirectional (MDI)/atraumatic or unidirectional/traumatic.

MDI is associated with generalized ligamentous laxity (e.g., Ehlers-Danlos or Marfan syndromes or idiopathic) and is treated primarily nonoperatively with physical therapy. Open or arthroscopic **capsular shift** is performed for recalcitrant cases. This involves “plication” of the capsule and/or labrum to decrease the capsular volume of the

shoulder. Patients with MDI, known as “voluntary dislocators,” frequently have psychiatric disorders and are very poor candidates for surgery.

Traumatic instability is usually anterior and is quite common in the young, active population. The shoulder is the most commonly dislocated joint. Recurrent dislocation in young, active patients is common (80–90%) and is associated with avulsion of the capsule/labrum from the anterior-inferior glenoid rim (Bankart lesion). The population undergoing a **Bankart repair** is almost invariably young and healthy. Older first-time dislocators (age > 50 yr) more commonly sustain rotator cuff (RC) tears or fractures, which do not result in chronic instability, but may require operative reduction and RC repair or fracture fixation. Posterior traumatic dislocation is much less common and is associated with high-energy trauma, seizures, or electrocution.

Instability surgery is often preceded by exam under anesthesia and arthroscopic examination, either in the beach chair or lateral decubitus position. The essential feature of instability surgery, whether arthroscopic or open, is the reattachment of the anterior-inferior capsulolabral complex back to the rim of the glenoid, thus reestablishing the normal “bumper” effect of the anterior-inferior labrum and decreasing the capsular volume of the shoulder. Nonanatomic procedures (reconstructive) are much less common but are still performed occasionally. These include transfer of the coracoid process to the anterior glenoid rim (**Bristow or Latarjet procedure**).

The **open Bankart repair** is performed in the beach chair position using the deltopectoral approach, with the interval between the deltoid and pectoralis major. The subscapularis (anterior RC muscle) lies just anterior to the joint capsule (Fig. 10.2-4), and this is either detached from its insertion or split. The capsule may then be opened to visualize the joint and rim of the glenoid. The glenoid rim is decorticated, providing bleeding bone to promote healing, and the anterior capsule is reattached through drill holes in the glenoid or with suture anchors. The capsule often is imbricated (overlapping folds) if it is redundant.

The shoulder and deltoid are highly vascular; however, bleeding is usually slight, with careful surgical technique. Major nerves are close but out of the plane of the operative field. The **musculocutaneous nerve** may be stretched by excessive medial retraction of the coracobrachialis (especially if a coracoid osteotomy is used) and the **axillary nerve** may be injured if the surgeon strays too far inferiorly.

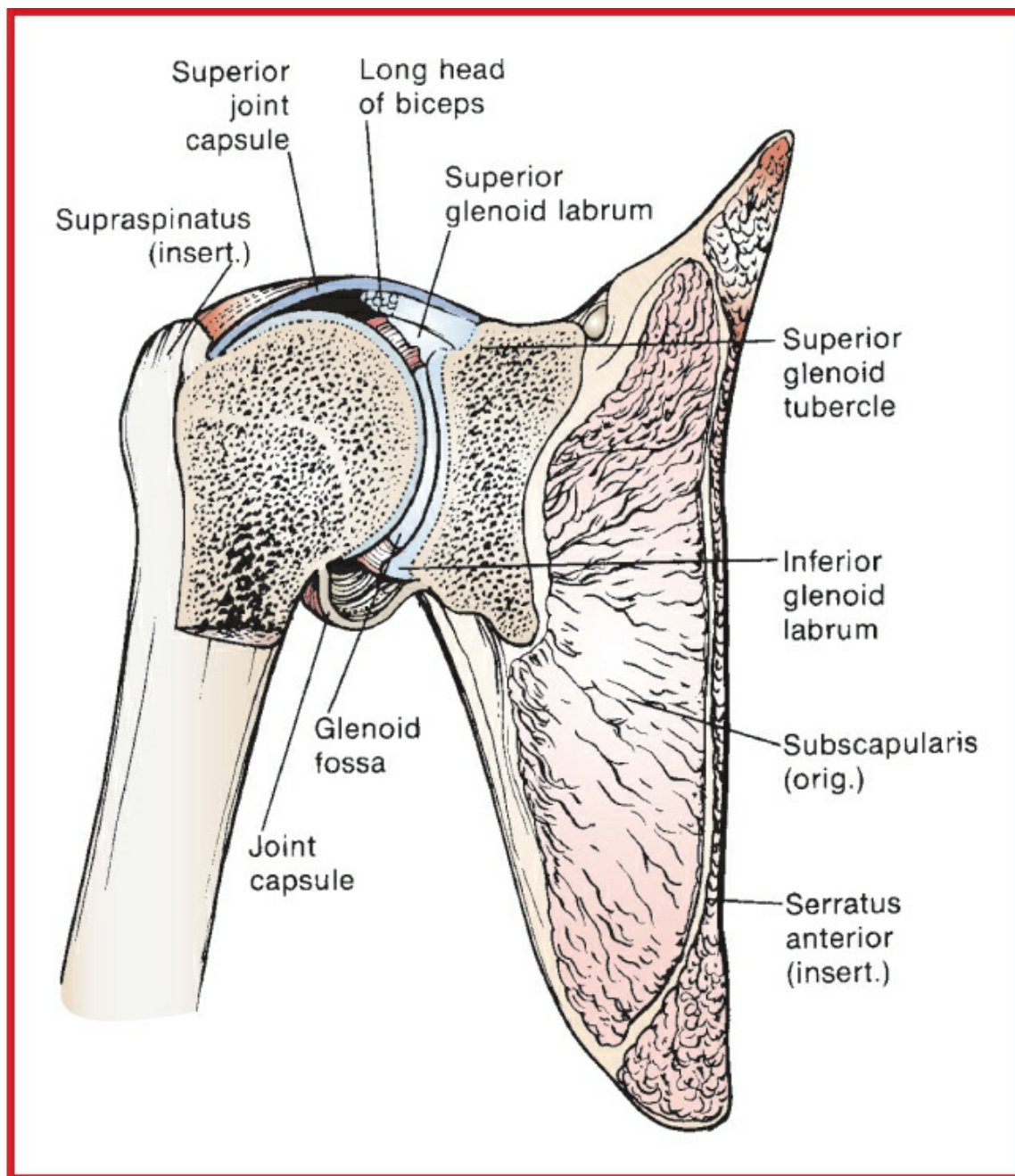


Figure 10.2-4 Cross-section of the joint: The joint capsule is redundant inferiorly to allow abduction. The long head of the biceps tendon traverses the joint. The tendon is surrounded by synovium and, therefore, is anatomically intracapsular but extrasynovial. (Reproduced with permission from Hoppenfeld S, de Boer P: *Surgical Exposures in Orthopaedics: The Anatomic Approach*, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 1994.)

If a **subscapularis-releasing technique** is used, the muscle is reattached and must be protected postop. External rotation of the shoulder is prevented for several weeks while the repair heals, and the surgeon prefers that the patient remain anesthetized until a shoulder immobilizer is applied.

The **arthroscopic Bankart repair** is similar to the open procedure but is performed

through two anterior portals with the scope coming in posteriorly. This procedure is less painful postop and allows for more rapid rehabilitation, because the subscapularis is not detached.

Open surgery for posterior dislocation is similar to the open Bankart repair, but it is done in the lateral position and utilizes the interval between the infraspinatus and teres minor. The RC attachment is preserved, but the posterior deltoid is detached and must be protected postop.

Usual preop diagnosis

Recurrent traumatic anterior or posterior instability; MDI; fracture dislocation

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plans (see [Appendix K, p. K-1](#)).

■ SUMMARY OF PROCEDURES		
	Open	Arthroscopic
Position	Beach chair for anterior; lateral decubitus for posterior	Lateral decubitus or beach chair
Incision	Deltopectoral (anterior); posterior approach (posterior)	Multiple small portals
Special instrumentation	Arthroscopic shaver; suture anchors; glenoid and humeral instrumentation	⇐ + arthroscopic instruments for suture passing and knot tying
Antibiotics	Cefazolin 2 g iv preop if bone work performed	⇐
Surgical time	2–4 h	⇐
Closing considerations	Continuous anesthesia until application of sling and swathe or abduction pillow	⇐

EBL	200–400 mL	Minimal
Postop care	No active motion for 6 wk	⇐
Mortality	Minimal	⇐
Morbidity	Axillary nerve palsy: 15% (may be preexisting) Suprascapular nerve injury	
Pain score	8	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–35 yr
Male:Female	2:1
Incidence	Common
Etiology	Trauma; hypermobility; Sz
Associated conditions	RC tears (>50 yr); hypermobility syndrome; Sz disorder; Hill-Sachs lesion (humeral head defect) may require bone grafting; superior labral tears; in most axillary nerve palsies from the injury; iatrogenic (rare)

Suggested Readings

1. Hohmann E, Tetsworth K, Glatt V: Open versus arthroscopic surgical treatment for anterior shoulder dislocation: a comparative systematic review and meta-analysis over the past 20 years. *J Shoulder Elbow Surg* 2017; 26(10):1873–80.
2. Levine WN, Sonnenfeld JJ, Shiu B: Shoulder instability: common problems and solutions. *Clin Sports Med* 2018; 37(2):161–77.

ANESTHETIC CONSIDERATIONS

(Procedures covered: shoulder arthroscopy; surgery for acromial impingement, RC tears, and AC disease; surgery for shoulder dislocations or instability)

PREOPERATIVE

Typically, three patient populations present for repair of RC tears or shoulder arthroscopy: (a) healthy posttrauma, (b) nonrheumatoid arthritic, and (c) rheumatoid

arthritic. Individuals presenting for repair of shoulder dislocations also may include those with a joint hypermobility syndrome (e.g., Marfan or Ehlers-Danlos) or Sz disorder patients.

Respiratory	Arthritic patients may exhibit Sx of pleural effusion or pulmonary fibrosis. Hoarseness may indicate cricoarytenoid joint (CAJ) involvement → difficult intubation. (See Anesthetic Considerations for Wrist Procedures.) Seizure disorder patients who suffer from recurrent shoulder dislocation as a result of frequent grand mal Sz also may suffer from occult aspiration pneumonia or pneumonitis. Tests: Consider CXR; PFTs; ABGs in debilitated rheumatoid patients
Cardiovascular	Arthritic patients may suffer from chronic pericardial effusions, valvular disease, and cardiac conduction defects. Patients presenting for shoulder stabilization because of joint hypermobility syndromes may have valvular dysfunction and are vulnerable to aortic dissection 2° HTN. These patients may require antibiotic prophylaxis for bacterial endocarditis. Tests: Consider ECG, ECHO in patients with severe rheumatoid arthritis. Recent ECHO to assess valve function and aortic root size indicated in most patients with Marfan syndrome
Neurologic	Arthritic patients may have cervical or lumbar radiculopathies that should be documented carefully preop. For example, head flexion may cause cervical cord compression. Patients with severe Sz disorders can suffer from recurrent shoulder dislocations 2° frequent violent grand mal Sz. Such patients should be treated maximally for Sz disorder prior to elective surgery. Be aware that as many as 15% of shoulder dislocations can be accompanied by axillary nerve palsy, which should be documented carefully preop. Tests: Consider C-spine radiographs to r/o occult subluxations in arthritic patients with neck complaints or upper extremity radiculopathy. Verify

	therapeutic levels of antiepileptic medication in Sz disorder patients.
Musculoskeletal	Arthritic patients may have limited neck and jaw ROM and may require fiberoptic intubation techniques. Bony deformities or muscle contractures may necessitate special attention to positioning. Patients with joint hypermobility syndromes presenting for shoulder surgery also may suffer other joint dislocations 2° positioning problems.
Hematologic	Virtually, all patients will be on some type of anti-inflammatory medication that may result in anemia or Plt inhibition. Ideally, patients should D/C NSAIDs at least 5 d preop; aspirin, 7 d. In addition, selected patients with Ehlers-Danlos are known to have severe coagulation defects that may preclude the use of regional anesthesia. Tests: A coag profile is mandatory in Ehlers-Danlos patients.
Endocrine	Rheumatoid patients may be on oral corticosteroids and may require supplemental perioperative steroids (e.g., 100 mg hydrocortisone q8h iv) to treat adrenal suppression, although the routine use of “stress-dose steroids” has been questioned.
Laboratory	Hb/Hct (in healthy patients); other tests as indicated from H&P. Patients with Ehlers-Danlos syndrome should always have banked blood available for surgery, except for minor procedures.
Premedication	Mild-to-moderate premedication (e.g., in adults, midazolam 1–2 mg iv, fentanyl 50–100 mcg iv, titrated to effect) is often desirable before placement of a regional block.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, ± celecoxib 200 mg po, ± gabapentin 300 mg po. Consider aprepitant 40 mg po for patients with h/o

severe PONV.

■ INTRAOPERATIVE

Anesthetic technique

GA or regional anesthesia (interscalene block), or a combination of the two techniques, can be used. A suprascapular block (when interscalene block is contraindicated) can be used for intraop → postop pain control in arthroscopic shoulder procedures. When logistically feasible, a combined technique is ideal. Unless contraindicated, a long-acting local anesthetic should be used in regional anesthesia for shoulder surgery to ameliorate postop pain.

General anesthesia

Induction	Standard induction. Arthritic patients may require awake fiberoptic intubation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Tranexamic acid (e.g., 10–15 mg/kg) iv before incision has been shown to ↓ EBL and may be requested by the surgeon.
Maintenance	Standard maintenance
Emergence	Management of emergence and extubation should be routine, except in difficult airway cases, which require awake extubation. Patients should remain anesthetized until the shoulder is immobilized. Consider iv acetaminophen if po dose not given <6 h ago. PONV prophylaxis should include ondansetron and dexamethasone.

Regional anesthesia

Local anesthetics	2% lidocaine or 1.5% mepivacaine have similar onset times (10–15 min) and similar duration (4–6 h). If extended postop pain control is desired, 0.5% bupivacaine or 0.5% ropivacaine can be used. Onset is usually within 30 min, with duration >6 h. Ropivacaine may be preferred for peripheral nerve
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	block due to its decreased cardiotoxicity.	
Interscalene block	0.5% bupivacaine or ropivacaine 20 mL The skin on the top of the shoulder (C3-C4) and the medial aspect of the upper arm (T2) may require separate subcutaneous field blocks. Phrenic nerve block → hemidiaphragmatic paralysis is an inevitable consequence of the interscalene block, which may not be tolerated by patients with significant preexisting respiratory compromise. Major complications, such as total spinal or pneumothorax resulting from interscalene block, are extremely rare. Interscalene block is contraindicated in patients with contralateral recurrent laryngeal nerve or phrenic nerve palsy (e.g., post CABG). If sedation is needed, midazolam (0.5–1.0 mcg boluses) and propofol (50–100 mg/kg/min by infusion), given initially in subanesthetic doses and thereafter titrated to effect, are good choices.	
Blood and fluid requirements	Minimal-to-moderate blood loss IV: 18 ga × 1 BSS @ 1.5–3 mL/kg/h	IV catheter placed in contralateral upper extremity. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors ± Precordial Doppler ± Arterial line	BP cuff should be placed on the arm for beach chair procedures. To help detect VAE, consider precordial Doppler monitoring when semisitting position used. Consider art line monitoring for high-risk patients (e.g., risk for aortic dissection).
Positioning	✓ and pad pressure points	Postural ↓ BP is the most common complication of

	✓ eyes ↑ VAE risk in semisitting position	the semisitting position. Changing to this position gradually can help prevent ↓ BP, as can the use of antiembolism stockings, plus fluid loading the patient. Marfan and Ehlers-Danlos patients require very gentle positioning to prevent joint dislocations.
Interscalene block complications	Total spinal Accidental epidural injection Local anesthetic toxicity (Sz/dysrhythmias) Stellate ganglion block (Horner syndrome) Laryngeal nerve block Phrenic nerve block Pneumothorax Persistent paresthesia	Resuscitative equipment, including airway management tools, should be immediately available. Nerve blocks should be performed in responsive and cooperative patients to minimize complications. May last ≤ 6 wk
Other complications	↓ BP during surgical prep and positioning Cardiac dysrhythmia VAE	↓ BP during long surgical preps normally can be avoided by using light inhalation anesthesia (e.g., 0.5 MAC) to ensure amnesia, with moderate muscle relaxation to prevent bucking on the ETT, and maintaining adequate hydration. Antiembolism stockings will help prevent venous pooling in lower limbs. Dysrhythmias may be 2° to irrigation fluids

containing epinephrine.

▼ POSTOPERATIVE

Multimodal pain management	PCA, regional block techniques including CPNB	Combined regional-general anesthetic techniques are excellent for shoulder procedures, especially with respect to postop pain management. Emphasize multimodal approach combining non-opioid and bolus/PCA opioid analgesics with regional anesthetics (e.g., continuous nerve block) whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6) Regional analgesia	Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	None indicated routinely	

Suggested Readings

1. Auyong DB, Allen CJ, Pahang JA, et al: Reduced length of hospitalization in primary total knee arthroplasty patients using an updated enhanced recovery after orthopedic surgery (ERAS) pathway. *J Arthroplasty* 2015; 30(10):1705–9.
2. Bambaren IA, Dominguez F, Elias Martin ME, et al: Anesthesia and Analgesia in the Patient with an Unstable Shoulder. *Open Orthop J* 2017; 11:848–60.
3. Borgeat A: Acute and nonacute complications associated with interscalene block and shoulder surgery. *Anesthesiology* 2001; 95(4):875–80.
4. Grauman S, Boethius J, Johansson J: Regional anaesthesia is associated with shorter postanaesthetic care and less pain than general anaesthesia after upper extremity surgery. *Anesthesiol Res Pract* 2016; 2016:6308371.
5. Zhu S, Qian W, Jiang C, et al: Enhanced recovery after surgery for hip and knee arthroplasty: a systematic review and meta-analysis. *Postgrad Med J* 2017;

GLENOHUMERAL SHOULDER ARTHROPLASTY

SURGICAL CONSIDERATIONS

Description

Shoulder replacement is performed for pain associated with end-stage arthritis. Primary osteoarthritis (wear-and-tear arthritis) is much less common in the shoulder than in the weight-bearing joints, such as the hip and knee. The most common causes of shoulder arthritis requiring **total shoulder arthroplasty (TSA)** are osteoarthritis, rheumatoid arthritis; avascular necrosis (AVN); posttraumatic arthritis; and rotator cuff tear arthropathy. Most patients are older than 65 yr.

TSA involves replacement of the humeral head with a stemmed prosthesis and resurfacing the glenoid with a polyethylene component. Both components may be cemented or uncemented, depending on the surgeon's preference. **Hemiarthroplasty** involves only replacement of the humeral side. This is indicated when the humeral head is involved primarily and the glenoid is in good condition, as in severe proximal humerus fractures, AVN of the humeral head, and some chronic dislocators. Replacement of the glenoid is contraindicated in the presence of an unreconstructable massive RC tear because eccentric forces lead to rapid loosening of the glenoid component. Some revision cases require glenoid bone grafting, which increases the complexity and potential blood loss.

Shoulder arthroplasty utilizes the beach chair position and the deltopectoral incision ([Fig. 10.2-5](#)). The deltopectoral interval is developed. The subscapularis insertion is incised, and the muscle is reflected medially. The capsule is incised and the joint exposed. The humeral head is dislocated anteriorly, and the head is removed with an oscillating saw. Reamers and broaches are used to prepare the proximal humerus for the prosthesis. If the glenoid is to be resurfaced, it is done before implantation of the final humeral component. The labrum is excised and a motorized reamer is used to remove the cartilage of the glenoid. Glenoid drill holes are made to conform to the back of the prosthesis. The glenoid prosthesis is cemented into place, with the component held in position manually until the cement hardens (~15 min). Trial humeral components are placed, and the appropriate sizing of the head and stem are assessed. The final humeral component is inserted with or without cement. The joint is reduced, and the subscapularis is repaired. Skin closure is followed by placement of a sling or shoulder

immobilizer. Postop management includes early passive ROM, limiting active internal rotation, and passive external rotation until the subscapularis has healed. Additional limitations may be instituted in revision situations or if RC repair is performed.

Variant approach: **Reverse shoulder arthroplasty (RSA)** was developed to overcome some of the shortcomings (e.g., rotator cuff stress, migration of the humeral component) of more conventional approaches by allowing the deltoid and other muscles to act as functional replacements for a deficient rotator cuff. This is accomplished by reversing the implants so that the socket component is attached to the head of the humerus and the ball component is attached to the native glenoid socket. As a result, RSA is inherently more stable because it creates a new medial center of rotation at the glenoid bone-implant interface, significantly reducing force on the damaged rotator cuff.

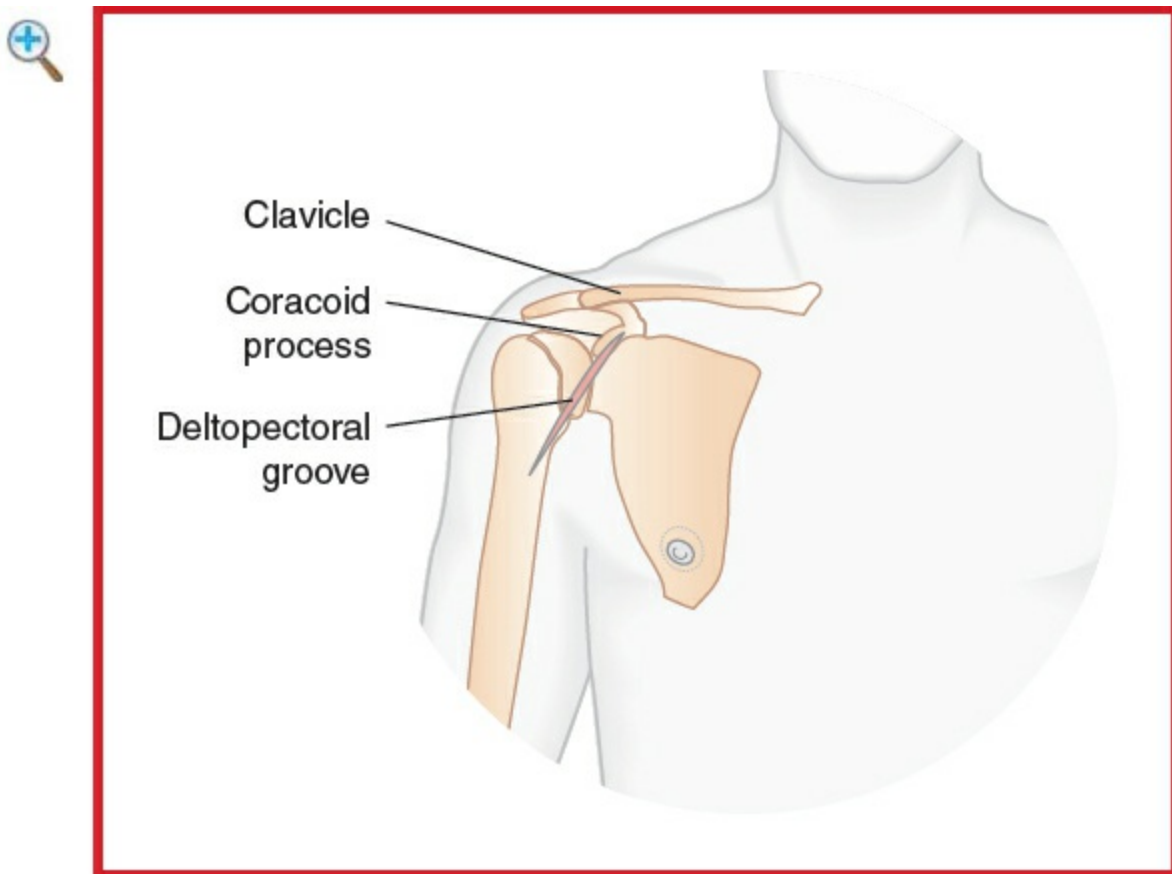


Figure 10.2-5 Incision in the deltopectoral groove. (Reproduced with permission from Hoppenfeld S, de Boer P: Surgical Exposures in Orthopaedics: The Anatomic Approach, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 1994.)

Usual preop diagnosis

Rheumatoid arthritis; posttraumatic arthritis; AVN; RC arthropathy; 4-part proximal humerus fracture

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plans (see [Appendix K, p. K-1](#)).

■ SUMMARY OF PROCEDURE

Position	Semisitting, beach chair
Incision	Deltopectoral incision (Fig. 10.2-5) or extended incision for complex revision
Special instrumentation	Glenoid and humeral instrumentation, in addition to usual shoulder instruments; mixing and introduction of cement
Unique considerations	Systemic illnesses of the patient; systemic complications of use of cement in the patient; precautions in pregnant staff working with methyl methacrylate; usage of laminar flow or UV lighting for total joint precautions, depending on the surgeon's preference and capabilities of the OR
Antibiotics	Cefazolin 2 g iv preop, and 48 h postop
Surgical time	2–5 h
Closing considerations	Drain; sling and swathe
EBL	200–1000 mL
Postop care	Immediate passive motion. No active internal rotation for 6 wk
Mortality	<1%
Morbidity	Blood loss Nerve injury: rare ↓ BP 2° cement: rare Infection
Pain score	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	45–80 yr
Male:Female	1:1
Incidence	Uncommon
Etiology	Osteoarthritis; inflammatory arthritis; trauma; avascular necrosis; systemic disease
Associated conditions	Inflammatory disease; systemic disease; alcoholism; RC pathology; arthritis and radiculopathy; cervical arthritis

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Typically, three patient populations present for shoulder arthroplasty: (a) healthy posttrauma, (b) nonrheumatoid arthritic, and (c) rheumatoid arthritic.

Respiratory	Rheumatoid arthritic patients may exhibit Sx of pleural effusion or pulmonary fibrosis. Hoarseness may indicate cricoarytenoid joint (CAJ) involvement → difficult intubation. (See Anesthetic Considerations for Wrist Procedures.) Tests: Consider CXR; PFTs; ABGs in debilitated rheumatoid patients.
Cardiovascular	Rheumatoid arthritic patients may suffer from chronic pericardial effusions, valvular disease, and cardiac conduction defects. Tests: Consider ECG and ECHO in patients with severe rheumatoid arthritis.
Neurologic	Arthritic patients may have cervical or lumbar radiculopathies that should be documented carefully preop. For example, head flexion may cause cervical cord compression. Tests: C-spine radiographs to r/o occult subluxations in rheumatoid patients with neck complaints or upper extremity radiculopathy
Musculoskeletal	Arthritic patients may have limited neck and jaw

	ROM that may require special intubation techniques. Bony deformities or muscle contracture may necessitate special attention to positioning.
Hematologic	Virtually, all nontrauma patients will be on some type of anti-inflammatory medication that may result in anemia or Plt inhibition. Ideally, patients should d/c NSAIDs at least 5 d preop. Immunosuppressive treatment for RA may ↑ risk of infection.
Endocrine	Rheumatoid patients may be on oral corticosteroids and may require supplemental perioperative steroids (e.g., 100 mg hydrocortisone q8h iv) to treat adrenal suppression, although the routine use of “stress-dose steroids” has been questioned.
Laboratory	Hb/Hct (healthy patients); other tests as indicated from H&P
Premedication	Mild-to-moderate premedication (e.g., in adults, midazolam 1–2 mg iv, fentanyl 50–100 mcg iv, titrated to effect) is often desirable before placement of a regional block.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, ± celecoxib 200 mg po, ± gabapentin 300 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA or a combination of GETA and regional anesthesia can be used. An interscalene brachial plexus block in combination with GA is excellent for surgical procedures on the shoulder. Unless contraindicated, a long-acting local anesthetic should be used in regional anesthesia for shoulder surgery to ameliorate postop pain.

General anesthesia

Induction	Standard induction. Rheumatoid patients may require awake fiberoptic intubation.
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ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Tranexamic acid (e.g., 10–15 mg/kg) iv before incision has been shown to ↓ EBL and may be requested by the surgeon.
Maintenance	Standard maintenance. Avoid N ₂ O in patients taking methotrexate. Because of the typically long duration of these cases, the opioid selected as part of the balanced anesthetic technique may be given by continuous infusion (e.g., iv sufentanil [0.25–1.0 mcg/kg/h]). Some surgeons prefer muscle relaxation beyond that provided by volatile anesthetic. The surgeon may request reduction in BP: maintain cerebral perfusion pressure.
Emergence	Management of emergence and extubation should be routine except in difficult airway cases that require awake extubation. Emergence from anesthesia should be delayed until patient's shoulder is securely immobilized in the sling and swathe to prevent undesired movement of the newly placed prosthesis. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include ondansetron and dexamethasone.

Regional anesthesia

Local anesthetics	Significant discomfort may be associated with this procedure; therefore, extended postop pain control is desirable. When a single-shot technique is used, 0.5% bupivacaine or 0.5% ropivacaine can be used. Onset is usually within 30 min, with duration up to 10–12 h. Ropivacaine may be preferred for peripheral nerve block, due to its decreased cardiotoxicity. For continuous catheter techniques, a postop infusion of 6–8 mL/h of 0.2% of any of these agents is appropriate.
Interscalene block	Typical anesthetics and doses

	0.5% bupivacaine or 0.5% ropivacaine, 20 mL. The skin on the top of the shoulder (C3-C4) and the medial aspect of the upper arm (T2) may require separate subcutaneous field blocks. Phrenic nerve block → hemidiaphragmatic paralysis is an inevitable consequence of the interscalene block, which may not be tolerated by patients with significant preexisting respiratory compromise. A continuous interscalene catheter (tunneled away from the surgical field) may be placed for postop pain management. Major complications, such as total spinal or pneumothorax resulting from interscalene block, are extremely rare. Interscalene block is contraindicated in patients with contralateral recurrent laryngeal nerve or phrenic nerve palsy. If sedation is needed, midazolam (0.5–1.0 mg boluses) and propofol (25–100 mcg/kg/min by infusion), titrated to effect, are good choices.	
Blood and fluid requirements	Moderate-to-significant blood loss IV: 16 ga × 1 BSS @ 1.5–3.0 mL/kg/h	Consider RBC recovery and reinfusion techniques (e.g., Cell Saver) if significant blood loss is anticipated. IV in nonoperative upper extremity. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Precordial Doppler	Consider invasive, hemodynamic monitoring in the debilitated or elderly patient. Because VAE is a possible complication of the semisitting position, consider using precordial Doppler for cases done in

Positioning	✓ and pad pressure points ✓ eyes ↑ VAE risk	this position. Postural ↓ BP is the most common complication of the semisitting position. Changing patient to this position gradually can help prevent ↓ BP, as can the use of antiembolism stockings and fluid loading the patient. Care with neck positioning. Check head position intraoperatively as shoulder traction may lead to injury.
Interscalene block complications	Total spinal Epidural anesthesia Local anesthetic toxicity (Sz/dysrhythmias) Stellate ganglion block (Horner syndrome) Laryngeal nerve block Phrenic nerve block Pneumothorax Persistent paresthesia	Resuscitative equipment, including airway management tools, should be immediately available. May last ≤6 wk
Other complications	Potential for embolic event ↓ BP during prep and positioning	Because of the increased risk of VAE during shoulder arthroplasty, N₂O may be d/c'd during placement of humeral component. Use of methyl methacrylate cement has been associated with the sudden onset of ↓ BP and even cardiac arrest, presumably due to profound vasodilation ±

associated VAE. ↓ BP during long surgical prep usually can be avoided by using light inhalation to ensure amnesia, with moderate muscle relaxation to prevent bucking on ETT, and maintaining adequate hydration.

POSTOPERATIVE

Multimodal pain management	PCA or regional block/catheter	Combined regional-general anesthetic techniques are excellent for shoulder procedures, especially for postop pain management. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics (e.g., continuous nerve block) whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6) Regional analgesia	Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	None indicated routinely	

Suggested Readings

1. Auyong DB, Allen CJ, Pahang JA, et al: Reduced length of hospitalization in primary total knee arthroplasty patients using an updated enhanced recovery after orthopedic surgery (ERAS) pathway. *J Arthroplasty* 2015; 30(10):1705–9.

2. Bohsali KI, Bois AJ, Wirth MA: Complications of shoulder arthroplasty. *J Bone Joint Surg Am* 2017; 99(3):256–69.
3. Borgeat A: Acute and nonacute complications associated with interscalene block and shoulder surgery. *Anesthesiology* 2001; 95(4):875–80.
4. Brolin TJ, Throckmorton TW: Outpatient shoulder arthroplasty. *Orthop Clin North Am* 2018; 49(1):73–9.
5. Conroy PH, Awad IT: Ultrasound-guided blocks for shoulder surgery. *Curr Opin Anaesthesiol* 2011; 24(6):638–43.
5. Pohl A, Cullen DJ: Cerebral ischemia during shoulder surgery in the upright position: a case series. *J Clin Anesth* 2005; 17(6):463–9.
7. Singhal K, Rammohan R: Going forward with reverse shoulder arthroplasty. *J Clin Orthop Trauma* 2018; 9(1):87–93.
8. Weinheimer KT, Smuin DM, Dhawan A: Patient outcomes as a function of shoulder surgeon volume: a systematic review. *Arthroscopy* 2017; 33(7):1273–81.
9. Zhu S, Qian W, Jiang C, et al: Enhanced recovery after surgery for hip and knee arthroplasty: a systematic review and meta-analysis. *Postgrad Med J* 2017; 93(1106):736–42.

SHOULDER GIRDLE PROCEDURES

SURGICAL CONSIDERATIONS

Description

Trauma about the shoulder girdle in young patients ranges from athletic injuries to life-threatening trauma. Some of these injuries include common athletic injuries, such as **acromioclavicular joint separations**, which rarely require surgery unless there are associated **acromial** or **clavicular fractures**. **Posterior sternoclavicular dislocations** may warrant surgical stabilization if the trachea is compressed. **Clavicle fractures**, frequently associated with **scapular fractures**, occasionally require open reduction.

Scapular fractures involving the glenoid also may require surgical stabilization. Extreme fractures involving the shoulder girdle (**scapulothoracic dissociations**) include scapular fracture, clavicle fracture, subclavian or axillary artery disruption, and brachial plexus injury. These may coexist with **proximal humerus fractures**, rib fractures, and pneumothorax. In the older, debilitated patient, the most common injury is proximal humeral fracture, which may be amenable to surgical stabilization or may be so comminuted as to warrant hemiarthroplasty.

A displaced proximal humerus fracture may require open reduction internal fixation

with a plate and screws or hemiarthroplasty through a deltopectoral approach utilizing a beach chair position (see Surgery for Shoulder Instability, and Glenohumeral Shoulder Arthroplasty). A displaced clavicle fracture may require open reduction internal fixation with a plate and screws utilizing a beach chair or supine position. A scapular body fracture would be stabilized with a plate and screws via a posterior approach utilizing a prone or lateral position (see Surgery for Shoulder Instability). As with other shoulder procedures, relaxation is necessary upon awakening the patient after the shoulder is placed into an immobilization device.

Usual preop diagnosis

Trauma about the shoulder girdle

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plans (see [Appendix K, p. K-1](#)).

■ SUMMARY OF PROCEDURES		
	Anterior	Posterior
Position	Semisitting or prone	Lateral decubitus (scapula)
Incision	Anterior, superior, or oblique for acromioclavicular; supraclavicular for clavicle; deltopectoral for proximal humerus and glenoid	Posterior lateral border or medial border of scapula, spinous scapula, depending on location
Special instrumentation	Plates and screws; tension band wiring; proximal humerus replacement for comminuted fractures	⇐
Unique considerations	Multiple traumas warrant early stabilization; may require vascular repair and brachial plexus exploration	⇐

Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	2–10 h	⇐
Closing considerations	Fracture dependent; most commonly requires application of sling and swathe	⇐
EBL	200–1200 mL or greater, depending on trauma	⇐
Postop care	May require ICU for multiple-trauma patients; otherwise, early mobilization with physical therapy	⇐
Mortality	Mortality dependent on associated conditions: Infection Neurologic injury Respiratory failure Massive blood loss Unrecognized pneumothorax Cardiac tamponade	⇐
Morbidity	Nerve injury (axillary, brachial plexus) Stiffness Poor healing	Axillary and suprascapular ⇐ ⇐
Pain score	6–10	6 (clavicle and AC joint) 8 (scapula and proximal humerus)

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–80 yr, depending on nature of trauma
Male:Female	5:1

Incidence	Common
Etiology	Trauma
Associated conditions	Axillary nerve palsy; musculocutaneous nerve palsy; brachial plexus injury; arterial disruption in high-energy trauma; brachial and great vessel injuries and posterior sternoclavicular dislocation; pneumothorax

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Brachial Plexus Surgery.

Suggested Readings

1. Cole PA, Gauger EM, Schroder LK: Management of scapular fractures. *J Am Acad Orthop Surg* 2012; 20(3):130–41.
2. Pandya NK, Namdari S, Hosalkar HS: Displaced clavicle fractures in adolescents: facts, controversies, and current trends. *J Am Acad Orthop Surg* 2012; 20(8):498–505.

BRACHIAL PLEXUS SURGERY

SURGICAL CONSIDERATIONS

Description

Brachial plexus injuries occur most commonly in two groups: traumatic birth injuries and high-energy trauma. Surgery ranges from exploration with neurolysis, to repairs, to cable nerve grafting. Typically, the latter requires grafting with the sural nerve and nerve pedicle transfer, such as transfer of the spinal accessory nerve to denervated paralyzed muscle, combined with muscle transfers. C5-C6 is most commonly injured in obstetrical (Erb) palsy. Injuries in adults are more commonly closed traction. Similar to obstetrical palsy, they occur with an outstretched, abducted arm with the neck rotated in the opposite direction. The most severe form includes complete avulsion at the preganglionic level, presenting with a Horner syndrome, winging of the scapula, and a flail arm. These are typically “supraclavicular” injuries and have a poorer prognosis. Surgical exposure may proceed above the clavicle similar to an anterior neck dissection or may require an extension below the clavicle. Occasionally, an osteotomy of the clavicle for extensive dissection is required. For axillary nerve dissection, a posterior approach also is used. Open injuries, such as gunshot or knife wounds, are typically

“infraclavicular” and have a better prognosis. (See diagram of brachial plexus, [Fig. 10.2-6](#).)

Usual preop diagnosis

Obstetrical palsy; adult trauma—most commonly motorcycle accident

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plans (see [Appendix K, p. K-1](#)).

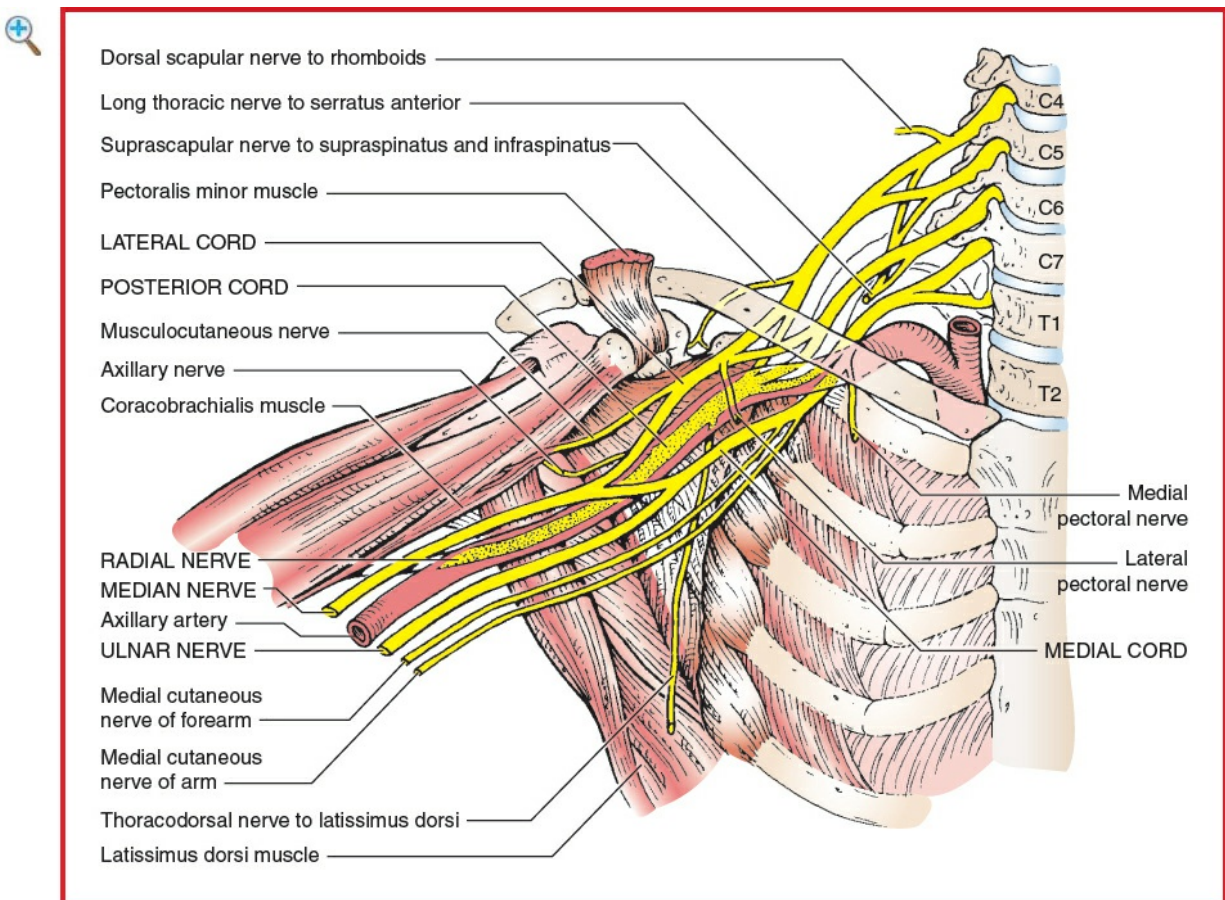


Figure 10.2-6 Brachial plexus: Its division into supraclavicular and infraclavicular portions is apparent. The proximal origin of the dorsal scapular nerve and the long thoracic nerve are demonstrated. The T1 spinal nerve arises below the head of the 1st rib. The relation of the cords of the plexus to the axillary artery at the level of the coracoid process (origin of the pectoralis minor and coracobrachialis muscles) is illustrated. The posterior cord and the radial nerve behind the axillary artery are stippled for clarity. (Reproduced with permission from Tindall GT, Cooper PR, Barrow DL. *The Practice of Neurosurgery*, Vol. III. Williams & Wilkins, Baltimore: 1996.)

■ SUMMARY OF PROCEDURE

Position

Lateral decubitus or semisitting

Incision	Supra- or infraclavicular, extensile to include deltopectoral incision. Clavicle osteotomy incision will provide for improved exposure. Supraclavicular incision used for supraclavicular brachial plexus
Special instrumentation	Nerve stimulator
Unique considerations	Associated trauma
Antibiotics	Cefazolin 2 g iv preop
Surgical time	4–10 h
Closing considerations	Other incisions if nerve grafts obtained
EBL	400–2000 mL
Mortality	Minimal
Morbidity	Bleeding Hematoma Pneumothorax Clavicular nonunion
Pain score	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Infants and children: 3 mo to 8 yr; adults: 20–40 yr
Male:Female	Adult: 5:1
Incidence	Infants: 0.3–8/1000 births; adults: commonly associated with motorcycle accidents
Etiology	Obstetrical palsy; trauma
Associated conditions	None known

Suggested Reading

1. Kaiser R, Waldauf P, Ullas G, et al: Epidemiology, etiology, and types of severe

adult brachial plexus injuries requiring surgical repair: systematic review and meta-analysis. Neurosurg Rev 2018.

ANESTHETIC CONSIDERATIONS

(Procedures covered: shoulder girdle procedures; brachial plexus surgery)

PREOPERATIVE

With the exception of traumatic birth injuries, most of these patients are healthy males who have suffered major blunt or penetrating trauma. For the acute and subacute trauma victim, the major anesthesia-related concerns center on associated traumatic injuries. Many adult trauma victims with brachial plexus injuries will be operated on in the first few days after their injury. For infants (usually operated on at 6–12 mo), the major anesthesia-related concerns are those routinely associated with pediatric anesthesia. Approximately half of all trauma victims are intoxicated. The anesthesia-related implications of ethanol intoxication include decreased anesthetic requirements, diuresis, vasodilation, and hypothermia.

Respiratory	As suggested by coexisting disease or acute trauma injuries. Look for evidence of occult chest injury, including pneumothorax (tachypnea, wheezing, ↓ BP, ↓ PaO ₂ , CXR changes) and pulmonary contusion (multiple rib fracture, ↓ PaO ₂). Tests: Consider CXR and ABGs in victims of significant trauma; other tests as indicated from H&P
Cardiovascular	As suggested by coexisting disease or acute trauma injuries. Look for evidence of occult cardiac or mediastinal injuries, such as myocardial contusion (e.g., ECG abnormalities typically consistent with ischemia) or great vessel rupture (e.g., widened mediastinum). Tests: Consider CXR (with NG tube in place to assess mediastinal widening) and ECG in victims of significant trauma; others as indicated from H&P
Neurologic	Victims of shoulder trauma are vulnerable to brachial plexus damage. Look for evidence of upper extremity nerve dysfunction and document any injuries preop.

	The possibility of closed-head injury also should be considered. Tests: Head CT prior to beginning a procedure under GA in a patient with evidence of head trauma
Musculoskeletal	As suggested by coexisting disease or acute trauma injuries. The amount of force necessary to produce a brachial plexus injury mandates a C-spine series to r/o C-spine fracture in all victims of brachial plexus trauma.
Laboratory	In general, most victims of significant trauma are best served by obtaining a wide variety of baseline lab studies to screen for unrecognized injury. These studies generally should include Hct, CBC, ABGs, UA, renal function tests, LFTs, and serum amylase.
Premedication	None
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, ± celecoxib 200 mg po, ± gabapentin 300 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA is preferred over regional techniques because of the unpredictable and prolonged length of these procedures and the need to evaluate brachial plexus function postop.

Induction	Rapid-sequence induction is mandatory in unscheduled cases, unless awake fiberoptic intubation is performed (see Anesthetic Considerations for Thoracolumbar Neurosurgical Procedures). C-spine fracture patients or those with facial injuries may require awake fiberoptic intubation or other special airway techniques, as indicated from H&P. Hemodynamically unstable, acute trauma patients can be induced more safely
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	with etomidate (0.3–0.4 mg/kg iv) or ketamine (1–3 mg/kg iv).	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Tranexamic acid (e.g., 10–15 mg/kg) iv before incision has been shown to ↓ EBL and may be requested by the surgeon.	
Maintenance	Balanced anesthesia with low-dose sevoflurane, iv fentanyl, and N ₂ O in O ₂ is suitable for stable patients. Hemodynamically unstable, acute trauma victims undergoing emergency surgery are not likely to tolerate this regimen and are better served by using a combination of medications designed to have minimal hemodynamic consequences (e.g., fentanyl for analgesia, rocuronium for muscle relaxation, and scopolamine or midazolam for amnesia). N ₂ O is best avoided in the trauma patient. For brachial plexus surgery, some surgeons prefer minimal muscle relaxation after tracheal intubation so that a nerve stimulator can be used to help identify surgical anatomy.	
Emergence	Management of emergence and extubation should be routine except in difficult airway or full-stomach cases, which require that extubation be delayed until the patient's airway reflexes have returned and the patient is fully awake. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include ondansetron and dexamethasone.	
Blood and fluid requirements	Significant blood loss IV: 14–16 ga × 1–2 BSS @ 1.5–3 mL/kg/h + replacement of blood loss @ 3 × volume Fluid warmer Airway humidifier	IV catheter placed in the nonoperative upper extremity is usually adequate in hemodynamically stable patients. Unstable, acute trauma victims require a minimum of 2 large-bore iv catheters or large-bore

		central lines. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors. ± SSEP ± TEE	Invasive hemodynamic monitoring and TEE should be considered in acute, multiple-trauma victims. Some surgeons request SSEP to make continuous assessment of preop intact brachial plexus possible. When using SSEP monitoring, high doses of volatile anesthetic agents should be avoided because they adversely affect SSEP readings.
Positioning	✓ and pad pressure points ✓ eyes VAE risk	Postural ↓ BP is the most common complication of the semisitting position, particularly during the surgical prep period. SCD or antiembolism stockings may be beneficial. VAE is a potential complication of this position.
Complications	Hemodynamic instability	Previously unrecognized injuries (e.g., pneumothorax, cardiac tamponade, intracranial bleeding) should be considered as a cause of unexplained intraop hemodynamic instability in all acute trauma

Possible VAE

victims.
VAE risk is increased with patient in semisitting position.

▼ POSTOPERATIVE

Complications	Sepsis ARDS	Many trauma victims survive the initial insult and subsequently develop sepsis or ARDS.
Multimodal pain management	PCA	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics (e.g., continuous nerve block) whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6) Regional analgesia	Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	Based on concurrent injuries	

Suggested Readings

1. Auyong DB, Allen CJ, Pahang JA, et al: Reduced length of hospitalization in primary total knee arthroplasty patients using an updated enhanced recovery after orthopedic surgery (ERAS) pathway. *J Arthroplasty* 2015; 30(10):1705–9.
2. Cao X, Pan F: Comparison of liposomal bupivacaine infiltration versus interscalene nerve block for pain control in total shoulder arthroplasty: a meta-analysis of randomized control trials. *Medicine (Baltimore)* 2017; 96(39):e8079.
3. Codding JL, Getz CL: Pain management strategies in shoulder arthroplasty. *Orthop Clin North Am* 2018; 49(1):81–91.
4. Koshy JC, Agrawal NA, Seruya M: Nerve transfer versus interpositional nerve graft reconstruction for posttraumatic, isolated axillary nerve injuries: a systematic

review. *Plast Reconstr Surg* 2017; 140(5):953–60.

5. Lee SK, Wolfe SW: Nerve transfers for the upper extremity: new horizons in nerve reconstruction. *J Am Acad Orthop Surg* 2012; 20(8):506–17.
6. Popinchalk SP, Schaffer AA: Physical examination of upper extremity compressive neuropathies. *Orthop Clin North Am* 2012; 43(4):417–30.
7. Zhu S, Qian W, Jiang C, et al: Enhanced recovery after surgery for hip and knee arthroplasty: a systematic review and meta-analysis. *Postgrad Med J* 2017; 93(1106):736–42.

ARM SURGERY

SURGICAL CONSIDERATIONS

Description

Surgical procedures on the arm are primarily for trauma or tumor surgery. Other procedures include extended approaches from the shoulder for significant trauma or tendon transfer. Exploration of peripheral nerves, most commonly of the radial nerve, is also included in this category, as are distal extensile approaches from the elbow for trauma or for lateral epicondylitis (“tennis elbow”). Depending on the lesion or fracture, the incision is developed through an internervous or intramuscular compartment. Procedures include **excisional biopsy** for soft tissue or bone tumors of the arm; **tumor excision**, which may be marginal, wide, or radical, depending on the tumor encountered; **tendon transfers**, such as pectoralis transfer to replace biceps function, used primarily for brachial plexus injuries; and **fractures and nonunion fractures of the humerus**. Positioning and location of incision are dependent on the level of the pathology. For example, for fractures involving the proximal half of the humerus, the standard **deltopectoral incision** may be extended distally along the interval between the biceps and triceps on the lateral aspect of the arm. This approach requires a beach chair ([Fig. 10.2-7](#)) or, occasionally, supine position. Distal third fractures are best approached posteriorly with the **triceps-splitting approach**. Distal fractures that extend into the elbow joint often require an olecranon osteotomy to visualize the fractured joint surface. Posterior approaches to the distal humerus are performed in either the lateral or prone position.

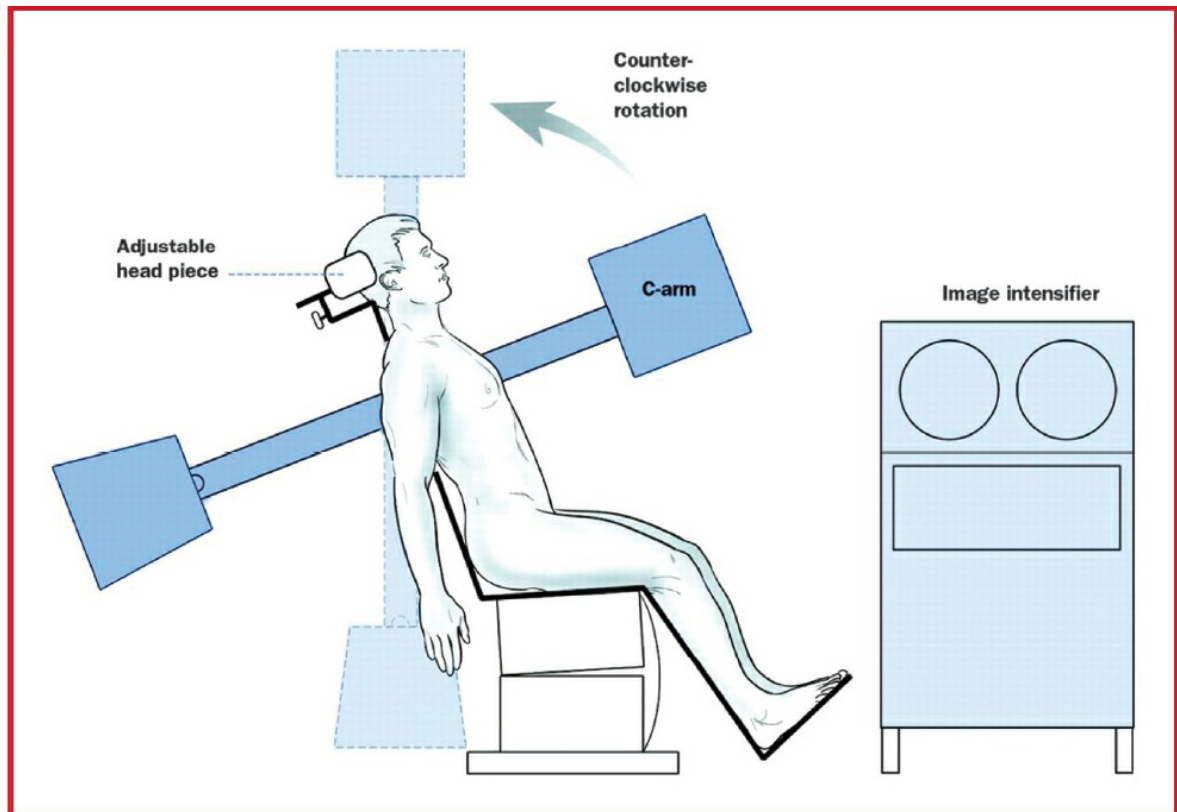


Figure 10.2-7 Beach chair positioning for shoulder surgery. Image intensification may be needed for intraoperative radiography (e.g., fracture surgery). (Reproduced by permission from Robinson CM, Page RS: Severely impacted valgus proximal humerus fractures. J Bone Joint Surg 2004; 86A:143–55.)

Usual preop diagnosis

Trauma; tumor

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plans (see [Appendix K, p. K-1](#)).

■ SUMMARY OF PROCEDURE

Position	Supine; semisitting position for extended deltopectoral; lateral decubitus or prone for distal humerus fractures
Incision	Anterolateral approach; posterior approach in the lateral decubitus or prone position
Special instrumentation	Plate and screws; external fixators; intramedullary rods; occasionally, methyl methacrylate cement for

	tumor surgery; sterile tourniquet for distal humerus; I.I. for fracture surgery
Unique considerations	CT-guided sclerotherapy preop for vascular tumors; longitudinal incisions for biopsy and tumor excisions (not violating fascial planes)
Antibiotics	Cefazolin 2 g iv preop
Surgical time	45 min to 6 h
Closing considerations	Drain frequently required
EBL	Minimal—500+ mL, depending on pathology
Mortality	Varies with pathology
Morbidity	Bleeding Shoulder stiffness Nerve injury
Pain score	4–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	Often relatively young
Male:Female	M > F
Incidence	Etiology dependent
Etiology	Fractures; nerve entrapment (radial) following trauma; tumors
Associated conditions	Radial or ulnar nerve injury with humeral fractures

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for arm procedures are often relatively young and otherwise healthy. Most of these patients present for elective repair of a traumatic injury. The preop workup is appropriate to patient's medical history. Some arm procedures, such as repair of a compound fracture, require immediate attention and necessitate emergency

surgery and full-stomach considerations.

Laboratory	Hb/Hct (healthy patients); other tests as indicated from H&P
Premedication	Mild-to-moderate premedication (e.g., in adults, midazolam 1–2 mg iv, fentanyl 50–100 mcg iv, titrated to effect) is often desirable before placement of a regional block.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, ± celecoxib 200 mg po, ± gabapentin 300 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

GA or regional anesthesia, or a combination of the two, can be used for surgical procedures on the arm. A brachial plexus block via the supraclavicular, infraclavicular, or axillary approach is excellent for procedures on the distal arm. The interscalene approach to the brachial plexus is suitable for more proximal humerus procedures. Regional anesthesia alone is a means of avoiding the risk of aspiration pneumonitis associated with GA in the patient with a full stomach requiring emergent surgery. Procedures longer than 3 h usually require general anesthesia in addition to regional. Regional anesthesia with sedation is usually well tolerated in shorter procedures.

General anesthesia

Induction	Standard induction except in acute trauma patients, where rapid-sequence induction is appropriate
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance
Emergence	Management of emergence and extubation should be routine, except in difficult airway cases, which

require awake extubation. Skin closure is frequently followed by application of a splint; patient should remain anesthetized during splinting procedure. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include ondansetron and dexamethasone.

Regional anesthesia

Local anesthetics

2% lidocaine or 1.5% mepivacaine have similar onset times (10–15 min) and similar duration (4–6 h). If extended postop pain control is desired, 0.5% bupivacaine or ropivacaine can be used. Onset is usually within 30 min, with duration of 8–12 h. If GA is planned, 0.25% ropivacaine or bupivacaine is sufficient.

Brachial plexus block

For proximal humerus surgery, an interscalene block is indicated with a catheter for postop pain management. Supraclavicular, infraclavicular, and axillary approaches are all suitable for more distal surgery. Supraclavicular or infraclavicular blocks are better tolerated in fracture patients, because the arm need not be moved. Ultrasound guidance can improve speed and comfort during block placement. A peripheral nerve catheter (infraclavicular approach) may be placed to prolong pain relief in appropriate patients.

- 2% lidocaine or 1.5% mepivacaine 30 mL for procedures lasting ≤ 2.5 h
- 0.5% bupivacaine or ropivacaine 30 mL for procedures lasting > 2.5 h

If sedation is needed, midazolam (0.5–1.0 mg boluses) or propofol (25–100 mcg/kg/min by infusion), titrated to effect, are good choices. Sensation to the skin of the upper inner arm is supplied by the intercostobrachial nerve (T2), which must be blocked separately.

Suggested Viewing

Links are available online to the following videos:

Shoulder Arthroscopy, Removal of Bone Spur (Subacromial Decompression):

<https://www.youtube.com/watch?v=NF1s1MYim30>

Brachial Plexus—Structure and Location—3D Anatomy Tutorial: https://www.youtube.com/watch?v=hrKesc_XSzo

Ultrasound-Guided Interscalene Brachial Plexus Nerve Block: <https://www.youtube.com/watch?v=Dg9BJ-32yLc>

Ultrasound-Guided Infraclavicular Brachial Plexus Nerve Block: <https://www.youtube.com/watch?v=hRyDtXrfYqc>

Ultrasound-Guided Supraclavicular Brachial Plexus Nerve Block: <https://www.youtube.com/watch?v=ztOIvfjsB-U>

CHAPTER 10.3

Spine Surgery

SURGEONS

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MINIMALLY INVASIVE POSTERIOR LUMBAR DISCECTOMY (MICRODISCECTOMY)

SURGICAL CONSIDERATIONS

Description

Since the mid-1990s, a number of techniques have been developed to allow the decompression of lumbar roots (removal of disc material) with as little trauma to the nerves and surrounding tissues as possible. In most instances, little or no bone is removed, and therefore, this is not technically a laminectomy or laminotomy. These minimally invasive procedures typically are carried out in healthy young or middle-aged adults with sciatica and are not done for more involved pathology, such as deformity, tumor, or infection. Transpedicular fixation and short-segment fusions may be attempted using modifications of these techniques.

Microdiscectomy approach

This can be done under general, regional (epidural or spinal), or local anesthesia. The patient is placed in a prone or kneeling position, and the posterior landmarks are palpated to identify the approximate level (e.g., L4-L5); then the overlying skin is

infiltrated with local anesthetic. A spinal needle is placed to the level of the lamina, and an x-ray or fluoroscopic image is taken to confirm the level. A 1-inch incision is made over the proposed interspace, and using either traditional or specialized retractors, the soft tissue is displaced to expose the ligamentum flavum. With the use of an operating microscope, the ligamentum flavum is removed, the nerve retracted, and the extruded disc excised. For a single level, this should take between 30 and 90 min, depending on the size of the patient and whether there is any scarring or adhesions from previous surgery.

Variant approach

Percutaneous discectomy through a posterolateral approach is usually reserved for “contained discs”—protrusions into, but not through, the outer annulus of the disc. These are usually done under MAC with local anesthetic. The percutaneous instruments may be positioned using fluoroscopic guidance with or without a fiberoptic light source and camera/monitor setup. The disc space is entered posterolaterally. The surgeon usually avoids anesthetizing the area around the nerve root so that the patient can alert the team if the root is struck by an instrument (quite painful). After the disc space is entered, fluoroscopic or camera images are used to guide the surgeon in the removal of the herniated disc. The disc material can be removed with specialized grabbers or automatic power-driven shavers.

Usual preop diagnosis

Chronic back pain 2° herniated lumbar disc; lumbar radiculopathy

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain meds, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, carb loading, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Microdiscectomy	Percutaneous Discectomy
Position	Prone or kneeling, with bolster or frame support. The abdomen must hang free to decompress the epidural veins.	Prone or lateral decubitus
Incision	Posterior midline or	About 8–12 cm lateral to

	slightly off midline, at the appropriate vertebral level	the midline at the appropriate vertebral level
Special instrumentation	Microscope; light source; specialized retraction and dissection instruments for working in a small, deep incision	Percutaneous instruments, including trocars, sounds, and arthroscopic-type grabbers and shavers; fluoroscopy; and, sometimes, camera/monitor and fiberoptic light setup
Unique considerations	Often outpatient procedure. Need to make room for microscope at head of bed	Patient must be alert enough to respond to pain if nerve roots are encountered.
Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	0.5–1.5 h	1–2 h
Closing considerations	Minimal suturing	Usually no closure (Band-Aids)
EBL	25–100 mL	Minimal
Postop care	PACU. Mobilization as soon as tolerated Usually discharged within 24 h	PACU → home. Mobilization as tolerated
Mortality	Very rare	⇐
Morbidity	Nerve injury Dural laceration Infection	Failure to decompress the nerve adequately — ⇐
Pain score	4	2

■ PATIENT POPULATION CHARACTERISTICS

Age range	16–60 yr
Male:Female	3:2

Incidence	Common
Etiology	Degenerative; trauma (rare)

MINIMALLY INVASIVE ANTERIOR LUMBAR INTERBODY FUSION THROUGH A TRANSPSOAS APPROACH

SURGICAL CONSIDERATIONS

Description

Traditional open techniques of lumbar fusion have shown long-term clinical efficacy. However, the morbidity of the approach, extensive blood loss, and length of postop recovery can offset the benefits of the intervention. Recently, minimally invasive techniques have decreased morbidity. One approach involves lateral approach via a small incision through retroperitoneal fat and the psoas major muscle for access to the lateral lumbar spine with the aid of neuromonitoring to avoid nerves of the lumbar plexus.

The orientation of neural structures of the lumbar plexus has a relatively high level of variability, necessitating the use of neuromonitoring for this approach. After induction of GETA, the surgeon places needle electrodes into muscles of the bilateral lower extremities to record the EMG response to lumbar nerve stimulation. The EMC threshold is the current (mA) required to stimulate the nerve, producing a muscle contraction. Direct nerve stimulation studies have shown that normal nerves elicit an EMG response with an average of 2 mA. Neuromuscular blockade should be avoided and the use of TIVA is encouraged. If neuromuscular blockade is required for intubation, succinylcholine should be used. If rocuronium is used, it is imperative that sugammadex be available for rapid reversal to permit adequate EMG monitoring. In either situation, use of a paralytic should be discussed with the surgeon and adequate reversal confirmed prior to the start of the procedure.

The patient is positioned in the lateral decubitus position and held in position with straps and tape. The arms should remain in a neutral position, and an axillary roll is placed just inferior to the axilla to support the area of the upper rib cage to relieve pressure on the axillary nerve and artery. The table is typically flexed to increase the distance between the iliac crest and the rib cage to gain access to the lateral lumbar

spine. Fluoroscopic imaging is required to determine true anterior/posterior and lateral axes of the desired spinal level. After the patient has been prepped and draped, an incision is made posteriorly at the border between the erector spinae muscles and the abdominal obliques to accommodate the surgeon's index finger and help identify the retroperitoneal space. With care, perforation of the peritoneum is avoided, and the finger is used to sweep this anteriorly. After identifying the psoas muscle, the index finger is swept up to a previously marked direct lateral target over the center of the affected intervertebral segment. An incision is made on the patient's side, through which dilators are placed, and the index finger is used to escort them safely to the psoas muscle, all while protecting the intra-abdominal contents (see [Fig. 10.3-1](#)). Careful passage of the dilators and EMG neuromonitoring avoid injury to the lumbar plexus. Once the self-retaining retractor system is positioned, care is taken to determine the course of the lumbar plexus in relation to the blades of the retractor. Once the retractor is positioned over the disc space, under direct vision, a discectomy is performed. Care is taken not to disrupt the anterior longitudinal ligament and risk injury to the main vascular structures coursing anteriorly.

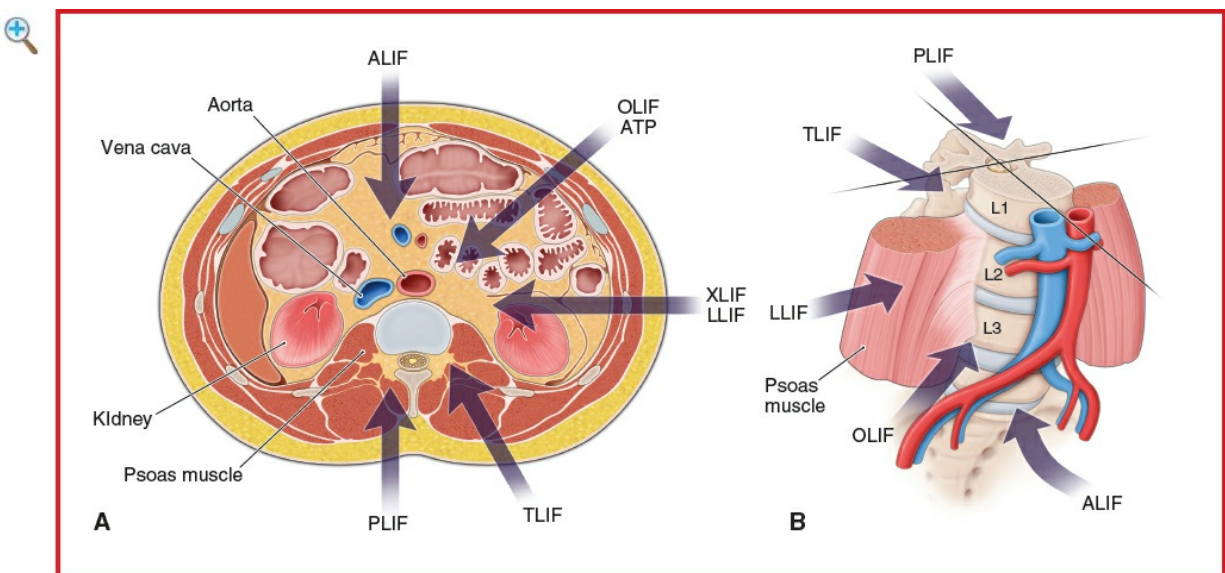


Figure 10.3-1 A and B: Surgical approaches to the lumbar spine for interbody fusion. ALIF, anterior lumbar interbody fusion; LLIF or XLIF, lateral or extreme lateral lumbar interbody fusion; OLIF/ATP, oblique lumbar interbody fusion/anterior to psoas; TLIF, transforaminal lumbar interbody fusion; PLIF, posterior lumbar interbody fusion.

Usual preop diagnosis

Chronic back pain; lumbar radiculopathy

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain meds, limiting NPO duration, alcohol and smoking

cessation, carb loading, postop pain management options, and possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Lateral decubitus with axillary roll. Table flexed
Incision	Posterior border between erector spinae and abdominal oblique muscles
Special instrumentation	Fluoroscopic imaging
Unique considerations	EMG monitoring: avoid muscle relaxation
Antibiotics	Cefazolin 2 g iv
Surgical time	2–3 h
Closing considerations	Minimal suturing
EBL	20–100 mL
Postop care	PACU. Mobilization as tolerated. Usually discharged within 24 h
Mortality	Rare
Morbidity	Reported complication rates: 6.2–52% Iliopsoas weakness Anterior thigh numbness Quadriceps weakness Radiculopathy Foot drop Durotomy
Pain score	4–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	30–80 yr
Male:Female	3:2
Incidence	Increasingly common

Suggested Readings

1. Adamus M, Hrabalek L, Wanek T, et al: Intraoperative reversal of neuromuscular block with sugammadex or neostigmine during extreme lateral interbody fusion, a novel technique for spine surgery. *J Anesth* 2011; 25(5):716–20.
2. Arnold PM, Anderson KK, McGuire RA Jr: The lateral transpsoas approach to the lumbar and thoracic spine: a review. *Surg Neurol Int* 2012; 3(Suppl 3):S198–215.
3. Rodgers WB, Gerber EJ, Patterson J: Intraoperative and early postoperative complications in extreme lateral interbody fusion: an analysis of 600 cases. *Spine (Phila PA 1976)* 2011; 36(1):26–32.
4. Tohmeh AG, Rodgers WB, Peterson MD: Dynamically evoked, discrete-threshold electromyography in the extreme lateral interbody fusion approach. *J Neurosurg Spine* 2011; 14(1):31–7.
5. Xu DS, Walker CT, Godzik J, et al: Minimally invasive anterior, lateral, and oblique lumbar interbody fusion: a literature review. *Ann Transl Med* 2018; 6(6):104.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients undergoing minimally invasive spine procedures are no more likely than the general population to have any specific comorbidity; they require normal preop evaluation.

Musculoskeletal

Because these patients may have chronic back pain with radiculopathy, they may not be suitable candidates for regional anesthetic techniques. Postop exacerbation of symptoms may be incorrectly ascribed to the anesthetic technique. A careful motor and sensory evaluation of the lower extremities should be documented. Patients commonly taking centrally acting analgesics or narcotics may require higher doses of sedative-hypnotics and analgesics in the periop period.

Hematologic

Patients should stop taking aspirin or NSAIDs at least 1 wk before surgery. In addition, an INR or PT and PTT should be checked preop in the presence of anticoagulation or liver disease.

	Tests: Hct; Plt; INR
Laboratory	Other tests as indicated from H&P
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon possible use of po celecoxib 400 mg po as a component of postop pain management. Consider preop acetaminophen 1000 mg po and, for patients with h/o severe PONV, aprepitant 40 mg po.

INTRAOPERATIVE

Anesthetic technique

Minimally invasive spine procedures are commonly done under GA; however, local or regional anesthetic techniques are suitable in selected patients. Percutaneous discectomies may require only MAC with sedation. These patients must be awake to alert the surgeon to inadvertent nerve root contact. In some centers, regional anesthesia (spinal or epidural) is the anesthetic of choice for elective uncomplicated spine surgery.

General anesthesia

Induction	Standard induction. Consider using wire-reinforced ETT to prevent kinking, with patient in prone position.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Dexamethasone (4–8 mg iv) may ↓ PONV. Sequential pneumatic compression devices should be used to ↓ VTE risk. Consider ketamine (0.25 mg/kg/h) and/or lidocaine (1–2 mg/kg/h) infusions as components of MMPM especially in chronic pain patients.
Maintenance	Standard maintenance. Usually 1- to 2-h operation. After exposure, further muscle relaxant use is unnecessary.
Emergence	No special considerations. Suggest wound infiltration

with LA. Consider iv hydromorphone (0.2–0.6 mg iv).

MAC

Regional anesthesia

Although theoretically advantageous, regional anesthesia is not commonly used for spinal surgery.

Spinal	Patient in sitting, lateral decubitus, or prone position for placement of subarachnoid block. Doses of local anesthetics should be adequate to provide a high lumbar level of sensory anesthesia (e.g., bupivacaine 6–10 mg with fentanyl 10 mcg).	
Epidural	Patient in sitting or lateral decubitus position for placement of epidural catheter. A test dose (e.g., 3 mL of 1.5% lidocaine with 1:200,000 epinephrine [5 mcg/mL]) is administered, and the patient is observed for development of subarachnoid block or Sx of an intravascular injection. Titrate lidocaine 2% with epinephrine (3–5 mL at a time) until desired surgical level is obtained.	
Blood and fluid requirements	IV: 18–20 ga × 1 BSS @ 5 mL/kg/h	Blood not likely to be required. Zero balance or Goal-Directed Fluid Management is recommended.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points	Lateral position: use axillary roll Prone position: Jackson table, Wilson frame, or bolsters to support shoulders/hips and optimize ventilation Knee-chest position: Andrews table

✓ **eyes**

Place head in cushioned holder with cutout for eyes, nose, and chin (e.g., ProneView®). Make certain eyes, nose, and chin are free and clear. Ensure that there is no posterior pressure on the ET tube, as this may result in injury to the uvula or soft palate. Pad elbows, knees, and other pressure points. Limit both upper arm abduction and elbow extension to <90°.

 POSTOPERATIVE

Complications	Urinary retention in the older patient Transient numbness/paresthesias, weakness	Rx: Consider catheterization. Due to nerve root irritation from operation. Rx with analgesics and/or muscle relaxants methocarbamol, cyclobenzaprine
Multimodal pain management	PCA Epidural analgesia Acetaminophen 1 g iv ± Fentanyl 25–50 mcg prn ± Ketorolac (30 mg iv) or celecoxib (200 mg po)	PO analgesics may be suitable: acetaminophen and codeine (Tylenol no. 3 1–2 tab q4–6 h) or oxycodone and acetaminophen (Percocet 1 tab q6h) Dose prior to the removal of an epidural catheter at the end of the case if it is already in place If not given preop Consult with surgeon regarding possible hemostasis concern with NSAIDs.
ERAS	Ondansetron 4 mg iv. ± Metoclopramide 10–20	Encourage early enteral nutrition, Foley catheter

	mg iv	removal, and mobilization
Tests	As indicated by patient status	with physiotherapy. Patient usually discharged from hospital within 24–48 h

Suggested Readings

1. Nielsen RV, Fomsgaard JS, Siegel H, et al: Intraoperative ketamine reduces immediate postoperative opioid consumption after spinal fusion surgery in chronic pain patients with opioid dependency: a randomized, blinded trial. *Pain* 2017; 158(3):463–70.
2. Wainwright TW, Immins T, Middleton RG: Enhanced recovery after surgery (ERAS) and its applicability for major spine surgery. *Best Pract Res Clin Anaesthesiol* 2016; 30(1):91–102.

SPINAL RECONSTRUCTION AND FUSION: THORACIC AND THORACOLUMBAR SPINE

SURGICAL CONSIDERATIONS

Description

Traditionally, most spinal procedures have been approached posteriorly. The advent of surgical treatment for vertebral TB and postpolio spinal deformities during the 1960s saw the development of surgical approaches to the anterior spine. These procedures initially were reserved for patients with significant deformities, especially kyphosis. More recently, the treatments of traumatic, neoplastic, and degenerative conditions have been included in the anterior approach. Regardless of the condition under treatment, the approach is similar for a given level. There are several more or less distinct types of surgical exposures, depending on the level.

Cervicothoracic approach

Most cephalad and difficult is the approach to the upper thoracic spine (T1-T3). This generally includes a modified anterior cervical exposure with a caudal extension, including a resection of the clavicle, part of the manubrium, and sometimes the rib at the thoracic outlet. Dangers in this exposure are to the great vessels at the thoracic outlet, trachea (rare) and esophagus (more common), lung parenchyma, sympathetic ganglia, lymphatic duct (on the left), and brachial plexus. Once the spine is exposed and the

discs and/or vertebrae are removed, the spinal cord is at risk. This procedure occasionally involves entering the thoracic cavity, in which case it is usually done intrapleurally—that is, through the parietal pleura. The lung needs to be collapsed at least partially. Spinal cord monitoring is usually performed; wake-up tests are not. Manipulation of the carotid artery and aortic arch may cause wide HR and BF fluctuations.

Transthoracic approach

Further down the spine, the levels from T5 to T10 are more easily reached via a transthoracic approach (Fig. 10.3-2). This involves a typical thoracotomy with the resection of a rib. The level of the rib resection is usually 1–2 levels above the highest vertebral level being approached. The great vessels and lung parenchyma are at risk, as is the thoracic duct (on the left). The patient is in the lateral decubitus position, and the mediastinum and heart usually fall to the opposite side, out of harm's way. Risk to the spinal cord depends on the difficulty and extent of the vertebral disease and the reconstruction. Spinal cord monitoring usually is performed intraop. The need for the lung to be deflated varies with the extent of the exposure. In centers where this procedure is frequently performed and the surgeons are accustomed to the respiratory motion during operation, DLTs are not routinely used. Because there is no (intended) violation of the lung parenchyma, air leaks and parenchymal repairs are not common.

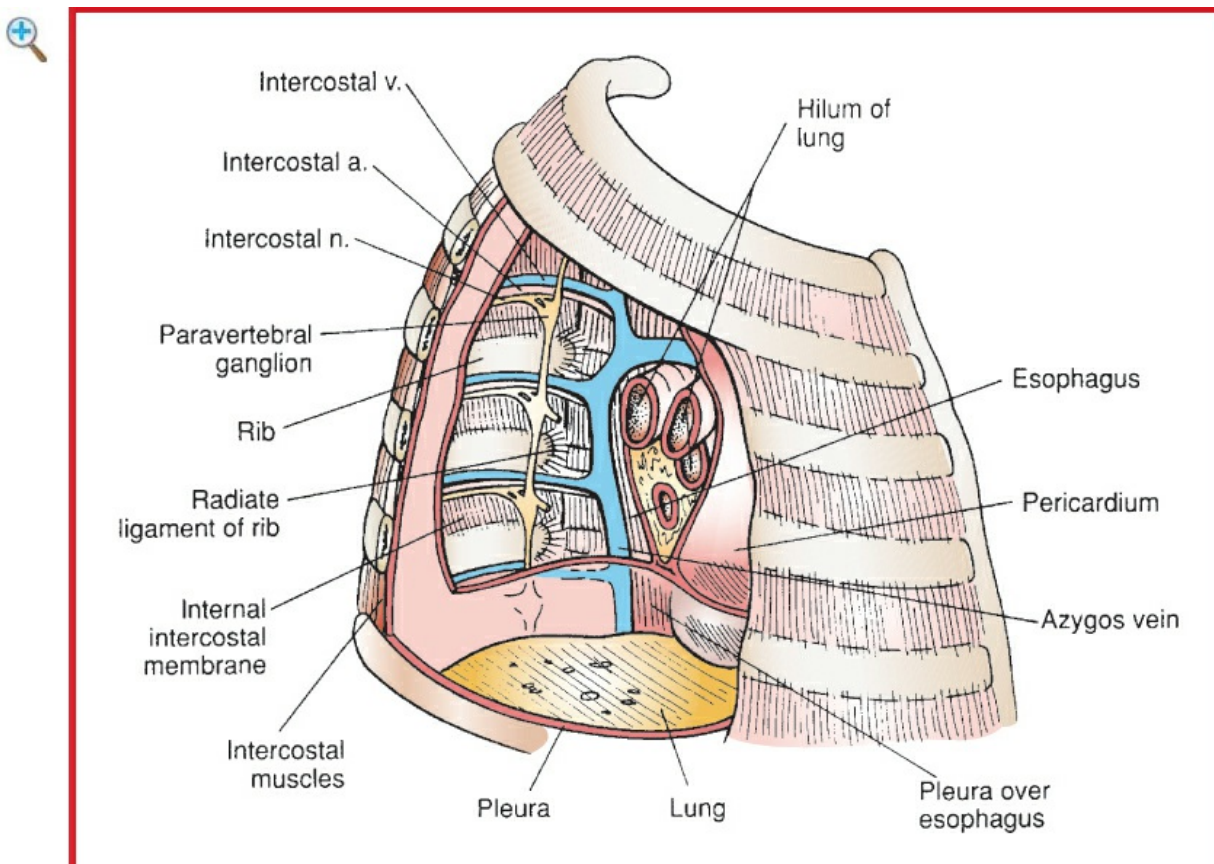


Figure 10.3-2 Surgical anatomy of the transthoracic approach. (Reproduced with permission from Hoppenfeld S, deBoer P: *Surgical Exposures in Orthopaedics*. JB Lippincott, Philadelphia: 1984.)

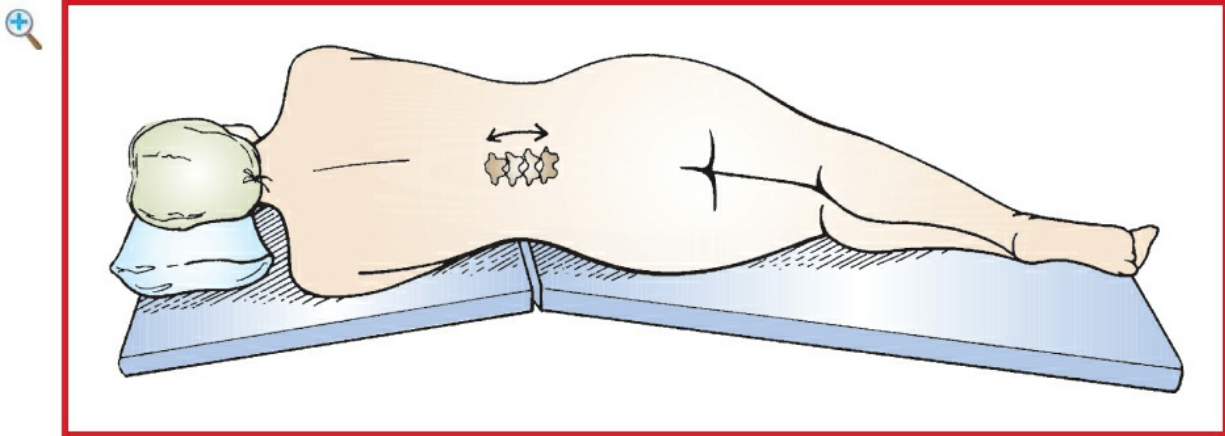


Figure 10.3-3 Patient position for transdiaphragmatic or retroperitoneal approach. (Reproduced with permission from Hoppenfeld S, deBoer P: *Surgical Exposures in Orthopaedics* JB Lippincott, Philadelphia: 1984.)

Transdiaphragmatic approach

When the exposure must transverse the diaphragm, a combined retroperitoneal and transthoracic approach is used. This requires the diaphragm to be sectioned circumferentially from the chest wall and spine. If only the very low segments of the thoracic spine (T10-T12) are exposed, the required deflation of the involved lung is minimal. The risks are the same as those encountered with the transthoracic or retroperitoneal approaches alone. Regardless of the level of exposure, the operating table may be used during the procedure to manipulate the spine for better exposure and to “lock in” implants, bone grafts, etc. Usually, the area of the spine to be exposed is centered above the “breaking” joint and kidney rests of the table ([Fig. 10.3-3](#)). After the initial exposure, the table is angled in the center with the head and legs pointing down and kidney rests elevated to “open up” the section of spine facing the surgeon. After removal of the disc, abscess, or tumor, a reconstruction—using bone graft, metal implants, bone cement, or a combination of these—is performed. In some cases, the operating table is straightened and, with the spine in neutral alignment, the stability of the reconstruction is tested.

The **morbidity** of these procedures depends primarily on the nature of the underlying disease. Obviously, bleeding and visceral injury are more likely when debriding a grapefruit-sized Potts abscess with several destroyed vertebrae than in removing a degenerated lumbar disc for fusion.

In some instances, the anterior procedure may be followed with a posterior fusion,

either immediately, the next day or after 5–7 d of convalescence. If done immediately after, the patient needs to be placed in the prone position, and the second procedure needs to be done through a midline exposure. Sometimes, anterior and posterior surgeries are performed simultaneously by two surgical teams. The most common reason for a staged anterior/posterior procedure (in the United States) is scoliosis; however, fractures at the thoracolumbar junction, after anterior decompression and reconstruction, are often instrumented and fused posteriorly.

Usual preop diagnosis

Fractures (usually at thoracic lumbar junction); idiopathic scoliosis; primary neoplasm or metastatic disease to the spine; pyogenic or TB osteomyelitis of the spine; Scheuermann kyphosis

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain meds, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, possibility of urinary catheter postop and rehabilitation plans. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Cervicothoracic Approach	Transthoracic Approach	Transdiaphragmatic Approach
Position	Supine; towel roll placed between shoulder blades, head slightly extended and turned away from operated side	Lateral decubitus (Fig. 10.3-3); intended segments over the table break and kidney rests; axillary roll	⇐
Incision	Inverted “L” longitudinally along manubrium to sternal notch, then transversely above clavicle	Along a rib, two levels above the highest segment to be exposed	⇐
Special	Instrumentation rarely	For	⇐ (More common to

instrumentation	used. Strut grafts—using rib, fibula, clavicle or cement—may be used to replace excised vertebrae. ± DLT	thoracolumbar scoliosis, a Zielke-type rod and screws may be used; more rigid instrumentation sometimes used in fractures. ± DLT	use instrumentation at the affected level than above)
Unique considerations	Aortic arch and carotid manipulation may cause BP/HR changes. Postop respiratory distress well described	During spinal reconstruction, manipulation of operating table may be essential to “lock in” graft or implant (see above).	⇐
Antibiotics	Cefazolin 2 g iv (+ gentamicin 80 mg iv, if indwelling bladder catheter)	⇐	⇐
Surgical time	2–6 h	⇐	⇐
Closing considerations	Patient usually transferred to bed prior to emergence; sudden jerking motions may dislodge graft or implant.	⇐	⇐
EBL	200–5000 mL. Blood loss is extremely variable; when bleeding occurs, it may be torrential	⇐	⇐

	from the aorta, vena cava, or iliac vessels and branches. In nontumor or infection cases, 200–400 mL is usual.		
Postop care	Chest drain; NG suction usually needed; short period of ICU observation is usual.	⇐	⇐
Mortality	<0.1%, except in cases of malignancy or sepsis	⇐	⇐
Morbidity	For elective degenerative cases: 5–10% overall	⇐	⇐
	DVT: 6%	⇐	⇐
	Neurologic: 3%	⇐	⇐
	Infection: 1%	⇐	⇐
	Sexual dysfunction	⇐	⇐
	For sepsis or tumor: 50–80% (overall)	⇐	⇐
	Cardiorespiratory failure	⇐	⇐
	Sepsis	⇐	⇐
Pain score	7–8 (if patient sensate at level of surgery)	7–8 (if patient sensate at level of surgery)	7–8 (if patient sensate at level of surgery)

■ PATIENT POPULATION CHARACTERISTICS

Age range	12–30 yr (scoliosis surgery); >40 (tumor and infection surgery); 15–35 yr (fractures)
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Male:Female	1:1, except more scoliosis surgery in females (1:4) and more fractures in males
Incidence	20,000/yr
Etiology	Scoliosis: idiopathic (50%), neuromuscular (15%), congenital (5%); trauma (20%); infections, tumors (10%)
Associated conditions	Pulmonary HTN and impaired pulmonary function; neuromuscular scoliosis (poliomyelitis, CP, muscular dystrophy, Friedreich ataxia); aspiration; cardiomyopathy

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Spinal Reconstruction and Fusion.

Suggested Readings

1. Chrastil J, Patel AA: Complications associated with posterior and transforamina lumbar interbody fusion. J Am Acad Orthop Surg 2012; 20(5):283–91.
2. Jain AK: Treatment of tuberculosis of the spine with neurologic complications. Clin Orthop 2002; 398:75–84.
3. Taft E, Francis R: Evaluation and management of scoliosis. J Pediatr Health Care 2003; 17(1):42–4.
4. Tay BK, Deckey J, Hu SS: Spinal infections. J Am Acad Orthop Surg 2002; 10(3):188–97.
5. Yeo JM, Vertinsky AT, Chew JB, et al: Imaging in adult scoliosis: preoperative assessment and postoperative complications. Semin Musculoskeletal Radiol 2011; 15(2):143–50.

SPINAL RECONSTRUCTION AND FUSION: LUMBOSACRAL SPINE

SURGICAL CONSIDERATIONS

Description

The same general considerations apply here as in thoracolumbar reconstruction segments. The thoracic cavity is not entered, nor is the diaphragm sectioned. Careful

preop assessment is needed, as these patients may have a wide range of coexisting diseases. The same operative procedure is performed for removal of a degenerative disc in a healthy patient as is carried out to decompress the cauda equina in a debilitated patient with metastatic breast carcinoma.

Retroperitoneal approach

Below the diaphragm, exposure of the lumbar spine (L2-S1) can be performed through a retroperitoneal approach (Fig. 10.3-4). This often involves a flank incision, often with resection of the 11th or 12th rib. The patient lies in a decubitus or partial decubitus position. At risk here are the great vessels above and below the bifurcation of the aorta (L4-L5). The ureter crosses the operative field and must be identified and protected. The sympathetic chain may be damaged along the vertebrae, but the consequences of this are minimal. The presacral plexus further down may be injured and result in persistent retrograde ejaculation.

A Pfannenstiel incision may be used to approach L5-S1 or L4-L5. Regardless of the level of exposure, the operating table is used during the procedure to manipulate the spine for better exposure and to “lock in” implants, bone grafts, etc. Usually, the area of the spine to be exposed is centered above the “breaking” joint and kidney rests of the table (Fig. 10.3-3). After initial exposure, the table is angled in the center with the head and legs pointing down and the kidney rests elevated to open up the section of spine facing the surgeon. After the removal of the disc, abscess, or tumor, a reconstruction—using bone graft, metal implants, bone cement, or a combination of these—is performed. The table is then straightened; with the spine in neutral alignment, the stability of the reconstruction is tested by this maneuver, which may need to be repeated several times.

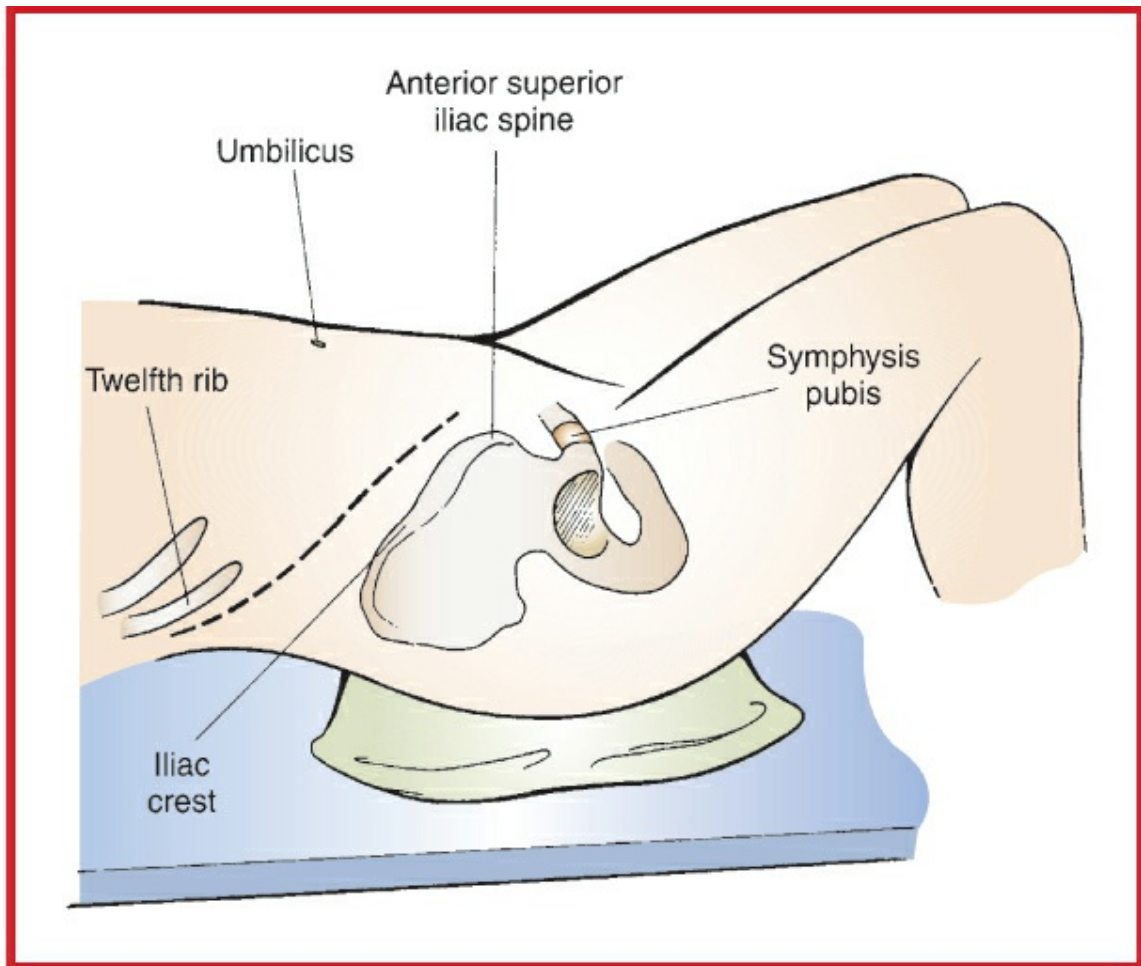


Figure 10.3-4 Retroperitoneal approach to lumbar spine. (Reproduced with permission from Hoppenfeld S, deBoer P: Surgical Exposures in Orthopaedics. JB Lippincott, Philadelphia: 1984.)

In some instances, the anterior reconstruction and fusion is followed with a posterior fusion, either immediately, the next day or after 5–7 d of convalescence. If done immediately after, the patient needs to be positioned prone on the operating table and the second procedure done through a midline exposure. Sometimes, anterior and posterior surgeries can be performed simultaneously by two surgical teams. The most common reason for a staged anterior/posterior procedure (in the United States) was scoliosis. Recent trends in degenerative lumbar disc surgery indicate that anterior and posterior fusion has become more common with some evidence suggesting better results in fusion and pain relief.

Variant procedure or approaches

When the L5 vertebra and sacrum need to be exposed widely, a transperitoneal approach may be needed. This involves a laparotomy or Pfannenstiel incision, displacement of the bowels out of the pelvis, and exposure of the lumbosacral junction (Fig. 10.3-5). The patient is supine for this procedure, and the surgical risks are similar, as with other intra-abdominal approaches.

In a recent modification to the retroperitoneal approach, the patient is placed in the supine position. In this supine retroperitoneal approach, an incision is made in either a longitudinal or transverse fashion between the umbilicus and the pubis. The rectus abdominis fascia is incised and the rectus abdominis is retracted medially or laterally, allowing access to the retroperitoneal space without violating the peritoneal cavity. The advantage of this approach over the transperitoneal approach includes decreased need for bowel manipulation, → ↓ third-space loss of fluid and ↓ heat loss.

Usual preop diagnosis

Degenerative disc disease; segmental instability; vertebral fractures requiring decompression; vertebral osteomyelitis or tuberculosis; neoplastic disease of the lumbar spine

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain meds, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, possibility of urinary catheter postop and rehabilitation plans. See [Appendix K, p. K-1](#).

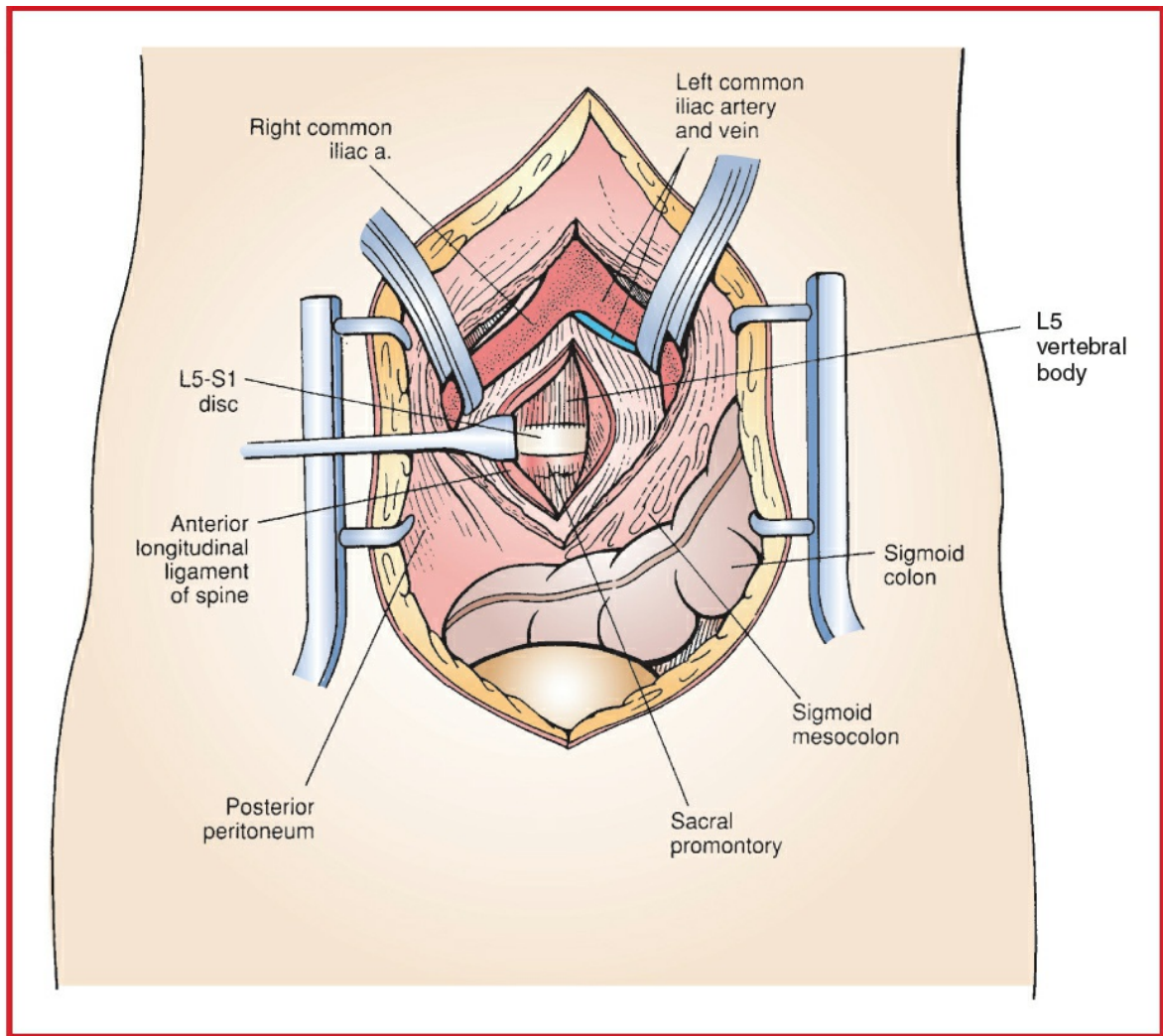


Figure 10.3-5 Transperitoneal approach to lumbosacral junction. (Reproduced with permission from Hoppenfeld S, deBoer P: Surgical Exposures in Orthopaedics. JB Lippincott, Philadelphia: 1984.)

■ SUMMARY OF PROCEDURES

	Retroperitoneal Approach (Lateral)	Transperitoneal Approach
Position	Lateral decubitus, affected side up. The up hip and knee are flexed to relax psoas muscle and allow its reflection to expose the lumbar vertebral bodies.	Supine
Incision	Flank incision curving anteriorly to the lateral margin of the rectus	Pfannenstiel above the pubis

	abdominis (Fig. 10.3-4)	
Special instrumentation	Occasional use of anterior instrumentation to stabilize fractures or to reconstruct the spine when entire vertebrae are removed	⇐
Unique considerations	In performing spinal reconstruction, manipulation of operating table is essential to “lock in” graft or implant (see above).	⇐ + a general bowel prep usually is performed preop.
Antibiotics	Cefazolin 2 g iv (+ gentamicin 80 mg iv, if indwelling bladder catheter already in place); exception is when infection is suspected and specific cultures are obtained intraop.	⇐
Surgical time	3–6 h	⇐
Closing considerations	The spine may be more or less stable after reconstruction, and patient usually transferred to bed prior to being awakened. Sudden jerking motions may dislodge graft or implant.	⇐
EBL	200–5000 mL. Blood loss extremely variable. When bleeding occurs, it may be torrential from the aorta, vena cava, or iliac vessels and branches. In nontumor/infection cases,	⇐

	200–400 mL is usual.	
Postop care	Patients with degenerative conditions and elective surgeries normally recover in PACU and return to ward. Patients with infections, fractures, and tumors are usually observed in ICU for 24 h postop. Ileus for 24–72 h is usual; NG suction usually continues until bowel sounds and passing flatus are present. Generally, mobilization depends on final stability.	⇐
Mortality	Malignancy or sepsis: 1–2%	⇐
	Elective: <0.1%	⇐
Morbidity	In patients with malignancy, sepsis or fractures with cauda equina compression, overall serious complications: 25–50%	⇐
	Cardiopulmonary failure	⇐
	Neurologic deficit	—
	Pneumothorax	—
	Sepsis	⇐
Pain score	6	6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Variable (infant-adult)
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Male:Female	1:1, except more scoliosis surgery in females (1:4)
Incidence	Uncommon
Etiology	Infection (osteomyelitis, TB); trauma; congenital; neoplasia; idiopathic

SPINAL RECONSTRUCTION AND FUSION: CERVICAL SPINE

SURGICAL CONSIDERATIONS

Description

Both an anterior or posterior approach can be used to decompress nerve roots (radiculopathy) and the spinal cord (myelopathy). Generally, involvement of one- or two-level pathology is best accomplished through an anterior approach, whereas the posterior approach may better serve multilevel pathology. An anterior approach is also preferred when the cervical spine has lost its normal lordotic contour. Posterior pathology, such as unilateral foraminal disc herniations, and ossification of the posterior longitudinal ligament (OPLL) often dictate a posterior approach. Complex pathology may necessitate combined approaches.

Anterior approach

This is done under GA. The patient is positioned supine on the operating table and a roll is placed between the scapulae to facilitate a neutral to slightly extended neck position. The arms are tucked in at the side, and gentle traction of the upper extremities may be applied to provide visualization of C7 using intraop fluoroscopy. Surface landmarks and fluoroscopy are used to locate the intended level. A transverse incision can be used up to four-level discectomies, whereas a longitudinal incision can be utilized for more levels. Generally, a left-sided approach is preferred to minimize recurrent laryngeal nerve (RLN) injury, but more recent data suggest no difference between a right-sided versus left-sided approach. After skin incision, the platysma muscle layer is exposed and divided either longitudinally or in line with the incision. A plane deep to the platysma is developed, and the medial border of the sternocleidomastoid (SCM) muscle is identified. Blunt dissection is initiated between the identified medial border of the SCM and the strap muscles. While staying medial to the carotid sheath, the trachea and esophagus are retracted to the contralateral side, and the dissection is carried deep to expose the spine. The desired disc levels are identified using fluoroscopy. The longus

colli muscle is identified and elevated carefully subperiosteally on either side to facilitate the insertion of deep-bladed, self-retaining retractors. Some data indicate that deflation of the ETT cuff with adjustments of the retractor may reduce the incidence of RLN and esophageal injury. Complete removal of the disc, other compressive pathology, or full versus complete corpectomy can be performed with the aid of curettes and a high-speed burr. Appropriately sized grafts are inserted and further stabilized with the aid of a plate and screws. The operative time for a single level is generally between 1 and 2 h, and the wound is typically closed over a drain to avoid a compression hematoma causing airway compromise. Cases involving three or more levels are at high risk for excessive soft tissue swelling and airway obstruction, and continued intubation may be a strong consideration.

Posterior approach

The patient is placed under GA in the supine position. Prior to turning the patient prone, a head holder (Mayfield vs. Gardner-Wells) is applied, and the surgeon generally controls the head while the anesthesiologist controls the airway in the process of log-rolling. As with the anterior approach, the arms are tucked in at the side, and gentle traction is applied for visualization of the entire cervical spine with fluoroscopy. The hips and knees are gently flexed in a sling while the remainder of the body is supported by bolsters allowing the abdomen to hang free on an open frame (Jackson) operating table. A reverse Trendelenburg position further aids in distribution of blood into the abdomen and legs. An extensive exposure can be created from the greater occipital protuberance to the thoracic spine. Dissection is carried down sharply through subcutaneous tissue and fat to the trapezius fascia with care taken to stay in the midline. A dry surgical dissection with electrocautery can be achieved in this manner as the ligamentum nuchae is relatively avascular. Subperiosteal dissection is then carried from the tip of the spinous processes over the laminae elevating the paraspinal muscles as a cuff. Depending on the intended surgery, procedures such as laminoforaminotomies, laminotomies, and laminectomies can be performed. Blood loss may be significant depending on the length of surgery as well as the extent of bony resection.

Usual preop diagnosis

Cervical radiculopathy; cervical myelopathy; spinal cord syndromes secondary to trauma

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain meds, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Anterior	Posterior
Position	Supine, roll between scapulae, tuck arms at sides with gentle traction	Prone, Jackson table, head holder, Gardner-Wells tongs/Mayfield head holder, tuck arms at sides with gentle traction
Incision	Transverse vs. longitudinal depending on number of levels	Longitudinal
Special instrumentation	Microscope; light source; fluoroscopy; specialized retractors and dissection instruments for working in a small, deep incision	⇐
Antibiotics	Cefazolin 2 g iv	Cefazolin 2 g iv ± gentamicin usually 80 mg iv
Surgical time	1.0–2.0 h/level	2–5 h
Closing considerations	Minimal suturing over a drain	Several layer closure over a drain
EBL	Minimal	Wide range
Postop care	PACU, mobilization as soon as tolerated. Multilevel procedures may require continued intubation postop.	⇐
Mortality	Very rare	⇐
Morbidity	Dysphagia Nerve injury Dural laceration Vertebral artery injury Esophageal injury	Nerve injury Dural laceration Vertebral artery injury Infection Air embolus

Airway obstruction
Dysphonia
Horner syndrome

Suggested Readings

1. Audu P, Artz G, Scheid S, et al: Recurrent laryngeal nerve palsy after anterior cervical spine surgery: the impact of endotracheal tube cuff deflation, reinflation, and pressure adjustment. *Anesthesiology* 2006; 105(5):898–901.
2. Beutler WJ, Sweeney CA, Connolly PJ: Recurrent laryngeal nerve injury with anterior cervical spine surgery risk with laterality of surgical approach. *Spine (Phila PA 1976)* 2001; 26(12):1337–42.
3. Fountas KN, Kapsalaki EZ, Nikolakakos LG, et al: Anterior cervical discectomy and fusion associated complications. *Spine (Phila PA 1976)* 2007; 32(21):2310–7.
4. Jeyamohan SB, Kenning TJ, Petronis KA, et al: Effect of steroid use in anterior cervical discectomy and fusion: a randomized controlled trial. *J Neurosurg Spine* 2015; 23(2):137–43.
5. Jung A, Schramm J, Lehnerdt K, et al: Recurrent laryngeal nerve palsy during anterior cervical spine surgery: a prospective study. *J Neurosurg Spine* 2005; 2(2):123–7.
5. Li Y, Shen H, Khan KZ, et al: Comparison of multilevel cervical disc replacement and multilevel anterior discectomy and fusion: a systematic review of biomechanical and clinical evidence. *World Neurosurg* 2018; 116:94–104. pii: S1878-8750(18)30957-4.

ANESTHETIC CONSIDERATIONS FOR SPINAL RECONSTRUCTION AND FUSION

(Procedures covered: anterior reconstruction and fusion of thoracic, thoracolumbar, and lumbosacral spine)

PREOPERATIVE

Patients presenting for spinal reconstruction commonly have either idiopathic or acquired scoliosis, a complex deformity involving both lateral curvature and rotation of the spine, as well as an associated deformity of the rib cage → ↓ pulmonary function. Types of scoliosis include idiopathic, congenital, neuromuscular, myopathic, traumatic, tumor-related, and mesenchymal disorders. The majority of cases are idiopathic, with a male:female ratio of 1:4. Normally, the cervical spine and lumbar spine are lordotic, whereas the thoracic spine is kyphotic. The instrumentation is intended to stabilize the spine until bony fusion of the spine has occurred. After the fusion is solid (6–12 mo), the instrumentation may be removed if it is broken, if it causes protrusions or lumps in

the back, or if the patient is having residual back pain. Nonscoliotic patients presenting for this surgery may have spinal instability as a result of trauma, metastatic cancer, or infection (e.g., TB). These patients are usually healthy, apart from their underlying pathology. The patients with metastatic cancer may need a careful workup with regard to respiratory, nutritional, and chemotherapeutic status. (See Anesthetic Considerations for Lobectomy, Pneumonectomy, or Mastectomy.)

Respiratory	Respiratory impairment proportional to angle of lateral curvature (Cobb angle; Fig. 10.3-6) <table><tr><th>Cobb angle</th><th>30°–60°</th><th>60°–90°</th><th>>90°</th></tr><tr><td>VC</td><td>↓ 25%</td><td>↓ 50%</td><td>↓ 70%</td></tr><tr><td>TLC</td><td>↓ 27%</td><td>↓ 37%</td><td>↓ 50%</td></tr></table> Restrictive pattern: ↓ TLC + ↓ ↓ VC • If VC > 70% predicted, respiratory reserve is adequate. • If VC < 40% predicted, postop ventilation usually is required. Expect further significant (~40%) ↓ VC immediately postop, requiring 7–10 d to resolve ↑ RR + ↓ TV → ↑ dead space + ↓ alveolar ventilation → V/Q mismatch → hypoxemia Patients with scoliosis of neuromuscular origin are more susceptible to aspiration and respiratory failure. Tests: CXR; ABG; PFT; assess exercise tolerance by Hx	Cobb angle	30°–60°	60°–90°	>90°	VC	↓ 25%	↓ 50%	↓ 70%	TLC	↓ 27%	↓ 37%	↓ 50%
Cobb angle	30°–60°	60°–90°	>90°										
VC	↓ 25%	↓ 50%	↓ 70%										
TLC	↓ 27%	↓ 37%	↓ 50%										
Cardiovascular	↑ PVR (<i>NB:</i> independent of severity of scoliosis). High incidence of CHD and mitral valve prolapse Tests: ECG; ECHO—consult cardiologist if apical systolic murmur or evidence of cardiovascular impairment is present												
Neurologic	SSEPs (posterior cord) and MEPs (anterior cord) will be monitored in most patients who are undergoing spinal reconstruction. Some surgeons, however, may request that the patient be awakened intraop, after completion of the instrumentation, but before closure of the wound, to ensure that no motor deficits have developed as a result of the correction. If an intraop wake-up test is planned, explain to the patient what the procedure entails, why it is necessary, and what to expect. Reassure the patient that the test will be brief, that the pain will be controlled, and that you will be there to keep them safe and as comfortable as possible. If preop focal neurologic lesions exist, they should be documented to distinguish them from changes associated with the surgical procedure.												
Musculoskeletal	When the Cobb angle is >25°, the degree of respiratory impairment will be significant, and the need for postop ventilatory support becomes more likely. Patients with muscular dystrophy may be more sensitive to myocardial depression from anesthetic agents and also may require postop ventilation 2° muscle weakness. Succinylcholine may cause severe rhabdomyolysis with hyperkalemia. These patients also may be at risk for MH.												
Hematologic	Avoid use of platelet inhibitors 1 wk before surgery. Have at least 2 U PRBCs available in the room at the start of the operation. Consider use of a red cell salvage device. Discuss use of controlled ↓ BP with surgeon. If anticipated blood loss is substantial and the patient is not anemic, consider encouraging autologous blood donation (time permitting) or performing isovolemic hemodilution intraop. Be aware that patients with previously placed spinal support (e.g., Milwaukee brace) for correction of scoliosis may have more blood loss than usual.												
Laboratory	✓ INR, PT, PTT, and Plt count preop. ✓ Hct the morning of operation or after induction of anesthesia if autologous donations have been made because, with modest blood loss and fluid therapy, the Hct may rapidly ↓ to <26%.												
Premedication	Standard premedication, if appropriate												
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon possible use of po celecoxib 400 mg po as a component of postop pain management. Consider preop acetaminophen 1000 mg po and aprepitant 40 mg po for patients with h/o severe PONV.												

INTRAOPERATIVE

Anesthetic technique

GETA: For pediatric cases, preheat room to 78°F. For operations in the prone position, use a wire-reinforced tube to prevent kinking and airway obstruction. If a transthoracic approach is used, lung isolation will be necessary. Double-lumen tubes provide the best method of achieving lung isolation overall and are available as small as 28 Fr (OD = 9.3 mm), which is typically suitable for children as young as 12 yr. If a wake-up test is required, consider the use of an endotracheal tube designed to allow laryngotracheal instillation of topical anesthetic (LITA[®] tube).

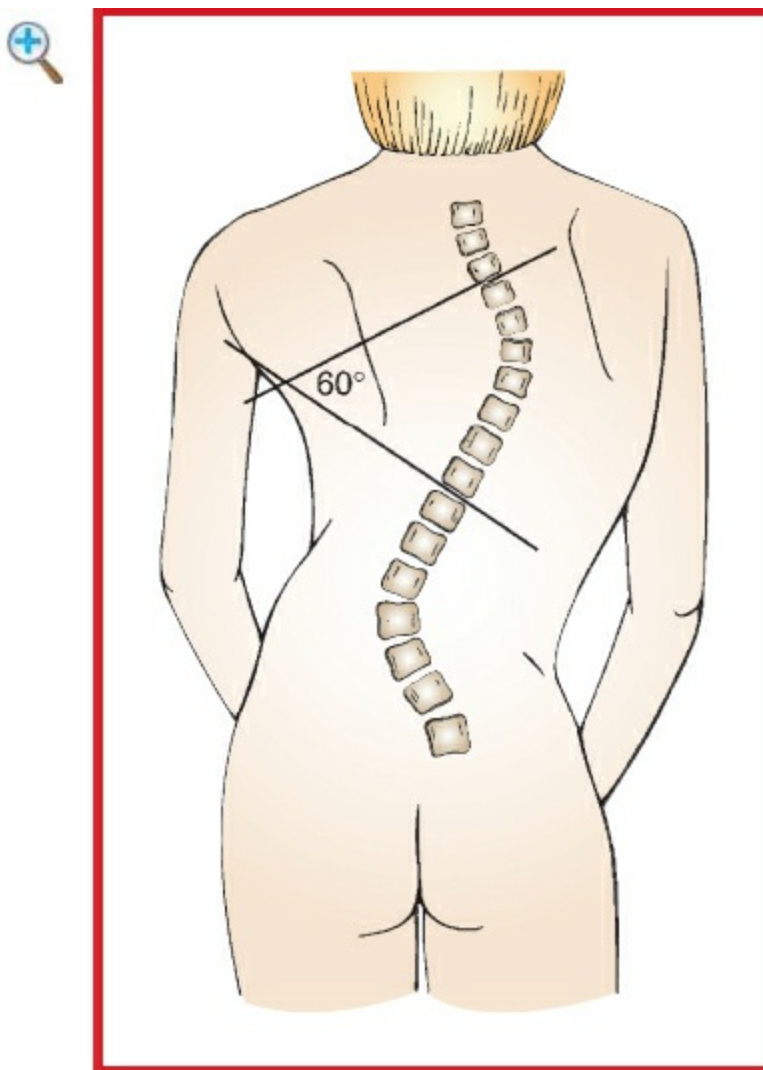


Figure 10.3-6 Cobb angle.

Induction

Standard induction. Assuming MEP monitoring will be used, consider succinylcholine vs. low-dose

	rocuronium (0.5 mg/kg) vs. deep anesthesia and remifentanil bolus (1.5 mcg/kg) to facilitate intubation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Dexamethasone (4–8 mg iv) may ↓ PONV. Sequential pneumatic compression devices should be used to ↓ VTE risk. Consider ketamine (0.25 mg/kg/h) and/or lidocaine (1–2 mg/kg/h) infusions as components of MMPM especially in chronic pain patients.
Maintenance	Neuromuscular blockers will completely abolish MEPs and must be avoided when using this monitoring technique. Normal doses of volatile anesthetics interfere with MEP and, to a lesser extent, SSEP monitoring. When monitoring MEPs, TIVA may be requested. When monitoring SSEPs only, a continuous infusion of a narcotic (ideally remifentanil) together with a reduced dose of inhaled anesthetic (0.5–1 MAC) and neuromuscular blockade usually produces satisfactory monitoring conditions. A BIS-type monitor may help with the timing of a wake-up test. The neurophysiology monitoring team must be updated if conditions that affect the evoked potentials change (depth of anesthesia, anesthetic agent, blood pressure, PCO ₂ , PO ₂).
Emergence	As soon as neuromonitoring is complete, switching to an inhaled anesthetic and stopping the propofol will allow for a timely emergence. Some surgeons may place an epidural catheter before closure for postop pain management or perform intercostal local anesthetic injections for the transthoracic approach. Prior to emergence, scoliosis patients may be fitted with a plaster mold to be used as a brace that will be worn several mo postop. Transfer from the operating table to bed must be smooth and gentle. The patient is typically extubated following transfer to the bed.

	Keep in mind that remifentanyl will not provide postop analgesia and that another longer-acting opioid must be administered before emergence. Suggest wound infiltration with LA. Consider iv acetaminophen 1 g if po dose given >4–6 h ago and hydromorphone (0.2–0.6 mg iv). Discuss possible postop use of ketorolac or celecoxib.	
Blood and fluid requirements	IV: 14–16 ga × 2 BSS @ 1–3 + mL/kg/h not to exceed 40 mL/kg T&C 2–4 U PRBCs ± Red cell salvage Albumin 5% prn Warm all fluids. Humidify gases.	Anticipate substantial blood loss (proportional to the number of levels) from extensive dissection of muscles from bony spine; less so with isolated thoracic fusion. It may vary from 25% to 100% of the patient's blood volume, although blood loss is usually less with the anterior approach. Goal-Directed Fluid Management is recommended. Colloid is preferred because it will remain intravascular until metabolized. Use Goal-Directed Fluid Management, p. K-4.
Control of blood loss	Position to prevent venous engorgement. ↓ MAP to 60 mm Hg Consider ↓ Hct to 25–28% (isovolemic hemodilution)	Although of questionable benefit, some surgeons may request deliberate hypotension. Decreasing MAP may ↓ blood loss. Generally, maintain MAP > 60 mm Hg in young, healthy patients and >80 mm Hg in older patients

		for spinal cord perfusion. Once dissection is complete, higher BP values acceptable. Maintain UO at 0.5–1 mL/kg/h during controlled hypotension.
Monitoring	Standard monitors Arterial line ±CVP line Urinary catheter SSEP, MEP ± BIS	CVP monitor helpful for continuous infusions and to assess fluid Rx. Trends in PPV or CVP will be more reliable than measurement of UO to assess changes in volume status in patients in the prone position. Frequent labs (ABG, lytes, coag) may be useful.
Positioning	✓ and pad pressure points ✓ eyes and neck Bite block (essential for MEP) Avoid posterior pressure on ETT.	Prone: Place head on cushioned head holder with cutout for eyes, nose, and chin. Make certain that none of the face rests on bed. ✓ elbows, knees, and feet. Lateral: Place head on donut with C-spine in neutral position. Pad and support both arms in neutral position. Ensure no pressure on eyes or ears.
Wake-up test (rarely necessary with intraop neuromonitoring)	30-min advance warning from surgeons is needed. • Decrease anesthetic agents. • Reverse muscle relaxants. • Target BIS in high 50 sec.	The wake-up test is an alternative to intraop neuromonitoring that assesses integrity of motor pathways in the

	• About 10 min before testing turn off anesthetics. • ↓ remifentanyl infusion to 0.05 mcg/kg/min or less. • If a LITA tube was used, administer 3–4 mL 1% lidocaine into the trachea. • Begin speaking to the patient. Request movement of all four limbs sequentially. Dangers: Air embolus Dislodgement of spinal instrumentation Accidental extubation	ventral cord. Uncontrolled patient movement during wake-up test can result in accidental extubation or dislodgement of the spinal instrumentation. Unrestrained inspiratory efforts may provoke venous air embolism. The anesthesiologist must be prepared to rapidly reanesthetize the patient.
SSEP and MEPs	SSEP: Dorsal cord function only MEPs: Ventral cord function	SSEP and MEP are used routinely. Avoid changes in anesthetic conditions (depth of anesthesia, agents, MAP, PCO₂, PO₂). An increase in the latency (>10%) or decrease in the amplitude (≥50%) usually indicates cord injury from hardware impingement, cord stretch or compression, or hypoperfusion. The surgeon and anesthesiologist should be notified immediately if the evoked potentials change. Unilateral changes to MEP/SSEP signals may indicate a positioning problem.
Complications	Spinal cord ischemia	SSEP indications of spinal

Massive blood loss
Fat embolism

cord ischemia should be treated by alerting the surgeon and ensuring optimal perfusion ($\uparrow$ BP, normalize PCO_2 , $\uparrow$ PO_2 , optimal Hct, $\downarrow$ PEEP, $\downarrow$ traction or pressure on the cord). Prompt transfusion may be necessary, and blood should be available in the room. Be prepared for the possibility of massive transfusion.

▼ POSTOPERATIVE

Complications

Bleeding

Postop bleeding is common. Rarely, the bleeding occurs in a space that is not drained and can result in complications such as epidural hematoma or hemothorax.

Pulmonary insufficiency
Hypothermia
Pneumothorax
Dislodgement of internal fixation

Postop ventilation may be required in patients with severe respiratory impairment (see Preoperative Considerations, above). In addition, thoracotomy, surgical trauma to the diaphragm, fat embolism, and inadequate analgesia may further $\uparrow$ the risk of postop pulmonary insufficiency. Careful

	Neurologic sequelae	handling of patient in transfer is mandatory. Neurologic sequelae probably remain the most feared complication, and it is important to document postop neurologic exam.
Multimodal pain management	PCA Intrathecal morphine 0.1–0.25 mg by surgeon intraop Thoracic/lumbar epidural opiates/local anesthetics IV acetaminophen ± Ketorolac (30 mg iv) or celecoxib (200 mg po)	Pain is a significant problem for postop scoliosis patients. Most need analgesia × 3–4 d. Preop consultation with patient (and parents) about different pain management techniques is important. 1 g q6h for adults, 15 mg/kg q6h if ≤50 kg or ≤12 Consult with surgeon regarding possible hemostasis concern.
ERAS	Ondansetron 4 mg iv ± Metoclopramide 10–20 mg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization with physiotherapy.
Tests	CXR; ABG; Hct	✓ for pneumothorax and line placement

Suggested Readings

1. Laratta JL, Ha A, Shillingford JN, et al: Neuromonitoring in spinal deformity surgery a multimodality approach. *Global Spine J* 2018; 8(1):68–77.
2. Mergeay M, Verster A, Van Aken D, et al: Regional versus general anesthesia for spine surgery. A comprehensive review. *Acta Anaesthesiol Belg* 2015; 66(1):1–9.
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pain patients with opioid dependency: a randomized, blinded trial. *Pain* 2017; 158(3):463–70.

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5. Sivaganesan A, Chotai S, White-Dzuro G, et al: The effect of NSAIDs on spine fusion: a cross-disciplinary review of biochemical, animal, and human studies. *Eur Spine J* 2017; 26(11):2719–28.
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7. Wainwright TW, Immins T, Middleton RG: Enhanced recovery after surgery (ERAS) and its applicability for major spine surgery. *Best Pract Res Clin Anaesthesiol* 2016; 30(1):91–102.
8. Willner D, Spennati V, Stohl S, et al: Spine surgery and blood loss: systematic review of clinical evidence. *Anesth Analg* 2016; 123(5):1307–15.
9. Zanirato A, Damilano M, Formica M, et al: Complications in adult spine deformity surgery: a systematic review of the recent literature with reporting of aggregated incidences. *Eur Spine J* 2018; 27:2272–84.

Suggested Viewing

Links are available online to the following videos:

Posterior Spinal Instrumentation & Fusion for Idiopathic Scoliosis:

<https://www.youtube.com/watch?v=TbQaT7z54E4>

Scoliosis Spinal Fusion Animation: <https://www.youtube.com/watch?v=WBIf4AQj5s0>

Minimally Invasive Tubular Microdiscectomy: 3-D Video:

https://www.youtube.com/watch?v=Cpkr_oy5Enc

Dr. Ali Bydon Performs a Minimally Invasive Lumbar Discectomy:

<https://www.youtube.com/watch?v=2j9QyiOxVF4>

CHAPTER 10.4

Hip, Pelvis, and Upper Leg Surgery

SURGEONS

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OPEN REDUCTION AND INTERNAL FIXATION (ORIF) OF THE PELVIS OR ACETABULUM

SURGICAL CONSIDERATIONS

Description

Pelvic fractures present several challenging treatment problems. Surgical management is complex and often difficult in nature. Specialized training and equipment are required for a successful outcome. Major trauma mechanisms produce pelvic ring injuries, and patients with pelvic ring disruptions frequently have associated systemic injuries, which may be life threatening (e.g., hemorrhagic shock). Pelvic stabilization and surgical control of hemorrhage may be performed acutely in the polytrauma patient who is hemodynamically unstable. This is in conjunction with an exploratory laparotomy performed by a trauma surgeon or interventional radiology embolization of vasculature. The majority of patients with pelvic fractures who are treated operatively are taken to the OR on a delayed basis, after they have been stabilized. Pelvic fractures that do not heal are “nonunions,” whereas those that heal in an unsatisfactory position are

“malunions.”

Anterior approaches to the pelvis include Pfannenstiel and ilioinguinal incisions, which are utilized for reduction and fixation of dislocations and fracture/dislocations of the symphysis pubis, fractures of the pubic rami, and access to the anterior aspect of the sacroiliac (SI) joint. **Posterior approaches** to the pelvis involve either vertical or curved incisions along the iliac crest and are used for reduction and fixation of SI joint dislocations, fracture/dislocations of the SI joint, and fractures of the iliac wing and of the sacrum. These procedures are often lengthy and are staged, requiring changes in patient position. Reductions are facilitated by neuromuscular paralysis. The posterior approach requires a large operative field, which may prevent the use of an epidural catheter. In addition, postop anticoagulation for DVT prophylaxis is used uniformly and may contraindicate the use of epidural catheters. The goal of pelvic reconstruction is to restore the anatomy and stability of the pelvis, which will decrease hemorrhage in the hemodynamically unstable patient, aid in mobilization of the multiply injured patient, and improve long-term function.

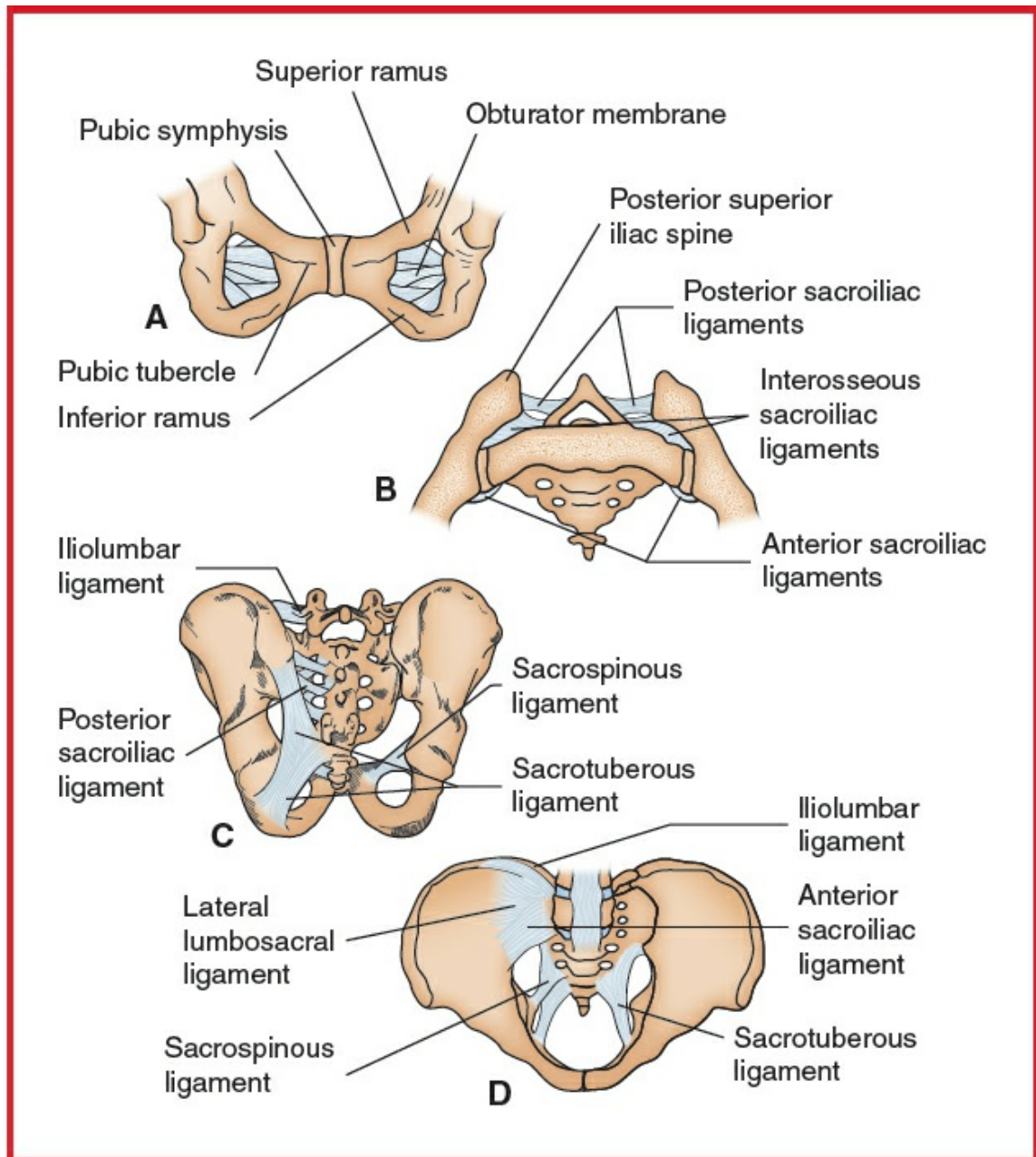


Figure 10.4-1 Schematic views of the pelvis with the principal ligamentous supports. **A:** Symphysis pubis fibrocartilage. **B:** Posterior SI ligaments. **C:** Posterior view. **D:** Anterior view. (Redrawn with permission from Chapman MW: Chapman's Orthopaedic Surgery, Vol. 1, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

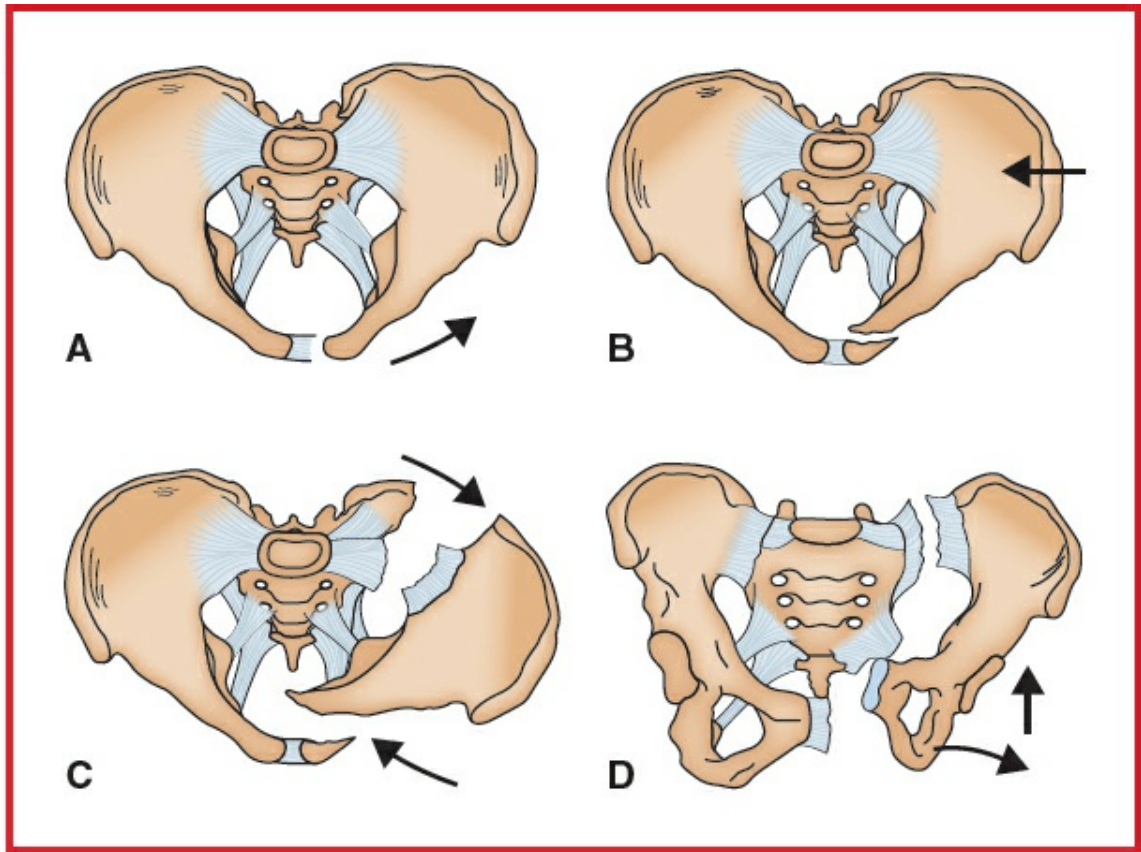


Figure 10.4-2 Schematic view of the principal pelvis injury patterns, as determined by the vector of the provocative blow. **A:** Anterior/posterior compression or external rotation injury. **B:** Stable lateral compression or internal rotation injury. **C:** Unstable lateral compression or internal rotation injury. **D:** Unstable vertical shear disruption. (Redrawn with permission from Chapman MW:Chapman's Orthopaedic Surgery, Vol. 1, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Usual preop diagnosis

Fractures of the pelvis/acetabulum; nonunion/malunion of the pelvis/acetabulum

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine (anterior); prone (posterior)
Incision	Pfannenstiel's, ilioinguinal (anterior); posterior, curving along the iliac crest (posterior)
Special	Radiolucent table; pelvic instruments and implants;

instrumentation	Cell Saver; I.I.
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)
Surgical time	1–6 h (anterior); 3–6 h (posterior)
Closing considerations	May require neuromuscular relaxation to aid reduction and closure; postop radiograph
EBL	≥1000 mL
Postop care	Multiple-trauma patient → ICU; others → PACU
Mortality	10%+, dependent on the extent of multiple trauma
Morbidity	Ileus: Common Neurologic injury: Common Genitourinary injuries: Not uncommon Failure of fixation: Rare Infection: Rare Malunion: Rare Nonunion: Rare
Pain score	9

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age, but predominance of males 30–60 yr old
Male:Female	5:1
Incidence	1–2%
Etiology	Motorcycle and motor vehicle accidents (60–80%); falls (10–15%); crush injuries (5%); other (5%)
Associated conditions	Frequently associated with trauma to other organ systems, including the head and neck, chest, abdomen, and extremities. These often will be addressed concurrently with pelvic or acetabular fracture.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Procedures About the Pelvis and Hip.

Suggested Readings

1. Burkhardt M, Kristen A, Culemann U, et al: Pelvic fracture in multiple trauma: are we still up-to-date with massive fluid resuscitation? *Injury* 2014; 45(Suppl 3):S70–5.
2. Halawi MJ: Pelvic ring injuries: surgical management and long-term outcomes. *J Clin Orthop Trauma* 2016; 7(1):1–6.
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4. Rice PL Jr, Rudolph M: Pelvic fractures. *Emerg Med Clin North Am* 2007; 25(3):795–802.
5. Ruatti S, Guillot S, Brun J, et al: Which pelvic ring fractures are potentially lethal? *Injury* 2015; 46(6):1059–63.
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7. Wang H, Robinson RD, Moore B, et al: Predictors of early versus late mortality in pelvic trauma patients. *Scand J Trauma Resusc Emerg Med* 2016; 24:27.

CLOSED REDUCTION AND EXTERNAL FIXATION OF THE PELVIS

SURGICAL CONSIDERATIONS

Description

This procedure entails manipulating the pelvis to obtain an acceptable reduction by closed means under GA and then applying an anterior external fixation device to maintain the reduction. The pins for the external fixator are inserted into the iliac crest either percutaneously or through small incisions. During this procedure, either radiographs or the I.I. is used to confirm that an acceptable reduction has been obtained. In some centers, this procedure is done in the emergency department as a lifesaving procedure.

Usual preop diagnosis

Displaced fracture of the pelvis; unstable fracture of the pelvis

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Percutaneously or through small incisions along the iliac crest
Special instrumentation	External fixation; often performed on a radiolucent table using I.I.
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic); combination antibiotics, if multiple severe, open fractures or other significant injuries are present
Surgical time	1–1.5 h
EBL	Negligible from surgical procedure; however, anticipate large blood losses (4+ U) from the pelvic fracture alone
Postop care	Multiple-trauma victim → ICU; others → PACU
Mortality	10% or more, depending on the extent of multiple trauma; 50% in open fractures
Morbidity	Ileus: Virtually 100% Sacroiliac (SI) pain: 15–30%+ Genitourinary problems, including bladder or urethral rupture: 13% Neurologic deficit to the lumbosacral plexus: 1–10% Malunion/severe deformity: 5% Leg-length discrepancy: 3–5% Impotence: 1–5% Residual instability: 1–3% Vascular complications: 1% Hypotension 2° to retroperitoneal hematoma: Common Respiratory distress: Common Gynecologic and colorectal injuries: More common with open fractures/dislocations Delayed union, nonunion: Not uncommon Osteomyelitis: Rare Rupture of the diaphragm: Rare

Pain score	7–10
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■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age, but predominance of males 30–60 yr old
Male:Female	5:1
Incidence	Common
Etiology	Motorcycle and motor vehicle accidents (60–80%); falls (10–15%); crush injuries (5%); other (5%)
Associated conditions	Frequently associated with trauma to other organ systems, including the head and neck, chest, abdomen, and extremities. Patient sustaining a pelvic fracture also has a probability of having other injuries, including musculoskeletal (85%), respiratory (60%), CNS (40%), GI (30%), urologic (12%), and CVS (6%). These often will be addressed concurrently with the pelvic fracture.

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Procedures About the Pelvis and Hip.

Suggested Readings

1. Prevezas N: Evolution of pelvic and acetabular surgery from ancient to modern times. *Injury* 2007; 38(4):397–409.
2. Rice PL Jr, Rudolph M: Pelvic fractures. *Emerg Med Clin North Am* 2007; 25(3):795–802.
3. Skitch S, Engels PT: Acute management of the traumatically injured pelvis. *Emerg Med Clin North Am* 2018; 36(1):161–79.

OPEN REDUCTION AND INTERNAL FIXATION (ORIF) OF ACETABULUM FRACTURES

■ SURGICAL CONSIDERATIONS

Description

Although the acetabulum is contained within the bony architecture of the pelvis, surgical management of acetabulum fractures is approached separately from pelvic fractures. The goal of surgical treatment of acetabulum fractures is to preserve the hip joint by accurately reconstructing the supporting bony anatomy. Surgical treatments of these challenging injuries are performed by surgeons who have undergone specialized training in orthopedic pelvic surgery. The mechanism of injury is usually high-energy trauma (e.g., motor vehicle, motorcycle accidents), industrial accidents, or a fall from a height that drives the femur into the acetabulum. Associated injuries to the pelvis are common, as are associated systemic injuries.

Optimal results are achieved when surgery is performed within 7 d. The approach is dictated mainly by the unique characteristics of the fractures. Essentially, three approaches are commonly used: the **ilioinguinal** (anterior; [Fig. 10.4-3](#)), the **extended iliofemoral** (lateral; [Fig. 10.4-4](#)), and the **Kocher-Langenbeck** (posterior; [Fig. 10.4-5](#)). The most difficult portion of the procedure is the reduction; it may be facilitated by neuromuscular relaxation, pelvic reduction instruments, and traction. The I.I. is used frequently throughout the procedure to assess the reduction and position of implants, which necessitates the use of lead aprons. A radiograph also is obtained at the end of the case to verify a satisfactory reduction and position of the implants. Patients are anticoagulated in the postop period to prevent thromboembolic complications. Weight-bearing restrictions are maintained until enough healing has occurred to permit functional ambulation.

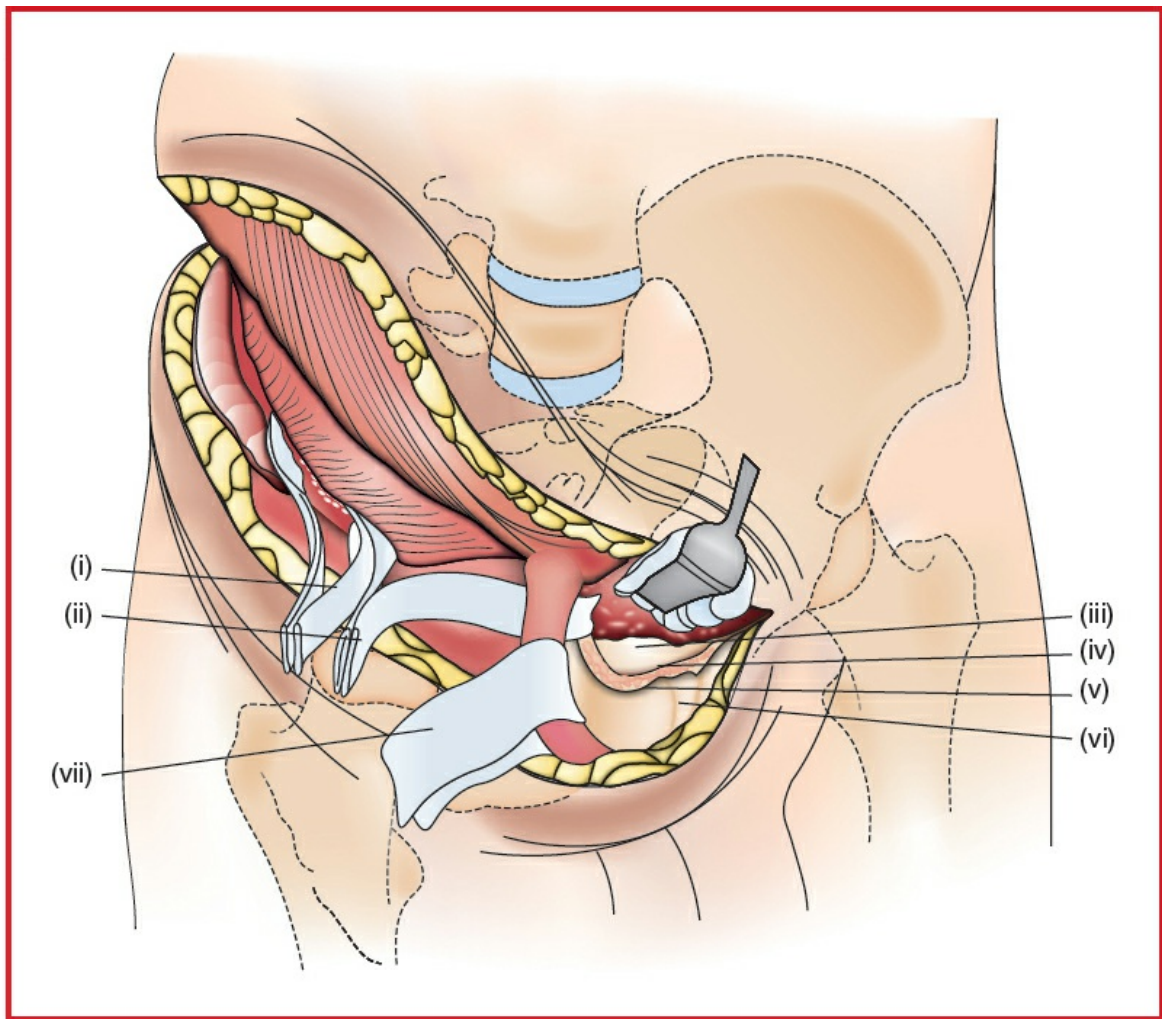


Figure 10.4-3 Ilioinguinal approach, right side: (i) Penrose drain around the iliopsoas, femoral nerve, and lateral femoral cutaneous nerve; (ii) Penrose drain around femoral vessels; (iii) bladder and space of Retzius; (iv) pubis; (v) pubic tubercle; (vi) symphysis pubis; (vii) Penrose drain around spermatic cord. (Redrawn from Sledge CB, ed: The Hip. Lippincott-Raven, Philadelphia: 1998.)

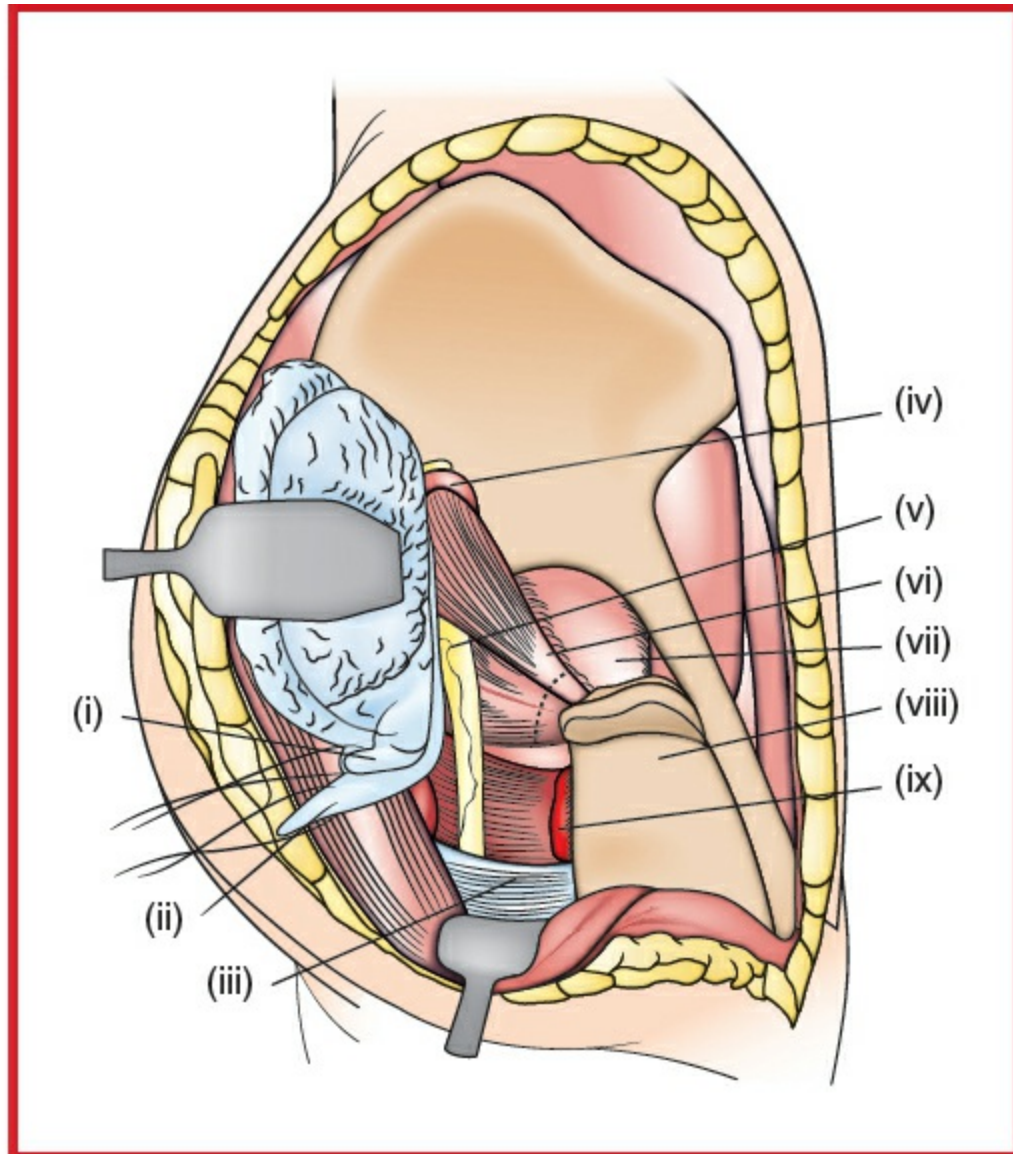


Figure 10.4-4 Extended iliofemoral approach: (i) Gluteus minimus tendon, (ii) gluteus medius tendon, (iii) gluteus maximus tendon, (iv) superior gluteal neurovascular bundle, (v) sciatic nerve, (vi) piriformis and conjoint tendons, (vii) hip joint capsule, (viii) greater trochanter, (ix) medial femoral circumflex artery overlying the quadratus femoris. (Redrawn from Sledge CB, ed: The Hip. Lippincott-Raven, Philadelphia: 1998.)

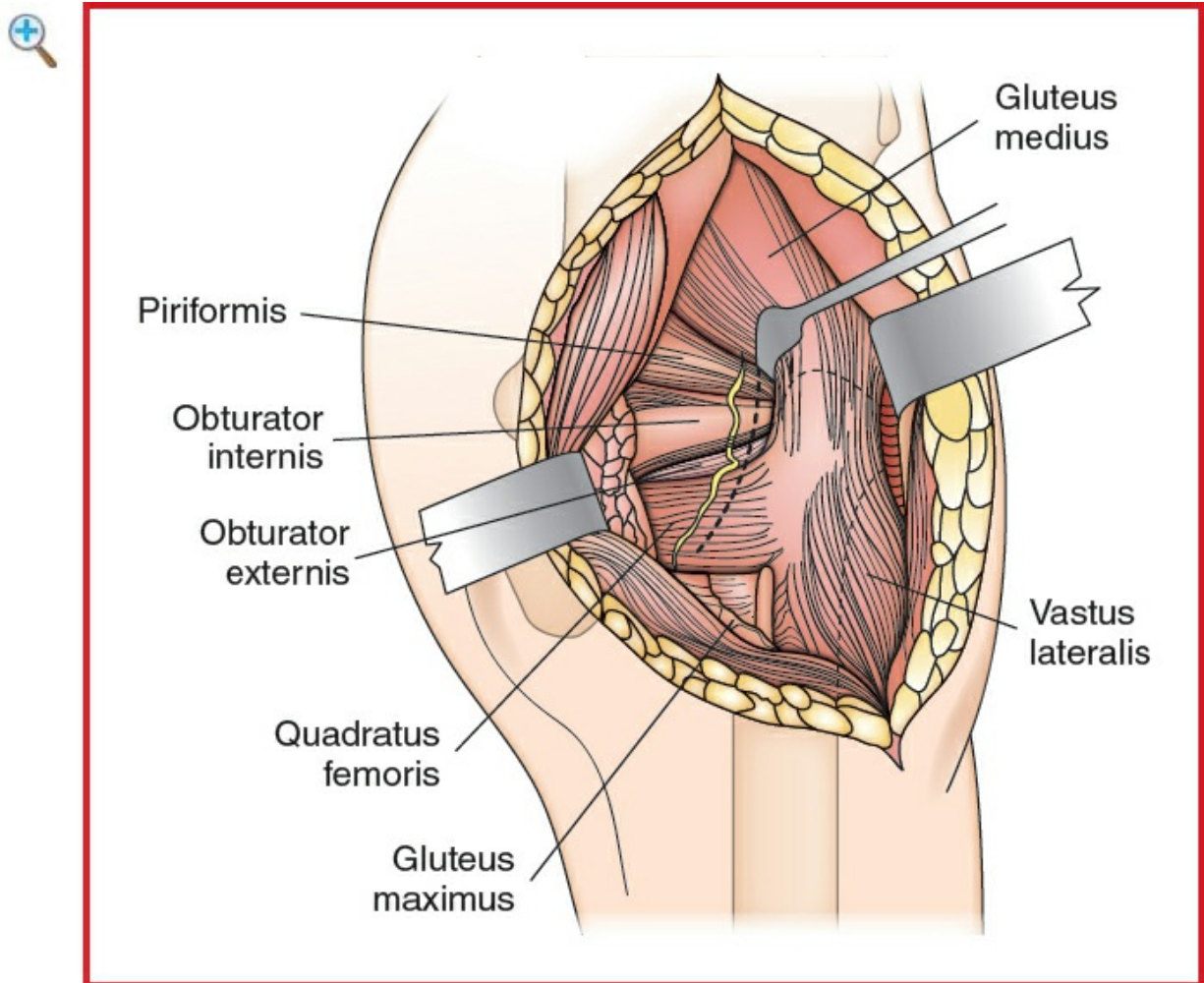


Figure 10.4-5 Kocher-Langenbeck approach. (Redrawn from Sledge CB, ed. *The Hip* Lippincott-Raven, Philadelphia: 1998.)

Usual preop diagnosis

Fracture of the acetabulum; nonunion/malunion of the acetabulum

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine (anterior); lateral decubitus (lateral); prone (posterior)
Incision	Ilioinguinal (anterior; Fig. 10.4-3); extended iliofemoral (lateral; Fig. 10.4-4), Kocher-Langenbeck (posterior;

Fig. 10.4-5)

Special instrumentation	Pelvic table; pelvic instruments and implants; I.I.; Cell Saver
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)
Surgical time	2–5 h
EBL	100–2000 mL
Postop care	Multiple trauma: → ICU; others → PACU
Mortality	~2%
Morbidity	Ectopic ossification: 24% Osteoarthritis after perfect reduction: 10.2%; after imperfect reduction: ~36% Injury to lateral cutaneous nerve of the thigh: 12% Sciatic nerve damage: 6% Infection: 4% Avascular bone necrosis: 4% DVT: 3% Wound hematoma: 1.5% Secondary displacement of fracture site: 1.5% PE: 1% Pseudoarthrosis: 0.7% Ileus after ilioinguinal approach: Common
Pain score	7–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	12–85 yr
Male:Female	66% (M); 34% (F)
Etiology	Motor vehicle accidents (70%); motor vehicle/pedestrian accidents (13%); falls (14%); other (3%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Procedures About the Pelvis and Hip.

Suggested Readings

1. Briffa N, Pearce R, Hill AM, et al: Outcomes of acetabular fracture fixation with ten years' follow-up. *J Bone Joint Surg Br* 2011; 93(2):226–36.
2. Gettys FK, Russell GV, Karunakar MA: Open treatment of pelvic and acetabular fractures. *Orthop Clin North Am* 2011; 42(1):69–83, vi.
3. Prevezas N: Evolution of pelvic and acetabular surgery from ancient to modern times. *Injury* 2007; 38(4):397–408.
4. Rice PL Jr, Rudolph M: Pelvic fractures. *Emerg Med Clin North Am* 2007; 25(3):795–802.
5. Skitch S, Engels PT: Acute management of the traumatically injured pelvis. *Emerg Med Clin North Am* 2018; 36(1):161–79.

OSTEOTOMY AND BONE GRAFT AUGMENTATION OF THE PELVIS

SURGICAL CONSIDERATIONS

Description

Acetabular insufficiency (acetabular dysplasia) is characterized by deficient anterior and lateral coverage of the acetabulum on the femoral head. This condition of the hip produces joint incongruity and instability, eventually leading to arthrosis and a dysfunctional hip joint. Treatment is aimed at reorienting the dysplastic acetabulum ([Fig. 10.4-6](#)). In children, bone grafting alone may be sufficient; in adults, however, pelvic osteotomy, to reorient or broaden the weight-bearing surface, is necessary. A supplemental bone graft to expand the weight-bearing surface may be added. In certain instances following pelvic osteotomy, incongruity of the hip may persist. In this situation, the pelvic osteotomy is combined with a proximal femoral osteotomy to restore congruence. Pelvic and proximal femoral osteotomies usually are fixed internally with screws and plates to allow early mobilization without displacement. Weight-bearing is permitted after healing of the osteotomy at ~8 wk.

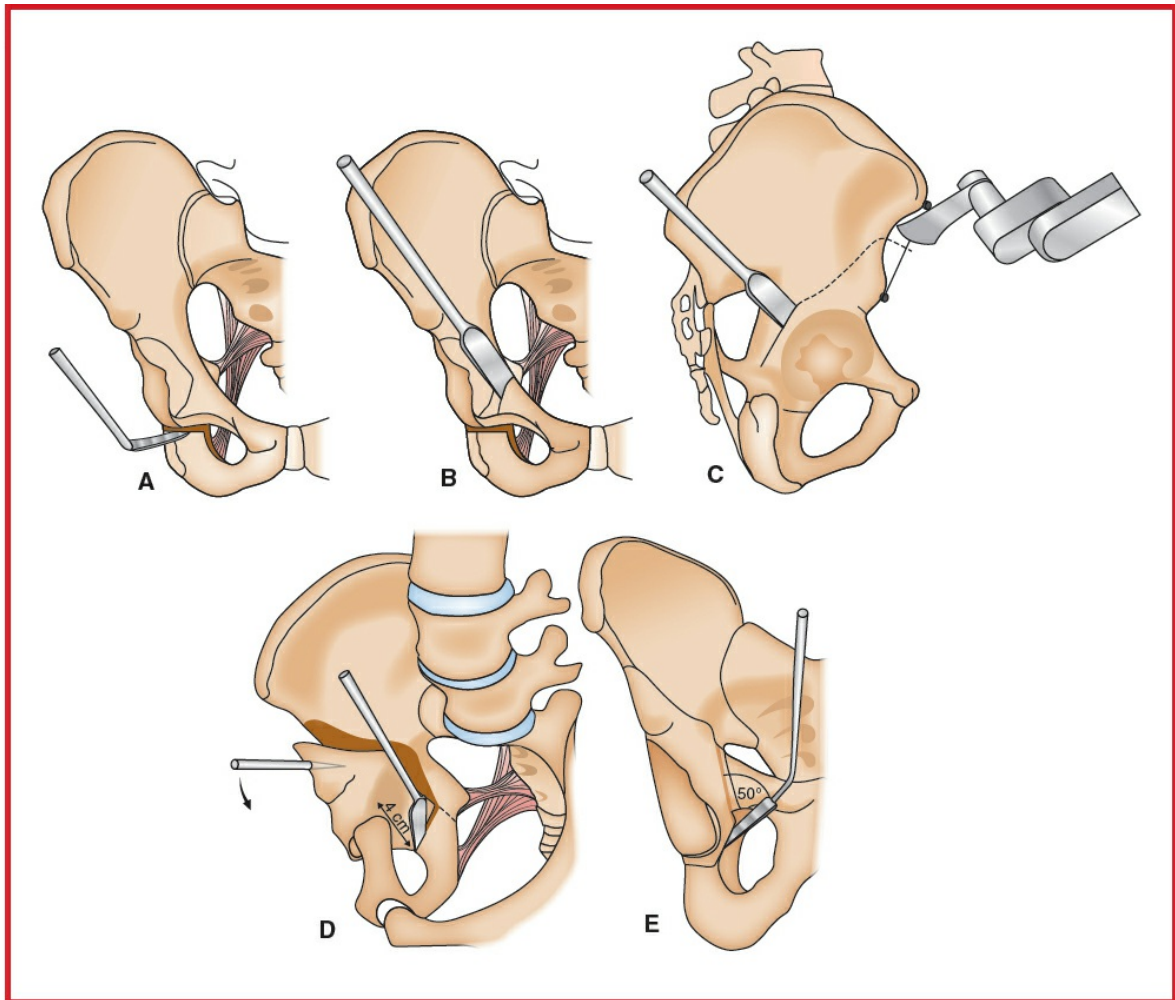


Figure 10.4-6 A–E: The Bernese periacetabular osteotomy. (Redrawn from Ganz R, Klaue K, Vinh TS, et al: A new periacetabular osteotomy for the treatment of hip dysplasias: technique and preliminary results. Clin Orthop 1988; 232:26–36.)

Usual preop diagnosis

Acetabular dysplasia; developmental dysplasia of the hip

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Anterior: ilioinguinal or iliofemoral and Smith-Petersen
Special	Pelvic instruments and implants; special osteotomes

instrumentation	and saws; I.I., Cell Saver
Unique considerations	Intraop radiographs and use of I.I.
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)
Surgical time	3 h
EBL	500+ mL
Postop care	PACU → room; usually on protected, weight-bearing walker or crutches × 8 wk
Mortality	Minimal
Morbidity	Ileus: 100% Leg-length discrepancy: Uniformly present after pelvic osteotomy Injury to lateral cutaneous nerve: Common (50%) Sciatic nerve: Uncommon (1%) Thromboembolism: 5–10% Wound infection: 1–5% Septic arthritis, osteomyelitis: 1–7% Delayed union, nonunion, malunion: 1–2% Genitourinary problems—urinary retention requiring catheterization: Common Hematoma: Common Hypotension 2° to retroperitoneal hematoma: Rare Vascular complications: Rare
Pain score	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–50 yr
Male:Female	3–4 times higher incidence in females for congenital hip dysplasia; equal incidence for other causes
Etiology	Congenital hip dysplasia; neuromuscular disorders (cerebral palsy, meningomyelocele); pediatric trauma to acetabular growth plate

Associated conditions

Depends on Dx

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Procedures About the Pelvis and Hip.

ARTHRODESIS OF THE SACROILIAC JOINT

SURGICAL CONSIDERATIONS

Description

In this procedure, a painful and/or unstable sacroiliac (SI) joint is fused, usually by excising the joint through an **anterior** or **posterior approach** and employing an iliac crest bone graft. Supplemental screw fixation of the joint is used. The incision follows the iliac crest from the anterior superior iliac spine past the convexity of the iliac tubercle; the aponeurosis of the external abdominal musculature is elevated from the iliac crest. The internal iliac fossa is exposed subperiosteally, posterior to the SI joint; then, the joint cartilage is excised and packed with cancellous bone strips. The SI joint is fixed with plates and screws. Alternatively, a posterior approach to the SI joint may be used. A straight vertical incision is made just lateral to the posterior superior iliac spine. The origin of the gluteus maximus is elevated from its origin off the posterior ilium and sacrum and reattached laterally. The SI joint is identified, débrided of cartilage, and packed with strips of cancellous bone. It is then fixed with screws.

Variant procedure or approaches

Anterior or posterior approach

Usual preop diagnosis

Arthritis or arthrosis of the SI joint; pelvic instability

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURE

Position	Supine (anterior) or prone (posterior)
Incision	Lateral portion of ilioinguinal (anterior); posterior approach to SI joint (posterior)
Special instrumentation	Radiolucent table; pelvic instruments and implants; I.I.; Cell Saver
Unique considerations	Intraop radiographs or use of I.I.
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)
Surgical time	2–3 h
EBL	250–500 mL
Postop care	Usually on protected, weight-bearing walker or crutches × 6–8 wk; anticoagulation for DVT prophylaxis
Mortality	Extremely low
Morbidity	Ileus: Virtually always Osteomyelitis: <1% Wound infection: <1% Genitourinary problems; urinary retention requiring catheterization: Common Delayed union, nonunion, malunion, leg-length discrepancy: Not uncommon Neurologic deficit; injury to lumbosacral plexus: Rare, unless present preop; L5 nerve root susceptible in anterior approaches Hypotension 2° to retroperitoneal hematoma: Rare Injury to the bowel: Rare Vascular complications; injury to iliac arteries: Rare Thromboembolism
Pain score	7

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–50 yr
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Male:Female	Increased incidence in males (trauma)
Incidence	Rare
Etiology	Trauma—post–pelvic fracture dislocation; painful septic arthritis

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Procedures About the Pelvis and Hip.

Suggested Readings

1. Rice PL Jr, Rudolph M: Pelvic fractures. *Emerg Med Clin North Am* 2007; 25(3):795–802.
2. Skitch S, Engels PT: Acute management of the traumatically injured pelvis. *Emerg Med Clin North Am* 2018; 36(1):161–79.
3. Wise CL, Dall BE: Minimally invasive sacroiliac arthrodesis: outcomes of a new technique. *J Spinal Disord Tech* 2008; 21(8):579–84.

AMPUTATIONS ABOUT THE HIP AND PELVIS: DISARTICULATION OF THE HIP AND HINDQUARTER AMPUTATION

SURGICAL CONSIDERATIONS

Description

These surgical procedures accomplish an excision of the entire lower extremity. In a **hip disarticulation**, the amputation is performed through the hip joint. An anterior, racquet-shaped incision is made, and all muscles crossing the hip joint are incised or detached. The femoral artery, vein, and nerve, obturator vessels, sciatic nerve, and deep vessels are isolated and ligated. The gluteal flap is brought anteriorly and sewn to the anterior portion of the incision. In a **hindquarter amputation**, excision of the lower extremity, hip joint, and a portion of the pelvis is performed. Anterior and posterior incisions are used, the iliac wing is divided posteriorly, and the symphysis pubis is disarticulated anteriorly. Either the common iliac or external iliac vessels are ligated, as are all nerves to the lower extremity. Usually, the gluteal flap is drawn anteriorly for closure. These procedures are performed very rarely—for severe trauma, tumor, or infection—and are often lifesaving surgeries. They often are performed in conjunction with a

general surgeon, and standard bowel prep is done. The operations are long and tedious, with extensive blood loss, in patients who are usually systemically ill.

Usual preop diagnosis

Malignant tumor of the femur, hip, or pelvis; traumatic amputation to the femur, hip, or pelvis; uncontrollable infection to the leg, hip, or pelvis (e.g., clostridia)

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Hip Disarticulation	Hindquarter Amputation
Position	Supine	Lateral decubitus; stabilized by beanbag and/or kidney rests
Incision	Anterior racquet type (rare)	Anterior and posterior
Unique considerations	Urinary catheter should be placed.	Urinary catheter; NG tube; scrotum strapped to the opposite thigh; anus stitched closed/sealed
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)	⇐
Surgical time	3–4 h	4–5 h
EBL	1000–2000 mL (intraop blood salvage system recommended, except for tumors)	2000–3000 mL
Postop care	ICU	⇐
Mortality	Rare in patients undergoing elective	⇐

	amputation for trauma or localized tumor; higher for patients with debilitated trauma, chronic infection, or extensive invasive malignant tumor; highest in clostridial infections: ~50%+	
Morbidity	Anemia: Common	⇐
	Electrolyte abnormalities: Common	⇐
	Hematoma: Common	⇐
	Neurologic injury to lumbosacral plexus or peripheral nerves: Common	⇐
	Paralytic ileus: Common	⇐
	Psychosocial problems: Common	⇐
	UTI: Common	⇐
	Flap necrosis: Not uncommon	
	Incomplete excision with recurrence of tumor or infection: Not uncommon	
	Injury to peritoneal or retroperitoneal contents, including bowel and bladder: Not uncommon	
	Vascular injury—iliac; other vessels: Not uncommon	
Pain score	10	10

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age
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Male:Female	Similar, except higher incidence in males for traumatic etiologies
Incidence	Uncommon
Etiology	Malignant tumor; trauma; infection—clostridial myonecrosis, chronic osteomyelitis, etc.

ANESTHETIC CONSIDERATIONS FOR PROCEDURES ABOUT THE PELVIS AND HIP

(Procedures covered: ORIF, pelvis and acetabulum; closed reduction, external fixation, pelvis; osteotomy and bone graft of the pelvis; arthrodesis of SI joint; amputations about the hip and pelvis: disarticulation of the hip and hindquarter amputation)

PREOPERATIVE

Patients presenting for pelvic surgery generally fall into two categories: (a) major trauma—pelvic fracture requires substantial force and seldom occurs alone. These patients require aggressive fluid resuscitation with large-bore ivs and invasive monitors (arterial line and CVP). If the patient can be made hemodynamically stable with volume resuscitation, a thorough evaluation for coexisting neurologic, thoracic, or abdominal trauma should be undertaken before anesthesia. (b) Tumor resection and amputation of the thigh, hip, and pelvis. Because of large intraop blood loss and third spacing of fluids, invasive hemodynamic monitoring is necessary. Although epidural anesthesia is seldom adequate for surgery, postop epidural analgesia is an effective means of controlling the tremendous pain caused by this type of surgery. Other patient populations covered in this section include otherwise healthy patients with congenital or acquired hip dysplasia presenting for augmentation procedures.

Respiratory	Trauma patients are at risk for hemothorax, pneumothorax, pulmonary contusion, fat embolism (preop and postop), impaired respiratory effort secondary to rib fractures, and aspiration. A chest tube will be needed before surgery if either a hemothorax or pneumothorax is present. Pulmonary fat embolus occurs in 10–15% of patients with long-bone fractures and can occur after isolated pelvic fractures. Sx include hypoxemia, ↑HR, ↑RR,
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	respiratory alkalosis, mental status changes, conjunctival petechiae, fat bodies in the urine, and diffuse pulmonary infiltrates. Sx of pulmonary aspiration are similar to those of fat embolism. Preop therapy for either should include supplemental O₂ to correct hypoxemia (may necessitate mechanical ventilation) and meticulous fluid management to prevent worsening of pulmonary capillary leak. The best predictor of aspiration/hypoventilation is reduced level of consciousness, usually measured using the Glasgow Coma Scale. Tests: CXR or others as indicated from H&P
Cardiovascular	Blunt chest trauma can produce both cardiac contusion and aortic tear. Preop ECG and CPK isoenzymes will help evaluate the presence of myocardial injury. A wide mediastinal silhouette suggests aortic tear, which requires evaluation with TEE or angiography. Tests: Consider ECG; CPK isoenzymes; others as indicated from H&P
Neurologic	The possibility of coexistent neurologic trauma necessitates a thorough preop mental status review and peripheral sensory exam. A CT scan of the head is indicated for any patient with loss of consciousness prior to anesthesia.
Musculoskeletal	For trauma patients, C-spine films will evaluate the stability of the C-spine before neck manipulation during ET intubation. Thoracic and lumbar x-rays also should be evaluated for the presence of traumatic spinal deformity or instability that would require special stabilization in the anesthetized patient. Tests: C-spine x-rays or others as indicated from H&P
Hematologic	Restore Hct to 25% prior to inducing anesthesia. Have available one blood volume (70 mL/kg) or one total erythrocyte mass (20 mL/kg) for intraop transfusion. Transfusions of more than one blood

	volume will require monitoring and possible replacement of Plts and coag factors. The incidence of DVT is very high in these patients and prophylaxis with SCDs or low-dose sc heparin is indicated whenever feasible.
Renal	Renal injury commonly results from trauma to the collecting system, myoglobinuria from rhabdomyolysis, and ischemic, acute, tubular necrosis from hypovolemia or aortic dissection. Foley catheters should be placed only after urologic consultation for possible urethral tear. Suprapubic catheters are often necessary. Monitoring of UO is mandatory to detect intraop compromise of the collecting system and to monitor adequacy of renal perfusion. Tests: Consider UA; BUN; Cr; others as indicated from H&P
Laboratory	Hct; electrolytes; other tests as indicated from H&P
Premedication	In hemodynamically stable patients, pain can be treated with morphine (1–2 mg iv q10min titrated to pain relief) prior to anesthesia.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, ± celecoxib 200 mg po, and ± gabapentin 300 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA is indicated due to the duration and extent of the surgery, as well as the varied positions that are necessary to accomplish pelvic fixation. Regional anesthesia is generally inadequate for major pelvic surgery; however, in elective surgeries, serious consideration should be given to postop epidural analgesia.

Induction	A rapid-sequence induction is necessary for trauma
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	patients to minimize aspiration risk. Elective cases can undergo a standard induction.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Tranexamic acid (e.g., 10–15 mg/kg) iv before incision has been shown to ↓EBL and may be requested by the surgeon.	
Maintenance	Standard maintenance. Ventilation: lung-protective strategy. A ventilatory strategy should be adopted with tidal volumes around 5–8 mL/kg and plateau airway pressures below 30 cmH ₂ O. A minimum PEEP of 5 cmH ₂ O should be applied and adjusted.	
Emergence	Extubate trauma patients when fully awake and protective airway reflexes have returned. Do not extubate patients with evolving pulmonary injuries (fat embolism, aspiration, or contusion). Monitoring in an ICU usually is indicated for trauma and cancer patients. Prolonged stays can be anticipated for patients with severe coexistent trauma. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include ondansetron and dexamethasone.	
Blood and fluid requirements	Large EBL IV: 14–16 ga × 2 BSS @ up to 12 mL/kg/h 2–4 U PRBC in OR Warm fluids. Humidify gases	Expect large blood losses (from 0.5 to 2 or more blood volumes) with all but augmentation procedures. Use goal-directed fluid management. Cell-s scavenging techniques are useful to reduce the requirement for blood. Care should be taken to ensure that cells have been adequately washed to minimize ↓ BP on reinfusion. See Goal-

Directed Fluid Management, [Appendix K, p. K-4](#).

Control of blood loss

Deliberate hypotension
Hemodilution

Patients with severe cardiovascular disease or carotid artery stenosis are not candidates for ↓ BP. Full replacement of any volume deficit is mandatory before inducing ↓ BP. Commonly used agents are isoflurane (1–3%) or esmolol (50–200 mcg/kg/min) ± SNP (0.25–3 mcg/kg/min). These agents are titrated to produce a 30% ↓ in preop MAP (but not <60 mm Hg).

In trauma patients

A cerebral perfusion pressure of >65 mm Hg is advocated. A case-by-case decision should be made, balancing the requirements of cerebral oxygenation against the risk of worsening hemorrhage. In isolated head injury, hypotension must not be tolerated: an arterial pressure of 85–90 mm Hg may be beneficial.

Monitoring

Standard monitors
Arterial line

Patients for shelf procedures may require only standard monitoring.

	CVP line ± PA catheter ± TEE UO	Patients with myocardial dysfunction should have fluid and inotropic/pressor therapy, guided by continuous central venous saturation monitoring, a PA catheter, and/or TEE.
Positioning	✓ and pad pressure points ✓ eyes	Meticulous padding of the chest, pelvis, and extremities is imperative to prevent nerve injury and ischemia of the extremities. ↓ BP → risk of neurovascular injury.
Complications	Hypothermia Damage to urinary collecting system Major blood loss Coagulopathy	Warming of hypothermic patient may unmask severe volume depletion that will increase fluid requirement to well above apparent losses.

POSTOPERATIVE

Complications	Nerve root damage Peripheral nerve damage	Preop or intraop damage to L4-S5 nerve roots and cauda equina, resulting in hemiplegia and bladder and bowel dysfunction. Neuropathy of the femoral, genitofemoral, and lateral femoral cutaneous nerves can result from pressure on the ilioinguinal ligament during surgery.
Multimodal pain	Acetaminophen	Emphasize multimodal

management	NSAIDs	approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
	IV morphine Spinal/epidural opiates	Epidural hydromorphone 50 mcg/mL infused at 100–250 mcg/h, ± bupivacaine 0.125–0.25% at 4–8 mL/h, provides excellent analgesia.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6) Regional analgesia	Encourage early enteral nutrition, Foley catheter removal, and mobilization, for example, continuous nerve block.
Tests	Hct CXR Coag profile, as indicated	

Suggested Readings

1. Auyong DB, Allen CJ, Pahang JA, et al: Reduced length of hospitalization in primary total knee arthroplasty patients using an updated enhanced recovery after orthopedic surgery (ERAS) pathway. *J Arthroplasty* 2015; 30(10):1705–9.
2. Harris T, Davenport R, Hurst T, Jones J: Improving outcome in severe trauma: trauma systems and initial management: intubation, ventilation and resuscitation. *Postgrad Med J* 2012; 88:588–94.
3. Jones J Jr, Southerland W, Catalani B: The importance of optimizing acute pain in the orthopedic trauma patient. *Orthop Clin North Am* 2017; 48(4):445–65.
4. Masursky D, Dexter F, McCartney CJ, et al: Predicting orthopedic surgeons' preferences for peripheral nerve blocks for their patients. *Anesth Analg* 2008; 106(2):561–7.
5. Nutescu EA: Assessing, preventing, and treating venous thromboembolism: evidence-based approaches. *Am J Health Syst Pharm* 2007; 64(11 Suppl 7):S5–13.
6. Rosencher N, Bonnet MP, Sessler DI: Selected new antithrombotic agents and neuraxial anaesthesia for major orthopaedic surgery: management strategies. *Anaesthesia* 2007; 62(11):1154–60.

7. Zhu S, Qian W, Jiang C, et al: Enhanced recovery after surgery for hip and knee arthroplasty: a systematic review and meta-analysis. *Postgrad Med J* 2017; 93(1106):736–42.

ARTHROPLASTY OF THE HIP

SURGICAL CONSIDERATIONS

Description

Total hip arthroplasty is one of the most successful procedures in orthopedic surgery. In this procedure, the hip joint ([Fig. 10.4-7](#)) is approached through one of several standard incisions. The femoral head is dislocated from the acetabulum, and the arthritic femoral head and a portion of the neck are excised. The acetabulum is reamed to accept a cemented or cementless cup made of metal and plastic. The femoral stem and head are usually modular, allowing for numerous shapes, sizes, lengths, etc. The metallic femoral component may be cemented or cementless. A hybrid total hip combines a cemented femoral stem and a cementless acetabular cup. After relocation of the new prosthetic hip joint and closure of the tissues, the patient may be given an abduction pillow to minimize the risk of dislocation. Mobilization takes place over the ensuing days.

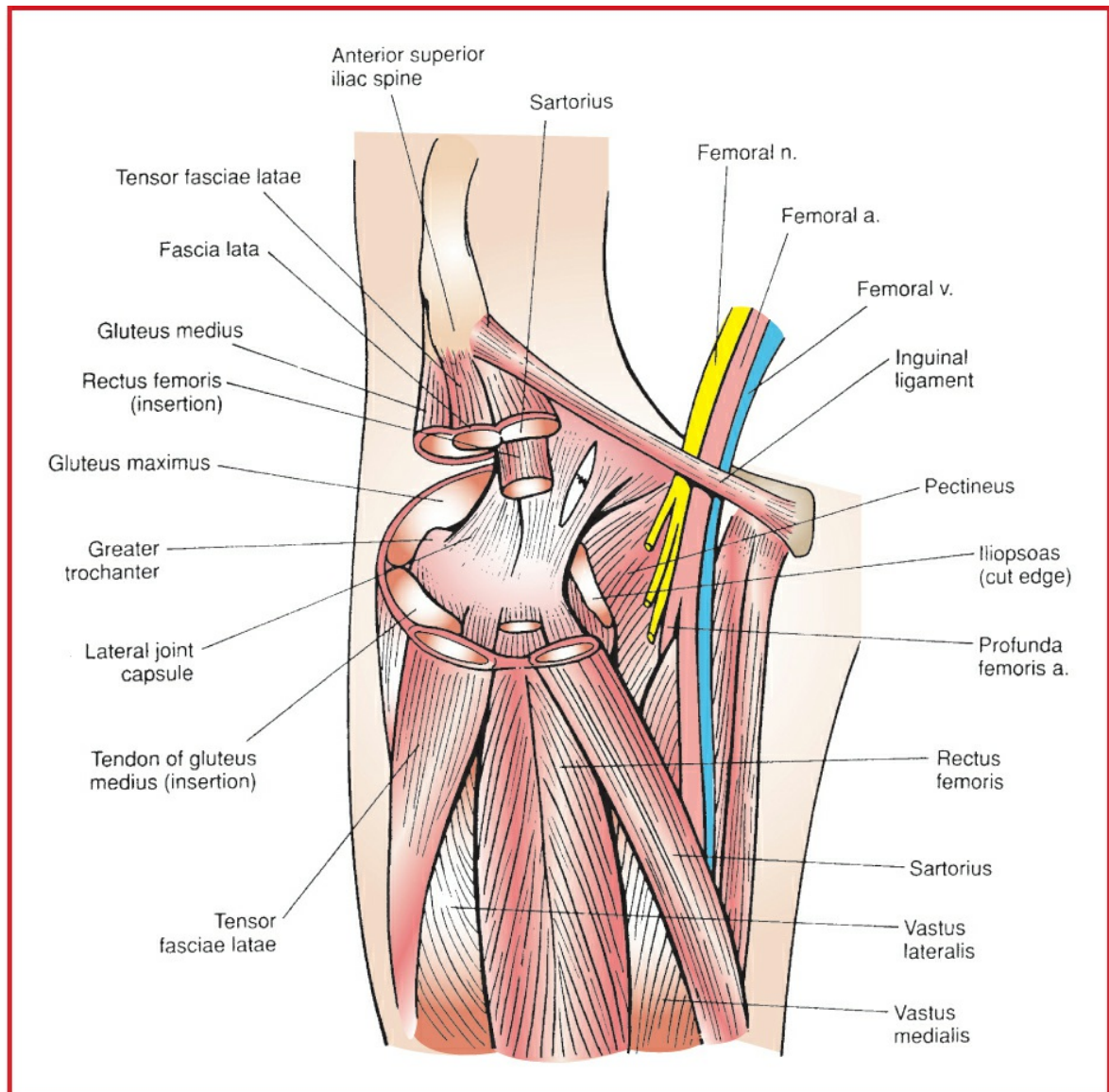


Figure 10.4-7 Surgical exposure of the hip joint. The hip joint may be exposed through a number of approaches. The relevant anatomical landmarks are shown here. (Reproduced with permission from Hoppenfeld S, deBoer P: Surgical Exposures in Orthopaedics. JB Lippincott, Philadelphia: 1984.)

Variant procedure or approaches

Unipolar (only the femoral side is replaced); **bipolar** (both the femoral side and the acetabular side are replaced; the acetabular cup is not fixed to the pelvis). **Revision procedures** are more arduous and time consuming, as the “failed” or loose component(s) must be removed and the bone prepared to accept new cemented or cementless components. These procedures require more specialized equipment for extracting prostheses and cement and rebuilding the femoral or acetabular bone stock (allografts, autografts, etc.). Often, special components are needed for implantation of a new prosthesis. In the **Girdlestone procedure (resection arthroplasty)**, the components are removed, but not replaced. This procedure is usually performed for

infection.

Usual preop diagnosis

Fracture of femoral neck; arthritis of the hip; arthrosis of the hip; loose (or malpositioned) hip prosthesis; chronic dislocation of hip arthroplasty; infected hip arthroplasty

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Unipolar, Bipolar, Total Hip Replacement	Revision, Total Hip Replacement	Girdlestone Resection Arthroplasty
Position	Supine (for anterior or anterolateral approaches); lateral decubitus position (for lateral or posterior approaches)	⇐	⇐
Incision	Anterior, lateral, or posterolateral over the hip joint	⇐	⇐
Special instrumentation	Appropriate prostheses and instrumentation	Special instruments for excising cement	⇐
Unique considerations	In lateral decubitus position, patient is usually stabilized by beanbag and/or kidney rests. SCDs used	A trochanteric osteotomy may be performed.	⇐
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if	⇐	⇐ after intraop cultures

	penicillin allergic)		
Surgical time	2–3 h	3–6 h or more	3 h or more
EBL	250–750 mL (intraop blood retrieval system may be useful)	≥1000 mL; intraop blood retrieval recommended	⇐
Postop care	Patient's legs immobilized between abduction wedge	⇐	⇐
Mortality	1–2% (increasing with age)	⇐	⇐
Morbidity	DVT:	⇐	⇐
	• Without prophylaxis: ≥50%	⇐	⇐
	• With chemoprophylaxis and mechanical prophylaxis: 2–3%	⇐	⇐
	Heterotopic ossification: 3–50% (average, 13%; significant, 4–5%)	>3–50%	⇐
	Intraop cementless fracture: 5–20%	⇐	⇐
	UTI: 7–14%	⇐	⇐
	Late aseptic loosening requiring revision: 5–10% (after 10 yr)	>5–10%	⇐
	Wound infection: 1%	3–10%	⇐
	• Primary psoriatic and diabetic patients: 5–10%		
	• Primary OA: 1%		
	Hematoma (major): <5%	⇐	5–10%

	Femoral/sciatic nerve injury: 0.7–3.5% PE: 1.8–3.4% (if no prophylaxis)	⇐	⇐
	Intraop cemented fracture: 1–3%	2–3%	1–3%
	Postop subluxation/dislocation: 0.5–3%	⇐	–
	Vascular injury to iliac vessels: <0.5%	>0.5%	⇐
	Urinary retention requiring catheterization: Rare	⇐	⇐
	GI bleed, MI, cholecystitis: Rare	⇐	–
			Neurologic injury: 3–10%
Pain score	7	8	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Hip fracture and cases of arthrosis of the hip joint: generally >60 yr Arthritis of the hip (e.g., rheumatoid arthritis or juvenile rheumatoid arthritis, traumatic arthritis): all ages
Male:Female	Dependent on disease etiology
Incidence	Common: ~500,000/yr in the United States
Etiology	Osteoarthritis; seropositive or seronegative arthritis; avascular necrosis; traumatic arthritis; congenital dislocation of the hip
Associated conditions	Dependent on primary conditions (e.g., rheumatoid arthritis patients may have numerous deformities, cardiorespiratory disease)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Hip Procedures.

Suggested Readings

1. Barrack RL: Current guidelines for total joint VTE prophylaxis: dawn of a new day. *J Bone Joint Surg Br* 2012; 94(11 Suppl A):3–7.
2. Berry DJ: Primary total hip arthroplasty. In: Chapman MW, ed. *Chapman's Orthopaedic Surgery*, Vol. 3, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001, 2769–94.
3. Johnson RL, Kopp SL, Hebl JR, et al: Falls and major orthopaedic surgery with peripheral nerve blockade: a systematic review and meta-analysis. *Br J Anaesth* 2013; 110(4):518–28.
4. Russo MW, Parks NL, Hamilton WG: Perioperative pain management and anesthesia: a critical component to rapid recovery total joint arthroplasty. *Orthop Clin North Am* 2017; 48(4):401–5.
5. Sugano N: Computer-assisted orthopaedic surgery and robotic surgery in total hip arthroplasty. *Clin Orthop Surg* 2013; 5(1):1–9.
6. Venker BT, Ganti BR, Lin H, et al: Safety and efficacy of new anticoagulants for the prevention of venous thromboembolism after hip and knee arthroplasty: a meta-analysis. *J Arthroplasty* 2017; 32(2):645–52.
7. Watts CD, Pagnano MW: Minimising blood loss and transfusion in contemporary hip and knee arthroplasty. *J Bone Joint Surg Br* 2012; 94(11 Suppl A):8–10.

ARTHRODESIS OF THE HIP

SURGICAL CONSIDERATIONS

Description

In adults, this procedure is accomplished by fusing the femur to the acetabulum. Some form of internal fixation is usually employed; a spica cast is sometimes placed immediately postop or a few days later. The patient is usually not a good candidate for total hip arthroplasty (e.g., a young, healthy male with unilateral traumatic arthritis). The hip usually is fused in 30° of flexion, 10–30° of external rotation, and neutral-to-slight adduction. The surgical procedure may be performed through anterior, lateral, or posterior incisions with the lateral being most common. A **trochanteric osteotomy** facilitates exposure. After excising the cartilage surfaces, internal fixation, using screws ± a plate, is performed ([Fig. 10.4-8](#)).

Usual preop diagnosis

Arthritis or arthrosis of the hip; previous septic arthritis of the hip; recurrent subluxation or dislocation of the hip

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

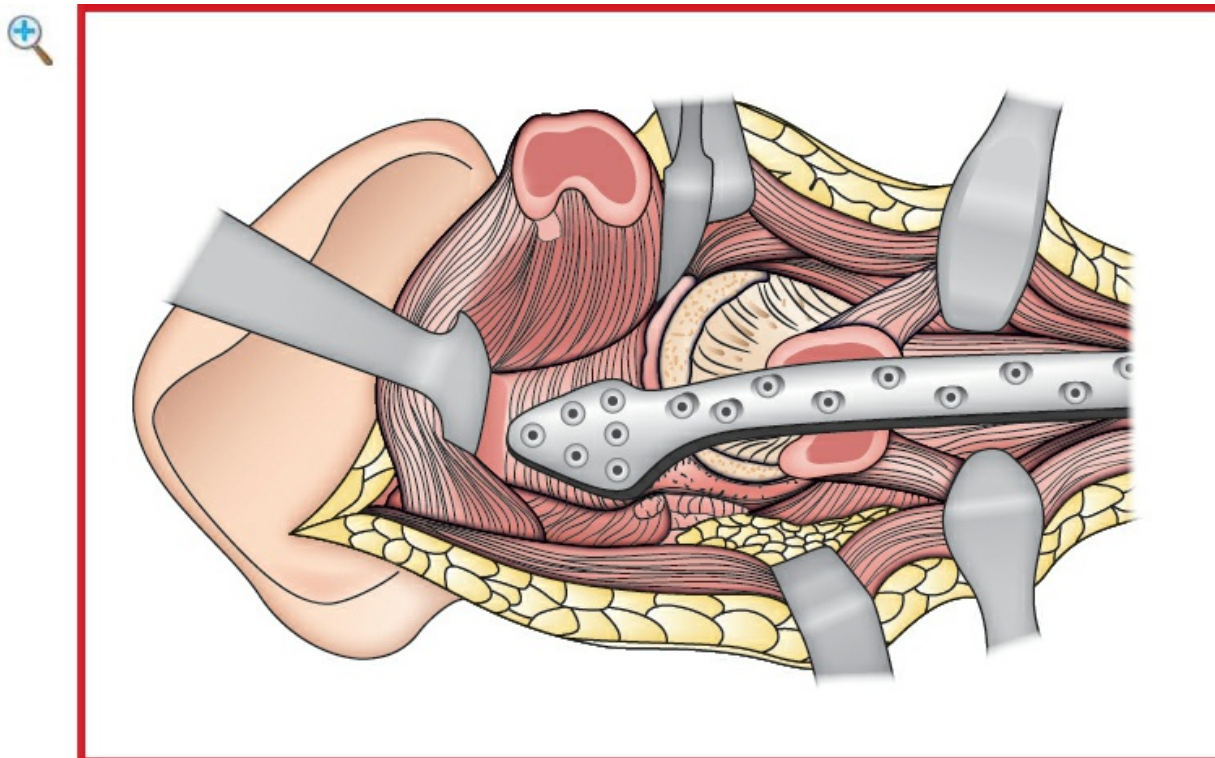


Figure 10.4-8 Application of cobra plate after it has molded to the shape of the acetabulum and femur and initial fixation with one proximal + distal outrigger compression screws. (Redrawn from Sledge CB, ed: The Hip. Lippincott-Raven, Philadelphia: 1998.)

■ SUMMARY OF PROCEDURE

Position	Usually supine; occasionally lateral decubitus
Incision	Anterior or lateral thigh
Special instrumentation	Plates and screws or other internal fixation; reamers from surface replacement arthroplasty set also may be useful; intraop x-ray; fracture table
Unique considerations	Intraop radiographs or I.I. used with patient on fracture table

Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)
Surgical time	3–4 h
EBL	500–1000 mL; Cell Saver recommended
Postop care	Spica cast
Mortality	Extremely low
Morbidity	Limb shortening: Some shortening is always present. Delayed union, nonunion, malunion: 10–15% Femoral shaft fracture: 5–10% Wound infection: <1% Genitourinary problems—urinary retention requiring catheterization: Common Ileus: Common Late degenerative arthritis of other lower extremity joints or back: Common, many years later Injury to iliac or femoral vessels: Rare Neurologic deficit; injury to lateral cutaneous nerve, sciatic nerve, or femoral nerves: Not uncommon Osteomyelitis: Rare Vascular complications: Rare Superior mesenteric artery syndrome causing duodenal obstruction: Extremely rare Thromboembolism: See Arthroplasty of the Hip.
Pain score	9

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–50 yr
Male:Female	Usually males > females
Incidence	Rare
Etiology	Trauma, general; neuromuscular disorders (cerebral palsy, meningomyelocele); trauma to acetabular growth plate; congenital hip dysplasia
Associated	Depends on etiology

conditions

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Hip Procedures.

Suggested Reading

1. Jain S, Giannoudis PV: Arthrodesis of the hip and conversion to total hip arthroplasty: a systematic review. J Arthroplasty 2013; 28(9):1596–602.

SYNOVECTOMY OF THE HIP

SURGICAL CONSIDERATIONS

Description

An **arthrotomy** of the hip joint is performed through one of several standard approaches (anterior, anterolateral, lateral, posterior). A **capsulotomy** is performed and is closed with reabsorbable sutures later in the case. Generally, the hip is not dislocated, but the cartilage surfaces are inspected and documented. The synovium, as well as any loose bodies, cartilage flaps, and osteophytes, is excised. Although weight bearing is usually protected, ROM and strengthening exercises are begun early.

Usual preop diagnosis

Chronic synovitis of the hip; loose bodies in the hip; juvenile and adult rheumatoid arthritis; pigmented villonodular synovitis; synovial chondromatosis

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURE

Position	Supine for anterior or anterolateral surgical approaches; lateral decubitus for posterior approaches
Incision	Overlying hip joint, depending on specific surgical approach

Unique considerations	Patient may have systemic disease (e.g., rheumatoid arthritis); careful positioning of limbs is necessary to avoid fracture or skin slough.
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)
Surgical time	2 h
EBL	<500 mL
Mortality	Rare
Morbidity	Thromboembolism: 10–50% Neurovascular injury—femoral or sciatic nerve or iliac vessels: <3% Wound infection or dehiscence: <3% Septic arthritis and osteomyelitis: <1% (unless synovectomy is performed for infection) Avascular necrosis of femoral head: Rare (if the hip is not dislocated) Hematoma: Rare (if drainage tubes are used) Inability to void, requiring urinary catheterization: Common
Pain score	7–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	<60 yr
Male:Female	Dependent on disease etiology (e.g., preponderance of females in rheumatoid arthritis)
Incidence	Rare
Etiology	Septic arthritis (very common); rheumatoid arthritis, juvenile/adult (rare); pigmented villonodular synovitis (rare); trauma (rare)
Associated conditions	See Etiology, above.

ANESTHETIC CONSIDERATIONS FOR HIP PROCEDURES

(Procedures covered: hip arthroplasty, arthrodesis, synovectomy)

PREOPERATIVE

Osteoarthritis is the most common indication for hip arthroplasty. These patients are usually elderly, and their anesthetic management is tailored to any concurrent disease. Rheumatoid and other inflammatory arthritides form another group of candidates for these procedures, and the special anesthetic considerations for these patients are outlined below. Avascular necrosis of the hip is seen in patients with sickle-cell disease and in heart transplant patients. A multimodal approach, aimed at reducing postoperative opioid usage, is commonly used. Consider preemptive analgesia with gabapentin 600 mg po, Celebrex 200 mg po, and acetaminophen 1 g po/iv to be given in the preop holding area.

Respiratory

Patients with rheumatoid arthritis frequently have associated pulmonary complications. SOB on performing activities of daily living or exercise (e.g., climbing a flight of stairs) warrants further evaluation with PFTs. Pulmonary effusions are common. Pulmonary fibrosis (rare) often manifests as cough and dyspnea. Rheumatoid arthritis involvement of the cricoarytenoid joints may produce glottic narrowing (requiring small ETT) and manifest as hoarseness. Arthritic involvement of the TMJ limits mouth opening and may necessitate special techniques (e.g., fiberoptic or light wand) for ET intubation.

Tests: As indicated from H&P

Cardiovascular

The severity of the arthritis often limits exercise; thus, a dobutamine stress ECHO and/or dipyridamole/thallium imaging may be necessary for adequate cardiac evaluation in patients with poor exercise tolerance. HTN and cardiovascular disease are common in elderly patients (dysrhythmias/TIAs → fall → hip fracture). Rheumatoid arthritis is associated with pericardial effusion, cardiac valve fibrosis, cardiac conduction abnormalities, and aortic regurgitation (AR). An ECG is indicated in all

	rheumatoid arthritis patients, and ECHO is indicated for patients with physical Sx suggestive of tamponade or cardiovascular disease. Test: As indicated from H&P
Neurologic	In patients with rheumatoid arthritis, a thorough neurologic exam preop often yields evidence of cervical nerve root compression. Patients with arthritis involving the cervical spine should have lateral neck films preop to determine the stability of the atlanto-occipital joint. After the stability of the spine has been established, full ROM of the neck should be evaluated for evidence of further nerve root compression or cerebral ischemia (suggesting vertebral artery compression). Evidence of cerebral ischemia mandates a neurovascular evaluation to plan intraop BP management. Tests: As indicated from H&P
Musculoskeletal	Pain and ↓ joint mobility make positioning and regional anesthesia difficult in patients with arthritis.
Hematologic	Rheumatoid arthritis patients often have anemia. Also, anemia may be 2° NSAID gastritis. Patients with Hb > 12 g/dL are candidates for preop autologous blood donation. Hip fracture → potential large-volume blood loss at fracture site. DVT is common after hip surgery, and prophylaxis for its occurrence reduces mortality. Effective preventive measures include SCDs and sc heparin. NSAID-induced coagulopathy may preclude the use of regional anesthesia. Test: As indicated from H&P
Renal	Estimation of renal function may be useful to predict drug clearance and the need for invasive monitoring in this elderly population. Test: As indicated from H&P
Laboratory	Other tests as indicated from H&P
Premedication	In the absence of limited pulmonary reserve or

	severe cardiac disease, a standard premedication is appropriate. Full-stomach precautions may be necessary for patients in acute pain. Consider preemptive analgesia (see above).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, ±celecoxib 200 mg po, and ±gabapentin 300 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA or regional anesthesia (may be difficult 2° pain on positioning).

General anesthesia

Induction	The lateral position may mandate ET intubation for patients undergoing GA. A careful preop airway evaluation will determine the need for special airway techniques (e.g., fiberoptic intubation, video laryngoscope, or light wand). Aggravation of cricoarytenoid arthritis that is common in rheumatoid arthritis patients can be minimized if a small ETT (6–7 mm cuffed) is used. For otherwise healthy patients, standard induction is appropriate.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Tranexamic acid (e.g., 10–15 mg/kg) iv before incision has been shown to ↓EBL and may be requested by the surgeon.
Maintenance	Standard maintenance. Neuromuscular blockade facilitates the placement and testing of the prosthesis. In otherwise healthy patients, induced hypotension (e.g., ↓ 20%) → ↓ blood loss. Maintain normothermia (forced-air blanket and fluid warmer).

Emergence

No special considerations. PONV prophylaxis (e.g., ondansetron 4 mg iv). Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include ondansetron and dexamethasone.

Regional anesthesia

Regional anesthesia offers the advantages of decreased periop DVT and pulmonary embolus, decreased intraop blood loss, and a drier surgical field (which may improve conditions for cementing). However, induction of regional anesthesia, with its attendant positioning requirements, can be uncomfortable in patients with limited joint mobility. Rheumatoid arthritis patients, however, rarely have involvement of the lumbar spine, and regional anesthesia offers the advantages of decreased periop DVT, decreased intraop blood loss, and no need for airway manipulation. Sedation or general anesthesia should be offered to supplement the regional technique. A typical dose for a subarachnoid block is 12.5–15 mg bupivacaine. 10–25 mcg of fentanyl and 100–200 mcg of morphine may be added for improved pain control up to 24 h postoperatively. Anesthesia to T10 is adequate. Full motor blockade is essential for placement of the prosthesis and assessment of the passive ROM. Lumbar epidural block (initial dose of 10–15 mL 2% lidocaine with epinephrine 1:200,000, administered over 10 min) has the advantage of slow onset, allowing time to treat the induced cardiovascular changes. Postop epidural opiates can provide excellent analgesia. Finally, a lumbar plexus block by the posterior approach (psoas compartment block) provides reliable postoperative analgesia, and hypotension and urinary retention are less common than with SAB or epidural analgesia. However, it is not sufficient for surgical anesthesia as the sole anesthetic. Continuous infusion is always initiated after the initial bolus of 15–20 mL of 0.125% bupivacaine or 0.2% ropivacaine. For patients with a high opioid tolerance, the use of a continuous epidural anesthetic (for either hip replacement or knee replacement) should be considered. For opioid-tolerant patients, low-dose iv ketamine (e.g., 250 mcg/kg/h) for all anesthetic techniques (not just GA) may be given intraop as an adjunct. In an effort to reduce adverse events associated with perioperative opioid consumption, there is emerging interest from the orthopedic community in eliminating the routine use of opioids in spinal anesthetics.

Blood and fluid requirements

Major blood loss
IV: 14–16 ga × 2
BSS @ 4–8 mL/kg/h

Cell scavenging helps reduce total transfusion requirement. Care should be taken to ensure that cells have been adequately washed to

		minimize ↓ BP on reinfusion. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Control of blood loss	Regional anesthesia Controlled hypotension	These techniques may be appropriate in selected patient populations.
Monitoring	Standard monitors ± CVP line ± Arterial line	Invasive monitoring is indicated in the presence of exercise-limiting cardiac or pulmonary disease.
Positioning	Axillary roll, beanbag ✓ and pad pressure points ✓ eyes	Meticulous padding of extremities and maintaining a neutral neck position are mandatory. A beanbag and axillary roll also are necessary to stabilize patient in the lateral position and to protect dependent arm from neurovascular compression injuries.
Complications	Methylmethacrylate: • ↓ BP 2° vasodilation • ↓ PaO₂ 2° embolization • Cardiovascular collapse VAE Major blood loss DVT (femoral vein: 80%) Nerve damage Femur fracture	Microemboli of air, fat, bone fragments, and cement may occur during pressurized cementing, which may lead to hypotension and hypoxia. Hypotension may be used to achieve a dry operating field, which improves cement penetration into the bone. However,

significant hypovolemia must be avoided because this predisposes to a severe reaction to pressurized cementing. Systemic ↓ BP and pulmonary HTN may occur. Care should be taken to ensure that patient is adequately hydrated before and during the procedure, and pressors may be necessary to maintain BP (ephedrine 5–20 mg iv or epinephrine 10–100 mcg iv and increasing the dose as necessary).

▼ POSTOPERATIVE

Complications	Nerve damage DVT Continued blood loss	Sciatic nerve injury is evidenced by foot drop and an inability to flex the knee.
Multimodal pain management	Neuraxial regimens: • SAB • Epidural analgesia	Intrathecal morphine 0.2–0.3 mg provides analgesia for up to 24 h after administration. May be administered along with bupivacaine for surgical anesthesia. Epidural bupivacaine 0.0125% 6–8 mL/h and hydromorphone 50 mcg/mL infused at 100–

		250 mcg/h provide good analgesia. Prior to removal of the epidural catheter, a 0.2-mg bolus of hydromorphone may be given. Epidural catheters are typically removed on the morning of postoperative day 2. Low molecular weight heparin may be started 2 h after catheter removal.
	Continuous lumbar plexus (psoas compartment) block	Infusion with 0.2% ropivacaine or 0.125% bupivacaine maintained at 6–12 mL/h
	Systemic regimens	Acetaminophen (1 g q6h) and celecoxib (200 mg q d) should be initiated immediately postop (if not contraindicated). Patient-controlled analgesia (PCA) with iv morphine or hydromorphone may be initiated as alternative to neuraxial blocks or to supplement a continuous lumbar plexus (psoas compartment) block. Consider pain service consult.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6) Regional analgesia	Encourage early enteral nutrition, Foley catheter removal, and mobilization, for example, continuous nerve block.

Tests

Hct
CXR, if CVP was placed.
Monitor UO.

Suggested Readings

1. Auyong DB, Allen CJ, Pahang JA, et al: Reduced length of hospitalization in primary total knee arthroplasty patients using an updated enhanced recovery after orthopedic surgery (ERAS) pathway. *J Arthroplasty* 2015; 30(10):1705–9.
2. Masursky D, Dexter F, McCartney CJ, et al: Predicting orthopedic surgeons' preferences for peripheral nerve blocks for their patients. *Anesth Analg* 2008; 106(2):561–7.
3. Miller AG, McKenzie J, Greenky M, et al: Spinal anesthesia: should everyone receive a urinary catheter?: a randomized, prospective study of patients undergoing total hip arthroplasty. *J Bone Joint Surg Am* 2013; 95(16):1498–503.
4. Ranawat AS, Ranawat CS: Pain management and accelerated rehabilitation for total hip and total knee arthroplasty. *J Arthroplasty* 2007; 22(7 Suppl 3):12–5.
5. Russo MW, Parks NL, Hamilton WG: Perioperative pain management and anesthesia: a critical component to rapid recovery total joint arthroplasty. *Orthop Clin North Am* 2017; 48(4):401–5.
6. Venker BT, Ganti BR, Lin H, et al: Safety and efficacy of new anticoagulants for the prevention of venous thromboembolism after hip and knee arthroplasty: a meta-analysis. *J Arthroplasty* 2017; 32(2):645–52.
7. Vieira EM, Goodman S, Tanaka PP: Anesthesia and rheumatoid arthritis. *Rev Bras Anesthesiol* 2011; 61:367–75.
8. Zhu S, Qian W, Jiang C, et al: Enhanced recovery after surgery for hip and knee arthroplasty: a systematic review and meta-analysis. *Postgrad Med J* 2017; 93(1106):736–42.

OPEN REDUCTION AND INTERNAL FIXATION (ORIF) OF PROXIMAL FEMORAL FRACTURES (FEMORAL NECK, INTERTROCHANTERIC, SUBTROCHANTERIC FRACTURES)

SURGICAL CONSIDERATIONS

Description

Fractures of the proximal femur are seen in two distinct populations: most commonly in elderly patients as the result of falls and in younger patients following trauma. In elderly patients, the fracture occurs through osteoporotic bone in the femoral neck, intertrochanteric, or subtrochanteric area (Fig. 10.4-9). Displaced femoral neck fractures are usually treated by **prosthetic replacement**. Nondisplaced or minimally displaced femoral neck fractures are usually treated by **closed reduction and percutaneous pinning** of the fracture. Intertrochanteric and subtrochanteric fractures, whether displaced or nondisplaced, are usually treated by **ORIF with a sliding hip screw and side plate, cephalomedullary nail, or blade plate** (Fig. 10.4-10). Prosthetic replacement is performed only rarely. Elderly patients frequently have numerous medical problems, which means that the fractures require prompt internal fixation/prosthetic replacement to facilitate early mobilization. In younger patients (16–40 yr), proximal femoral fractures are almost always treated by ORIF with screws, plates and screws, or intramedullary devices. These are normally much higher-energy fractures, often associated with multiple traumas.

Variant procedure or approaches

Variants include **percutaneous pinning** of nondisplaced femoral neck fracture, **ORIF of displaced femoral neck fracture** (also see Arthroplasty of the Hip), and **ORIF of intertrochanteric or subtrochanteric fracture**.

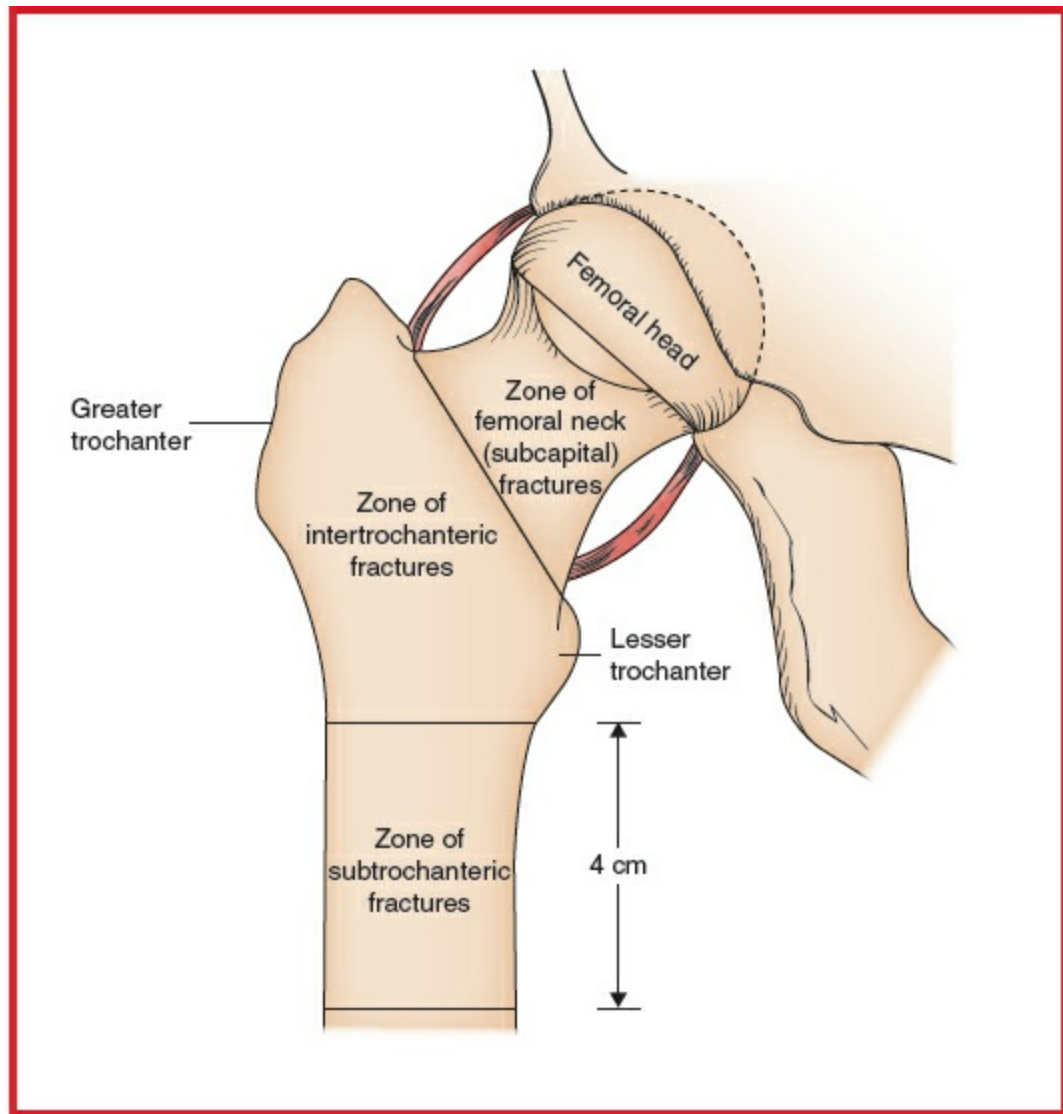


Figure 10.4-9 Anatomical classification of fractures of the proximal femur. (Redrawn with permission from Hardy JD: Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

Usual preop diagnosis

Nondisplaced femoral neck fracture; displaced femoral neck fracture (those not requiring prosthetic replacement); intertrochanteric $\pm$ subtrochanteric fracture

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

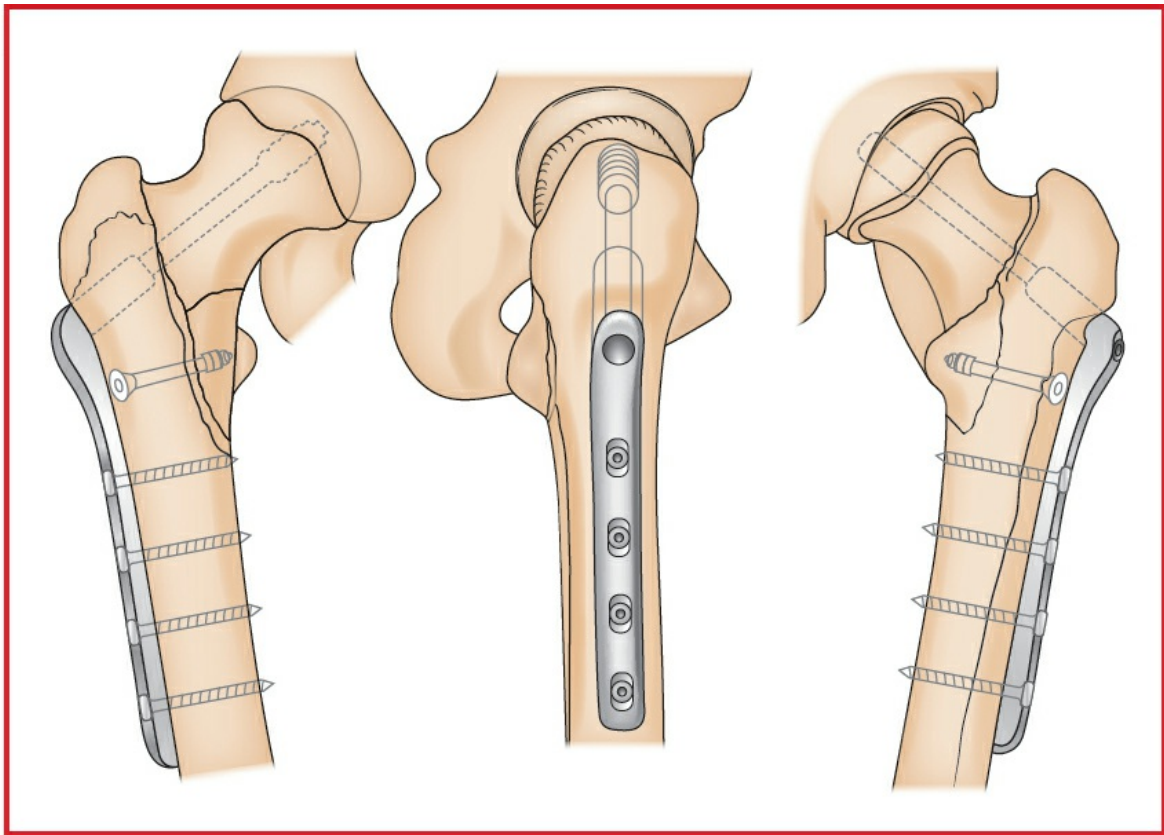


Figure 10.4-10 Intertrochanteric hip fracture treated with dynamic hip screw. (Redrawn from Sledge CB, ed: The Hip. Lippincott-Raven, Philadelphia: 1998.)

■ SUMMARY OF PROCEDURES

	Nondisplaced Femoral Neck Fracture	Displaced Femoral Neck Fracture	Intertrochanteric ±Subtrochanteric Fracture
Position	Supine, on fracture table	⇐	⇐
Incision	Proximal lateral thigh	⇐	⇐
Special instrumentation	Usually multiple percutaneous pins	Multiple screws or other devices	Screw-plate device or intramedullary device
Unique considerations	Fracture table and I.I. used. Percutaneous pinning may be	⇐	⇐

	accomplished with local anesthesia only.		
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)	⇐	⇐
Surgical time	1 h	1.5–2 h, including placing patient on fracture table and obtaining adequate reduction of fracture	1.5–3 h
EBL	<100 mL	250–500 mL	500+ mL
Postop care	Generally PACU → room; if medically unstable → ICU	⇐	⇐
Mortality	10–30% in the first 12 mo postop in elderly; in younger patients, depends on other involved trauma	⇐	⇐
Morbidity	Dysrhythmias: ~50%	⇐	⇐
	MI: ~50%	⇐	⇐
	Respiratory failure: ~50%	⇐	⇐
	Urinary retention requiring catheterization: ~50%	⇐	⇐

	UTI: ~50%	⇐	⇐
	Thromboembolism: 40%+	⇐	⇐
	Avascular necrosis and late segmental collapse: 10–20%	≥15–35%	1%
	Infection, deep: 2–17%	⇐	⇐
	Infection, superficial: 2–17%	⇐	⇐
	Septic arthritis: 2–17%	⇐	⇐
	Nonunion: 5–15%	20–30%	2%
	Malunion: <10%	>10%	10–20%
	Loss of reduction: <5%	>10%	10%
	Hematoma	—	—
	Intraop comminution of the fracture	—	—
	Neurologic injury: Rare	⇐	⇐
	Vascular injury: Rare	⇐	⇐
Pain score	5–6	7	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Usually >60 yr (patients with an intertrochanteric fracture average 65–70 yr); occasionally, younger patients, 16–35 yr (as part of multiple-trauma situation)
Male:Female	Elderly 1:4–5
Incidence	Extremely common—about 5–100/100,000; femoral neck fractures are about twice as common as intertrochanteric fractures.
Etiology	Accidents and falls (may be 2° TIA, stroke, MI,

Associated conditions

dysrhythmia); pathologic fracture; multiple trauma (younger patients); stress fracture

Numerous serious medical conditions often present in elderly; senile dementia; multiple trauma often

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower Extremity Procedures.

Suggested Readings

1. Kearns RJ, Moss L, Kinsella J: A comparison of clinical practice guidelines for proximal femoral fracture. *Anaesthesia* 2013; 68(2):159–66.
2. Moja L, Piatti A, Pecoraro V, et al: Timing matters in hip fracture surgery: patients operated within 48 hours have better outcomes. A meta-analysis and meta-regression of over 190,000 patients. *PLoS One* 2012; 7(10):e46175.
3. Palmer JS, Huber CP: Operative management of hip fractures: a review of the NICE guidelines. *Br J Hosp Med (Lond)* 2012; 73(9):C141–44.
4. Pivec R, Johnson AJ, Mears SC, et al: Hip arthroplasty. *Lancet* 2012; 380(9855):1768–77.

OPEN REDUCTION AND INTERNAL FIXATION (ORIF) OF DISTAL FEMUR FRACTURES

SURGICAL CONSIDERATIONS

Description

ORIF of the distal femur fracture involves a longitudinal incision along the femoral shaft, obtaining reduction by direct visualization of the fracture fragments, and applying plates and screws along the femur for rigid internal fixation. An iliac crest bone graft may be necessary. Some intramedullary devices are also available for fixation of these fractures.

Usual preop diagnosis

Fracture of the distal femur; nonunion/malunion of the distal femur; degenerative arthritis of the knee, with deformity

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO

duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine. Patient usually arrives at OR in balanced traction if fracture is acute.
Incision	Anterior knee, lateral or medial thigh along the femoral shaft
Special instrumentation	Special plates, screws, rods, reduction clamps; radiolucent table; intraop blood salvage
Unique considerations	Usually requires intraop radiographs; tourniquet
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)
Surgical time	3 h (or more, depending on difficulty)
EBL	750 mL or more
Postop care	Multiple-trauma victim → ICU; others → PACU
Mortality	~3–4%, depending on the extent of multiple trauma
Morbidity	Nonunion: 4–33% Malunion: 4–31% Infection, osteomyelitis, septic arthritis, closed/open: Grade I: 1–5% Grade II: 5–20% Grade III: >20% Delayed union: 0–17% Vascular complications: 2–3% Neurologic deficit to peripheral nerves, peroneal nerve: ~3% Compartment syndrome: Rare Hypotension: Rare Leg-length discrepancy: Rare Respiratory distress and fat embolism: Rare
Pain score	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age; predominance of males <40 yr (trauma); degenerative arthritis of the knee <60 yr. Special rare case is elderly patient with a supracondylar fracture above a total knee replacement.
Male:Female	5:1
Incidence	Common in trauma center patients; rare in cases of degenerative arthritis of the knee (osteotomy) or elderly patient with a supracondylar fracture above a total knee replacement
Etiology	Motorcycle and motor vehicle accidents; falls; industrial injury; degenerative arthritis of the knee
Associated conditions	Frequently associated with trauma to other organ systems

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower Extremity Procedures.

OPEN REDUCTION AND INTERNAL FIXATION (ORIF) OF THE FEMORAL SHAFT WITH PLATE

■ SURGICAL CONSIDERATIONS

Description

ORIF of the femoral shaft involves obtaining a reduction by open means, usually through a longitudinal lateral incision along the length of the femur, and applying plates and screws along the femur to maintain the reduction. An iliac crest bone graft may be necessary.

Usual preop diagnosis

Fracture of the femur

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO

duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine or lateral decubitus
Incision	Lateral thigh along the length of the femur ± iliac crest incision
Special instrumentation	Special plates, screws; reduction clamps; blood salvage device
Unique considerations	Fracture or radiolucent table; I.I. Patient usually arrives at OR in balanced traction.
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)
Surgical time	3 h or more, depending on difficulty
EBL	750 mL; Cell Saver recommended
Postop care	Multiple-trauma victims: ICU Others: PACU
Mortality	Dependent on the extent of multiple trauma
Morbidity	Knee stiffness: 20–30% Delayed union, nonunion, malunion: 5–21% Leg-length discrepancy: 0–11% Failure of fixation: 5–10% Infection; osteomyelitis: <5% Hypotension: Not uncommon Respiratory distress and fat embolism: Not uncommon, often subclinical Compartment syndrome: Rare Neurologic deficit to peripheral nerves: Rare Vascular complications: Rare
Pain score	9

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age, but predominance of males <30 yr
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Male:Female	5:1
Incidence	Unknown
Etiology	Motorcycle and motor vehicle accidents; falls; industrial injuries
Associated conditions	Frequently associated with trauma to other organ systems

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower Extremity Procedures.

Suggested Reading

1. Nahm NJ, Vallier HA: Timing of definitive treatment of femoral shaft fractures in patients with multiple injuries: a systematic review of randomized and nonrandomized trials. *J Trauma Acute Care Surg* 2012; 73(5):1046–63.

INTRAMEDULLARY NAILING OF FEMORAL SHAFT

SURGICAL CONSIDERATIONS

Description

Intramedullary nailing of the femur is the standard procedure for fractures of the femoral shaft ([Fig. 10.4-11](#)). It is indicated for virtually any fracture, from the lesser trochanter to the distal femur, within 7 cm of the articular surface. The procedure also is used for the treatment of nonunions and malunions of the femoral shaft. Typically, the nail is placed in an antegrade fashion from proximal to distal. There are, however, indications in which the nail is inserted in a retrograde fashion from distal to proximal (e.g., bilateral femur fractures, ipsilateral femur and tibia fractures, distal femur fractures, and multiple-trauma, obese, and pregnant patients). Early fixation of femoral shaft fractures in severe polytrauma has several benefits. The advantages of early fixation of long bones include improved pain control, early mobilization, improved pulmonary function, and decreased morbidity and mortality.

Femoral nails are inserted after reaming the intramedullary canal, which allows a larger-diameter implant and improves the mechanical properties of the bone implant interface. Reaming may produce systemic effects by embolic showering of medullary contents to the pulmonary vasculature, a phenomenon that has been documented by TEE.

This situation may be exacerbated in the polytrauma patient with pulmonary injury and may produce posttraumatic pulmonary failure. Patients with femur fractures may have associated injuries.

Patients may arrive from the ICU in skeletal traction or from the emergency department especially in the case of an open fracture. Hemorrhage up to 1 L may be contained in the thigh following a femur fracture; therefore, patients may be hypovolemic at the start of the procedure. Because the procedure is essentially percutaneous, apparent blood loss may be underestimated because of the hemorrhaged blood contained in the thigh.

The patient is placed in the supine or lateral decubitus position on either a radiolucent table or a fracture table. Antegrade insertion of the nail requires a lateral incision several cm in length proximal to the greater trochanter. The hip abductors are split, and portal into the femoral canal is created in the piriformis fossa. The femoral canal is reamed over a guide wire. The intramedullary nail is then inserted into the intramedullary canal with gentle taps, using a hammer.

Retrograde insertion of the nail is performed through an incision several cm long over the anterior aspect of the knee. The knee joint is entered and the portal to the intramedullary canal is made in the non-weight-bearing portion of the intercondylar notch. The nail is inserted in the same fashion as the anterior nail. Cross-locking screws are commonly used, and they are inserted through the rod.

Variant procedure or approaches

The application of femoral nailing has been expanded to treat nonunions, malunions, posttraumatic deformities of the femur, and leg-length differences. Specialized additional equipment, such as an intramedullary saw or an external fixator, may be required for these procedures.

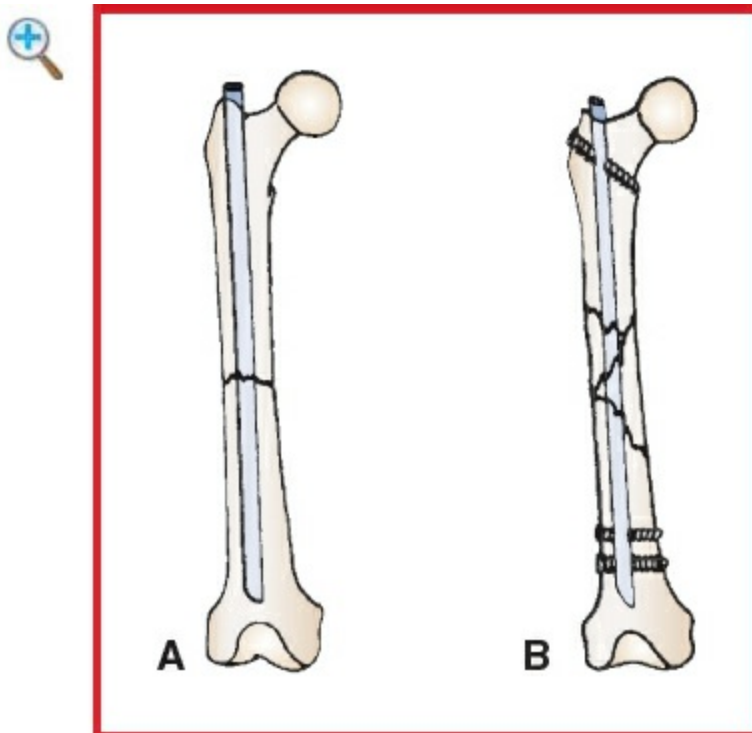


Figure 10.4-11 Simple (A) and locked (B) intramedullary fixation of the femoral shaft. (Reproduced with permission from Hardy JD: Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

Usual preop diagnosis

Femur fracture; nonunion/malunion of the femur; leg-length discrepancy

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine or lateral decubitus
Incision	Proximal lateral thigh or anterior knee
Special instrumentation	Intramedullary nails; intramedullary saw for closed femoral shortening of the femur; fracture or radiolucent table; I.I.
Unique considerations	Patient may have multiple traumas with other associated injuries. Procedure is heavily dependent on I.I.

Antibiotics	Cefazolin 2 g iv q 6–8 h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)
Surgical time	2–3 h; 3+ h for nonunion/malunion of the femur or closed femoral shortening of the femur
EBL	250–500 mL; ≥750 mL for nonunion/malunion of the femur or closed femoral shortening of the femur (Cell Saver recommended)
Postop care	If surgery is performed acutely, patient is usually a multiple-trauma victim with numerous injuries and extensive blood loss; usually goes to ICU.
Mortality	Dependent on the extent of trauma
Morbidity	Respiratory distress and fat embolism: 10% Malunion: 5–10% Infection, osteomyelitis: Open technique: 1–10% Closed technique: 0–1% Neurologic deficit to peripheral nerves: 2% Vascular complications: ~2% (up to 15% have occult vascular abnormalities) Delayed union: 1% Nonunion: 1% Hypotension: Common in multiple-trauma situations Knee stiffness: Common Leg-length discrepancies: Not uncommon Compartment syndrome: Rare Failure of fixation: Rare
Pain score	6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age, but predominance of males <30 yr
Male:Female	5:1
Incidence	Very common
Etiology	Motorcycle and motor vehicle accidents (60–80%); falls (5–10%); industrial injuries (5–10%); previous

Associated conditions

trauma (rare)

Frequently associated with trauma to other organ systems

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower Extremity Procedures.

Suggested Readings

1. Rhorer AS: Percutaneous/minimally invasive techniques in treatment of femoral shaft fractures with an intramedullary nail. J Orthop Trauma 2009; 23(5 Suppl):S2–5.
2. Ricci WM, Gallagher B, Haidukewych GJ: Intramedullary nailing of femoral shaft fractures: current concepts. J Am Acad Orthop Surg 2009; 17(5):296–305.

REPAIR OF NONUNION/MALUNION OF PROXIMAL THIRD OF THE FEMUR, PROXIMAL FEMORAL OSTEOTOMY FOR OSTEOARTHRITIS

SURGICAL CONSIDERATIONS

Description

Operations for nonunion/malunion of the proximal femur entail realigning the bones with a femoral osteotomy (as necessary), stabilizing the reduction with internal fixation (using a nail/plate-nail/rod device), and supplementing this with a bone graft. In young patients (<50 yr) in whom early osteoarthritis of the hip spares some of the cartilage, the hip may be realigned with **proximal femoral osteotomy**. This entails cutting the bone at the level of the lesser trochanter, realigning the hip, and stabilizing the osteotomy with internal fixation.

Variant procedure or approaches

Osteotomy of proximal 1/3 of the femur for degenerative arthritis of the hip.

Usual preop diagnosis

Nonunion/malunion of proximal 1/3 of the femur; early degenerative arthritis of the hip

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO

duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Repair Nonunion/Malunion	Proximal Femoral Osteotomy
Position	Supine or lateral decubitus	⇐
Incision	Proximal lateral thigh	⇐
Special instrumentation	Plates, screws; reduction clamps; occasionally, an intramedullary device. Cell Saver recommended. Some surgeons use fracture or radiolucent table with I.I.	⇐
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)	⇐
Surgical time	2 h for bone grafting alone; 3 h or more if difficult malunion	3 h
EBL	500–750 mL or more	500 mL +
Mortality	Rare: <1%	⇐
Morbidity	Leg-length discrepancy: Common Technical complications/fixation failure: 1–5% Superficial, deep infection/osteomyelitis: 1% Malunion: Not uncommon Compartment syndrome: Rare	– ⇐ ⇐ – ⇐

	Neurologic deficit to peripheral nerves: Rare Vascular complications: Rare	⇐ ⇐
		Progressive pain/arthritis (by 5–10 yr): 40–50% Delayed union: 5–10% Nonunion: 1–5%
Pain score	8	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age	⇐
Male:Female	1:1	⇐
Incidence	Rare	⇐
Etiology	Previous surgery (rare); previous trauma (rare)	Early degenerative arthritis of the hip (not uncommon)
Associated conditions	May accompany multiple traumas	⇐

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower Extremity Procedures.

CLOSED REDUCTION AND EXTERNAL FIXATION OF THE FEMUR

■ SURGICAL CONSIDERATIONS

Description

This procedure entails manipulating the femur to obtain an acceptable reduction by closed or limited-open means and then applying an external fixation device to maintain the reduction. The pins for the external fixator are inserted percutaneously or through small incisions. This method of treatment may be used for severe open fractures (e.g., grade III) with extensive bone and soft-tissue injury.

Usual preop diagnosis

Displaced, open fracture of the femur

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine or lateral
Incision	Done percutaneously or through small incisions
Special instrumentation	External fixation device of surgeon's choice; radiolucent table with I.I.
Unique considerations	Fracture is usually an open, extremely comminuted fracture in a multiple-trauma patient.
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if penicillin allergic)
Surgical time	1 h
EBL	Operative blood loss usually 100–200 mL; however, blood loss may be extensive before surgery.
Postop care	Multiple-trauma victims → ICU; others → PACU
Mortality	Dependent on the extent of trauma
Morbidity	Refracture: 2–12% Hypotension 2° blood loss and other injuries: Common Stiffness of the knee: Common Delayed union, nonunion, malunion: More common in comminuted, open fractures Leg-length discrepancy: More common in severely comminuted fractures Osteomyelitis: More common in open fractures Respiratory distress: More common in multiple-trauma situations Amputation: Rare Compartment syndrome: Rare

	Neurologic deficit to peripheral nerves: Rare Vascular complications: Rare
Pain score	9–10 (due to extensive open fracture)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age, but predominance of males <30 yr
Male:Female	5:1
Incidence	Common
Etiology	Motorcycle and motor vehicle accidents; falls; industrial injury
Associated conditions	Frequently associated with trauma to other organ systems, including the head and neck, chest, abdomen, and extremities; these will often be treated simultaneously with the femur fracture.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower Extremity Procedures.

Suggested Viewing

Links are available online to the following videos:

Hip Fractures, Types and Fixation—Everything you Need to Know—Dr. Ebraheim: <https://www.youtube.com/watch?v=6wok5JGEpTQ>

Hip Replacement Surgery: <https://www.youtube.com/watch?v=BYwVaKkRdF4>

Femoral Shaft Fracture Intramedullary Nailing: <https://www.youtube.com/watch?v=vOCf8MhnQBs>

Ultrasound-Guided Femoral Nerve Block: https://www.youtube.com/watch?v=5ht_N8j2KL8

Ultrasound-Guided Subgluteal Sciatic Nerve Block: <https://www.youtube.com/watch?v=rl8rZOEMveE>

CHAPTER 10.5

Knee Surgery

SURGEONS

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ARTHROPLASTY OF THE KNEE

SURGICAL CONSIDERATIONS

Description

In this procedure, an arthrotomy of the knee joint is performed, and metallic and plastic components are used for replacement of the knee joint surfaces (total knee replacement). See [Figure 10.5-1](#). The femur, patella, and tibia are exposed; cartilage and minimal bone are excised with a saw. The new components may be cemented or uncemented. Alternatively, arthroplasty may be performed on only one compartment of the knee (i.e., medial/lateral unicompartmental knee replacement). In revision procedures, one or more components of the old joint are removed, and new components are placed. In resection or excision arthroplasty of the knee (usually for infection of the prosthesis), the components are removed but not replaced.

Usual preop diagnosis

Arthritis of the knee; arthrosis of the knee; loose (or malpositioned) knee prosthesis; infected knee

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plans. See [Appendix K: Enhanced Recovery After Surgery](#), p. K-1.

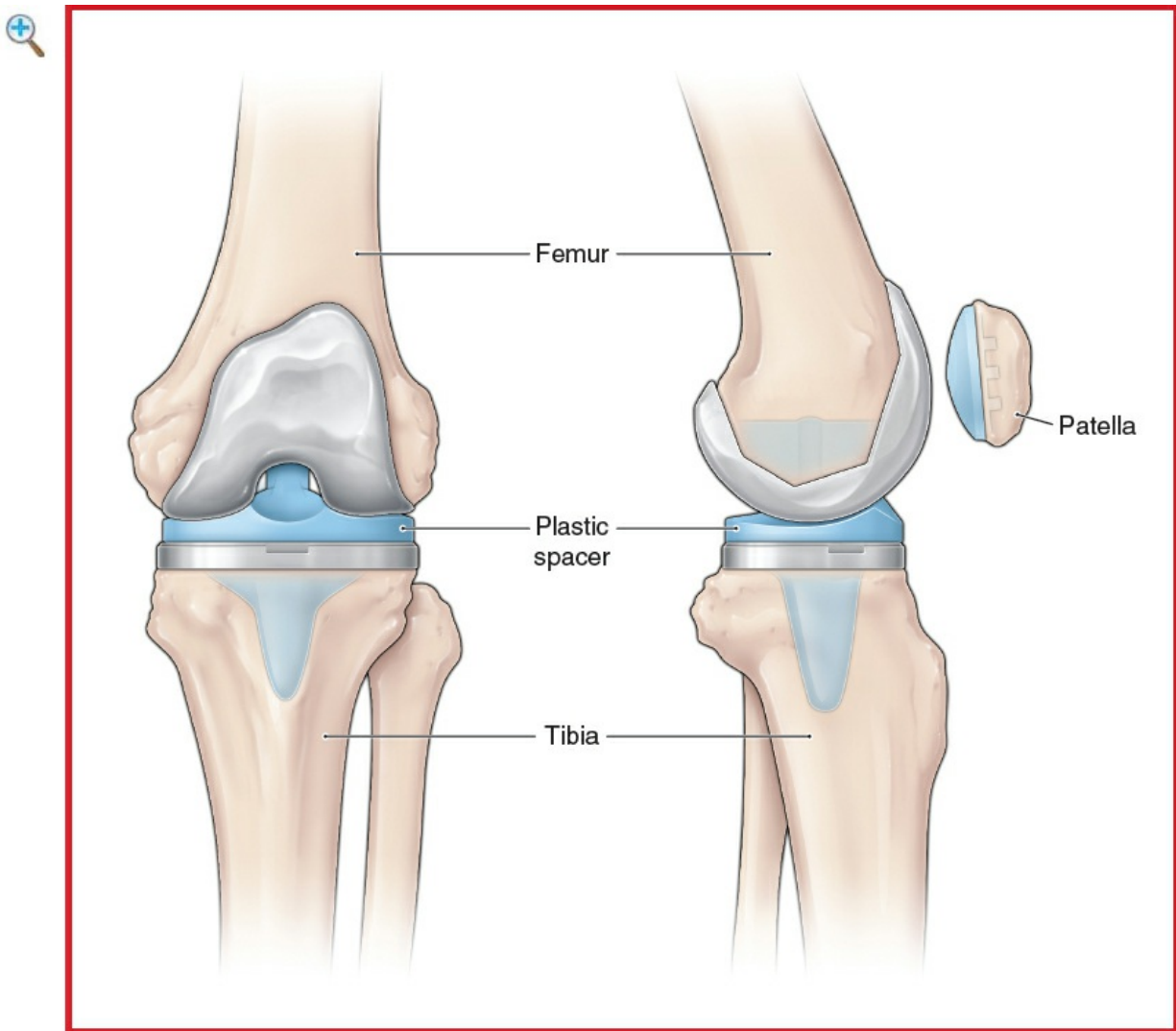


Figure 10.5-1 AP and lateral views of the components used for TKR.

■ SUMMARY OF PROCEDURES			
	Knee Replacement	Revision	Resection/Excision
Position	Supine	⇐	⇐
Incision	Anterior or anteromedial over patella	⇐	⇐

Special instrumentation	Appropriate prostheses and instrumentation	Special instruments for excising cement	⇐
Unique considerations	± Tourniquet; ± SCD	⇐	⇐
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if true penicillin allergic)	⇐	⇐
Surgical time	2 h	3–4 h or more	3 h
Closing considerations	In infected or complex revision cases (rare), a local or free flap is required.	⇐	⇐
EBL	300–500 mL	500–1000 mL	⇐
Postop care	Bulky dressing or splint; continuous passive motion (CPM) may begin in the PACU or ward.	⇐	Splint/cast
Mortality	Rare	⇐	⇐
Morbidity	DVT, without prophylaxis: 50–75%	⇐	⇐
	DVT, with prophylaxis (e.g., low molecular weight heparin, Coumadin, SCD, antiembolism stockings): 2–3%	⇐	⇐
	Postop subluxation/dislocation of patella: 20%	>30%	—

Superficial wound necrosis: 10–15%	>10–15%	≥10–15%
Wound infection: primary rheumatoid or psoriatic arthritis, diabetes: 5–10%	>5–10%	Rare
Primary osteoarthritis (OA): 1%		
PE: 1–7%	⇐	⇐
Postop subluxation/dislocation of knee joint: 1–6%	≥1–6%	–
Late aseptic loosening requiring revision after ~10 yr: 5%	–	–
Peroneal nerve injury: 1–5%	>1–5% (more common in difficult revisions)	1–5%
Urinary retention requiring catheterization: common	–	–
Hematoma requiring reoperation: rare	–	–
Hypotension	–	–
Knee stiffness	–	–
Intraoperative fracture: rare	⇐	⇐
Wound dehiscence: rare	–	–
Fat embolism: rare	–	–
Vascular injury to popliteal vessels: rare	–	–

Pain score

7

8

9

■ PATIENT POPULATION CHARACTERISTICS

Age range	Generally, >60 yr. Arthritis of the knee (e.g., rheumatoid arthritis or juvenile rheumatoid arthritis); hemophilia, ≥18 yr
Male:Female	1:1
Incidence	Common (~400,000/yr in the United States)
Etiology	Arthrosis of the knee (degenerative joint disease [DJD] or OA); seropositive or seronegative arthritis; traumatic arthritis; hemophilic arthropathy of the knee
Associated conditions	Dependent on primary condition (e.g., osteoarthritis)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Knee Procedures.

Suggested Readings

1. Johnson RL, Kopp SL, Hebl JR, et al: Falls and major orthopaedic surgery with peripheral nerve blockade: a systematic review and meta-analysis. *Br J Anaesth* 2013; 110(4):518–28.
2. Lieberman JR, Heckmann N: Venous thromboembolism prophylaxis in total hip arthroplasty and total knee arthroplasty patients: from guidelines to practice. *J Am Acad Orthop Surg* 2017; 25(12):789–98.
3. Lu Q, Peng H, Zhou GJ, et al: Perioperative blood management strategies for total knee arthroplasty. *Orthop Surg* 2018; 10(1):8–16.
4. Tzatzairis T, Fiska A, Ververidis A, et al: Minimally invasive versus conventional approaches in total knee replacement/arthroplasty: A review of the literature. *J Orthop* 2018; 15(2):459–66.
5. Yu ZX, Yang ZZ, Yao LL: Effectiveness of liposome bupivacaine for postoperative pain control in total knee arthroplasty: a PRISMA-compliant meta-analysis of randomized controlled trials. *Medicine (Baltimore)* 2018; 97(13):e0171.

ARTHRODESIS OF THE KNEE

SURGICAL CONSIDERATIONS

Description

In this procedure, the femur is fused to the tibia, obliterating the knee joint. Through a midline incision and anterior or median parapatellar arthrotomy, the cartilage surface and a small amount of bone are excised. The cut ends are opposed and aligned in 0°–20° of flexion and 5–10% of valgus. The bones are stabilized with plates, screws, an intramedullary rod, or an external fixator.

Usual preop diagnosis

Arthritis or other arthrosis of the knee; previous septic arthritis of the knee; failed or infected knee arthroplasty

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Usually supine
Incision	Anterior midline over knee
Special instrumentation	External fixator; internal fixation with plates and screws or intramedullary nail
Unique considerations	Intraop radiographs or I.I.; tourniquet
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if true penicillin allergic)
Surgical time	3 h (+1 h, if necessary, to excise total knee arthroplasty)
Closing considerations	Cast or splint while anesthetized
EBL	<100 mL, if tourniquet and local fixation used 500–1000 mL, if no tourniquet used, or if intramedullary procedures are used
Mortality	Rare but depends primarily on age and medical condition of patient

Morbidity	Thromboembolism $\geq$ incidence following total knee replacement DVT (without prophylaxis): 50–75% DVT (if prophylaxis used): 10–20% PE (if no prophylaxis; reduced if anticoagulation or SCDs used): 1–7% Failure of fusion (nonunion), malunion: 10% After failed knee replacement: 19–44% With Charcot joint: as high as 50% Pin tract infection: $\geq$1–10% Wound infection: 5% Deep infection and osteomyelitis Urinary retention requiring catheterization, UTI: common Breakage or failure of internal or external fixation: rare Fat embolism: rare GI bleed, MI: rare Hematoma: rare Hypotension: rare Intraop femoral or tibial fracture: rare Neurologic injury, usually popliteal nerve or peroneal nerve: rare Superficial wound necrosis and wound dehiscence: rare Vascular injury to popliteal vessels: rare Amputation: extremely rare (usually 2° acute arterial occlusion or uncontrollable local sepsis)
Pain score	9

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age
Male:Female	1:1
Incidence	Rare
Etiology	Failed or infected total knee replacement (probably most common etiology); trauma to knee—

unreconstructable, intra-articular fractures; total unstable knee or failed ligament repairs with severe DJD in a young patient

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Knee Procedures.

Suggested Reading

1. Schwarzkopf R, Kahn TL, Succar J, et al: Success of different knee arthrodesis techniques after failed total knee arthroplasty: is there a preferred technique? J Arthroplasty 2014; 29(5):982–8.

OPEN REDUCTION AND INTERNAL FIXATION (ORIF) OF PATELLAR FRACTURES

SURGICAL CONSIDERATIONS

Description

In ORIF of patellar fractures, a short incision over the patella is used to perform a reduction by direct visualization of the fracture fragments of the patella. Because this is generally an intra-articular fracture, the fragments should be reduced precisely. The torn quadriceps retinaculum is also repaired. Part or the entire patella may be excised; pins, wires, and/or screws are normally used to fix the patellar fragments together internally. Thereafter, the knee is casted, or early motion of the knee is started.

Usual preop diagnosis

Fracture of patella; severe degenerative arthritis of patellofemoral joint

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURES

Position	Supine
Incision	Anterior over patella

Special instrumentation	Wire, pins, screws as necessary
Unique considerations	Intraop radiographs may be obtained; tourniquet
Antibiotics	Cefazolin 2 g iv q6–8h × 24 h (vancomycin or clindamycin for 24 h if true penicillin allergic)
Surgical time	1.5–2 h
Closing considerations	Splint or cast usually applied
EBL	<100 mL
Mortality	<1%
Morbidity	Late degenerative arthritis of patellofemoral joint: ~50–60% DVT: ~5% Wound infection, septic arthritis, osteomyelitis: ~5% Delayed union, nonunion, malunion: ~2–5% Knee stiffness: common Weakness: common Avascular necrosis: rare Complex regional pain syndrome: rare Following patellectomy—quadriceps strength: ~75% of normal
Pain score	7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age; frequently seen in young, active, healthy adults
Male:Female	1:1
Incidence	~1% of all skeletal injuries
Etiology	Trauma: falls (60%); motorcycle and motor vehicle accidents (25–35%); industrial injury (6%); degenerative arthritis of patellofemoral joint (rare)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Knee Procedures.

Suggested Readings

1. Callaghan JJ, O'Rourke MR, Saleh KJ: Why knees fail: lessons learnedJ Arthroplasty 2004; 19(4 Suppl 1):31–4.
2. Dy CL, Little MT, Berkes MB, et al: Meta-analysis of re-operation, nonunion, and infection after open reduction and internal fixation of patella fractures. J Trauma Acute Care Surg 2012; 73(4):928–32.
3. Kakazu R, Archdeacon MT: Surgical management of patellar fractures. Orthop Clin North Am 2016; 47(1):77–83.

REPAIR OR RECONSTRUCTION OF KNEE LIGAMENTS

SURGICAL CONSIDERATIONS

Description

Collateral ligaments usually are repaired by direct suture or by stapling the torn ligaments to bone. Cruciate tears are generally repaired only if bone is avulsed at one end of the ligament, again with direct suture, staples, or screws. For collateral ligament repair, a longitudinal incision is made directly over the ligament medially or laterally. The ligament is exposed by deep dissection and elevation of skin flaps. The torn ligament is repaired by direct suture or by fixing it to bone with a screw or staple. Following closure, the knee is immobilized with a long leg splint or cast. Cruciate ligaments are repaired in similar fashion, except for the approaches: medial parapatellar (with anterior arthrotomy) for the anterior cruciate ligament (ACL) and posteromedial (with posterior arthrotomy) for the posterior cruciate ligament (PCL). Cruciate ligament reconstruction is performed for instability 2° intrasubstance tears of these ligaments. Homografts, such as a portion of the patellar tendon or semitendinosus tendon, normally are used, but allografts or synthetics also are available. (The ligaments of the knee are illustrated in [Figs. 10.5-2](#) and [10.5-3](#).)

Usual preop diagnosis

Trauma

ERAS: Preop counseling by the surgeon/anesthesiologist should include patient

education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plans. See [Appendix K](#), p. K-1.

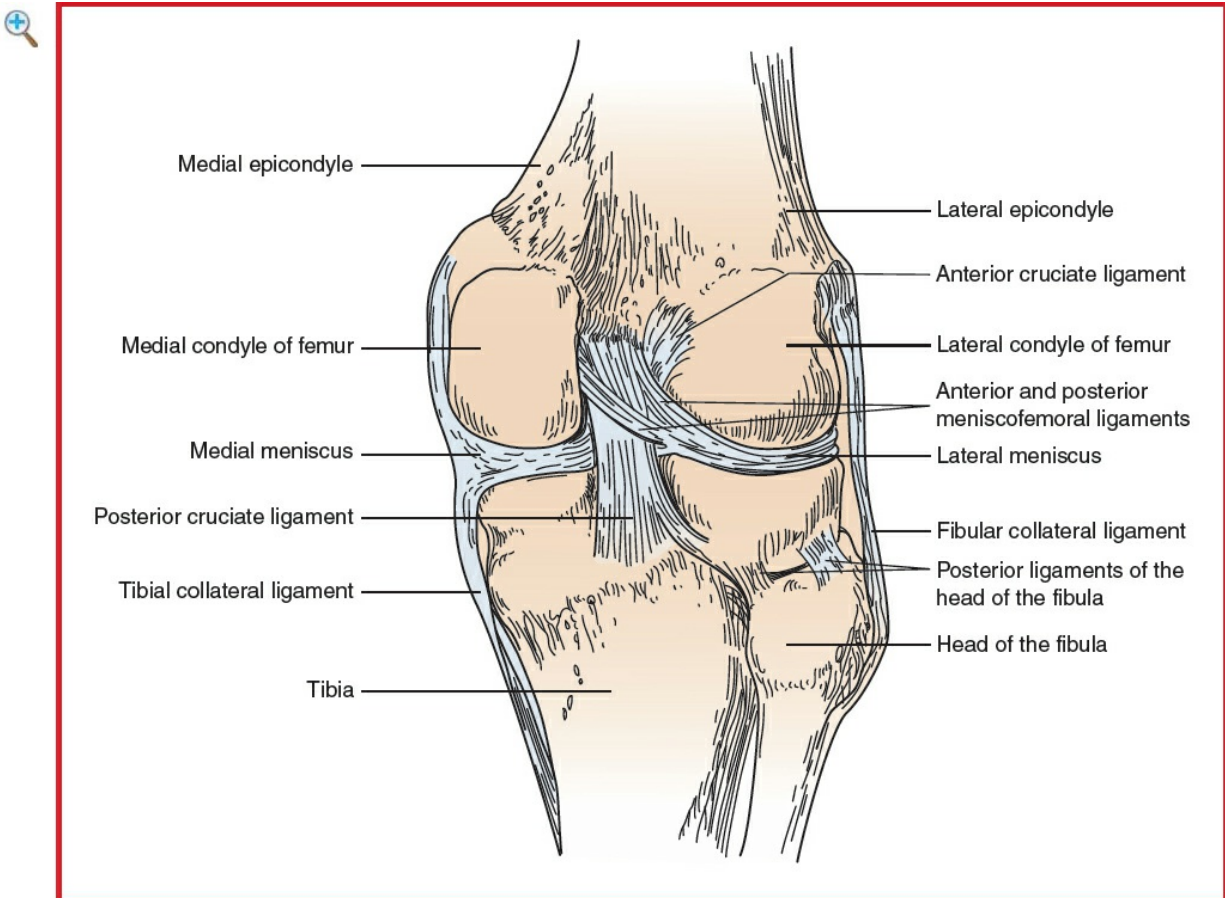


Figure 10.5-2 The cruciate ligaments (posterior view). (Reproduced with permission from Clemente CD: Clemente's Dissector, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2007, 270.)

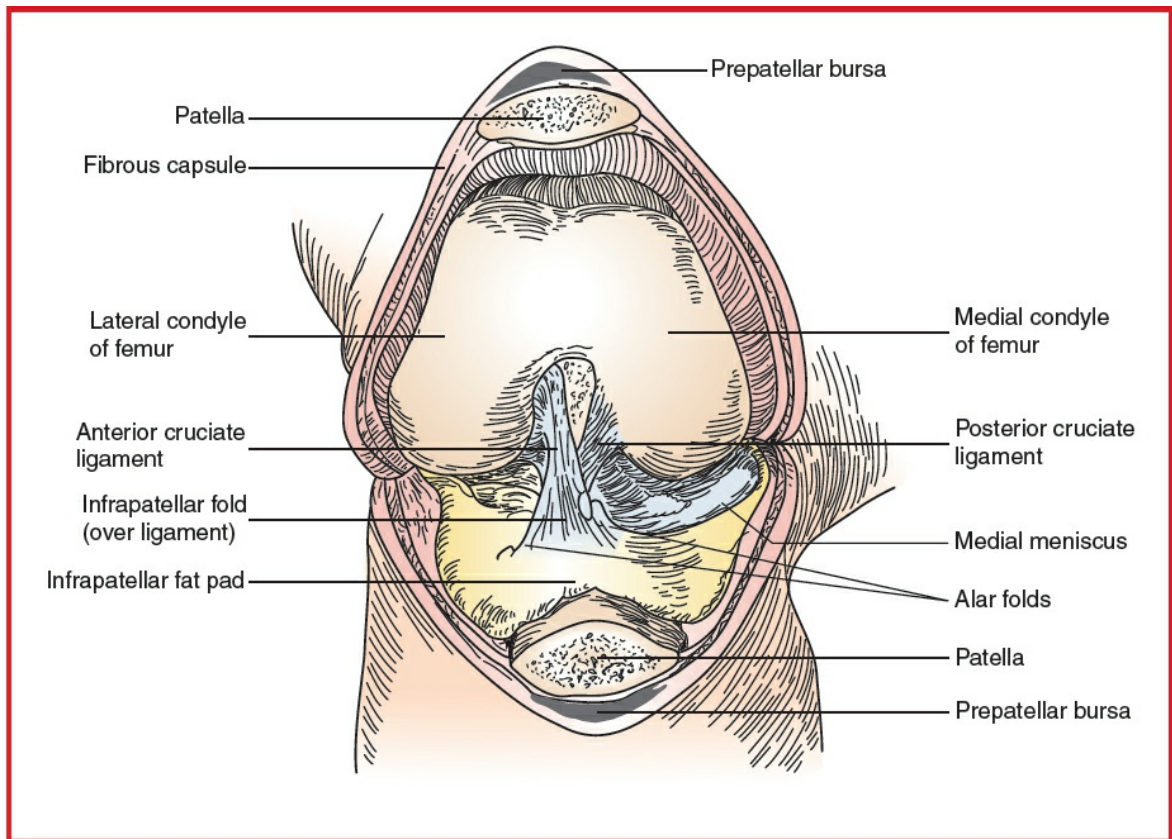


Figure 10.5-3 The knee joint opened anteriorly. (Reproduced with permission from Clemente CD: Clemente's Dissector, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2007, 269.)

■ SUMMARY OF PROCEDURES

	Repair or Collateral Reconstruction	Repair or Cruciate Reconstruction
Position	Supine	⇐
Incision	Over collateral ligament	Anterior and lateral ACL or medial PCL
Special instrumentation	Staples	Drill guides, staples, screws
Unique considerations	Often arthroscopically assisted; tourniquet	⇐
Antibiotics	Cefazolin 2 g iv (vancomycin or clindamycin if penicillin allergic)	⇐
Surgical time	2 h	⇐

Closing considerations	Splint or cast while anesthetized	⇐
EBL	100 mL	⇐
Postop care	PACU → room or home	⇐
Mortality	Minimal	⇐
Morbidity	Infection: <1%	⇐
	Thrombophlebitis: <5%	⇐
Pain score	4	7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Young adult
Male:Female	2:1
Incidence	Common
Etiology	Trauma: 100%

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Knee Procedures.

Suggested Readings

1. Cole BJ, Cotter EJ, Wang KC, et al: Patient understanding, expectations, outcomes and satisfaction regarding anterior cruciate ligament injuries and surgical management. *Arthroscopy* 2017; 33(5):1092–6.
2. Nadarajah V, Roach R, Ganta A, et al: Primary anterior cruciate ligament reconstruction: perioperative considerations and complications. *Phys Sportsmed* 2017; 45(2):165–77.
3. Siegel L, Vandenakker-Albanese C, Siegel D: Anterior cruciate ligament injuries anatomy, physiology, biomechanics, and management. *Clin J Sport Med* 2012; 22(4):349–55.
4. Tjoumakaris FP, Herz-Brown AL, Bowers AL, et al: Complications in brief: anterior cruciate ligament reconstruction. *Clin Orthop Relat Res* 2012; 470(2):630–6.

PATELLAR REALIGNMENT

SURGICAL CONSIDERATIONS

Description

The goal of this procedure is prevention of chronic subluxation or dislocation of the patella. Soft tissue components of the surgery include incision (release) of the lateral patellar retinaculum and reefing or tightening of the medial retinaculum ([Fig. 10.5-4](#)). In cases of severe malalignment of the extensor mechanism, the insertion of the patellar tendon may be moved to a new, more medial location (**tibial tubercle transfer**). In this procedure, the tibial tubercle generally is detached with a saw or osteotomes, leaving a bone pedicle attached distally. The tubercle is then rotated medially on the pedicle and fixed in its new position with a screw. Many surgeons routinely perform an anterior compartment fasciotomy to prevent postop compartment syndrome.

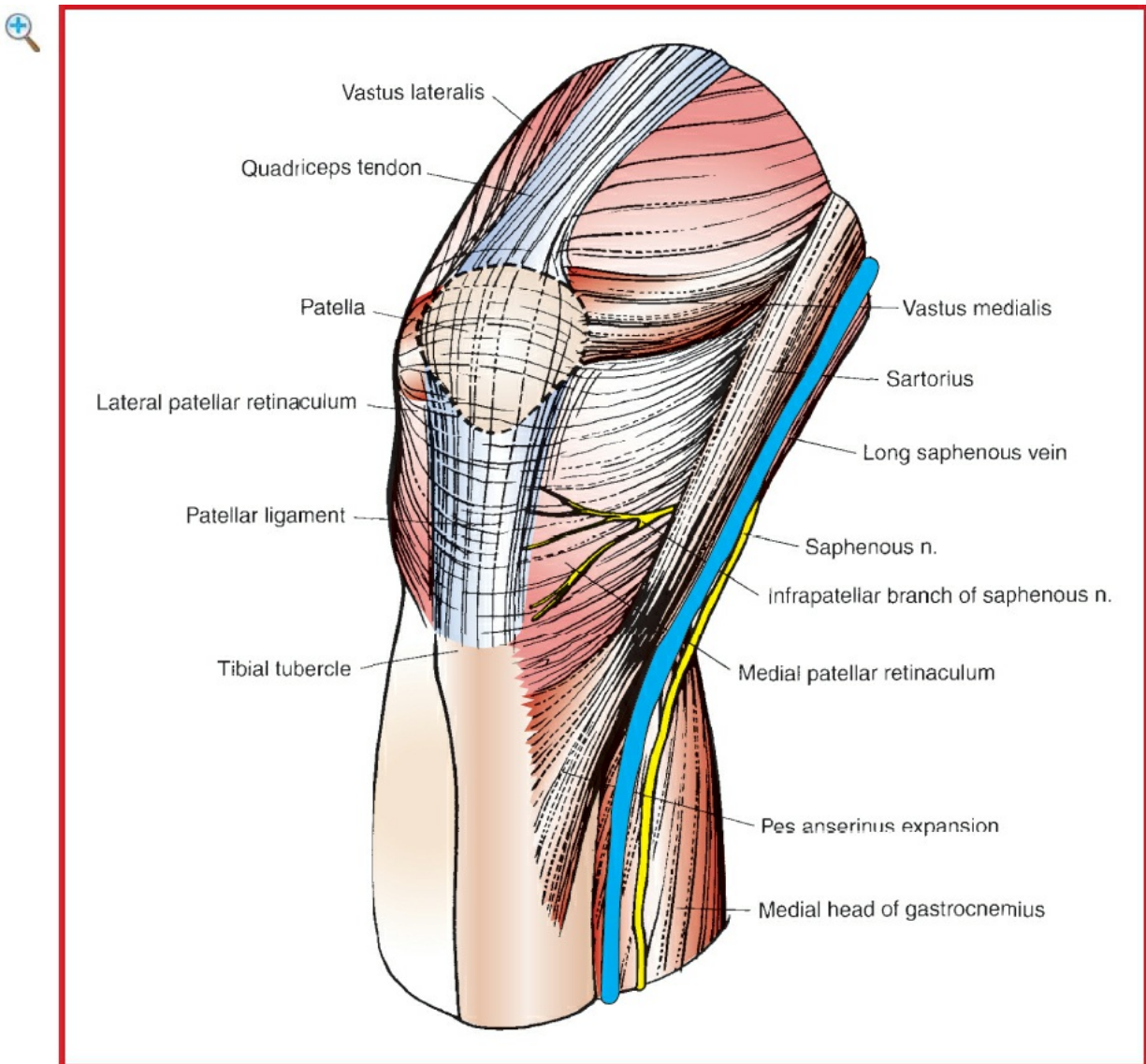


Figure 10.5-4 Outer layer of anteromedial aspect of the knee joint. Shows anatomy of the patellar retinaculum. (Reproduced with permission from Hoppenfeld S, de Boer P. *Surgical Exposures in*

Usual preop diagnosis

Chronic patellar subluxation or dislocation

ERAS: Preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Patellar Realignment	Tibial Tubercle Transfer
Position	Supine	⇐
Incision	Anteromedial or anterolateral to the knee	⇐
Special instrumentation	None	Screws or staples
Unique considerations	Tourniquet	⇐
Antibiotics	Cefazolin 2 g iv (vancomycin or clindamycin if true penicillin allergic)	⇐
Surgical time	1 h	1.5 h
Closing considerations	None	Splint or cast while anesthetized
EBL	50 mL	100 mL
Postop care	PACU → room or home	⇐
Mortality	Minimal	⇐
Morbidity	Hemarthrosis: 100% Redislocation: 20% Thrombophlebitis: 10–20%	5% 25% ⇐

	Compartment syndrome:	⇐
	Infection: <1%	⇐
Pain score	6	7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Usually young adult
Male:Female	1:2
Etiology	Trauma (70%); congenital (30%)
Associated conditions	Patellofemoral dysplasia (60–70%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Knee Procedures.

Suggested Reading

1. Koh JL, Stewart C: Patellar instability. *Orthop Clin North Am* 2015; 46(1):147–57.

ARTHROSCOPY OF THE KNEE

SURGICAL CONSIDERATIONS

Description

Knee arthroscopy is used to diagnose and treat intra-articular problems, most commonly torn meniscus, but the procedure is also used for ligament injuries (Figs. 10.5-2 and 10.5-3), osteochondral fractures, loose bodies, arthritis, and infections. In knee arthroscopy, multiple portals or entry points for the arthroscope and instruments generally are used. The most common portals are anteromedial and anterolateral adjacent to the patellar ligament. Other portals may be suprapatellar, parapatellar, and posterior. Portals are made by making a stab wound with a knife and then entering the joint with a combination of sharp and blunt trocars. A diagnostic inspection from one of the anterior portals is normally performed at the outset. A second portal is used with a nerve hook to manipulate intra-articular tissues. If resection or repair is performed, the appropriate instruments are inserted through one of the portals. Meniscus repair and cruciate reconstruction may require separate longitudinal incisions, which are usually

posteromedial or posterolateral, for placement of sutures and/or drill holes.

Meniscectomy and/or **debridement** often are performed in conjunction with arthroscopy. **Cruciate ligament reconstruction** usually is performed with arthroscopic assistance. At the end of the procedure, the knee joint is copiously irrigated with NS or LR solution through one of the portals. Portals are closed with a single suture and Steri-Strips®; compression bandages are applied; and often a knee immobilizer is used.

Usual preop diagnosis

Torn meniscus; cruciate ligament tear; arthritis

ERAS: Preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Arthroscopy	Meniscectomy/Debridement	Cruciate Reconst
Position	Supine	⇐	⇐
Incision	1- to 2-cm portals	⇐	⇐ + ante midline a lateral
Special instrumentation	Arthroscopic video system; small biters and graspers	⇐ + shaver	⇐ + drill g and drills fixation s
Unique considerations	Thigh holder; foot of table 90°; ± tourniquet	⇐	⇐
Antibiotics	Cefazolin 2 g iv (vancomycin or clindamycin if true penicillin allergic)	⇐	⇐
Surgical time	0.5 h	1–2 h	2–3 h

Closing considerations	No splint; local anesthetic injected	⇐	⇐
EBL	Minimal	⇐	50 mL
Postop care	PACU → home	⇐	⇐ or over
Mortality	<0.1%	⇐	⇐
Morbidity	Hemarthrosis: 5–20%	5%	⇐
	Thrombophlebitis: <2%	⇐	⇐
	Infection: 0.1%	⇐	⇐
	Stiffness: <0.1%	<4%	⇐
Pain score	3	4	6

■ PATIENT POPULATION CHARACTERISTICS

Age range	10–70 yr (usually 20–40 yr)
Male:Female	2:1
Incidence	Most common arthroscopic procedure (85% of total)
Etiology	Trauma (~85%); arthritis (~10%); infection (~5%)
Associated conditions	Usually healthy; systemic arthritis (<5%)

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Knee Procedures.

Suggested Readings

1. Bollen SR: Is arthroscopy of the knee completely useless? Meta-analysis—a reviewer's nightmare. *Bone Joint J* 2015; 97-B(12):1591–2.
2. Gwathmey FW Jr, Golish SB, Diduch DR: Complications in brief: meniscus repair. *Clin Orthop Relat Res* 2012; 470(7):2059–66.
3. Minas T: A primer in cartilage repair. *J Bone Joint Surg Br* 2012; 94(11 Suppl A):141–6.

KNEE ARTHROTOMY

SURGICAL CONSIDERATIONS

Description

Arthrotomy of the knee is the opening of the joint for drainage, excision of intra-articular tissue (synovium, meniscus, loose bodies), ligament repair/reconstruction, or fracture fixation. The knee generally is opened with a parapatellar incision, either medial or lateral, and the joint capsule is incised just adjacent to the patella. After the intra-articular pathology is addressed, a tight capsular closure is performed, followed by subcutaneous tissue and skin closure.

Variant procedure or approaches

Arthrotomy with debridement may be used for infection or arthropathy, which produces debris. In both cases, **synovectomy** may be necessary.

Usual preop diagnosis

Infection; trauma (fracture, sprain, torn meniscus); arthritis

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Arthrotomy	Arthrotomy with Debridement	Arthrotomy with Synovectomy
Position	Supine	⇐	⇐
Incision	Medial or lateral parapatellar	⇐	⇐
Special instrumentation	Tourniquet	⇐	⇐
Antibiotics	Cefazolin 2 g iv (vancomycin or clindamycin if true penicillin allergic)	⇐	⇐

Surgical time	1 h	2 h	2 h
Closing considerations	Compressive dressing; may be splinted; suction drain	⇐	⇐
EBL	100 mL	⇐	⇐
Postop care	PACU → room	⇐	⇐
Mortality	Minimal	⇐	⇐
Morbidity	Hemarthrosis: 100%	⇐	⇐
	Degenerative arthritis: 5–20%	⇐	⇐
	Stiffness: 5%	⇐	⇐
	Thrombophlebitis: 5%	⇐	⇐
	Infection: 1%	10%	20%
Pain score	7	7	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Infant–elderly (usually young adult)
Male:Female	1:1
Incidence	Common
Etiology	Infection; trauma; arthritis
Associated conditions	Inflammatory arthritis (20%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Knee Procedures.

REPAIR OF TENDONS—KNEE AND LEG

SURGICAL CONSIDERATIONS

Description

Acute ruptures of tendons in the lower limb are repaired by **direct suture** and sometimes reinforced with part of another tendon. At the knee, patellar tendon ruptures are most common; at the ankle, Achilles tendon ruptures are most common. A longitudinal incision generally is made directly over the tendon. The tendon sheath is opened, and tendon ends are reapproximated with a nonabsorbable tendon stitch. If necessary, the repair may be augmented by synthetic tape or fascia or protected with a wire that takes tension off the repair. The tendon sheath is closed separately from the skin incision, and a cast or splint is applied. **Achilles tendon repair** and **posterior tibial tendon repair** require different positioning. For an Achilles tendon repair, the patient is placed prone, and a longitudinal incision is made just medial to the tendon, spanning the rupture. The tendon sheath is incised and carefully protected. Torn ends of the tendon are approximated with multiple tendon stitches and may be protected with a fascial flap developed from the gastrocnemius fascia. The tendon sheath is closed carefully, followed by skin wound closure. A splint or cast is applied with the foot in equinus (plantar flexion).

Usual preop diagnosis

Tendon rupture

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and rehabilitation plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Posterior Tendon Repair	Achilles Tendon Repair
Position	Supine	Prone
Incision	Over tendon	⇐
Special instrumentation	Wire or synthetic tape for augmentation	⇐
Unique considerations	Tourniquet	⇐
Antibiotics	Cefazolin 2 g iv (vancomycin or clindamycin if penicillin)	⇐

	allergic)	
Surgical time	1 h	⇐
Closing considerations	Splint or cast while anesthetized	⇐
EBL	Minimal	⇐
Postop care	PACU → room or home	⇐
Mortality	Minimal	⇐
Morbidity	Weakness: ~10%	⇐
	Wound slough: 5%	⇐
	Adhesions: <1%	⇐
	Infection: <1%	⇐
		Rerupture: 5–10%
Pain score	3	3

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age
Male:Female	1:1
Incidence	Uncommon
Etiology	Trauma (90%); chronic tendinitis (10%)
Associated conditions	Obesity; diabetes mellitus (DM); inflammatory arthritis

■ ANESTHETIC CONSIDERATIONS FOR KNEE PROCEDURES

(Procedures covered: arthroplasty; arthrodesis; ORIF of patellar fractures; repair/reconstruction of ligaments; patellar realignment; arthroscopy; arthrotomy; tendon repair—knee and leg)

■ PREOPERATIVE

Trauma and osteoarthritis (OA) are the most common indications for these procedures. Trauma patients (e.g., those with sports injuries) are often young and healthy, whereas arthritic patients are often elderly, and anesthetic management must be tailored to any concurrent disease. Patients with rheumatoid and other inflammatory arthritides form

another group of candidates for these procedures; the special anesthetic considerations for these patients are described in Anesthetic Considerations for Hip Procedures. A final group of patients undergoing these procedures are hemophiliacs, who develop arthritis from recurrent bleeding into their joints. The hematologic management of these patients is discussed below.

Respiratory	These patients often have rheumatoid arthritis and associated pulmonary conditions. For example, pulmonary effusions are common. Limited respiratory reserve warrants further evaluation. Pulmonary fibrosis (rare) often manifests as a cough and dyspnea. Rheumatoid arthritis involving the cricoarytenoid joints may manifest as hoarseness, glottic narrowing, and difficult intubation. Arthritic involvement of the TMJ and cervical spine may further complicate airway management. Test: As indicated from H&P
Cardiovascular	The severity of the arthritis often limits exercise and makes assessment of cardiovascular status difficult. Dobutamine stress ECHO and dipyridamole thallium imaging may be necessary for an adequate cardiac evaluation. Rheumatoid arthritis is associated with pericardial effusion, cardiac valve fibrosis, cardiac conduction abnormalities, and aortic regurgitation (AR). Tests: ECG and others as indicated from H&P
Neurologic	In arthritic patients, a thorough preop neurologic exam often yields evidence of cervical nerve root compression. After the stability of the neck has been established, the full range of neck motion should be evaluated for evidence of nerve compression or cerebral ischemia (suggesting vertebral artery compression). Consider preop lateral neck films to determine stability of atlanto-occipital joint and evidence of vertebral spurs that may interfere with intubation. Test: As indicated from H&P

Musculoskeletal	Pain and ↓ joint mobility may make positioning and regional anesthesia difficult in this patient population.
Hematologic	Hemophiliacs require restoration of clotting factors preop. Administer 1 U of factor concentrate/kg body weight for each 2% increase necessary to achieve clotting factor activity of 40% normal. FFP contains 1 U/mL and cryoprecipitate 20 U/mL. Hemophilia B (factor IX deficiency), but not hemophilia A (factor VIII deficiency), can be treated with prothrombin complex concentrate; however, these products can activate clotting factors and → DIC. Approximately 10% of hemophiliacs develop antibodies to exogenous clotting factors, and the care of these patients should be guided by a hematologist. Tests: Hct; other tests as indicated from H&P
Laboratory	Other tests as indicated from H&P
Premedication	Standard premedication. Preop patellar pain is treated effectively with a femoral nerve block at the inguinal ligament, using 10 mL of lidocaine 1.5% with epinephrine 1:200,000. Consider preemptive analgesia (depending on anticipated postop pain) with gabapentin 300 mg po, celecoxib 200 mg po, and acetaminophen 1 g po/iv to be given in the preop holding area.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

For many of these patients, regional anesthesia may be the preferred technique, offering the advantages of ↓ blood loss, ↓ DVT, minimal respiratory impairment, and effective postop analgesia. Patients with rheumatoid arthritis rarely have involvement of the lumbar spine. Because rheumatoid arthritis frequently affects the C-spine, however,

these patients may have limited range of neck motion, an unstable atlanto-occipital joint, and cricoarytenoid and TMJ arthritis. Careful airway evaluation, therefore, is important to determine the appropriateness of special intubation techniques (e.g., fiberoptic).

Regional anesthesia

A continuous peripheral nerve block (CPNB), adductor canal, or common femoral nerve blocks provide similar effect on postop length of hospital stay and rehabilitation compared to an epidural pain management but have a lower incidence of side effects (urinary retention, hypotension, and dysesthesia). A combined femoral and sciatic nerve block provides superior pain control in the first 36 postoperative hours over a single, femoral nerve block. In addition to the nerve block, either a GA or an SAB is needed for the intraoperative phase because a CPNB does not reliably provide surgical anesthesia. Both nerves can be localized conventionally using a nerve stimulator or with ultrasound guidance. A typical initial local anesthesia dose for each nerve is 20 mL of 0.5% bupivacaine or 0.75% ropivacaine. As an alternative, a continuous adductor canal (lower femoral nerve) block can be used to provide good analgesia with minimal motor blockade during repair of tendons. A bolus of 30 mL of 0.5% ropivacaine with epinephrine 1:200,000 is used, and a continuous infusion (0.2% at 10 mL/h) started in the recovery room. There are emerging data supporting the safe and efficacious use of adductor canal blocks/indwelling catheters and local infiltration of analgesics in total knee arthroplasty. Currently this is our preferred approach, in combination with a general anesthetic, for total knee arthroplasty.

An epidural block provides both intraop surgical anesthesia and postop pain control, but it is contraindicated in patients receiving Coumadin postop. If the patient prefers not to receive a peripheral nerve or an epidural block, a subarachnoid block provides a useful alternative regional anesthesia technique, depending on the patient population (e.g., younger patients may be at ↑ risk of spinal headache following SAB). Anesthesia extending from S2 to T12 (T8, if tourniquet is used) is adequate for knee surgery. Full motor blockade is essential for fixation of the patella or placement of the joint prosthesis and assessment of the passive ROM of the prosthesis. Typical drugs and doses include subarachnoid: 12.5–15 mg of 0.75% bupivacaine with morphine 0.2 mg, and epidural: 15–20 mL 2% lidocaine with epinephrine 1:200,000 in divided doses.

Tranexamic acid (TXA) reduces intraop and postop blood loss through inhibition of fibrin-clot degradation by plasmin. This results in stable clot formation with less fibrinolysis. Furthermore, TXA has been proven to be safe in wide range of patient populations without increasing the incidence of DVT or PE. Investigations are still underway in patients with coronary stents to prove TXA's safety in this population.

Dosing is 10–15 mg/kg given prior to incision and again upon closure—with data showing total blood loss (throughout hospitalization) being reduced by 500 mL.

General anesthesia

Induction	Standard induction is appropriate for patients with normal airways.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Tranexamic acid (e.g., 10–15 mg/kg) iv before incision has been shown to ↓EBL and may be requested by the surgeon.	
Maintenance	Standard maintenance. Neuromuscular relaxation facilitates the placement of the prosthesis. Hemophiliacs will require infusion of clotting factors. For hemophilia A and von Willebrand disease, 1.5 U/kg/h; for hemophilia B, 0.75 U/kg/h. Maintain normothermia (forced air blanket), and use fluid warmer.	
Emergence	The tourniquet is deflated around the time of emergence. In patients with moderate-to-severe lung disease, controlled ventilation should be continued until after the lactic acid that has accumulated in the leg has been metabolized (3–5 min) because these patients may be unable to increase ventilation to buffer this acid load. Provide PONV prophylaxis with ondansetron 4 mg iv and dexamethasone (4 mg iv).	
Blood and fluid requirements	IV: 14–16 ga × 1 BSS @ maintenance during the case, and 5–10 mL/kg bolus prior to tourniquet deflation	A tourniquet blocks intraop blood loss. When it is deflated, prepare for a 1–2 U blood loss over the ensuing hour, more if the posterior tibial artery has been damaged in the dissection. Avoid under-resuscitation. See Goal-Directed Fluid

Management, [Appendix K, p. K-4](#).

Control of blood loss	Tourniquet	Inflation pressure is typically 100 mm Hg + systolic pressure. Maximum tourniquet time is 2 h, followed by a 30 min reperfusion interval, if further tourniquet time is necessary.
	Tranexamic acid (TXA)	10–15 mg/kg prior to incision and upon surgical closure. Caution in those with cardiac stents
Monitoring	Standard monitors ± CVP line	A CVP line is indicated if monitoring the CVP trend is expected to affect anesthetic care.
	± Arterial line Additional monitoring	CVO ₂ Sat, PA cath, TEE may be indicated in special cases.
Positioning	✓ and pad pressure points ✓ eyes	In rheumatoid arthritis patients, meticulous padding of the extremities is mandatory.
Complications	Posterior tibial artery trauma Peroneal nerve palsy	A 20% ↓ MAP is common on tourniquet deflation. Additional crystalloid (5–10 mL/kg) may be necessary to replace edema fluid and blood loss to the leg.

Complications	Hemorrhage from the posterior tibial artery Peroneal nerve palsy → foot drop Tourniquet-related nerve injury, posttourniquet syndrome (PTS)	✓ surgical drain output Examine patient for evidence of neurologic dysfunction, and notify surgeons as necessary. PTS is a self-limiting condition in which the affected limb is edematous, pale, and weak.
Multimodal pain management	Neuraxial regimens: • Epidural anesthesia • SAB	Epidural bupivacaine 0.0125% infused at 6–8 mL/h with hydromorphone 50 mcg/mL infused at 100–250 mcg/h provides good analgesia. Prior to removal of the epidural catheter, a 0.2-mg bolus of hydromorphone may be given. Epidural catheters are typically removed on the morning of postop day 2. Low molecular weight heparin may be started 2 h after catheter removal. Intrathecal morphine 0.2–0.3 mg provides analgesia for up to 24 h. May be administered along with bupivacaine for surgical anesthesia

Peripheral regimens:

- Single-shot peripheral nerve block (femoral or combined femoral/sciatic block)

- CPNB

Systemic regimens:

These nerves may be blocked preop or postop as a pain rescue measure. A typical dose for each nerve is 20 mL of 0.5% bupivacaine or 30 mL of 0.5% ropivacaine. Continuous infusion of bupivacaine 0.125% through standard or portable infusion pumps at 6–10 mL/h. Anticoagulation may continue with a peripheral nerve catheter in place. Catheter is removed on POD#2 at the latest. Foley catheters are not required with peripheral nerve catheters.

Oral pain management with acetaminophen (1 g every 6 h if not contraindicated) should be continued immediately postop. Celecoxib (Celebrex®) 200 mg once a day. Oxycodone 5–20 mg po q2h prn. Morphine 0.5–2 iv mg q3h prn. Consider pain service consult as well as the use of controlled-release oxycodone (OxyContin®) and PCA with iv morphine or hydromorphone in opioid-dependent patients, as an

		alternative to neuraxial blocks or to supplement peripheral nerve blocks.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6) Regional analgesia	Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	Hct; other studies as indicated	Patients with coagulopathies require replacement therapy for 6–10 d.

Suggested Readings

1. Auyong DB, Allen CJ, Pahang JA, et al: Reduced length of hospitalization in primary total knee arthroplasty patients using an updated enhanced recovery after orthopedic surgery (ERAS) pathway. *J Arthroplasty* 2015; 30(10):1705–9.
2. Bono JV, Robbins CE, Mehio AK, et al: Pharmacologic pain management before and after total joint replacement of the hip and knee. *Clin Geriatr Med* 2012; 28(3):459–70.
3. Caparrini C, Miniati I, Ponti M, et al: Perioperative pain management in fast-track knee arthroplasty. *Acta Biomed* 2017; 88(2S):139–44.
4. Ganapathy S: Wound/intra-articular infiltration or peripheral nerve blocks for orthopedic joint surgery: efficacy and safety issues. *Curr Opin Anaesthesiol* 2012; 25(5):615–20.
5. Gibbs DM, Green TP, Esler CN: The local infiltration of analgesia following total knee replacement: a review of current literature. *J Bone Joint Surg Br* 2012; 94(9):1154–9.
5. Hebl J: Clinical pathways for total joint arthroplasty: essential components for success. *Refresher Course Lect Anesthesiol* 2012; 324:1–10.
7. Jæger P, Zaric D, Fomsgaard JS, et al: Adductor canal block versus femoral nerve block for analgesia after total knee arthroplasty: a randomized, double-blind study. *Reg Anesth Pain Med* 2013; 38(6):526–32.
8. Johnson RL, Kopp SL, Hebl JR, et al: Falls and major orthopaedic surgery with peripheral nerve blockade: a systematic review and meta-analysis. *Br J Anaesth* 2013; 110(4):518–28.
9. Kuper M, Rosenstein A: Infection control in total knee and total hip arthroplasties. *Am J Orthop* 2008; 37(1):E2–5.
9. Lieberman JR, Heckmann N: Venous thromboembolism prophylaxis in total hip

arthroplasty and total knee arthroplasty patients: from guidelines to practice. J Am Acad Orthop Surg 2017; 25(12):789–98.

1. Lin ZX, Woolf SK: Safety, efficacy, and cost-effectiveness of tranexamic acid in orthopedic surgery. Orthopedics 2016; 39(2):119–30.
2. Melvin SJ, Stryker LS, Sierra RJ: Tranexamic acid in hip and knee arthroplasty. J Am Acad Orthop Surg 2015; 23(12):732–40.
3. Webb CA, Mariano ER: Best multimodal analgesic protocol for total knee arthroplasty. Pain Manag 2015; 5(3):185–96.
4. Zhu S, Qian W, Jiang C, et al: Enhanced recovery after surgery for hip and knee arthroplasty: a systematic review and meta-analysis. Postgrad Med J 2017; 93(1106):736–42.

Suggested Viewing

Links are available online to the following videos:

Total Knee Replacement Surgery by Dr. Eric W. Janssen:

<https://www.youtube.com/watch?v=GRSWV57u9eg>

Knee Arthroscopy: Portal Placement and Examination Course: <https://www.youtube.com/watch?v=y0WDWZX9WLk>

Anterior Cruciate Ligament (ACL) Reconstruction Animation: <https://www.youtube.com/watch?v=q96M0jRqn7k>

Knee Ligament Anatomy Animation: <https://www.youtube.com/watch?v=RTV5Yo3E7VQ>

Continuous Adductor Canal (Saphenous Nerve) Block: <https://www.youtube.com/watch?v=jPJISpkf4rE>

Continuous Femoral Nerve Block: <https://www.youtube.com/watch?v=UVMxkZH4CNc>

CHAPTER 10.6

Lower Leg, Ankle, Foot, and Other Lower-Extremity Procedures

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OPEN REDUCTION AND INTERNAL FIXATION (ORIF) OF THE TIBIAL PLATEAU FRACTURE

SURGICAL CONSIDERATIONS

Description

ORIF of the tibial plateau or proximal tibia fracture involves making a longitudinal incision along the proximal leg, lateral to the knee, obtaining a reduction by direct visualization of the fracture fragments, and applying plates and screws along the tibia for rigid internal fixation. An additional posteromedial incision may also be utilized. Occasionally, surgeons will also perform a knee arthroscopy to assess for intra-articular reduction. An iliac crest bone graft may be necessary. Patients may have initial stabilization with an external fixator based on the extent of the fracture and the soft tissue envelope. A **proximal tibial osteotomy** involves correcting malalignment (valgus and varus) of the lower extremity by excising a wedge of bone from the tibia and correcting the mechanical axis or by adding a bony wedge for an opening wedge

osteotomy based on the deformity and surgeon preference.

Usual preop diagnosis

Tibial plateau or proximal tibial fracture; nonunion/malunion of the tibial plateau or proximal tibia; degenerative arthritis of the knee, with varus or valgus deformity

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	ORIF Tibial Plateau Fracture	Proximal Tibial Osteotomy
Position	Supine	⇐
Incision	Lateral to knee, usually; posteromedial, occasionally; ± arthroscopy	Transverse or lateral incision
Special instrumentation	Special plates, screws; reduction clamps; radiolucent table; possible bone graft	⇐
Unique considerations	Intraop radiographs or I.I.; tourniquet	⇐
Antibiotics	Cefazolin 2 g iv q6–8h × 48 h	⇐
Surgical time	~2.5–3 h; more, depending on difficulty	⇐
Closing considerations	Splint, cast while anesthetized	⇐
EBL	<200 mL	⇐
Postop care	Multiple-trauma victim → ICU; others → PACU; ±	PACU → ward

Mortality	CPM Rare, except in severe multiple trauma	None
Morbidity	Compartment syndrome: 10–20% Wound infection: 7–15% DVT (symptomatic): 3–5% Delayed union, nonunion, malunion: <5% Peripheral nerve damage: 3% Intra-articular fracture: 2% Hypotension (multiple trauma) Leg-length discrepancy Osteomyelitis, septic arthritis Respiratory distress and fat embolism Vascular complications	<25% ~2% 2% ⇐ 0.2%
Pain score	7	7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age; fracture most common in younger trauma patients and elderly Degenerative arthritis of knee, <50 yr
Male:Female	1:1
Incidence	Common
Etiology	Trauma: Falls, motorcycle and motor vehicle accidents, industrial injuries Degenerative: Arthritis of knee

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

Suggested Readings

1. Atesok K, Doral MN, Whipple T, et al: Arthroscopy-assisted fracture fixation. *Knee Surg Sports Traumatol Arthrosc* 2011; 19(2):320–9.
2. Berkson EM, Virkus WW: High-energy tibial plateau fractures. *J Am Acad Orthop Surg* 2006; 14(1):20–31.
3. Kugelman D, Qatu A, Haglin J, et al: Complications and unplanned outcomes following operative treatment of tibial plateau fractures. *Injury* 2017; 48(10):2221–9.
4. McNamara IR, Smith TO, Shepherd KL, et al: Surgical fixation methods for tibial plateau fractures. *Cochrane Database Syst Rev* 2015; (9):CD009679.
5. Schuster P, Geblein M, Schlumberger M, et al: Ten-year results of medial open-wedge high tibial osteotomy and chondral resurfacing in severe medial osteoarthritis and varus malalignment. *Am J Sports Med* 2018; 46(6):1362–70.

INTRAMEDULLARY NAILING, TIBIA

SURGICAL CONSIDERATIONS

Description

In intramedullary nailing of the tibia, a metal nail is placed into the medullary canal of the tibia to stabilize (or prevent) a fracture. Following the incision, an awl may be used to make an entry hole in the proximal metaphysis of the tibia, through which a guide wire is introduced. The guide wire is placed across the aligned fracture, and the nail is introduced and driven over the guide wire. Before nail insertion, the medullary canal often is reamed to allow use of a larger nail. Most nails are interlocked both proximally and distally with screws that pass from the bone through holes in the nail.

Usual preop diagnosis

Fracture, nonunion or malunion of the tibia

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURE

Position	Supine, on fracture table. Consider inducing anesthesia before moving patient.
Incision	Proximal longitudinal incision over the patellar tendon;

	stab wounds for interlocking screws
Special instrumentation	Nails, screws, and insertion instruments; intramedullary reamers; I.I.
Antibiotics	Cefazolin 2 g iv preop
Surgical time	2 h
Closing considerations	No splint or cast
EBL	200 mL
Postop care	PACU → room
Mortality	Minimal
Morbidity	Compartment syndrome: <5% Infection: <2% Neurapraxia: <1%
Pain score	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	>16 yr
Male:Female	5:1
Etiology	Trauma (95%); tumor (5%)
Associated conditions	Multiple trauma (50%); compartment syndrome (5%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

Suggested Reading

1. Duan X, Al-Qwbani M, Zeng Y, et al: Intramedullary nailing for tibial shaft fractures in adults. Cochrane Database Syst Rev 2012; (1):CD008241.

EXTERNAL FIXATION, TIBIA

SURGICAL CONSIDERATIONS

Description

Fractures of the tibia are fixed with percutaneous pins that are clamped to an external frame. Stainless steel pins are drilled into the proximal and distal fragments of the fracture through stab wounds in the skin and subcutaneous tissues. Usually 2–3 pins are placed on either side of the fracture. Pin clamps and an external frame are attached and the fracture is aligned with the assistance of the I.I. or under direct vision. Following fracture alignment, the pin clamps and frames are tightened to hold fracture alignment. External fixation is often used with open fractures. **Small-pin ringed fixators** (e.g., Ilizarov) are used for fracture fixation, leg lengthening, and treatment of bony defects. Wound irrigation and debridement often accompany application of the fixation frame.

Usual preop diagnosis

Tibial fracture; tibial nonunion or malunion; tibial shortening

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Stab wounds. Small-pin fixator may require metaphyseal incision for osteotomy.
Special instrumentation	Pins; fixation frame; I.I.
Antibiotics	Cefazolin 2 g iv preop
Surgical time	0.5–1 h Small-pin fixator: 3–5 h
Closing considerations	May be open fracture (usually left open)
EBL	50 mL; small-pin fixator, 100 mL
Postop care	PACU → room
Mortality	Minimal

Morbidity	Infection: 15% Compartment syndrome: <2% Neurapraxia: <1%
Pain score	2–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages
Male:Female	5:1
Incidence	Common
Etiology	Trauma (95%); shortened limb (<2%); ununited or malunited fracture (<2%)
Associated conditions	Open fracture (95%); compartment syndrome (<2%); congenital anomaly (<1%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

OPEN REDUCTION AND INTERNAL FIXATION (ORIF) OF DISTAL TIBIA, ANKLE, AND FOOT FRACTURES

SURGICAL CONSIDERATIONS

Description

ORIF is nearly always required for displaced fractures involving the ankle or joints in the foot. A longitudinal incision is made over the fractured medial and/or lateral malleoli or affected area of the foot. Dissection is carried directly down to the bone, and the fracture is identified and reduced under direct vision.

Open fractures may require **irrigation and debridement**. The fractures are realigned under direct vision and fixed and stabilized with pins, plates, and/or screws. An intraop radiograph is obtained to confirm reduction and placement of hardware. The incisions are closed and a splint or cast is applied.

Usual preop diagnosis

Fracture of the distal tibia, ankle, or foot

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	ORIF Ankle	With Irrigation and Debridement
Position	Supine; lateral (calcaneus fractures)	⇐
Incision	Longitudinal over fracture site	⇐ + extension of existing wound
Special instrumentation	Pins, plates, and screws; tourniquet; x-ray or I.I.	⇐
Antibiotics	Cefazolin 2 g iv preop	⇐
Surgical time	2 h	2–3 h
Closing considerations	Splint or cast while anesthetized	Splint or cast; may leave wound open
EBL	50 mL	100 mL
Postop care	PACU → room	⇐
Mortality	Minimal	⇐
Morbidity	Wound dehiscence: 10% Loss of reduction: 7% Infection: 3%	⇐ ⇐ 15%
Pain score	4	4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Infant-elderly
Male:Female	1:1

Incidence	~250,000 cases/yr in the United States
Etiology	Trauma: 100%
Associated conditions	Alcohol abuse; obesity; diabetes mellitus (DM)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

Suggested Readings

1. Assal M, Ray A, Stern R: Strategies for surgical approaches in open reduction internal fixation of pilon fractures. *J Orthop Trauma* 2015; 29(2):69–79.
2. Belmont PJ Jr, Davey S, Rensing N, et al: Patient-based and surgical risk factors for 30-day postoperative complications and mortality after ankle fracture fixation. *J Orthop Trauma* 2015; 29(12):e476–82.
3. Crist BD, Khazzam M, Murtha YM, et al: Pilon fractures: advances in surgical management. *J Am Acad Orthop Surg* 2011; 19(10):612–22.
4. Hsu AR, Anderson RB, Cohen BE: Advances in surgical management of intra articular calcaneus fractures. *J Am Acad Orthop Surg* 2015; 23(7):399–407.

REPAIR NONUNION/MALUNION, TIBIA

SURGICAL CONSIDERATIONS

Description

This procedure is used to treat a fracture that has not healed or was misaligned upon healing. The fracture is mobilized, usually grafted with autogenous or allograft bone, and realigned. With an anterior approach, a longitudinal incision is made anteromedial or anterolateral to the shaft of the tibia. Dissection is carried directly down to the bone and the nonunion identified. If the tibia is approached with a posterolateral incision, the patient is turned prone or lateral, and a longitudinal incision is made just posterior to the fibula. Dissection is carried down posteriorly to the interosseous membrane, to the tibia, and the procedure becomes identical to the anterior approach. Tissue interposed between the bone ends may or may not be debrided. The cortex of the bone adjacent to the nonunion is roughened with an osteotome. Autogenous or allograft bone is placed adjacent to or in the nonunion site. In the case of a malunion, the bone may be osteotomized with a saw or osteotomes to allow realignment. For skeletal fixation, a

plate may be attached to the bone through the same incision. Alternatively, an intramedullary nail may be placed through an incision anterior to the tibial tubercle. If an intramedullary device is used, the canal may be reamed with intramedullary reamers prior to placement of the nail. A third type of **skeletal fixation** is the external fixator that stabilizes the nonunion via percutaneous pins placed into the proximal and distal tibia, which are then spanned by a device with pin clamps at both ends. An intraop x-ray or I.I. is used to confirm fixation and placement of devices and alignment.

Variant procedure or approaches

Autogenous **bone grafting from the iliac crest** is commonly used to stimulate healing. An incision is made directly over the iliac crest, and muscles are stripped from the crest and table of the ilium. Osteotomes and gouges are used to remove either the inner or outer table of the ilium and cancellous bone between the two tables.

Usual preop diagnosis

Ununited or malunited fracture

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Basic Repair with Skeletal Fixation	With Iliac Graft
Position	Supine (prone with posterior lateral graft)	⇐
Incision	Anteromedial or posterolateral to shaft of tibia	Anteromedial; parallel to iliac crest
Special instrumentation	Tourniquet; x-ray or I.I., pins, plates, screws, rods, external fixator; tourniquet	⇐
Antibiotics	Cefazolin 2 g iv preop. (If infected nonunion anticipated, antibiotics are withheld until cultures are	⇐

	obtained.)	
Surgical time	2–3 h	2.5 h
Closing considerations	Splint or cast applied while anesthetized. ⇐	
EBL	100 mL	200–300 mL
Postop care	PACU → room	⇐
Mortality	Minimal	⇐
Morbidity	Thrombophlebitis: 5%	⇐
	Compartment syndrome: 1%	⇐
	Infection: 1%	⇐
	Hematoma: <1%	5%
Pain score	5–8	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	10–80 yr (usually 20–40 yr)
Male:Female	5:1
Incidence	5–10% of tibia fractures; 50–75% of open fractures
Etiology	Trauma: 100%
Associated conditions	Poor nutrition (50%); infection (10%); metabolic disease (10%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

Suggested Readings

1. Lark RK, Lewis JS Jr, Watters TS, et al: Radiographic outcomes of ring external fixation for malunion and nonunion. *J Surg Orthop Adv* 2013; 22(4):316–20.
2. Mechrefe AP, Koh EY, Trafton PG, et al: Tibial malunion. *Foot Ankle Clin* 2006; 11(1):19–33, vii.
3. Ng A, Barnes ES: Management of complications of open reduction and internal fixation of ankle fractures. *Clin Podiatr Med Surg* 2009; 26(1):105–25.

ARTHROSCOPY OF THE ANKLE

SURGICAL CONSIDERATIONS

Description

Ankle arthroscopy is usually a diagnostic procedure, although it may be used for debridement or removal of loose bodies. The ankle joint generally is inspected through anterolateral and anteromedial portals. A posterolateral portal also may be used. Each portal is made via a 5-mm stab wound in the skin (Fig. 10.6-1); then instrumentation is placed, using trocars. Based on the procedure and surgeon preference, a mechanical distractor consisting of a strap that wraps around the ankle and attaches to the bed frame may be used. The portals are closed with sterile tape or a single suture. **Debridement** may be used to treat local or generalized articular damage. Bone marrow aspirate or platelet-rich plasma may be injected to promote healing.

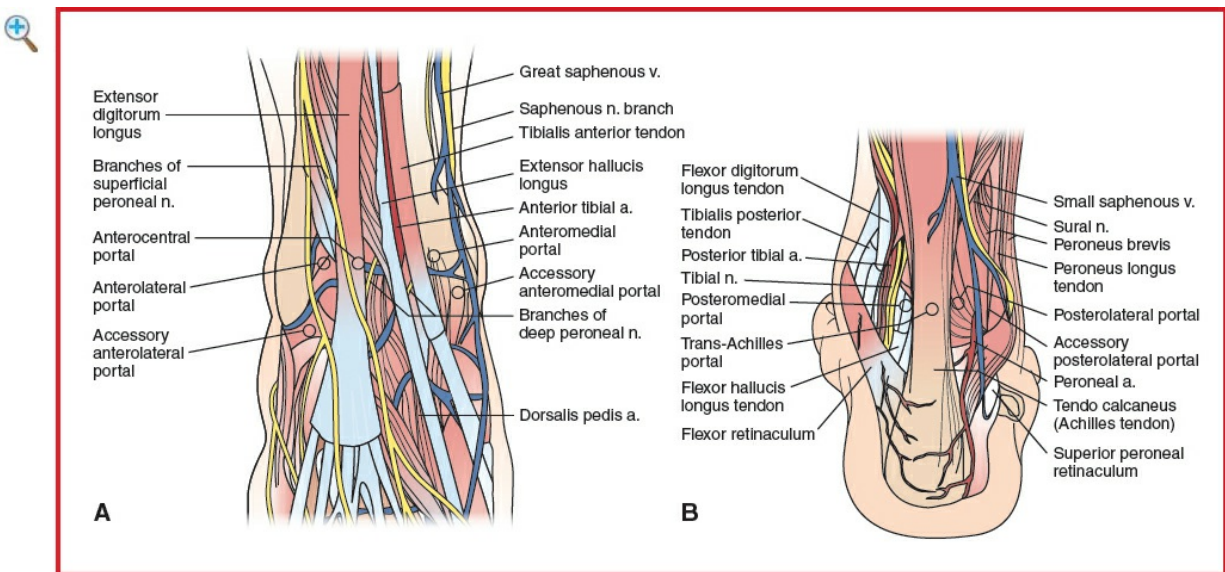


Figure 10.6-1 Portals for ankle arthroscopy. **A:** Anterior anatomy and portals. The anterolateral and anteromedial portals are used routinely. **B:** Posterior anatomy and portals. The posterolateral portal also is used routinely. (Redrawn with permission from Ferkel RD: Arthroscopic Surgery: The Foot and Ankle Lippincott-Raven, Philadelphia: 1996.)

Usual preop diagnosis

Trauma; infection; arthritis

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain

management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

	Arthroscopy
Position	Supine
Incision	0.5 cm portals (incisions)
Special instrumentation	Arthroscopic video system; arthroscopy-specific instruments, leg positioner and ankle distractor
Unique considerations	± Tourniquet. May use ankle distractor
Antibiotics	Cefazolin 2 g iv preop (optional)
Surgical time	1–2 h
Closing considerations	± Splint; incisions injected with local anesthetic.
EBL	Minimal
Postop care	PACU → home
Mortality	<0.01%
Morbidity	Hemarthrosis: 5% Thrombophlebitis <2% Infection: <1% Superficial peroneal nerve irritation: <1%
Pain score	2–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	12–70 yr (usually 20–40 yr)
Male:Female	1:1
Incidence	Uncommon
Etiology	Trauma (70%); arthritis (20%); infection (5%)
Associated conditions	Usually healthy; may have systemic arthritis

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

Suggested Readings

1. Glazebrook MA, Ganapathy V, Bridge MA, et al: Evidence-based indications for ankle arthroscopy. *Arthroscopy* 2009; 25(12):1478–90.
2. Hsu AR, Gross CE, Lee S, et al: Extended indications for foot and ankle arthroscopy. *J Am Acad Orthop Surg* 2014; 22(1):10–9. doi: 10.5435/JAAOS-22-01-10. Review.
3. Lui TH: Arthroscopy and endoscopy of the foot and ankle: indications for new techniques. *Arthroscopy* 2007; 23(8):889–902.

ANKLE ARTHROTOMY

SURGICAL CONSIDERATIONS

Description

Arthrotomy of the ankle is the opening of the joint for drainage, debridement, or fracture treatment. The joint usually is opened with an anterolateral midline or anteromedial longitudinal incision. Tendons and neurovascular structures are carefully retracted to expose the joint capsule, which is then opened in line with the skin incision. After intra-articular pathology is addressed, careful closure of the capsule is performed, taking care to obtain good hemostasis.

Usual preop diagnosis

Infection; trauma; arthritis

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURE

Position	Supine
Incision	Anterior midline or anteromedial longitudinal
Special	Tourniquet

instrumentation	
Antibiotics	Cefazolin 2 g iv preop unless infection is suspected
Surgical time	1–2 h
Closing considerations	± Splint while anesthetized; may have suction drain
EBL	Minimal
Postop care	PACU → room
Mortality	Minimal
Morbidity	Hemarthrosis: 20% Thrombophlebitis: 5% Infection: 1%
Pain score	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Infant-elderly
Male:Female	1:1
Incidence	Rare
Etiology	Trauma (70%); arthritis (20%); infection (10%)
Associated conditions	Inflammatory arthritis; multiple trauma; immunosuppression

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

ANKLE ARTHRODESIS

SURGICAL CONSIDERATIONS

Description

An **ankle fusion** may need to be performed for severe pain 2° arthritis of the ankle. Fusion can be performed via an anterior, lateral, or posterior approach. It can also be

performed arthroscopically; for this, please refer to the “Arthroscopy of the Ankle” section. For the lateral approach, the fibula is osteotomized. The ankle joint is exposed, and the surfaces of the joint are débrided. Cancellous bone is exposed on the distal tibia and talus, and the joint is compressed with bone screws from the distal tibia into the talus with or without a plate (see [Fig. 10.6-2](#)) or sometimes with an external fixation device with pins going through the distal tibia and talus. The wound is closed, sometimes over a drain, and a splint is applied.

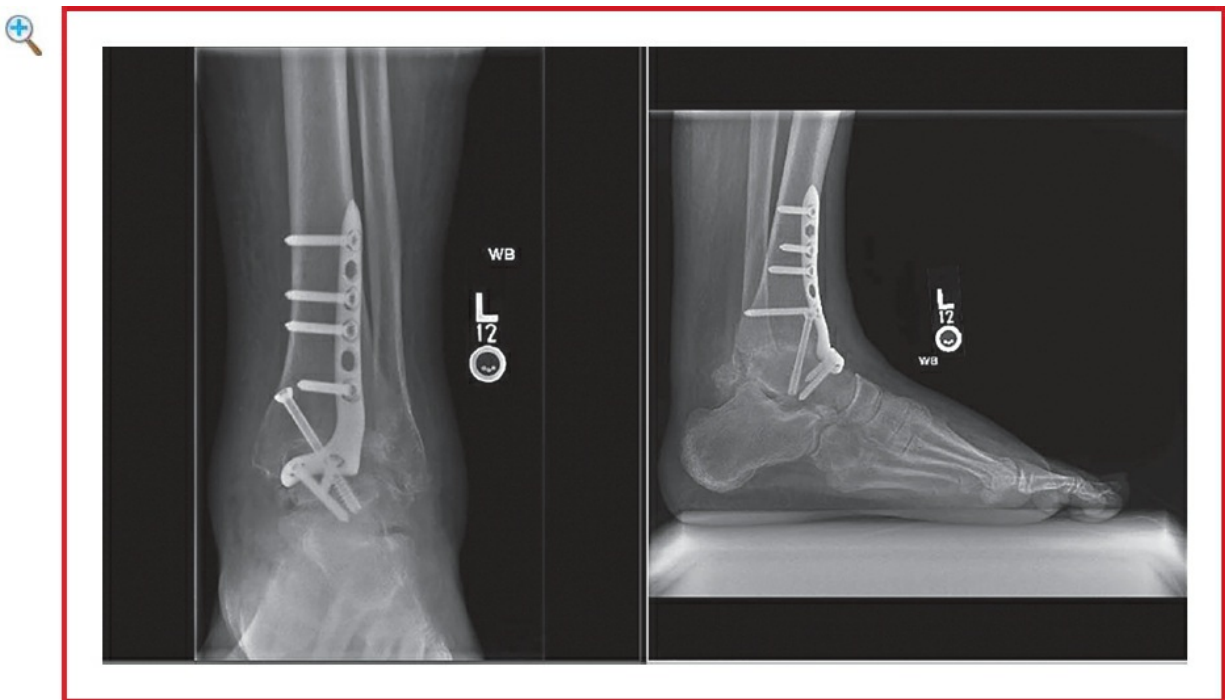


Figure 10.6-2 Tibiotalocalcaneal (TTC) arthrodesis.

Usual preop diagnosis

Arthritis of the ankle

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Anterior midline, lateral, or posterior over ankle joint
Special	Tourniquet; bone plate and screws or external fixator

instrumentation	
Unique considerations	Intraop radiographs; tourniquet use
Antibiotics	Cefazolin 2 g iv preop
Surgical time	2 h
Closing considerations	Splint; suction drain
EBL	100 mL
Postop care	PACU → room
Mortality	Minimal
Morbidity	Nonunion (late): 15% Thrombophlebitis: 10% Hematoma: 5% Wound dehiscence: 5% Infection: 1%
Pain score	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	All adult
Male:Female	1:1
Etiology	Degenerative arthritis; trauma; avascular necrosis of talus; septic arthritis
Associated conditions	Inflammatory arthritis; any disease requiring steroids

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

TOTAL ANKLE REPLACEMENT

SURGICAL CONSIDERATIONS

Description

A **total ankle replacement or arthroplasty** may need to be performed for severe pain 2° arthritis of the ankle joint. An anterior approach is utilized. The tibiotalar joint is exposed, and the surfaces of the joint are débrided. A series of different jigs are utilized in order to make the proper bony cuts to align the joint. These are performed with the assistance of x-ray. The final implants are sized and selected, and then press fit to the tibia and the talus (see [Fig. 10.6-3](#)). A polyethylene spacer is then placed between the metal components. Additional procedures may be needed to better align the joint. The wound is closed, sometimes over a drain, and a splint is applied.

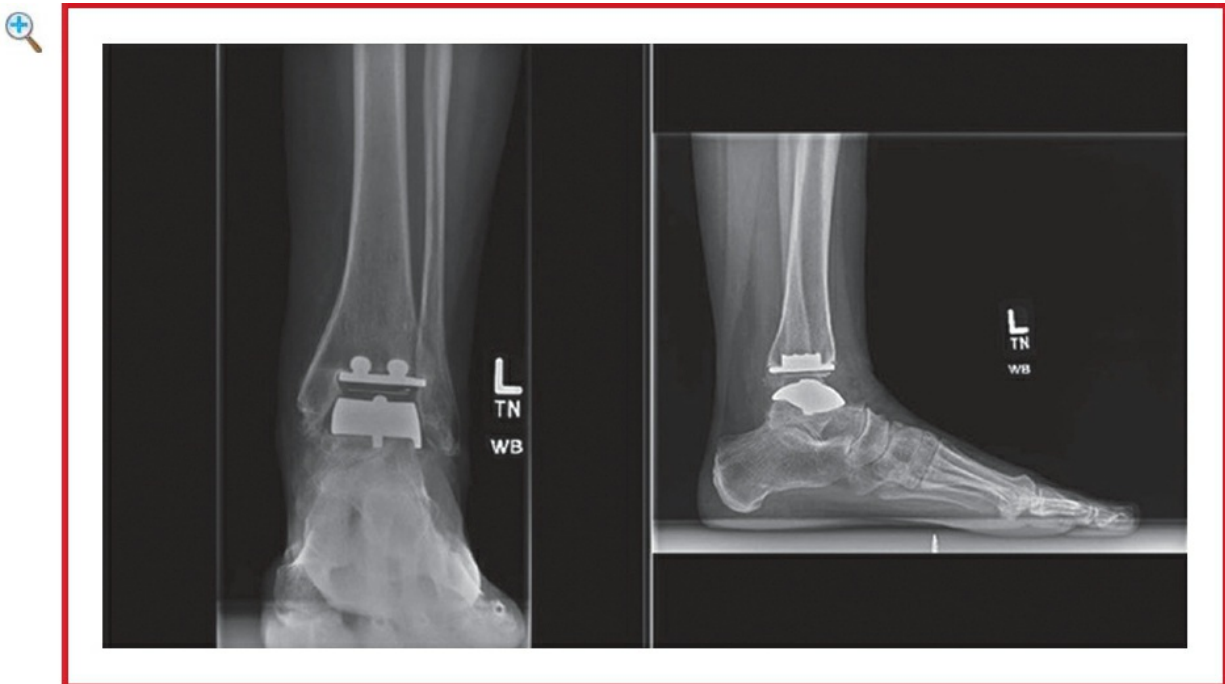


Figure 10.6-3 AP and lateral views following total ankle replacement.

Usual preop diagnosis

Arthritis of the ankle

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURE

Position	Supine
Incision	Anterior midline over ankle joint
Special instrumentation	Tourniquet; alignment jigs and final implants
Unique considerations	Intraop radiographs; tourniquet use
Antibiotics	Cefazolin 2 g iv preop
Surgical time	2–3 h
Closing considerations	Splint; suction drain
EBL	100 mL
Postop care	PACU → room
Mortality	Minimal
Morbidity	Aseptic loosening: 15% Thrombophlebitis: 10% Wound dehiscence: 5% Infection: 1%
Pain score	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adult > 50 yo
Male:Female	1:1
Etiology	Degenerative arthritis; trauma; rheumatoid arthritis
Associated conditions	Inflammatory arthritis; hemophilia; posttraumatic arthritis

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

Suggested Readings

1. Chou LB, Coughlin MT, Hansen S Jr, et al: Osteoarthritis of the ankle: the role o

- arthroplasty. J Am Acad Orthop Surg 2008; 16(5):249–59.
2. DeOrto JK, Gadsden J: Total ankle arthroplasty and perioperative pain. J Surg Orthop Adv 2014; 23(4):193–7.

REPAIR/RECONSTRUCTION OF ANKLE LIGAMENTS

SURGICAL CONSIDERATIONS

Description

Lateral ankle ligaments are repaired in the setting of chronic ankle instability. An incision is made posterior to the distal fibula, curving around the lateral malleolus and ending in the anterolateral ankle. The lateral ankle ligaments are incised and imbricated. The extensor retinaculum is also advanced to add strength to the repair. Occasionally, the peroneus brevis tendon is also used, usually in cases of revision surgery. The brevis is identified and detached from its musculotendinous junction in the leg, and one end is sutured to the peroneus longus tendon. A hole is drilled from anterior to posterior in the distal lateral malleolus; then the detached end of the peroneus brevis tendon is threaded through the hole. It is then attached to either the calcaneus or the talus, anterior to the lateral malleolus, with a staple or by suturing into a hole in the bone. The skin and subcutaneous tissues are closed and a splint or cast is applied.

Usual preop diagnosis

Lateral instability of the ankle

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURE

Position	Supine or lateral decubitus
Incision	Posterolateral aspect of ankle
Special instrumentation	Tourniquet; suture tacks or suture tape

Antibiotics	Cefazolin, 2 g iv preop
Surgical time	2 h
Closing considerations	Splint or cast while still anesthetized.
EBL	Minimal
Postop care	PACU → room or home
Mortality	Minimal
Morbidity	Infection: <1% Rerupture: <1% Wound dehiscence: <0.1%
Pain score	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Young adults
Male:Female	2:1
Etiology	Ankle sprain
Associated conditions	Peroneal tendon tears; osteochondral lesions of the talus

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

AMPUTATION THROUGH ANKLE (SYME)

SURGICAL CONSIDERATIONS

Description

Syme amputation (Figs. 10.6-4 and 10.6-5) is ankle disarticulation with closure, using a posterior flap, including the heel pad. It is more functional than below-knee amputation because patients can bear weight on the end of the stump; however, success is poor in patients with vascular disease or peripheral neuropathy. The posterior flap is

dissected directly from the calcaneus, carefully preserving the tough heel pad and its blood supply. The heel pad is sutured directly to the distal tibia to prevent migration and to cover the bone end. The posterior flap is then sutured to the anterior flap with interrupted sutures, and a compression dressing or cast is applied.

Usual preop diagnosis

Trauma; infection

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

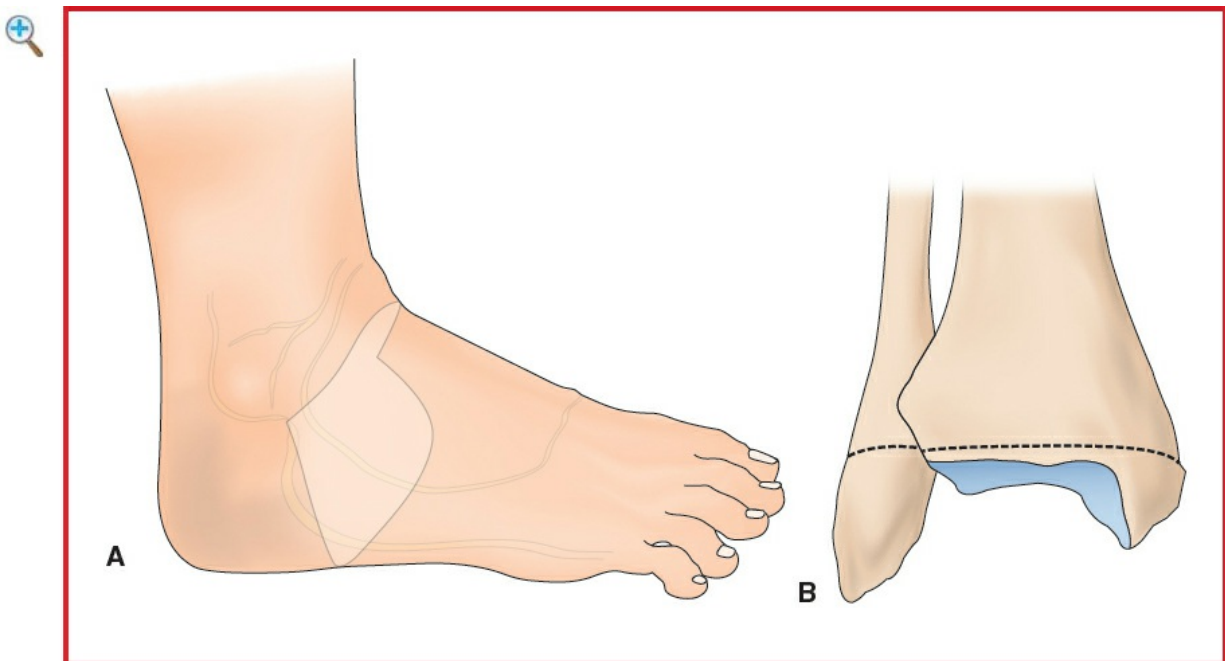


Figure 10.6-4 Syme amputation. **A:** Skin incisions. **B:** Level of bone transection in adults. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al, eds: *Surgery: Scientific Principles and Practice*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

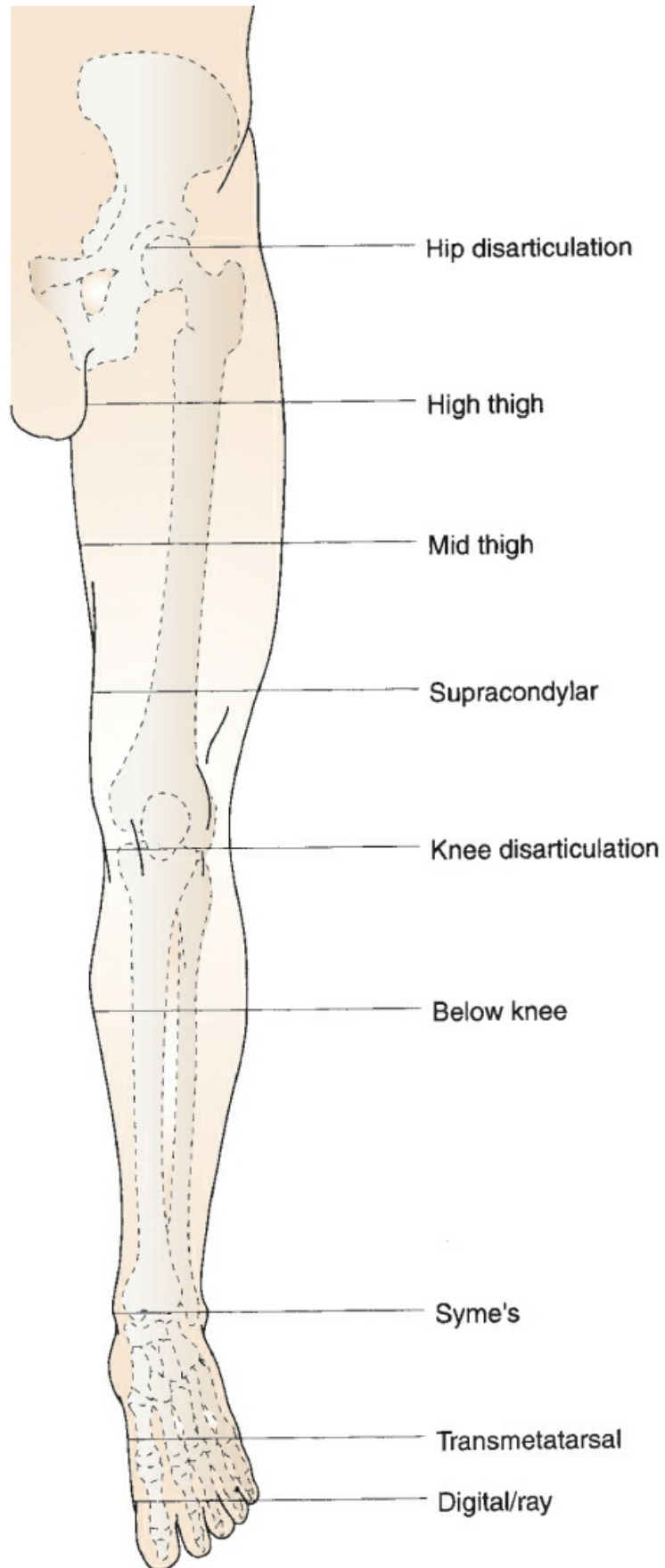


Figure 10.6-5 Common amputation levels for the lower extremity. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al, eds. *Surgery: Scientific Principles and Practices* 3rd edition. Lippincott Williams & Wilkins: 2001.)

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Anterior and posterior flaps
Special instrumentation	Tourniquet
Antibiotics	Cefazolin 2 g iv preop
Surgical time	1.5–2 h
Closing considerations	Bulky dressing or cast; if infection present, wound may be left open.
EBL	100 mL
Postop care	PACU → room
Mortality	Minimal
Morbidity	Phantom pain: 90% Infection: 10–15% Wound breakdown: 10–15% Pneumonia: 12% MI: 7% PE: 6% Hematoma: 5% Stroke: 5%
Pain score	5

■ PATIENT POPULATION CHARACTERISTICS	
Age range	Typically, >60 yr, though also children with congenital deformities
Male:Female	3:1
Incidence	20,000–30,000/yr
Etiology	Trauma (50%); infection (30%); congenital anomaly

	(5%)
Associated conditions	Peripheral vascular disease (30–40%); diabetes mellitus (DM) (<20%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

AMPUTATION, TRANSMETATARSAL

SURGICAL CONSIDERATIONS

Description

This amputation, usually for infection or ischemic necrosis of the toes, is performed at the midmetatarsal level, leaving the patient able to walk without a prosthesis. A transverse dorsal incision is made at the transmetatarsal level, and a plantar incision is made beginning at the corners of the dorsal incision and extending distally to the metatarsal heads to create a long plantar flap. The plantar flap is reflected proximally to the midmetatarsal level and tapered distally. The metatarsals are sectioned with a saw, and nerves and tendons are sectioned proximal to the osteotomies. The plantar flap is then brought over the ends of the bones and sutured with interrupted sutures to the dorsal flap. A compression dressing or splint is applied.

Variant procedure or approaches

Other partial-foot amputations, such as **midtarsal** and **ray amputation**, are much less common. They are managed in a fashion similar to that of the transmetatarsal amputation.

Usual preop diagnosis

Gangrene of the toes; infection

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURE

Position	Supine
Incision	Dorsal and plantar flaps
Special instrumentation	Tourniquet
Antibiotics	Cefazolin 2 g iv preop
Surgical time	1–2 h
Closing considerations	Bulky dressing
EBL	50 mL
Postop care	PACU → room
Mortality	Minimal
Morbidity	Phantom pain: 90% Infection: 10–15% Wound breakdown: 10–15% Hematoma: 5%
Pain score	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	>60 yr
Male:Female	3:1
Incidence	20,000–30,000 total amputations/yr
Etiology	Vascular disease (70%); infection (25%); trauma (<5%); congenital anomalies (<1%)
Associated conditions	Vascular disease (70%); diabetes mellitus (30%); pulmonary disease (30%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

LENGTHENING OR TRANSFER OF TENDONS,

ANKLE, AND FOOT

SURGICAL CONSIDERATIONS

Description

In cases of motor imbalance from neuromuscular disease or trauma, tendons are lengthened or transferred to a new insertion to partially restore balance or normalize joint motion. For **tendon lengthening**, a longitudinal incision generally is made directly over the tendon. Subcutaneous tissues and tendon sheath are incised to expose the tendon, which is transected with a Z-type incision. The tendon is placed in its lengthened position, and the ends of the Z are closed with suture. If present, the tendon sheath is closed separately from the skin closure. In a **tendon transfer**, the tendon usually is cut close to its insertion and transferred to a new bony insertion, which often requires a separate incision. The tendon is attached to the bone either with a suture anchor or by a bone tunnel and fixation by a screw or sutures.

Variant procedure or approaches

Achilles tendon lengthening is used to bring the ankle out of equinus. A **posterior tibial tendon lengthening** and/or **posterior ankle capsulotomy** may accompany the procedure.

Usual preop diagnosis

Contracture of muscle

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Tendon Lengthening	Achilles Tendon Lengthening
Position	Supine	Prone
Incision	Over tendon; sometimes multiple incisions	Over tendon
Special instrumentation	Tourniquet	←

Antibiotics	If young, none; in elderly or infirm, cefazolin 2 g iv preop	⇐
Surgical time	2 h	1 h
Closing considerations	Splint or cast while anesthetized	⇐
EBL	10 mL	⇐
Postop care	PACU → room	PACU → room or home
Mortality	Minimal	⇐
Morbidity	Infection: <1%	⇐
Pain score	4	3

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age
Male:Female	1:1
Incidence	Rare
Etiology	Neuromuscular disease (80%); trauma (20%)
Associated conditions	Static encephalopathy/cerebral palsy (75%); other neuromuscular disease (25%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

AMPUTATION ABOVE THE KNEE

SURGICAL CONSIDERATIONS

Description

In above-knee amputations, the distal part of the lower extremity is excised, starting just above the knee at the level of the distal third of the femur (Fig. 10.6-5). A stump is fashioned and will require prosthetic fitting at a later time. The most commonly

performed stumps incorporate anterior and posterior flaps of equal length. The underlying muscles (hamstrings and quadriceps) are either sewn to each other (**myoplasty**) or to bone (**myodesis**). In a **guillotine**, or **open amputation**, the stump is not fashioned (tissues are not closed) until later. This is a multistage procedure used for dirty, traumatic amputations, infection, or above-knee amputations with questionable survival and usually is done as a lifesaving measure. Internal fixation of part of the remaining femur may be indicated in traumatic amputations. The patient returns to the OR every 1–3 d for redebridement until closure of the clean stump can be performed.

Usual preop diagnosis

PVD or gangrene of lower extremity; trauma to lower extremity; open-femur fracture with traumatic amputation; tumor of lower extremity

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, rehabilitation plans, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Anterior and posterior on thigh
Special instrumentation	Amputation saw and rasp; drill for myodesis
Unique considerations	Patient often very ill from sepsis, chronic disease, or trauma
Antibiotics	Cefazolin 2 g iv q6–8h), ± gentamicin (80 mg iv q8h); adjust dosage for renal status, ± penicillin (1–2 million U iv q4h).
Surgical time	1–2 h
Closing considerations	Compressive dressing ± special stump sock
EBL	250 mL or more; higher for traumatic amputations
Postop care	Generally PACU → room (if medically unstable → ICU)

**Mortality
Morbidity**

~10–20%; higher in PVD (10–39%)

Phantom limb: 85–95%

Phantom pain: 2–15%

Wound infection ± deep infection: <15% in PVD

Respiratory failure or pneumonia: 10–15%

MI: 7–10%

Thromboembolism: 6–10%

Cerebrovascular accident: 5–10%

Contractures—flexion and abduction: Common

Urinary retention requiring catheterization: Common

Failure to heal ± wound dehiscence: Uncommon

Hematoma: Rare

Neuromas: Rare

Reamputation: Rare

UTI: Rare

Contralateral amputation, especially in diabetics and those with PVD Postop depression

Pain score

7–10

■ PATIENT POPULATION CHARACTERISTICS**Age range**

PVD, diabetic gangrene: 70–90% >60 yr

Multiple trauma with traumatic amputation, tumor of lower extremity: 18–35 yr

Male:Female

Overall, 3:1

Elderly, predominance of males

Multiple trauma, 4–5:1

Tumor 1:1

Incidence

Common for PVD patients; rare for trauma or tumor

Etiology

PVD and diabetic gangrene (70–90%); multiple trauma (younger patients) (rare—usually with severe grade IIIC injuries with neurovascular severance); tumor (rare); uncontrollable infection (e.g., gas gangrene) (rare)

**Associated
conditions**

Diabetes (70–80% of patients presenting for this procedure); numerous other serious medical conditions; multiple trauma in younger patients

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Above- and Below-Knee Amputation.

AMPUTATION BELOW THE KNEE

SURGICAL CONSIDERATIONS

Description

Below-knee amputation is ablation of the lower limb, usually at the level of the midleg. A long, posterior flap normally is used to cover the stump. The condition of the soft tissues may dictate the level and/or type of flaps used. The procedure begins with an anterior transverse incision made over the midtibia. A long posterior flap, which is 2–3 times the diameter of the leg in length, is then made. The bone is exposed anteriorly, and the anterolateral neurovascular structures and muscles are transected and ligated as appropriate ([Fig. 10.6-6A](#)). The bone is then transected with a bone saw, and the posterior structures are transected and ligated as appropriate. The amputated leg and foot are then removed from the table, and the posterior flap is tapered and shaped for closure ([Fig. 10.6-6B](#)). Deep sutures are placed to secure the posterior muscles to the anterior tibia. The skin opening and subcutaneous tissues are closed with interrupted sutures. Finally, a drain is placed (sometimes), and either a compression dressing or an immediate postop cast is applied.

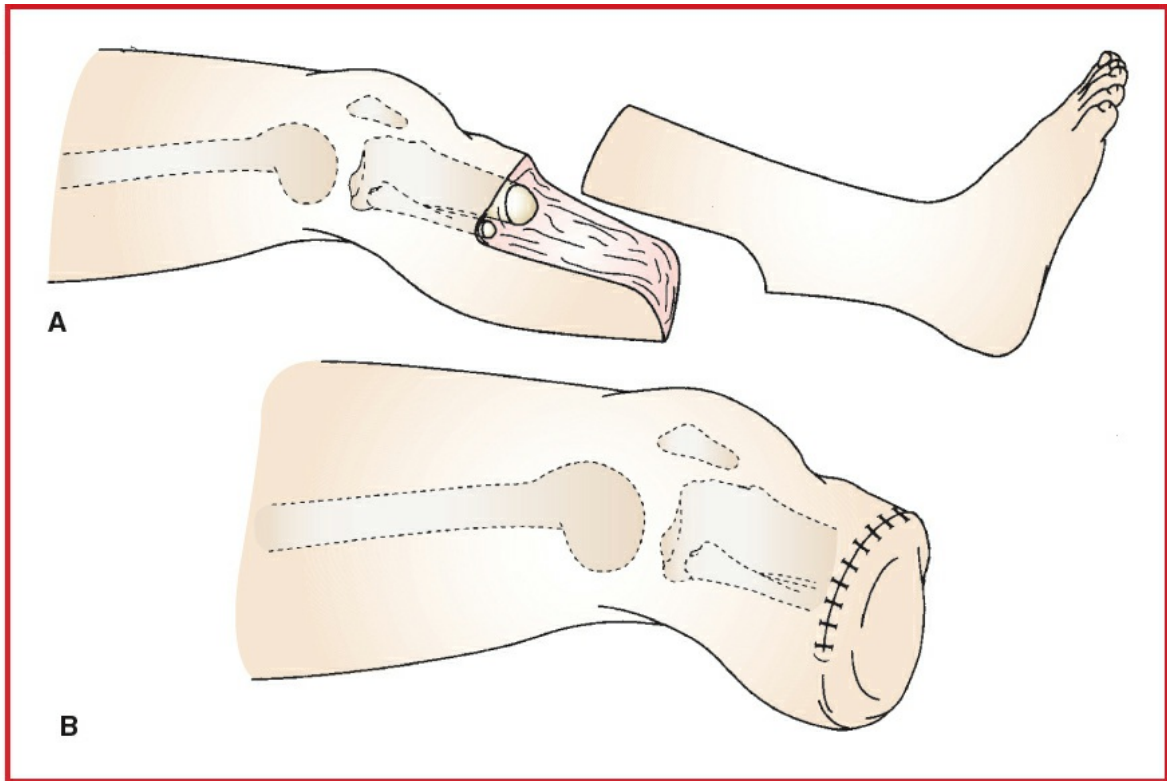


Figure 10.6-6 Below-knee amputation. **A:** The tibia is transected 1 cm proximal to the skin incision, and the fibula is transected an additional 1 cm proximal to the level of the tibial transection. The posterior calf muscles are incised along the plane of the skin incision. **B:** The posterior flap is rotated anteriorly and approximated. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al, eds: *Surgery: Scientific Principles and Practice* 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Variant procedure or approaches

Guillotine amputation may be used as the first of a two-stage procedure in infected or contaminated cases. With a guillotine amputation, the bone and soft tissues are transected very quickly in guillotine fashion at the midtibial level. Neurovascular structures are ligated as appropriate. These wounds are usually left open and a compression dressing applied.

Usual preop diagnosis

Dysvascular limb; infection; trauma

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, rehabilitation plans, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Anterior and posterior flaps; for guillotine amputation, circumferential incision
Special instrumentation	Bone saw; tourniquet, if traumatic (tourniquet contraindicated if infected or avascular)
Antibiotics	Cefazolin 2 g iv preop
Surgical time	1.5 h; for guillotine amputation, 0.5 h
Closing considerations	May use cast; drain. Bulky dressing needed for guillotine amputation.
EBL	200 mL
Postop care	PACU → room
Mortality	10%
Morbidity	Phantom pain: 90% Infection: 10–15% Wound breakdown: 10–15% Pneumonia: 12% MI: 7% PE: 6% Hematoma: 5% Stroke: 5%
Pain score	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Usually, >60
Male:Female	3:1
Incidence	20,000–30,000 total amputations/yr
Etiology	Dysvascular limb (70%); trauma (20%); infection (5%); tumor (5%); congenital anomaly (<1%)
Associated conditions	Vascular disease (70–80%); malnutrition (50%); diabetes mellitus (30%); pulmonary disease (30%)

ANESTHETIC CONSIDERATIONS FOR ABOVE- AND BELOW-KNEE AMPUTATION

PREOPERATIVE

Vascular disease and tumors are the two most common indications for these surgeries. Patients presenting for amputations often have severe systemic vascular disease. Their inability to perform exercise limits the usefulness of preop Hx in evaluating cardiopulmonary reserve and often necessitates invasive studies for full evaluation.

Respiratory	Smoking is a risk factor common to both vascular and pulmonary diseases. Chronic bronchitis or COPD patients should have maximum medical therapy (e.g., inhaled bronchodilators, theophylline, and steroids, when appropriate) prior to anesthesia. Regional anesthesia is an excellent choice for patients with severe pulmonary disease. Tests: As indicated from H&P
Cardiovascular	Significant cardiovascular disease is present in ~30% of these patients. Particularly in diabetics, CAD is often silent. Dipyridamole thallium imaging of the heart can reveal the preop myocardium at risk of ischemia; however, therapy of stenotic coronary arteries usually can be undertaken only after the amputation. Medical management, to include β-blockers when tolerated, will reduce perioperative MIs. Tests: ECG; others as indicated from H&P
Neurologic	Peripheral and autonomic neuropathies may be present in the diabetic patient. Hence, these patients may be more susceptible to injury from malpositioning and are less able to tolerate the hemodynamic changes associated with regional anesthesia. Preexisting neurologic deficits should be carefully documented. Autonomic neuropathy is diagnosed by any of the following: postural $\downarrow$ BP (in patient not bedridden); loss of sinus arrhythmia; resting $\uparrow$ HR; miosis-abnormal response to darkness.

Musculoskeletal	Rhabdomyolysis can occur in the presence of partial ischemia. (See Renal, below.)
Hematologic	Often a trial of heparin, warfarin, or thrombolytic therapy will have been undertaken before amputation. Coag studies, including PT, PTT, and bleeding time, are, therefore, often necessary to determine the appropriateness of epidural or intrathecal anesthesia. Warfarin-induced elevation of the PT can be reversed preop with FFP 5–10 mL/kg body weight. This therapy may induce fluid overload in patients with poor cardiac reserve. For these patients, diuretics should be administered to maintain normovolemia. Tests: As indicated from H&P
Renal	Limb ischemia can result in myoglobinemia from rhabdomyolysis. Evidence of progressive renal failure or rising CPK-MM fractions should be treated with hydration, forced alkaline diuresis, and prompt amputation. Tests: Consider Cr; BUN; CPK enzymes; urine myoglobin.
Laboratory	Diabetic patients require preop control of blood glucose and periop glucose monitoring.
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po, ±celecoxib 200 mg po, and ±gabapentin 600 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Either regional or GA may be appropriate.

Regional anesthesia

Both SAB and epidural blocks are useful techniques. Subarachnoid anesthesia has the advantage of limited spread of the block above the level of surgery while obtaining adequate blockade of the sacral roots that are resistant to low-dose epidural techniques. Epidural anesthesia allows for extending the duration of anesthesia and for the administration of postop epidural analgesia. Anesthesia from T12 (T8 with tourniquet) is adequate. Full motor blockade is not necessary. Typical drugs and doses include subarachnoid, 75 mg of 5% lidocaine in 5% dextrose (controversial) with morphine 0.2 mg, and epidural, 12–15 mL 2% lidocaine with epinephrine 1:200,000 in divided doses.

General anesthesia

Induction	Standard induction is appropriate for patients with normal airways. Intubation is indicated for diabetic patients with gastroparesis. Beware of difficult airway in long-standing insulin-dependent diabetics.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance	
Emergence	No special considerations. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include ondansetron and dexamethasone.	
Blood and fluid requirements	Moderate blood loss IV: 16 ga × 1 BSS @ 4–6 mL/kg/h	Expect 100–200 mL blood loss, mostly during cleaning of the wound made while developing a flap. See Restrictive Fluid Management, Appendix K , p. K-4.
Control of blood loss	Tourniquet may be used.	Inflation pressure is typically 100 mm Hg + systolic pressure. Maximum “safe” tourniquet time is 1.5–2

		h, followed by a 5- to (preferably) 15-min reperfusion interval, if further tourniquet time is necessary.
Special considerations	Tourniquet deflation and limb reperfusion	Mild ↓ BP is common. In patients with moderate-to-severe lung disease, continue controlled ventilation until after the lactic acid accumulated in the ischemic leg is metabolized (3–5 min) because these patients may be unable to increase ventilation adequately to buffer this acid load.
Monitoring	Standard monitors ± CVP line ± Arterial line	Invasive monitoring is indicated in the presence of severe cardiac or pulmonary disease. Serial blood glucose determination should be made in the diabetic patient.
Positioning	✓ and pad pressure points ✓ eyes	Meticulous padding of the extremities is necessary to prevent ischemic skin ulceration in patients with vascular insufficiency.

POSTOPERATIVE

Complications	Hematoma Bleeding	✓ drains

Multimodal pain management	Spinal opiates Epidural analgesia	Epidural hydromorphone 50 mcg/mL infused at 50–200 mcg/h provides excellent analgesia. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional analgesia (e.g., continuous nerve block) whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6) Regional analgesia	Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	CXR if CVP was placed.	Other studies as indicated

Suggested Readings

1. De Jong R, Shysh AJ: Development of a multimodal analgesia protocol for perioperative acute pain management for lower limb amputation. *Pain Res Manag* 2018; 2018:5237040.
2. Salinas FV: Evidence basis for ultrasound guidance for lower-extremity peripheral nerve block: update 2016. *Reg Anesth Pain Med* 2016; 41(2):261–74.
3. Sites BD, Brull R: Ultrasound guidance in peripheral regional anesthesia philosophy, evidence-based medicine, and techniques. *Curr Opin Anaesthesiol* 2006; 19(6):630–9.

FASCIOTOMY OF THE THIGH

SURGICAL CONSIDERATIONS

Description

There are three compartments in the thigh. Increased intracompartmental pressure in the thigh requires surgical release of tight skin and fascial structures ([Fig. 10.6-7](#)). This usually occurs after severe trauma to the thigh (e.g., crush injury, comminuted fracture),

after prolonged vascular surgery (with ischemia to the thigh), or with infection. Compartment syndrome is a true emergency and must be treated within minutes of recognition. Failure to do so may result in loss of limb. Conventional devices may be used to measure intracompartmental pressure (normal ≤ 30 mm Hg), or diastolic pressure minus compartment pressure should be >30 mm Hg). **Fasciotomy of the thigh** involves incising the skin and fascia typically through an anterolateral incision to release the anterior and posterior compartment and to débride any necrotic tissue. A medial incision can be done to release the adductor compartment. The wound is left open for later redebridement, delayed primary closure, or skin grafting. Typically, a negative pressure dressing is applied at the time of surgery. Thus, the fasciotomy begins a multistage procedure of incision and debridement with subsequent attempt at closure.

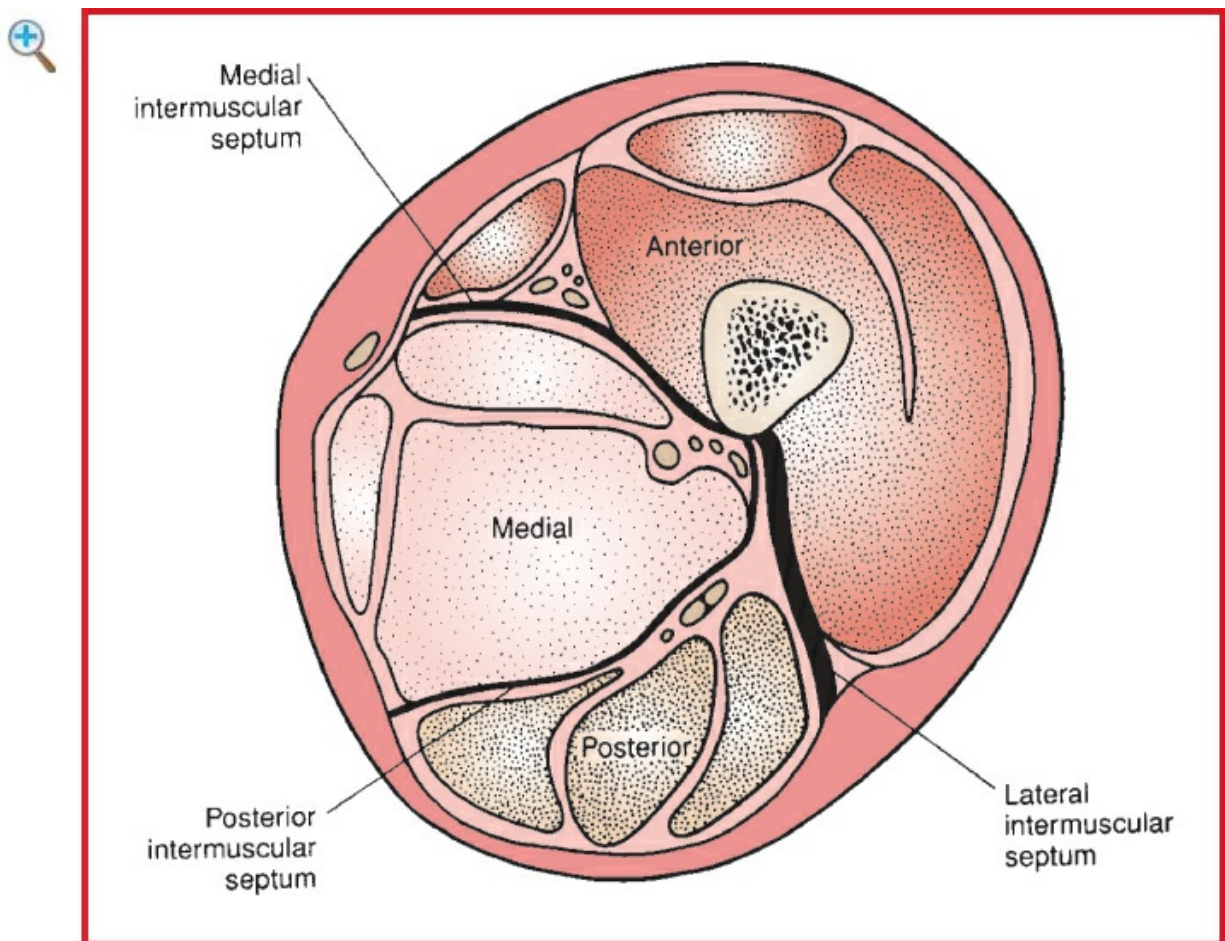


Figure 10.6-7 Cross section of the thigh showing the three major compartments.

Usual preop diagnosis

Compartment syndrome of thigh; crush injury to thigh; necrotizing fasciitis

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist

should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine or lateral decubitus
Incision	Lateral thigh, medial thigh
Unique considerations	Patient may be very ill. If an ipsilateral femoral fracture is present with a compartment syndrome, the surgeon may want to perform ORIF, intramedullary nailing, or external fixation of the fracture.
Antibiotics	Cefazolin 2 g iv q6h
Surgical time	1.5–2 h for fasciotomy alone
Closing considerations	Wound left open and covered by sterile dressings
EBL	250–500 mL
Postop care	If surgery is performed acutely, patient frequently will be a multiple-trauma victim with numerous injuries and extensive blood loss; usually goes to ICU.
Mortality	Dependent on extent of multiple trauma
Morbidity	Hypotension and fluid loss: Common Neurologic deficit to peripheral nerves: Common, if decompression delayed Respiratory distress and fat embolism: Not uncommon, if concomitant femur fracture Vascular complications: Not uncommon Amputation: Rare, if decompression prompt Systemic sepsis: Rare Wound infection: Rare New compartment syndrome; insufficient fasciotomy: Rare
Pain score	7–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age, but predominance of males <30 yr
Male:Female	5:1
Incidence	Extremely rare
Etiology	Trauma—motorcycle and motor vehicle accidents, falls, industrial injury, crush injuries; postop—local hematoma and swelling; thrombosis or disruption of blood supply to thigh (e.g., failed proximal vascular bypass surgery, aortic dissection); massive infection of thigh compartment (e.g., gas gangrene)
Associated conditions	Burns; drug and alcohol overdose; frequently associated with trauma to other organ systems

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Fasciotomy of the Leg.

Suggested Readings

1. Rush RM Jr, Arrington ED, Hsu JR: Management of complex extremity injuries: tourniquets, compartment syndrome detection, fasciotomy, and amputation care. *Surg Clin North Am* 2012; 92(4):987–1007, ix.
2. von Keudell AG, Weaver MJ, Appleton PT, et al: Diagnosis and treatment of acute extremity compartment syndrome. *Lancet* 2015; 386(10000):1299–310.

FASCIOTOMY OF THE LEG

■ SURGICAL CONSIDERATIONS

Description

This procedure is the surgical decompression of fascial compartments for treatment or prevention of compartment syndrome. Patients are often very ill and unstable with other injuries or disease. Compartment syndrome is a true emergency and must be treated within minutes of recognition. Failure to do so may result in loss of limb. There are four compartments in the lower leg: anterior, lateral, deep posterior, and superficial posterior (Fig. 10.6-8). Generally, all four compartments are released during the

procedure. A **four-compartment fascial decompression** can be performed through two incisions—medial and lateral. A medial longitudinal incision is made just posterior to the tibia; through this incision, the superficial and deep posterior compartments are identified and the fascia incised in longitudinal fashion. A straight, lateral, longitudinal incision is made, and the deep fascia overlying the anterior and lateral compartments is identified. The fascia of each compartment is then incised longitudinally. Skin incisions are rarely closed because of swelling and need for possible repeat debridement. A negative pressure dressing is typically applied along with a splint postop. Multistage procedure may be needed after primary release for repeat debridement, delayed primary closure, or skin grafting.

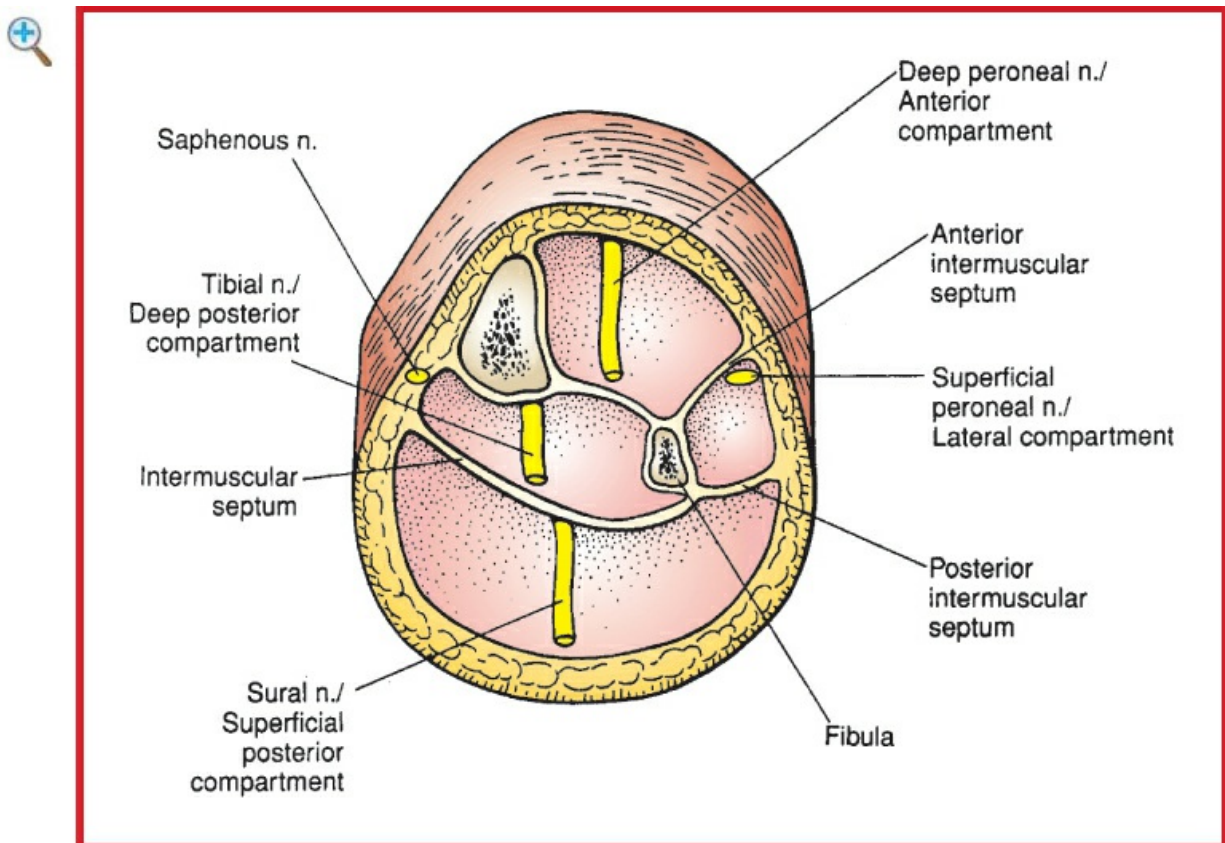


Figure 10.6-8 Cross section of the left leg, middle lower third, showing the four compartments with associated peripheral nerves.

Usual preop diagnosis

Compartment syndrome; vascular trauma

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Medial and lateral parallel to tibia
Unique considerations	Often associated with fracture; may require fixation
Antibiotics	Cefazolin 2 g iv preop
Surgical time	30 min+
Closing considerations	Wounds left open; splint may be required.
EBL	100 mL
Postop care	PACU → room; vascular monitoring is carried out clinically via a pulse oximeter on toes.
Mortality	Minimal
Morbidity	Myonecrosis: 50% Thrombophlebitis: 10–20% Infection: 10–15%
Pain score	3

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages
Male:Female	1:1
Incidence	~5% of tibia fractures
Etiology	Trauma—blunt fracture, vascular (90%); drug overdose (10%); burns (<5%); revascularization (<5%)
Associated conditions	Multiple trauma (60%); vascular disease (15%)

Suggested Readings

1. Cone J, Inaba K: Lower extremity compartment syndrome. *Trauma Surg Acute Care Open* 2017; 2(1):e000094.

2. von Keudell AG, Weaver MJ, Appleton PT, et al: Diagnosis and treatment of acute extremity compartment syndrome. Lancet 2015; 386(10000):1299–310.

ANESTHETIC CONSIDERATIONS FOR FASCIOTOMY OF THIGH AND LEG

PREOPERATIVE

Compartment syndromes and necrotizing fasciitis are the indications for these procedures (necrotizing fasciitis also may cause compartment syndrome). Patients with compartment syndrome often have no systemic disease, whereas patients with necrotizing fasciitis have a rapidly life-threatening infection that requires prompt surgical debridement and often is complicated by rhabdomyolysis and DIC.

Respiratory	Usually no special considerations, unless massive sepsis.
Cardiovascular	Sepsis is uniformly present in patients with necrotizing fasciitis. Management includes antibiotics and hemodynamic support with dopamine (5–15 mcg/kg/min) or epinephrine (0.02–0.25 mcg/kg/min), with therapy guided by invasive hemodynamic monitoring, which may include PA catheter.
Neurologic	If fasciotomy is for compartment syndrome, there may be compromise of distal nerves and blood flow. Perform a thorough neurologic exam of the involved extremity to document preop deficits.
Hematologic	If infection is the indication for the fasciotomy, DIC is likely. Evaluate for pathologic bleeding. Administer factors necessary to correct coagulopathy during the procedure. Tests: CBC; PT; PTT; fibrinogen; fibrin split products; Plt count
Renal	Both necrotizing fasciitis and compartment syndrome often cause myoglobinuria and rhabdomyolysis. Myoglobinuria can be inferred from urine that is dipstick-positive for occult blood but microscopically free of RBCs in the absence of hemolysis (therefore,

	no free Hb in the urine).
Laboratory	Hct; serial K ⁺ levels if there is an active diuresis; other studies as indicated from H&P
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Regional techniques

Generally, most surgeons do not want regional anesthesia for compartment release in order to monitor neurologic status. In the absence of surgeon preference, regional anesthesia can be used for compartment syndrome decompression, unless there is evidence of DIC or systemic infection. These surgeries are usually of short duration (<1 h). Sepsis and hemodynamic instability usually mandate GETA for fasciotomy in patients with necrotizing fasciitis.

Regional anesthesia

Subarachnoid anesthesia has the advantage of adequate blockade of the sacral roots that are resistant to low-dose epidural techniques. Anesthesia from T10 to S2 is adequate. Typical drugs and dosages include subarachnoid, 12.5–15 mg of 0.5% bupivacaine, and epidural, 12–15 mL 2% lidocaine with epinephrine 1:200,000 in divided doses.

General anesthesia

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance
Emergence	Consider postop ventilation for patients with impaired oxygenation or ongoing hemodynamic instability; otherwise, no special considerations.

	Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include ondansetron and dexamethasone.	
Blood and fluid requirements	IV: 16 ga × 1 (compartment syndrome) 14–16 ga × 2 (necrotizing fasciitis) BSS @ 4–6 mL/kg/h	To prevent renal damage, insure adequate circulatory volume; induce osmotic diuresis with mannitol 0.25 g/kg iv. Furosemide 10–100 mg also may be necessary to maintain diuresis. Replace UO with BSS as guided by invasive monitoring. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors UO ± Arterial line (± ABG) ± CVP or PA catheter ± $S_{cv}O_2/S_vO_2$	Fluid losses (third-spacing, bleeding) may be significant. Patients with necrotizing fasciitis require an arterial line with PPV and either a CVP or PA catheter with $S_{cv}O_2/S_vO_2$ to guide fluid and inotropic/pressor therapy.
Positioning	✓ and pad pressure points ✓ eyes	

POSTOPERATIVE

Complications	DIC Renal failure 2° rhabdomyolysis
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	Hypo/hyperkalemia Sepsis syndrome, including ARDS	
Multimodal pain management	PCA or epidural analgesia	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional analgesia (e.g., continuous nerve block) whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6) Regional analgesia	Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	Hct Electrolytes UA (dipstick and microscopic)	For patients with sepsis: coag profile, including PT/PTT, fibrin split products, and Plt count

BIOPSY, LEG AND FOOT

SURGICAL CONSIDERATIONS

Description

Biopsy is performed to excise tissues for pathologic evaluation, usually through a small, longitudinal wound. For **incisional biopsy**, a longitudinal incision is made over the mass. Overlying soft tissues are incised with minimal undermining. The area in question is incised and the biopsy removed, with care being taken to prevent spillage into the adjacent tissues. The pathologist often is asked to perform a frozen section to determine whether diagnostic tissue is present. The wound is closed with interrupted sutures and a compression dressing applied. A splint or cast may be used if a significant amount of bone has been removed. If the lesion is small, x-ray control or image intensification may be necessary for localization. **Needle biopsy** may be used for distinct osseous lesions to obtain small amounts of tissue for culture or histology. **Excisional biopsy** may be used for benign lesions like exostoses or lipomas.

Usual preop diagnosis

Tumor; infection

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, regional/local anesthetic options, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Incisional Biopsy	Needle Biopsy	Excisional Biopsy
Position	Supine	⇐	⇐
Incision	Short longitudinal	Stab wound	Stab wound
Special instrumentation	Bone-cutting instruments; x-ray or I.I.	Trephine (e.g., Craig needle); x-ray or I.I.	X-ray or I.I.
Unique considerations	Tourniquet	⇐	⇐ (Bone graft may be necessary.)
Antibiotics	None (may be given postop)	⇐	⇐
Surgical time	1 h	0.5 h	0.5–2 h
Closing considerations	May be splinted	Usually no splint	May be splinted
EBL	50 mL	Minimal	100–200 mL
Postop care	PACU → room or home	⇐	PACU → room
Mortality	Minimal	⇐	⇐
Morbidity	Hematoma: 5%	⇐	⇐
	Tumor spread: <5%	⇐	⇐
	Infection: <1%	⇐	⇐
Pain score	3	2	3

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages
Male:Female	1:1
Incidence	Rare
Etiology	Tumor (75%); infection (25%)
Associated conditions	Metastatic disease; immune compromise

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lower-Extremity Procedures.

BIOPSY OR DRAINAGE OF ABSCESS/EXCISION OF TUMOR

■ SURGICAL CONSIDERATIONS

Description

This procedure involves obtaining a piece of tissue for histologic and/or bacteriologic Dx by closed, percutaneous means or by open biopsy. Subsequently, the area may be drained (abscess or infection) or an open excision of a tumor may follow ($\pm$ internal fixation). For excision of tumors of the pelvis, acetabulum, or femur, please consult the appropriate section describing fractures of the area. Each case must be individualized.

Variant procedure or approaches

Excision of infection or tumor of proximal or distal femur, femoral shaft, pelvis, or acetabulum

Usual preop diagnosis

Femur: biopsy of mass; infection; osteomyelitis. Pelvis or acetabulum: biopsy of pelvis or acetabulum; drainage of abscess or infection of pelvis or acetabulum; osteomyelitis of pelvis or acetabulum; septic arthritis of acetabulum

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include patient education, local/regional anesthetic options, limiting NPO

duration, alcohol and smoking cessation, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine or lateral decubitus
Incision	Percutaneous, short or long; location depends on site of lesion.
Special instrumentation	Biopsy needles and instruments to take a core biopsy; bone cement may be used to plug biopsy site. Some surgeons use fracture table or radiolucent table with I.I.
Unique considerations	May require intraop frozen section and Gram stain
Antibiotics	After tissue has been obtained, cefazolin 2 g iv q6h × 48 h. A Gram stain will help decide immediate antibiotic coverage, but cultures and sensitivities are ultimately necessary.
Surgical time	1 h for simple procedures; much longer (up to 12 h) for more extensive excisional procedures ± further reconstruction
EBL	100–1000 mL or more
Postop care	If procedure is extensive with much blood loss, or the patient is unstable or ill from chronic sepsis or invasive tumor, it is prudent to send patient to ICU.
Mortality	Dependent on extent of procedure. Biopsy or drainage of a small, localized abscess in soft tissue or bone is rarely life-threatening. Wide/radical excision of a malignant tumor in the pelvis or extremities is frequently life- and/or limb-threatening.
Morbidity	The following are dependent on site and procedure: Intraop fracture Nonunion Chronic osteomyelitis Compartment syndrome

- Residual instability of pelvis or hip joint
- Fracture of pelvis or acetabulum, nonunion
- Chronic osteomyelitis or septic arthritis
- Hypotension 2° to blood loss
- Respiratory distress
- Neurologic injury to lumbosacral plexus, sciatic nerve, or other peripheral nerves
- Vascular injury to iliac or other vessels
- Injury to GI, genitourinary, or gynecologic organs

Pain score 2–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age; predominance of elderly patients with tumors
Male:Female	1:1
Incidence	Rare
Etiology	Benign and malignant tumors (common); infection (rare); previous surgery (rare); previous trauma (rare)
Associated conditions	Metastatic disease or other foci of infection

■ ANESTHETIC CONSIDERATIONS FOR LOWER-EXTREMITY PROCEDURES

(Procedures covered: ORIF of femur, tibia, ankle, and foot; intramedullary nailing of femur and tibia; closed reduction and external fixation of femur and tibia; distal tibia, ankle, and foot procedures; repair nonunion/malunion of femur and tibia; ankle arthroscopy, arthrotomy, arthrodesis; repair/reconstruction, ankle ligaments; Syme amputation; transmetatarsal amputation; tendon (ankle, foot) lengthening; biopsy of leg and foot; biopsy or drainage of abscess/excision of tumor)

■ PREOPERATIVE

Trauma victims constitute the largest group of patients for these procedures. Early administration of antibiotics and minimizing the time between fracture and surgery for open wounds significantly reduce the incidence of wound infection. Evaluations for

other injury, adequacy of fluid resuscitation, and preexisting conditions need to be undertaken promptly and used as a guide for anesthetic management. Patients with bone cancer form another subset of patients and often have concurrent medical conditions and have undergone chemotherapy or radiation therapy preop.

Respiratory	Pulmonary fat embolus occurs in 10–15% of patients following bone fracture. Sx include hypoxemia, ↑ HR, tachypnea, respiratory alkalosis, mental status changes, and conjunctival petechiae. Lab analysis may reveal fat in the urine. Preop therapy for this condition should include supplemental O₂ with mechanical ventilation, to correct hypoxemia, and meticulous fluid management to prevent worsening pulmonary capillary leak. Tests: Consider CXR; others as indicated from H&P
Cardiovascular	Cardiac contusion or tamponade are possible if blunt chest trauma has occurred during the injury. A large volume of blood can be hidden around a long bone fracture site. ↑ HR, orthostasis, or ↓ BP indicate hypovolemia, and this should be corrected with crystalloid (10–40 mL/kg) or blood if Hct < 24%. In patients with a tibial or distal femur fracture, and who are presenting with hemodynamic instability and ongoing blood loss, consider applying a tourniquet to the thigh prior to induction. Tests: Consider ECG; CPK enzyme levels and ECHO will help evaluate the presence of cardiac injury.
Neurologic	Perform a thorough neurologic evaluation, including mental status and peripheral sensory exams. A CT scan of the head is indicated for any patient with prolonged loss of consciousness prior to anesthesia. Drug abuse is common in trauma patients, and they should be asked specifically about any drug use. Tests: Patients with inappropriate behavior or a positive drug abuse Hx should undergo a urine and plasma drug screen.
Musculoskeletal	Consider cervical instability, and obtain spine films if

	mechanism of injury included rapid deceleration or trauma to the head or neck. Myoglobinemia and $\uparrow$ K ⁺ may result from crush injury.
Hematologic	Patients with cancer who have undergone chemotherapy and multiple transfusions often develop sensitivities to blood products and may require specialized blood products, such as leukocyte-poor PRBC or red cells negative for a particular antigen. The availability of these blood products should be confirmed before surgery. Tests: Hct and others as indicated from H&P
Renal	Tests: UA
Laboratory	Other tests as indicated from H&P
Premedication	Due to the risk of gastric aspiration, minimal or no premedication is given to trauma victims. For other patients, standard premedication. Narcotic premedication (morphine 1–2 mg iv q10min titrated to effect) is appropriate for patients experiencing pain with movement.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

For trauma patients, regional anesthesia permits evaluation of mental status, provides intact airway reflexes, and $\downarrow$ blood loss. Combative patients and those requiring multiple concurrent surgical procedures or prolonged (>2 h) procedures are often managed with GETA.

Regional anesthesia

Either subarachnoid or epidural blocks are useful techniques. Subarachnoid anesthesia has the advantage of adequate blockade of the sacral roots that are resistant to low-dose epidural techniques. Epidural anesthesia allows for the administration of postop

epidural analgesia. Anesthesia from T12 (T8 with tourniquet) to S2 is adequate. Full motor blockade is desirable. Typical drugs and doses include subarachnoid, 12.5–15 mg of 0.5% bupivacaine with morphine 0.2 mg (omit if outpatient), and epidural, 12–15 mL 2% lidocaine with epinephrine 1:200,000 in divided doses (Na bicarbonate 0.1 mg/mL will speed onset of block).

General anesthesia

Induction	Standard induction is appropriate for patients with normal airways. Trauma patients require a rapid-sequence induction and intubation with cricoid pressure to prevent gastric aspiration.	
ERAS	Verify pre-incision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance. Trauma patients are often cold and require active warming if <35°C (convection blanket and active humidifier). Warming the patient may unmask severe hypovolemia that should be corrected.	
Emergence	Trauma patients should have full return of protective airway reflexes and, given the possibility of fat embolism, ensure that oxygenation is adequate on 50% O ₂ before extubation. PONV prophylaxis, should include ondansetron and dexamethasone.	
Blood and fluid requirements	IV: 14–16 ga × 2 BSS @ 4–8 mL/kg/h Warm fluids. Humidify gases.	Some fractures can involve large (30 mL/kg) blood losses that are hidden in the leg or thigh. Clinical signs of hypovolemia and serial Hct determination should guide fluid therapy. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Control of blood loss	Tourniquet	Inflation pressure is

		typically 100 mm Hg + systolic pressure. Maximum “safe” tourniquet time is 1.5–2 h, followed by a 5- to (preferably) 15-min reperfusion interval, if further tourniquet time is necessary.
Monitoring	Standard monitors ± Arterial line ± CVP line	Arterial/CVP lines indicated for patients with ↓ BP not readily correctable with crystalloid infusion, massive blood loss (>1 blood volume), or the need for postop ventilation.
Positioning	✓ and pad pressure points ✓ eyes	
Special considerations	Release of tourniquet	A 20% ↓ MAP is common on tourniquet deflation. Additional crystalloid (5–10 mL/kg) may be necessary to replace edema fluid and blood loss to the leg.
Complications	Fat embolism Myoglobinemia	

POSTOPERATIVE

Complications	Hypoxemia VTE (DVT)	May be 2° fat embolism.
Multimodal pain management	Spinal opiates: Epidural anesthesia	Epidural hydromorphone 50 mcg/mL infused at

	Spinal anesthesia	100–250 mcg/h provides excellent analgesia. Intrathecal morphine 0.2–0.3 mg provides analgesia for up to 24 h after administration. (Monitor for delayed respiratory depression.) Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional analgesia (e.g., continuous nerve block) whenever possible (see Appendix K, p. K-6).
	For elective cases, consider multimodal analgesia.	Gabapentin, 300 mg, given preop can significantly reduce postop pain and opioid consumption.
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6) Regional analgesia	Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	Hct CXR, if CVP placed or oxygenation is impaired.	Other studies as indicated

Suggested Readings

1. Crisologo PA, Monson EK, Atway SA: Gabapentin as an adjunct to standard postoperative pain management protocol in lower extremity surgery. *J Foot Ankle Surg* 2018; 57(4):781–4.
2. Fraser TW, Doty JF: Peripheral nerve blocks in foot and ankle surgery. *Orthop Clin North Am* 2017; 48(4):507–15.
3. Ganapthy S: Wound/intra-articular infiltration or peripheral nerve blocks for orthopedic joint surgery: efficacy and safety issues. *Curr Opin Anaesthesiol* 2012;

25(5):615–20.

4. Johnson RL, Kopp SL, Hebl JR, et al: Falls and major orthopaedic surgery with peripheral nerve blockade: a systematic review and meta-analysis. *Br J Anaesth* 2013; 110(4):518–28.
5. McDonald E, Winters B, Nicholson K, et al: Effect of postoperative ketorolac administration on bone healing in ankle fracture surgery. *Foot Ankle Int* 2018; 39(10):1135–40.
5. Patterson JT, Morshed S: Chemoprophylaxis for venous thromboembolism in operative treatment of fractures of the tibia and distal bones: a systematic review and meta-analysis. *J Orthop Trauma* 2017; 31(9):453–60.
7. Sites BD, Brull R: Ultrasound guidance in peripheral regional anesthesia: philosophy, evidence-based medicine, and techniques. *Curr Opin Anaesthesiol* 2006; 19(6):630–9.

Suggested Viewing

Links are available online to the following videos:

Tibial Plateau Fracture Surgery (Arthroscopically Assisted):

<https://www.youtube.com/watch?v=mfbgeC5H9To>

Treatment of Ankle Fractures: <https://www.youtube.com/watch?v=fKDAhnwy0ec>

Ultrasound-Guided Popliteal Sciatic Nerve Block: <https://www.youtube.com/watch?v=kzhSiQBPE7s>